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Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

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Abstract

High temperature superconductivity in copper oxides (cuprates) represents one of the most striking phenomena in condensed matter physics. These materials host superconducting states with the highest known critical temperatures, emerging upon doping an antiferromagnetic insulator compound. The antiferromagnetic (AF) insulating and superconducting (SC) phases defy standard descriptions in terms of band picture and conventional phonon-mediated superconductivity. In contrast, these phenomena are genuine signatures of strong electronic correlations which hinder effective single-particle descriptions of the many-electron system.

After 40 years since their discovery, it is widely accepted that the Hubbard model on the square lattice provides a minimal description of the low-energy electronic properties of cuprates. The model describes the competition between the electronic kinetic energy, modelled by nearest-neighbours hopping t, and the Coulomb repulsion, modelled by a purely local interaction potential U. At half-filling, this competition results in the so-called "Mott localisation" phenomenon, which distinguishes the metallic phase for $U/t \ll 1$ from an insulating phase for $U/t \gg 1$. The latter is characterised by the almost complete localisation of electrons due to the high energetic cost of hopping towards occupied sites. Interestingly, this basic competition is intimately related to AF ordering and, upon doping, to the formation of the SC state with characteristic d-wave symmetry.

The fact that a purely repulsive interaction U is responsible for the formation of a SC state is remarkable, being superconductivity typically associated with an effective attractive interaction. In fact, a d-wave SC state can also be obtained, in general, by considering an attractive potential between electrons of neighbouring sites. Motivated by this observation, in this thesis, we studied a modified version of the Hubbard model, dubbed as the "extended Hubbard model", in which we include a nearest-neighbours attractive potential $-V$. The general goal of this work is to explore the interplay between the purely local repulsion and the non-local attraction. It is worth mentioning that this is not only motivated by the fact that both interactions are, in principle, responsible for d-wave superconductivity, but also by recent experimental observations which strongly suggest that the term $-V$ needs to be included in order to accurately describe the magnetic spectra of some specific compounds.

In this thesis, we study the extended Hubbard model using the Hartree-Fock method. Namely, we approximate the ground state by implementing a variational search in the subspace of the many-body Hilbert space of gaussian states. Despite its simplicity, the Hartree-Fock method provides a powerful framework to study competition among different phases, originating from different symmetry breaking mechanisms. In particular, we study the competition between three different phases: the normal phase, characterised by no symmetry breaking, the AF phase, characterised by a lowering of SU(2) symmetry, and the SC phase, characterised by the breaking of U(1) charge conservation.

This thesis work led to a number of original results, which can be summarised by the general observation that the non-local attraction $-V$ acts cooperatively with the local repulsion U for the stabilisation of the various phases hosted by the Hubbard model.

For the normal phase, we find that the non-local attraction acts as a source of Mott localisation, leading to an effective reduction of the electronic bandwidth. In particular, for values of V larger than a critical value and close to half-filling, we find a strong Mott localisation characterised by the complete flattening of the electronic bands. Interestingly, this result shows an example of Mott localisation that can be described at the level of the Hartree-Fock approximation, whereas Mott localisation in the conventional Hubbard model usually requires more sophisticated approximations.

For the AF phase, we restrict our analysis to the half-filled case. We find that the pairing-induced Mott localisation leads to a significant enhancement of the magnetisation, which primarily originates from the local repulsion U.

Finally, we focus on the superconducting phase and derive self-consistency equations in both singlet and triplet Cooper pairing channels, showing how Cooper pairing can be mediated simultaneously by both the local repulsion and the non-local attraction. Therefore, we numerically investigate the singlet channel and find evidence of "symmetry mixing", namely, the formation of a superconducting gap that is neither symmetric nor anti-symmetric under rotations of $\pi/2$ in the plane. We show that symmetry mixing is connected to the emergence of a finite nematic order parameter, that signals the breaking of planar rotational symmetry. This nematic symmetry breaking results in the fact that the renormalisation of bands becomes anisotropic, producing a difference in electronic kinetics in the x and y directions.

To summarise, the original results obtained in this thesis show that the non-local pairing provides a different route towards the formation of key strongly correlated phenomena, ranging from Mott localisation to nematic superconductivity, and call for future investigation using more sophisticated methods such as, for example, Dynamical Mean Field Theory.

Contents

List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
DoF	Degree of freedom
DoS	Density of states
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
PHS	Particle-hole symmetry
RMP(s)	Renormalized model parameter(s)
SC	Superconducting
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Chapter 1

Introduction

The discovery of superconductivity in copper oxide compounds, commonly referred to as cuprates, is one of the major breakthroughs in condensed matter physics in the last 50 years. In 1986, Bednorz and Muller [1] discovered that the compound BaLaCuO exhibits superconductivity below a critical temperature around 30K. Until then, superconductivity had been observed up to critical temperatures (T_c) of few Kelvins. This new class of superconductors marked the beginning of the era of high-T_c superconductivity. Today, the highest observed value is $T_c \approx 151$ K at ambient pressure on the cuprate $HgBa_2Ca_2Cu_3O_{8+\delta}$ [2].

Cuprates are the prototypical example of quantum materials characterised by rich phase diagrams, hosting very different phases upon changing external parameters. In Fig. 1.1 we sketch a typical example of the cuprates phase diagram as a function of temperature T and doping δ, with respect to the parent compound, at $\delta = 0$ [3]. The parent compounds are antiferromagnetic (AF) insulators below a Néel temperature $T_N \approx 300$K. Upon doping with either electron or holes, the Néel temperature decreases, until antiferromagnetism is eventually suppressed. Increasing doping, the system undergoes a superconducting (SC) phase transition with a critical temperature T_c that shows a dome-like structure, and is maximised at an optimal doping δ^*. The optimal doping discriminates an underdoped region ($\delta < \delta^*$), where the system shows a bad metallic behaviour, and an overdoped region ($\delta > \delta^*$), where it obeys conventional Fermi liquid theory. The full microscopic understanding of such complex phase diagrams and the interplay between the different phases have been subjects of an intense research activity in the last 40 years, and remain, to date, an open question.

Today, it is generally accepted that the physics of cuprates is highly determined by strong electronic correlations, namely the fact that many observed phenomena cannot be described by standard Fermi liquid theory. The simplest theoretical framework to describe strong electronic correlations in cuprates is the single-band Hubbard model which, as we are going to explain, is able to capture the formation of the AF state at zero doping, as well as the emergence of a d-wave SC state upon doping.

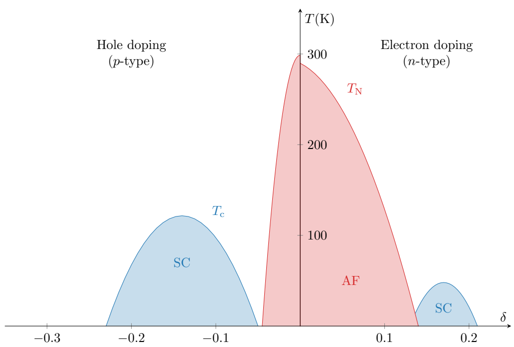


Figure 1.1 Sketch of the phase diagram of cuprates. These curves result from graphic elaboration and do not refer to a specific physical dataset.

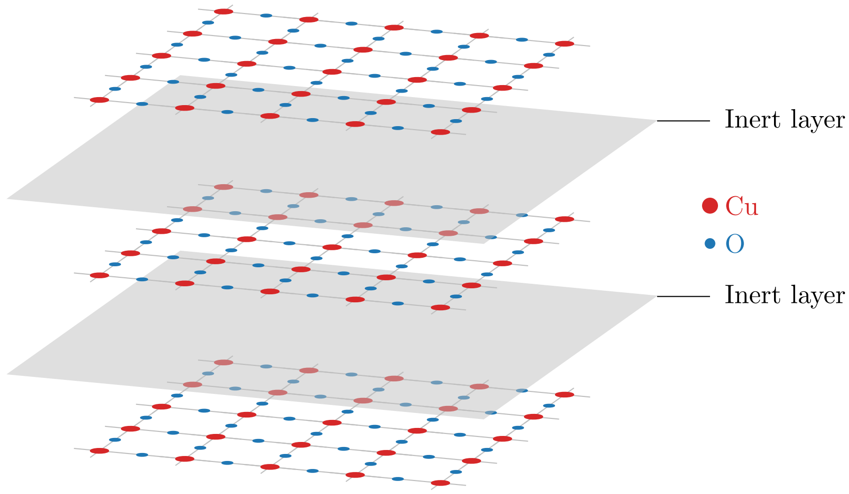


Figure 1.2 Schematics of a typical layered copper oxide compound. The inter layers are represented as solid planes, while the electrically active layers as a square grid hosting copper on its nodes and oxygen on its edges.

1.1 Cuprates and the square Hubbard model

The crystal structure of copper oxide superconductors is given by a stacking of 2D CuO_2 sheets separated by inert layers of insulating materials, see Fig. 1.2. The intermediate layers act as reservoirs of electrons (or holes) and dope the active CuO_2 sheets [4]. Due to the presence of these intermediate inert layers, the CuO_2 sheets are weakly coupled to each other, and can be treated as independent. In these sheets, copper and oxygens are arranged as a checkerboard lattice: oxygen atoms form a square lattice, and in the center of each cell is placed a copper atom. At zero doping, due to the high electronegativity of oxygens, copper atoms are ionised and left with one hole in their d shell, while oxygen atoms reach a closed shell configuration. Therefore, at low energy the kinetics of electrons is given by the hopping among the d orbitals of Cu^{2+} ions, arranged in a planar square lattice [5]. We notice that the number of electrons per unit cell is odd. Therefore, according to band theory, the system should be an half-filled metal, while instead it exhibits an insulating behaviour. Band theory relies on the assumption that electrons are nearly-free, and its failure in this context strongly suggests that the AF insulating phase originates from a strong Coulomb repulsion between electrons.

The simplest model capable of describing this situation is the single-band Hubbard model [6],

$$\hat{H}_{\mathrm{HM}} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \qquad \text{where } t, U > 0, \tag{1.1}$$

being t the hopping energy and U the Coulomb repulsion, simplified to a local on-site interaction. Sum is intended on the spin index $\sigma \in \{\uparrow, \downarrow\}$ and the sites i, j of a planar square lattice $\mathcal{L}$. The notation $\langle ij \rangle$ indicates nearest neighbours (NNs).

The Hubbard model captures some of the low-energy properties of copper oxides [5, 7] and is capable of showing a rich variety of physical phenomena [8, 9]. Indeed, despite its simplicity, it can describe the Mott insulating state, antiferromagnetism and the formation of the d-wave SC state. These are three key phenomena for the the physics of cuprates.

1.1.1 The Mott insulating state in the Hubbard model

Conventional band theory describes the electronic properties of materials on the assumption that electron-electron interaction is negligible. Within this framework, a material forms a band insulator whenever the conduction band fills up. Due to Pauli principle, electrons fill all states of the band

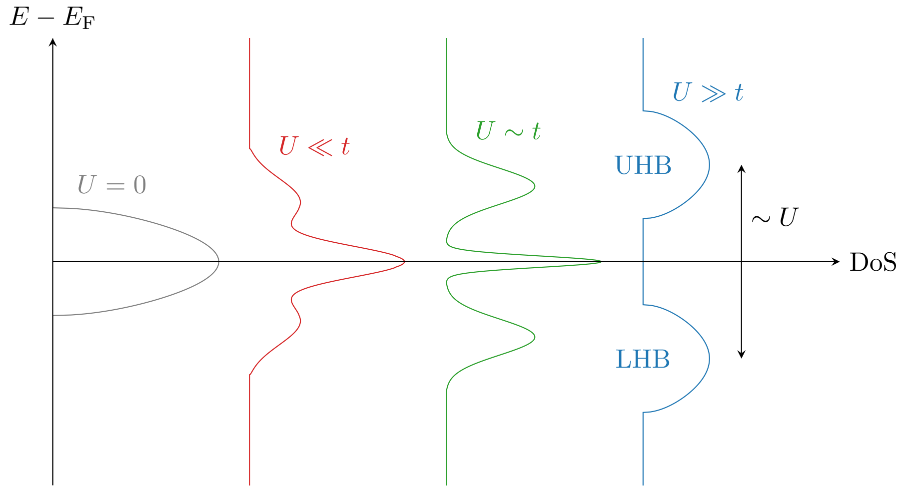


Figure 1.3 | Sketch of the Hubbard DoS as we vary the ratio U/t. For $U = 0$ (gray plot on the left) we have a free electronic band. As we increase U, two peaks appear while the central coherent peak shrinks. At $U \gg t$ (blue plot on the right), the DoS is separated in upper and lower Hubbard bands (UHB, LHB). E_{F} is the Fermi energy.

and have no room left for low-energy scattering, hence the filled band is completely ineffective for conduction. On the other hand, if the conduction band is partially filled, the material is a metal [4]. In general, band theory predicts a metallic behaviour if the number of electrons per unit cell is odd; an insulating behaviour if is even.

In the Mott insulating state, the insulating properties do not derive from a single-particle band being filled up and separated by a gap from another band at higher energy; instead, from the effect of the Coulomb repulsion that "freezes" electrons on the lattice sites [3]. One can picture this situation as a sort of traffic jam, where electrons do not move because the energetic cost of having two electrons on the same site would be too high [10]. This produces an insulating behaviour: electrons are completely localised and conductivity is low.

The Hubbard model describes naturally Mott physics [8]. Consider for instance the two-sites Hubbard model,

$$\hat{H} = -t\left(\hat{c}^{\dagger}_{1\uparrow}\hat{c}_{2\uparrow} + \hat{c}^{\dagger}_{2\downarrow}\hat{c}_{1\downarrow}\right) + U\left(\hat{n}_{1\uparrow}\hat{n}_{1\downarrow} + \hat{n}_{2\uparrow}\hat{n}_{2\downarrow}\right) , \tag{1.2}$$

where the number subscript indicates the site index. In the strong coupling limit $U \gg t$, we can treat the kinetic terms perturbatively. Indeed, let E_i be the energy given by having i electrons on one site. At order zero in perturbation theory, the difference in energy between the state with one site doubly occupied and the other empty, and the state with two singly occupied sites is

$$E_2 + E_0 - 2E_1 = U . \tag{1.3}$$

The zeroth order ground-state, then, is a combination of states where both sites are singly occupied. Insertion of the kinetic terms as a perturbative correction does not change the scenario (see App. A for details).

In the Hubbard model, the transition from a metallic state to a Mott insulator is controlled by the strength of the Coulomb repulsion U, and can be described by the means of Dynamical Mean Field Theory (DMFT) [11]. The Mott transition impacts visibly the density of states (DoS), which undergoes significant changes [12], as we sketch in Fig. 1.3. Let us describe the general behaviour: to start, let $U = 0$ and consider a lattice with one electron per site (half-filling). Electrons are free and do not interact, therefore move freely across the lattice. Due to Pauli exclusion principle, they form a Fermi sea that occupies the lower half a metallic band with a width $W = \mathcal{O}(t)$ (gray plot on the left of Fig. 1.3). As we increase the value of U, it becomes increasingly harder for electrons to move. When U becomes sufficiently large (with respect to t), the electronic density of states splits in two bands (blue plot on the right of Fig. 1.3). The lower Hubbard band (LHB), filled, accommodates all completely localised electrons; in the upper Hubbard band (UHB), empty, are all the states where a site gets doubly occupied. This splitting is of order $\sim U$, a fact that follows from extending the Eq. (1.3) from a simple two-sites toy model to a lattice. At intermediate values of U (central plots of Fig. 1.3), the DoS presents a peculiar three-peaks shape, which is the result of a mixture of the two Mott bands, and a residual peak of the original free band around the Fermi energy E_F. One of the signatures of a Mott transition, thus, is the flattening of the bare single-particle bands.

1.1.2 Antiferromagnetism in the Hubbard model

In the Hubbard model, AF ordering can rise from the competition among the kinetic and the Coulomb terms, and is connected to the previously discussed Mott physics. Indeed, in the strong coupling regime $U \gg t$, the Hubbard hamiltonian is effectively approximated by the Heisenberg hamiltonian with AF coupling [13],

$$\hat{H}_\mathrm{HM} \simeq \frac{4t^2}{U} \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j \, ,$$

where $\hat{\mathbf{S}}_i$ is the spin operator on site i. Antiferromagnetism is established through the "super-exchange mechanism", which takes advantage of the fact that electrons are localised on-site. This is best explained on the two-sites Hubbard model toy model of Eq. (1.2): as we explained in Sec. 1.1.1, the kinetic term can be treated as a perturbation to the fully-localised ground-state, which favours the formation of a spin singlet between the two sites (see App. A for details).

1.1.3 Unconventional superconductivity

Injection of charge (electrons or holes) in copper oxide compounds disrupts AF ordering and establishes a variety of peculiar phases [3]. Among these, the high temperature SC phase is significantly different from that the conventional Bardeen-Cooper-Shrieffer (BCS) theory. Because of this, is denominated "unconventional". In BCS theory, couples of electrons are "glued" together by an effective attraction that is mediated by lattice phonons. This mechanism is referred to as "Cooper pairing". The resulting state is a condensate of Cooper pairs, with an energy spectrum that is fully gapped and thus no low-energy excitations. The gap $\Delta_\mathbf{k}$, as a function of momentum, is isotropic (s-wave) and represents the energy cost of breaking a Cooper pair glued by phonons.

In copper oxides, superconductivity is established with significantly different features. Pairing is still active, but is likely not mediated by phonons. Most notably, the SC gap is not isotropic but changes sign under rotations of $\pi/2$ on the plane [8]. Because of this, we talk about $d_{x^2-y^2}$-wave superconductivity (from now on, we will omit the subscript "$x^2 - y^2$") A consequence is the appearance of gapless quasiparticle excitations in the energy spectrum. Ideed, a d-wave gap has nodal lines (continuous sets of points, in reciprocal space, where it is null) which, in a square lattice geometry, result in a set of nodes around which dispersion is linearised (Dirac cones) [14]. Moreover, Mott physics and d-wave superconductivity are related. Indeed, such a spatial structure of the gap function is understood to be a manifestation of the fact that electrons are spatially correlated in such a way to avoid occupying the same site [8].

The source of pairing in cuprates is still object of debate. A many-body state resulting from a superposition of spin singlets across different sites has been proposed by Anderson [15, 16] to capture the relation between antiferromagnetism and Mott physics. In the phase diagram of copper oxides, a dome-like d-wave SC region "emerges" from the AF region, as doping is increased at low temperature. Hence, it has been proposed that the same Mott-induced spin fluctuations that lead to the formation of the AF state could act as an effective Cooper pairing [14, 17–19]. The Hubbard model at finite doping can be investigated using DMFT extensions, and numerical simulations have confirmed the presence of a dome-like d-wave SC region in the phase diagram [20].

Electron-phonon coupling has been studied too, but recent developments suggest that it could act as a source of additional interactions – namely, a nearest neighbours attraction mediated by phonons [21]. This idea leads to the next section, where we present and motivate the extension to the Hubbard model that we studied.

1.2 The Extended Hubbard model

The Hubbard model of Eq. (1.1) has been long considered a fair description of the main features of copper oxides [8]. There are evidences, however, that the model could be missing some interactions that must be accounted to describe d-wave superconductivity in cuprates, for instance a nearest-neighbours attraction [21]. Moreover, recent developments have shown numerically that inclusion of a non-local attraction between neighbouring sites is necessary to reproduce some features of the excitation spectrum of some doped cuprates [22].

Moreover, it has been known since early Hartree-Fock (HF) calculations that adding a non-local attraction contributes to d-wave superconductivity on the square lattice [23]. With these motivation, in this thesis, we follow this plan:

1. We insert "artificially" a non-local attraction in the Hubbard model, knowing that it leads to d-wave superconductivity;
2. We study the resulting model, searching for manifestations of Mott physics and understanding how the newly added interaction relates to it;
3. We analyse how antiferromagnetism and superconductivity are impacted by the findings of the step above.

In fact, this thesis work focuses on the following question: we know that the Hubbard U leads to Mott insulation and, in turn, to AF and d-wave SC phases. What is the fate of Mott physics if, instead, we start from a model with an attractive interaction which gives rise to a conventional d-wave superconductor, independently of U?

We define the extended Hubbard model (EHM) by adding a non-local attraction to the Hubbard model of Eq. (1.1),

$$\hat{H}_{\text{EHM}} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} \qquad \text{where } t, U, V > 0. \tag{1.4}$$

The newly added interaction is independent of spin index and acts on nearest neighbours.

1.3 Summary of the main results

We will study the ground state of this model by the means of theoretical and numerical HF methods, varying the strength of the interactions U, V and electronic density. The HF method, a constrained variational search for the best approximation of the true ground-state, provides a simple but powerful framework to explore competition among phases. Specifically, we investigate three phases (namely: normal, AF and SC), varying the model parameters. We limit to finite square lattices and approximately zero temperatures. We obtain various original results, ranging from the insurgence of Mott localisation phenomena induced by the non-local attraction, to the cooperation among the two interactions to stabilise AF and SC phases.

For the normal phase, we observe both theoretically and numerically the emergence of a self-consistent Mott localisation effect, induced by the non-local attraction $-V$, which narrows electronic bandwidth. In particular, we identify the presence of a critical boundary in the (δ, V) plane which delimits a region of complete bands flattening. Inside that region, electronic mobility is null and Mott localisation is most effective. The flattening of electronic bandwidth reminds of the shrinking of the central coherent peak in the DoS of Fig. 1.3, with the substitution $U \to -V$. Most notably, these results are completely independent of U. The local repulsion is generally regarded as the main source of Mott localisation, but our findings show that the same effect can emerge from a non-local pairing $-V$.

For the AF phase, we limit to half-filling and commensurate antiferromagnetism, and find that the non-local pairing $-V$ contributes to the insurgence of a macroscopic magnetisation by acting cooperatively with the local repulsion U. The latter, as we specified in Sec. 1.1.2, is the primary source of antiferromagnetic correlations in the Hubbard model. This cooperative effects emerges from a similar bands shrinking effect observed for the normal phase, that can be explained as follows: as t is effectively lowered by the non-local attraction, the ratio U/t grows. The latter, in the Hubbard model, determines the magnetisation. Therefore, the non-local pairing $-V$ effectively enhances antiferromagnetic ordering.

Finally, we study the translationally invariant SC phase, and derive a self-consistent framework in both singlet and triplet Cooper pairing channels. We investigate numerically the singlet channel as we vary U, V and δ, finding evidence of a number of effects. First, we observe the insurgence of a SC order parameter which is sustained by both the local repulsion and the non-local attraction. We study the spatial symmetry structure of this order parameter, and identify numerically the boundaries of the regions of parametric space where it is purely symmetric or antisymmetric under $\pi/2$ rotations, and most importantly a region where it is neither of the two. In this "symmetry mixing region", we observe the emergence of a nematic order parameter, which manifests the fact that planar rotational symmetry breaks, and the hopping amplitude becomes anisotropic in directions x and y. This effect is originated by the non-local attraction and coexists with the bands shrinking effect already observed for the normal and AF phases, which is also present for the SC phase.

Our original findings show that a non-local attraction can act, in the Hubbard model at HF level, as a source of many interconnected physical phenomena. The interplay of Mott localisation, AF and SC orderings and the insurgence of nematicity gives rise to a rich scenario which can be addressed by the means of more advanced numerical techniques.

1.4 Outline of this thesis

In this section, we give an outline of next chapters.

Chap. 2 We introduce the HF method for a generic fermionic quartic hamiltonian, and discuss how it is used to find a variational approximation to its ground state. Then, we explain how this problem can be re-stated in terms of a finite set of self-consistency equations. In the end, we propose an iterative algorithmic scheme to solve these equations.

Chap. 3 We discuss the symmetries of the extended Hubbard model of Eq. (1.4). Then we give a definition of the three phases (normal, AF, SC) and describe each based on which of the original symmetries is spontaneously broken.

Chap. 4 We apply the HF method to the normal phase and derive the self-consistency equations. Then we present numeric results that compose evidence of V-induced Mott localisation.

Chap. 5 We apply the HF method to the AF phase and derive the self-consistency equations. First, we discuss antiferromagnetism in the conventional Hubbard model of Eq. (1.1), hence setting $V = 0$. Then we let $V > 0$ and discuss how the HF solution changes. In the end, we present numeric results that show how AF ordering is enhanced (at half-filling) by V-induced Mott localisation.

Chap. 6 We apply the HF method to the SC phase and derive the self-consistency equations. We discuss how the presence of a non-local attraction gives rise to a stable SC solution, while at $V = 0$ the the local repulsion cannot produce superconductivity at HF level. In the end, we discuss numeric results that show:

a) The competition among (extended) s-wave and d-wave superconductivity, and how the two symmetries can mix in a region of parameters space;

b) The insurgence of nematic effects (namely, breaking of planar rotational symmetry) related to the mixing of gap symmetries;

c) The presence of Mott localisation in SC phase.

Chap. 7 We summarise our findings and *thank for all the fish.*

Appendices are organised as follows:

App. A We discuss the superexchange mechanism already introduced in Sec. 1.1.2.

App. B We define the set of gaussian states, which are used in the HF method discussed in Chap. 2.

App. C We explain how HF methods and mean-field theory are related.

App. D We give more details about antiferromagnetism in the conventional Hubbard model.

App. E We extend the theoretical derivation of Chap. 5, carried out for the conventional Hubbard model, including the non-local pairing. We retrieve the set of self-consistency equations for the AF phase, the expressions of entropy and free energy densities.

App. F We derive a set of theoretical constraints on the HF solution for the normal and AF phase, discussed respectively in Chaps. 4 and 5.

App. G We derive analytically the HF approximation for the SC phase, following a procedure very similar to Chap. 5 and App. E. We retrieve the set of self-consistency equations for the SC phase, the expressions of entropy and free energy densities.

Chapter 2

Introduction to Hartree-Fock methods

In this chapter, we give a general introduction to the Hartree-Fock (HF) variational method and its numerical implementation. In Quantum Mechanics, variational methods allow to approximate the ground state of a many-body hamiltonian by minimising the energy computed for a specific subspace of physical states. In particular, in the HF method we approximate the real ground-state with the best possible many-body state that is an eigenstate of a non-interacting fermionic hamiltonian.

Despite its simplicity, the method is very powerful for describing various types of quantum many-body phenomena and, in particular, spontaneous symmetry breaking (SSB). The description of HF method in this chapter is largely based on [4, 45].

2.1 Variational principle and gaussian states

Let $\hat{H}$ be a generic hamiltonian acting on the Hilbert space $\mathcal{H}$. For any many-body state $|\Psi\rangle \in \mathcal{H}$, the energy is given by the expectation value (EV)

$$E[\Psi] = \langle\Psi|\hat{H}|\Psi\rangle \,.$$

Let $|\Psi_0\rangle$ be the ground state of $\hat{H}$, and $E_0 \equiv E[\Psi_0]$. Then

$$\forall\,|\Psi\rangle \in \mathcal{H} \quad : \quad E[\Psi] \geq E_0 \,,$$

since the ground-state minimises energy, by definition. Let $\mathcal{S} \subset \mathcal{H}$ be a subspace of states, ad let $|\Psi\rangle$ be a state inside that locally minimises energy,

$$|\Psi\rangle \in \mathcal{S} \qquad \frac{\delta E[\Psi]}{\delta\,|\Psi\rangle} = 0 \,.$$

If the true ground state belongs to $\mathcal{S}$, the found local minimum coincides with $|\Psi_0\rangle$; otherwise, approximates it. In Fig. 2.1 we give a pictorial representation of this idea: the state $|\Psi\rangle \in \mathcal{S}$ is an approximation, found by constrained minimisation, of the true ground-state $|\Psi_0\rangle$.

In HF methods, we identify $\mathcal{S}$ with the set of so-called "gaussian states". These are all the ground-states of quadratic hamiltonians, namely the hamiltonians that are bilinear in the fermionic creations operators (see App. B for details). The restriction to gaussian states significantly simplifies the computation of expectation values. Indeed, thanks to Wick's theorem, the expectation value of quartic operators exactly decomposes as

$$\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \rangle = \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\delta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\gamma \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\gamma \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\delta \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \langle \hat{c}_\gamma \hat{c}_\delta \rangle}_{\text{Cooper}} \,. \tag{2.1}$$

The subscripts indicate the standard names to the different types of contractions, that we adopt throughout the text.

Operatively, one can think of the HF method as an approximation of the the hamiltonian with a non-interacting hamiltonian

$$\hat{H} \overset{\text{HF}}{\simeq} \hat{H}_{\text{HF}} = \sum_\alpha E_\alpha \hat{\gamma}^\dagger_\alpha \hat{\gamma}_\alpha \,,$$

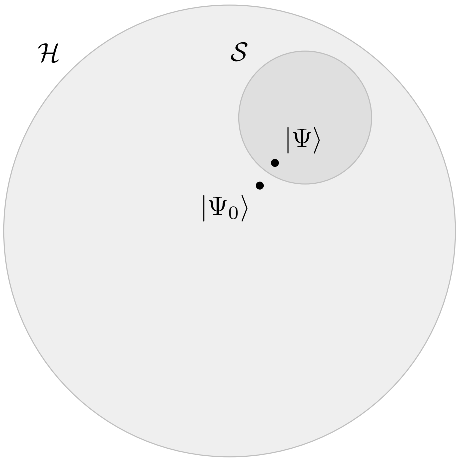


Figure 2.1 | Pictorial representation of the search for the best approximation $|\Psi\rangle$ to the true ground-state $|\Psi_0\rangle$ inside the subspace $\mathcal{S} \subset \mathcal{H}$.

where $\hat{\gamma}_\alpha$ are fermionic operators determined by the minimisation procedure. These new operators satisfy Fermi anticommutation rules

$$\left\{\hat{\gamma}_\alpha, \hat{\gamma}^\dagger_\beta\right\} = \delta_{\alpha\beta} \, ,$$

and are connected to the conventional electronic operators $\hat{c}_\alpha$ through an unitary map. See App. B for details.

Let us stress that the operatorial approximation $\stackrel{\text{HF}}{\simeq}$ is a shorthand notation that must be intended as follows: $\hat{H}_{\text{HF}}$ is that non-interacting operator whose ground-state is the best approximation to the ground-state of $\hat{H}$. This does not mean that in general the two operators are related.

2.2 The HF variational method

We now dicuss the implementation of the variational HF method for the generic quartic hamiltonian,

$$\hat{H} = \sum_{\alpha,\beta} A_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \, , \tag{2.2}$$

where

$$A_{\alpha\beta} = A^*_{\beta\alpha} \quad \text{and} \quad B_{\alpha\beta\gamma\delta} = B^*_{\delta\gamma\beta\alpha} \, , \tag{2.3}$$

to ensure hermiticity. Moreover, the following equation must hold:

$$B_{\alpha\beta\gamma\delta} = B_{\beta\alpha\delta\gamma} \, , \tag{2.4}$$

due to the property $\hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta = \hat{c}^\dagger_\beta \hat{c}^\dagger_\alpha \hat{c}_\delta \hat{c}_\gamma$, which follows from Fermi anticommutation rules.

Our aim is to approximate the above hamiltonian with a non-interacting hamiltonian. The most general quadratic hamiltonian is given by

$$\hat{H} \stackrel{\text{HF}}{\simeq} \hat{H}_{\text{HF}} \equiv E_0 + \sum_{\alpha,\beta} k_{\alpha\beta} \hat{\gamma}^\dagger_\alpha \hat{\gamma}_\beta \, . \tag{2.5}$$

As is specified in Sec. B.3, $\hat{\gamma}_\alpha$ and $\hat{c}_\alpha$ operators must be linked by an unitary map. To find the best approximation to the original hamiltonian means to identify:

- the offset E_0;
- the fermionic operators $\hat{\gamma}_\alpha$, $\hat{\gamma}^\dagger_\alpha$;

- the $k_{\alpha\beta}$ coefficients.

Note that, in terms of the algebraic solution, E_0 is irrelevant. To determine it correctly is only relevant in order to get the correct approximation of the actual ground-state energy. Let us discuss briefly the features of $\hat{H}_{\mathrm{HF}}$ and then propose a solution.

2.2.1 Properties of the target quadratic hamiltonian

In this section we enumerate some useful properties of $\hat{H}_{\mathrm{HF}}$. We will use also the following notation:

$$\hat{a}^{-}_{\alpha} \equiv \hat{c}_{\alpha} \quad \text{and} \quad \hat{a}^{+}_{\alpha} \equiv \hat{c}^{\dagger}_{\alpha}\,.$$

Due to linearity of the unitary mapping that connects $\hat{\gamma}_\alpha$ and $\hat{c}_\alpha$ operators, the operator $\hat{H}_{\mathrm{HF}}$ of Eq. (2.5) can also be expressed as the most general quadratic hamiltonians for electronic operators:

$$\hat{H}_{\mathrm{HF}} = E_0 + \sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\hat{a}^{p}_{\alpha}\hat{a}^{q}_{\beta}\,. \tag{2.6}$$

By changing basis from $\hat{\gamma}_\alpha$ to $\hat{c}_\alpha$ operators, we moved all our ignorance into the new $h^{pq}_{\alpha\beta}$ matrix elements. These satisfy the relations:

$$h^{++}_{\alpha\beta} = \left(h^{--}_{\beta\alpha}\right)^*\,, \tag{2.7}$$

$$h^{+-}_{\beta\alpha} = -h^{-+}_{\alpha\beta}\,. \tag{2.8}$$

Proof 2.1 — Proofs of Eqs. (2.7), (2.8): symmetry properties of the tensor $h^{pq}_{\alpha\beta}$.

Let us start by proving Eq. (2.7). Let the Nambu spinor

$$\hat{\Phi} \equiv \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^{\dagger} \end{bmatrix}\,,$$

being $\hat{\mathbf{c}}$ the vector of all annihilation operators $\hat{c}_\alpha$ and $\hat{\mathbf{c}}^\dagger$ of all creation operators $\hat{c}^\dagger_\alpha$. The HF hamiltonian can be represented as

$$\begin{aligned}\hat{H}_{\mathrm{HF}} &= E_0 + \hat{\Phi}^\dagger \mathbf{h}\hat{\Phi} \\ &= E_0 + \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^{\dagger} \end{bmatrix}^\dagger \begin{bmatrix} h^{+-} & h^{++} \\ h^{--} & h^{-+} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^{\dagger} \end{bmatrix}\,,\end{aligned}$$

where h^{pq} are matrices. Our hamiltonian must be hermitian. We conclude that

$$\hat{H}^\dagger_{\mathrm{HF}} = \hat{H}_{\mathrm{HF}} \quad \Longrightarrow \quad h^{++} = \left[h^{--}\right]^\dagger\,,$$

which reads, element-wise:

$$h^{++}_{\alpha\beta} = \left(h^{--}_{\beta\alpha}\right)^*\,.$$

This proves Eq. (2.7).

Eq. (2.8) is proven as follows. We can set, without loss of generality,

$$\mathrm{Tr}\, h^{+-} + \mathrm{Tr}\, h^{-+} = 0\,.$$

This passage is guaranteed by the fact that if $\mathbf{h}$ is not traceless, we can redefine it by subtracting its trace and including the latter in the energy shift E_0. Then

$$\begin{aligned}\hat{H}_{\mathrm{HF}} &= E_0 + \sum_{\alpha,\beta}\left[h^{+-}_{\alpha\beta}\hat{c}^\dagger_\alpha\hat{c}_\beta + h^{-+}_{\alpha\beta}\hat{c}_\alpha\hat{c}^\dagger_\beta\right] + \text{(same sign terms)}\\ &= E_0 + \sum_{\alpha,\beta}\left[h^{+-}_{\alpha\beta}\left(\delta_{\alpha\beta} - \hat{c}_\beta\hat{c}^\dagger_\alpha\right) + h^{-+}_{\alpha\beta}\left(\delta_{\alpha\beta} - \hat{c}^\dagger_\beta\hat{c}_\alpha\right)\right] + \text{(same sign terms)}\\ &= E_0 - \sum_{\alpha,\beta}\left[h^{+-}_{\beta\alpha}\hat{c}_\alpha\hat{c}^\dagger_\beta + h^{-+}_{\beta\alpha}\hat{c}^\dagger_\alpha\hat{c}_\beta\right] + \text{(same sign terms)}\,.\end{aligned}$$

In the second line we got rid of the traces sum and exchanged indices $\alpha \leftrightarrow \beta$. Confronting last and first row we conclude:

$$h^{+-}_{\beta\alpha} = -h^{-+}_{\alpha\beta}\,,$$

which proves Eq. (2.8).

2.2.2 Self-consistent variational solution

This section gives the operational method to obtain the variational solution. We follow and extend the argument proposed by Fabrizio [45]. Let

$$\Delta^{pq}_{\alpha\beta} \equiv \left\langle \hat{a}^p_\alpha \hat{a}^q_\beta \right\rangle,$$

thus, explicitly

$$\Delta^{+-}_{\alpha\beta} = \left\langle \hat{c}^\dagger_\alpha\hat{c}_\beta\right\rangle, \qquad \Delta^{-+}_{\alpha\beta} = \left\langle \hat{c}_\alpha\hat{c}^\dagger_\beta\right\rangle, \qquad \Delta^{++}_{\alpha\beta} = \left\langle \hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\right\rangle, \qquad \Delta^{--}_{\alpha\beta} = \left\langle \hat{c}_\alpha\hat{c}_\beta\right\rangle. \tag{2.9}$$

The first two are "normal densities"; the second two are "anomalous densities", since their values are non-zero if total charge conservation is broken. Conjugation of the last two gives a symmetry relation similar to Eq. (2.7),

$$\left(\Delta^{++}_{\alpha\beta}\right)^\dagger = \Delta^{--}_{\beta\alpha}\,. \tag{2.10}$$

Here $\Delta^{pq}_{\alpha\beta}$ are functions of the fields $h^{pq}_{\alpha\beta}$. These quantities satisfy the equation:

$$\sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\frac{\partial \Delta^{pq}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} = 0\,. \tag{2.11}$$

Proof 2.2 — Proof of Eq. (2.11).

The HF ground-state energy is given by

$$\begin{aligned}E_{\mathrm{HF}} &= \left\langle \hat{H}_{\mathrm{HF}}\right\rangle\\ &= E_0 + \sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\Delta^{pq}_{\alpha\beta}\,,\end{aligned}$$

where the expectation value is taken over the (up to now unknown) HF ground-state. Upon taking the derivative, we obtain

$$\begin{aligned}\frac{\partial E_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}} &= \frac{\partial}{\partial h^{rs}_{\mu\nu}}\sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\Delta^{pq}_{\alpha\beta}\\ &= \Delta^{rs}_{\mu\nu} + \sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\frac{\partial \Delta^{pq}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\,.\end{aligned} \tag{2.12}$$

From Hellman-Feynman theorem [4], we know that for a parametric hamiltonian it holds:

$$\begin{aligned}\frac{\partial E_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}} &= \langle\Psi|\frac{\partial \hat{H}_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}}|\Psi\rangle \\ &= \langle\Psi|\frac{\partial}{\partial h^{rs}_{\mu\nu}}\sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\hat{a}^p_\alpha \hat{a}^q_\beta|\Psi\rangle \\ &= \langle \hat{a}^r_\mu \hat{a}^s_\nu\rangle \\ &= \Delta^{rs}_{\mu\nu}\,. \end{aligned} \tag{2.13}$$

Eqs. (2.12) and (2.13), combined, imply

$$\sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\frac{\partial \Delta^{pq}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} = 0\,,$$

which proves Eq. (2.11).

Recall the most general quartic hamiltonian defined in Eq. (2.2). The HF approximation to its ground state is obtained as the ground-state of a quadratic hamiltonian, as given in Eq. (2.6), whose coefficients $h^{pq}_{\alpha\beta}$ satisfy the following relations:

$$h^{+-}_{\alpha\beta} = \frac{1}{2}\left[A_{\alpha\beta} + \sum_{\gamma,\delta}\left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\Delta^{+-}_{\gamma\delta}\right]\,, \tag{2.14}$$

$$h^{++}_{\alpha\beta} = \sum_{\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{--}_{\gamma\delta}}{2}\,. \tag{2.15}$$

These equations, operationally, give the instructions to build the best possible fermionic non-interacting hamiltonian to reproduce our original hamiltonian.

Proof 2.3 — Proof of Eqs. (2.14), (2.15).

We use Eq. (2.11) to minimise the variational energy $E = \langle \hat{H}\rangle$. From Eq. (2.1), performing Wick's contractions, the variational energy becomes

$$\begin{aligned} E &= \langle \hat{H}\rangle \\ &= \sum_{\alpha,\beta} A_{\alpha\beta}\langle \hat{c}^\dagger_\alpha \hat{c}_\beta\rangle + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta\rangle \\ &= \sum_{\alpha,\beta} A_{\alpha\beta}\langle \hat{a}^+_\alpha \hat{a}^-_\beta\rangle + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\left(\langle \hat{a}^+_\alpha \hat{a}^+_\beta\rangle\langle \hat{a}^-_\gamma \hat{a}^-_\delta\rangle - \langle \hat{a}^+_\alpha \hat{a}^-_\gamma\rangle\langle \hat{a}^+_\beta \hat{a}^-_\delta\rangle + \langle \hat{a}^+_\alpha \hat{a}^-_\delta\rangle\langle \hat{a}^+_\beta \hat{a}^-_\gamma\rangle\right) \\ &= \sum_{\alpha,\beta} A_{\alpha\beta}\Delta^{+-}_{\alpha\beta} + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\left(\Delta^{++}_{\alpha\beta}\Delta^{--}_{\gamma\delta} - \Delta^{+-}_{\alpha\gamma}\Delta^{+-}_{\beta\delta} + \Delta^{+-}_{\alpha\delta}\Delta^{+-}_{\beta\gamma}\right) \\ &= \sum_{\alpha,\beta} A_{\alpha\beta}\Delta^{+-}_{\alpha\beta} + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta}\left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\Delta^{+-}_{\alpha\beta}\Delta^{+-}_{\gamma\delta} + \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}\Delta^{--}_{\gamma\delta}\,, \end{aligned} \tag{2.16}$$

where in the second line, we used the relations (2.9). Upon taking the derivative, we obtain

$$\begin{aligned}\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta} A_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} &+ \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta}\left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\left(\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\Delta^{+-}_{\gamma\delta} + \Delta^{+-}_{\alpha\beta}\frac{\partial \Delta^{+-}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}\right) \\ &+ \frac{1}{2}\sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta}\left(\frac{\partial \Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\Delta^{--}_{\gamma\delta} + \Delta^{++}_{\alpha\beta}\frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}\right)\,,\end{aligned}$$

which, by using Eq. (2.4), becomes:

$$\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta}\left[A_{\alpha\beta} + \sum_{\gamma,\delta}\left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\Delta^{+-}_{\gamma\delta}\right]\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \\ + \sum_{\alpha,\beta,\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{--}_{\gamma\delta}}{2}\frac{\partial \Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + \sum_{\alpha,\beta,\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}}{2}\frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}\,. \tag{2.17}$$

Let us define

$$C_{\alpha\beta} \equiv \frac{1}{2}\left[A_{\alpha\beta} + \sum_{\gamma,\delta}\left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\Delta^{+-}_{\gamma\delta}\right] \quad \text{and} \quad D_{\alpha\beta} \equiv \sum_{\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{--}_{\gamma\delta}}{2}\,. \tag{2.18}$$

From Fermi anticommutation rules, we have

$$\Delta^{+-}_{\alpha\beta} = \delta_{\alpha\beta} - \Delta^{-+}_{\beta\alpha} \quad \Longrightarrow \quad \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} = -\frac{\partial \Delta^{-+}_{\beta\alpha}}{\partial h^{rs}_{\mu\nu}}\,,$$

which implies

$$\begin{aligned}\sum_{\alpha,\beta} 2C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} &= \sum_{\alpha,\beta}\left(C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\right) \\ &= \sum_{\alpha,\beta}\left(C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} - C_{\beta\alpha}\frac{\partial \Delta^{-+}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\right)\,.\end{aligned}$$

Moreover, using Eqs. (2.3), (2.10),

$$\sum_{\alpha,\beta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}}{2} = \sum_{\alpha,\beta}\left(\frac{B_{\delta\gamma\beta\alpha}\Delta^{--}_{\beta\alpha}}{2}\right)^*\,,$$

and, recognising $D_{\gamma\delta}$ from Eq. (2.18), we conclude

$$\sum_{\alpha,\beta,\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}}{2}\frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}} = \sum_{\gamma,\delta} D^*_{\delta\gamma}\frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}\,.$$

We recollect everything and impose the solution to be a stationary point of energy,

$$\frac{\partial E}{\partial h^{rs}_{\mu\nu}} \stackrel{!}{=} 0\,.$$

The notation "$\stackrel{!}{=}$" indicates, throughout the thesis, that we are imposing some condition. Eq. (2.17) becomes

$$\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta}\left(C_{\alpha\beta}\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} - C_{\beta\alpha}\frac{\partial \Delta^{-+}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + D_{\alpha\beta}\frac{\partial \Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + D^*_{\beta\alpha}\frac{\partial \Delta^{--}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\right) \stackrel{!}{=} 0\,. \tag{2.19}$$

Comparing this equation with Eq. (2.11), we conclude that it must hold

$$h^{+-}_{\alpha\beta} = C_{\alpha\beta} \quad \text{and} \quad h^{++}_{\alpha\beta} = D_{\alpha\beta}\,.$$

Recalling the explicit form of $C_{\alpha\beta}$ and $D_{\alpha\beta}$ from Eq. (2.18), we prove Eqs. (2.14), (2.15).

We notice that the right-hand side of both Eqs. (2.14), (2.15) depends on the fields themselves, as the quantities $\Delta^{pq}_{\alpha\beta}$ are defined as expectation values of two-points fermionic operators over the ground-state $|\Psi\rangle$. For a given set of fields $h^{pq}_{\alpha\beta}$, the latter is determined (up to some unitary transformation) as the ground-state of a fermionic non-interacting model. In other words, being $\mathbf{h}$ the tensor containing all fields,

$$\Delta^{pq}_{\alpha\beta}[\mathbf{h}] = \langle\Psi[\mathbf{h}]|\hat{a}^p_\alpha \hat{a}^q_\beta|\Psi[\mathbf{h}]\rangle \ . \tag{2.20}$$

The three Eqs. (2.14), (2.15) and (2.20) define the solution of the HF variational method. Specifically, we need to determine $\Delta^{pq}_{\alpha\beta}$ such that, for given fields, Eq. (2.20) is respected.

The list of parameters $\{\Delta^{pq}_{\alpha\beta}\}$ to be determined self-consistently may be not that short: by looking at Eq. (2.20), the indices α, β, p, q are spanning a large Hilbert space. However, this equation can be manipulated and transformed, reducing the set of parameters to be determined self-consistently to a very manageable number. For each phase we define these last as "Hartree-Fock parameters" (HFPs): a set of objects self-consistently determined by a set of equivalent, but transformed, version of Eq. (2.20).

The great advantage of this scheme is that, with an appropriate choice of the Hilbert space basis (for example, moving to reciprocal space for a translationally-invariant model), we can implement easily model symmetries.

2.2.3 Operative hamiltonian approximation

In this section we give an alternative expression for Eqs. (2.14), (2.15), by using an equivalent (but more straightforward) operatorial approximation. For a generic two-body quartic operator, the HF approximation is given by the decomposition

$$\begin{aligned}
\hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\beta \hat{c}_\alpha \overset{\text{HF}}{\simeq} & \, \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \langle \hat{c}_\beta \hat{c}_\alpha \rangle + \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \hat{c}_\beta \hat{c}_\alpha - \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \langle \hat{c}_\beta \hat{c}_\alpha \rangle & \text{(Cooper)} \\
& - \hat{c}^\dagger_\alpha \hat{c}_\beta \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle - \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \hat{c}^\dagger_\beta \hat{c}_\alpha + \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle & \text{(Fock)} \\
& + \hat{c}^\dagger_\alpha \hat{c}_\alpha \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle + \langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \hat{c}^\dagger_\beta \hat{c}_\beta - \langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle \, . & \text{(Hartree)}
\end{aligned}$$

The proof of this statement is given in App. C, where we also argue the interpretation of HF method as a mean-field theory (MFT). Let us summarise this section through a practical rule to decompose the quartic part of the local repulsion $\hat{H}_U$ and the non-local attraction $\hat{H}_V$ in the EHM, by applying Eq. (C.9).

Local repulsion. Applying the HF method, the local repulsion $\hat{H}_U$ is approximated by the quadratic operator:

$$\begin{aligned}
\hat{H}_U &= U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} & (2.21) \\
&\overset{\text{HF}}{\simeq} U \sum_i \Big(\hat{c}^\dagger_{i\uparrow} \hat{c}_{i\uparrow} \langle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\downarrow} \rangle + \langle \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\uparrow} \rangle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\downarrow} - \langle \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\uparrow} \rangle \langle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\downarrow} \rangle & \text{(Hartree)} \\
&\qquad - \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\downarrow} \langle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\uparrow} \rangle - \langle \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\downarrow} \rangle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\uparrow} + \langle \hat{c}^\dagger_{i\uparrow} \hat{c}_{i\downarrow} \rangle \langle \hat{c}^\dagger_{i\downarrow} \hat{c}_{i\uparrow} \rangle & \text{(Fock)} \\
&\qquad + \hat{c}^\dagger_{i\uparrow} \hat{c}^\dagger_{i\downarrow} \langle \hat{c}_{i\downarrow} \hat{c}_{i\uparrow} \rangle + \langle \hat{c}^\dagger_{i\uparrow} \hat{c}^\dagger_{i\downarrow} \rangle \hat{c}_{i\downarrow} \hat{c}_{i\uparrow} - \langle \hat{c}^\dagger_{i\uparrow} \hat{c}^\dagger_{i\downarrow} \rangle \langle \hat{c}_{i\downarrow} \hat{c}_{i\uparrow} \rangle \Big) \, . & \text{(Cooper)}
\end{aligned}$$

Non-local attraction. Applying the HF method, the local repulsion $\hat{H}_V$ is approximated by the quadratic operator:

$$\begin{aligned}
\hat{H}_V &= -V \sum_{\langle ij \rangle} \sum_{\sigma,\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} & (2.22) \\
&\overset{\text{HF}}{\simeq} -V \sum_{\langle ij \rangle} \sum_{\sigma,\sigma'} \Big(\hat{c}^\dagger_{i\sigma} \hat{c}_{i\sigma} \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{j\sigma'} \rangle + \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{i\sigma} \rangle \hat{c}^\dagger_{j\sigma'} \hat{c}_{j\sigma'} - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{i\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{j\sigma'} \rangle & \text{(Hartree)} \\
&\qquad - \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \rangle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} + \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma'} \rangle \langle \hat{c}^\dagger_{j\sigma'} \hat{c}_{i\sigma} \rangle & \text{(Fock)} \\
&\qquad + \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \langle \hat{c}_{j\sigma'} \hat{c}_{i\sigma} \rangle + \langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \rangle \hat{c}_{j\sigma'} \hat{c}_{i\sigma} - \langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \rangle \langle \hat{c}_{j\sigma'} \hat{c}_{i\sigma} \rangle \Big) \, . & \text{(Cooper)}
\end{aligned}$$

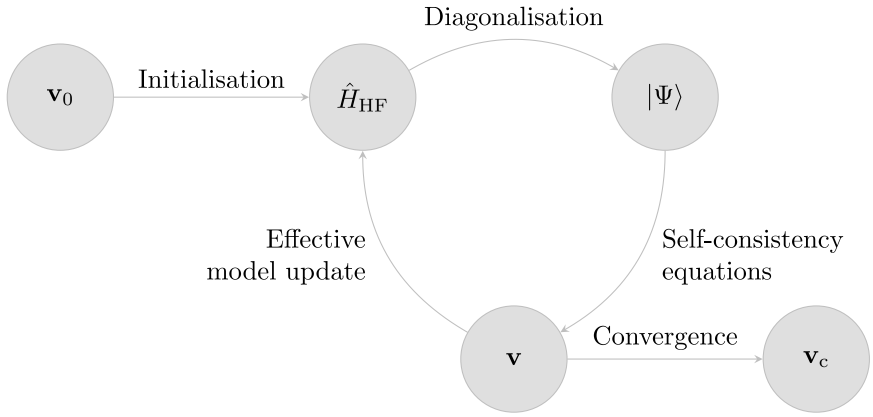


Figure 2.2 Schematisation of the self-consistent Hartree Fock method described in Sec. 2.3. The effective model is solved, the solution is used to estimate self-consistently the Hartree Fock vector $\mathbf{v}$, which is then used to update iteratively the model itself. The notation $\mathbf{v}_0$ indicates the initialiser HFPs vector, $\mathbf{v}_{\mathrm{c}}$ the converged HFPs vector.

We will use these decomposition referring to each sector in dedicated chapters.

2.3 The iHF algorithm

In this section we describe the algorithm we used to compute Eq. (2.20). We refer to this technique as "iterative Hartree Fock" (iHF). Let $\mathbf{v}$ be the vector containing all HFPs. Suppose we have N HFPs v_i,

$$\mathbf{v} \in \mathbb{C}^N \quad \text{with} \quad \mathbf{v} \equiv \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix} .$$

In order to get a physical solution, each HFP must satisfy a self-consistency equation similar to Eq. (2.20). Each equation has the structure:

$$v_i = (\text{Some non-linear function of } \mathbf{v}) .$$

Let A be the non-linear operator collecting all these functions. The problem of finding a self-consistent solution can be reformulated as follows: $\mathbf{v}$ represents a physical solution if it is a fixed point of the map A,

$$\mathrm{A}\colon \mathbb{C}^N \to \mathbb{C}^N \quad : \quad \mathrm{A}(\mathbf{v}) = \mathbf{v} \quad \implies \quad \mathbf{v} \text{ is a solution.} \tag{2.23}$$

We implement the search for a self-consistent solution as follows: first, we obtain an analytic expression for A. Then we choose an initialiser $\mathbf{v}_0$ and apply A to it, obtaining a new HFPs vector. Iterating this procedure, we "follow" the non-linear map A until we converge. Specifically:

1. We start with a generic quartic hamiltonian:

$$\hat{H} = \sum_{\alpha\beta} A_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta + \sum_{\alpha\beta\gamma\delta} B_{\alpha\beta\gamma\delta} \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta .$$

   We choose the model parameters and set the electronic target density;

2. Following the discussion of Sec. 2.2, we approximate analytically $\hat{H}$ with a quadratic hamiltonian

$$\hat{H} \simeq \hat{H}_{\mathrm{HF}} \equiv \sum_{\alpha\beta} \left[h^{+-}_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta - h^{+-}_{\beta\alpha} \hat{c}_\alpha \hat{c}^\dagger_\beta + h^{++}_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta + \left(h^{++}_{\beta\alpha} \right)^* \hat{c}_\alpha \hat{c}_\beta \right] ,$$

   with the fields given by Eqs. (2.14), (2.15);

3. We impose selection rules on the fields $h^{+-}_{\alpha\beta}$, $h^{++}_{\alpha\beta}$, eventually eliminating those not respecting the symmetry sector we are targeting. For example, if we choose not to break total charge conservation, we set $\forall\alpha,\beta\colon h^{++}_{\alpha\beta}=0$;
4. We choose a starting value for the HFPs vector $\mathbf{v}_0$;
5. We set $\mathbf{v}=\mathbf{v}_0$ and enter the iterative scheme of Fig. 2.2:
   a) Get a numeric approximation of the hamiltonian for the current value of $\mathbf{v}$, $\hat{H}_{\mathrm{HF}}[\mathbf{v}]$;
   b) Determine the value $\mu[\mathbf{v}]$ giving the desired density;
   c) Diagonalise the effective hamiltonian $\hat{H}_{\mathrm{HF}}[\mathbf{v}]-\mu[\mathbf{v}]\hat{N}$, and obtain the approximated ground-state $|\Psi[\mathbf{v}]\rangle$;
   d) Compute $\mathrm{A}(\mathbf{v})$ using self-consistency equations;
   e) If the algorithm converged, exit; otherwise, update the value of $\mathbf{v}$ repeat from point a).

In the following subsections we describe technical details concerning the chemical potential fixing and the convergence criteria used for the practical implementation of the above iterative scheme.

2.3.1 Fixed density simulations and the role of μ

In this thesis, all derivations and simulations are carried out at fixed target density. This is done by fixing the chemical potential μ during iterations of the algorithm. This must be done iteratively: changing at each step the HFPs vector, we are actively modifying the approximate hamiltonian. Chemical potential must be determined accordingly to maintain the required density.

For a given choice of HFPs vector and chemical potential, the ground state $|\Psi[\mathbf{v},\mu]\rangle$ is uniquely determined; then we compute density as

$$n(\mathbf{v},\mu)\equiv\frac{1}{\mathcal{V}}\sum_{\alpha}\langle\Psi[\mathbf{v},\mu]|\hat{c}^{\dagger}_{\alpha}\hat{c}_{\alpha}|\Psi[\mathbf{v},\mu]\rangle\ ,$$

being $\mathcal{V}$ the volume. Suppose we fix density to a value $n_0\in[0,1]$. At step i, naming the current HFPs vector $\mathbf{v}_i$, μ_i (the chemical potential at current step) is determined as

$$\mu_i\equiv\min_{\mu}|n(\mathbf{v}_i,\mu)-n_0|\ . \tag{2.24}$$

The density n must be a monotonically increasing function of μ: then we can translate the above equation, which would inspire a minimisation procedure, to a much simpler node-search rule:

$$\mu_i=\mu\,|_{\delta n(\mathbf{v}_i,\mu)=0}\quad\text{being}\quad\delta n(\mathbf{v}_i,\mu)\equiv n(\mathbf{v}_i,\mu)-n_0\,.$$

In order to find where the function changes sign (i.e. the position of μ_i) we use a basic bisection algorithm:

1. Set an arbitrary tolerance $\Delta n>0$ and a shifting parameter $\Delta\mu>0$;
2. Choose starting lower and upper boundaries, $\mu_{\mathrm{L}}<\mu_{\mathrm{U}}$;
3. Compute $\delta n(\mathbf{v}_i,\mu_{\mathrm{L}})$. Evaluate:
   - if $|\delta n(\mathbf{v}_i,\mu_{\mathrm{L}})|\le\Delta n$, set $\mu_i=\mu_{\mathrm{L}}$ and halt;
   - if $\delta n(\mathbf{v}_i,\mu_{\mathrm{L}})<-\Delta n$ do nothing;
   - if $\delta n(\mathbf{v}_i,\mu_{\mathrm{L}})>\Delta n$, shift down the lower boundary, $\mu_{\mathrm{L}}\to\mu_{\mathrm{L}}-\Delta\mu$ and repeat this step;
4. Compute $\delta n(\mathbf{v}_i,\mu_{\mathrm{U}})$. Evaluate:
   - if $|\delta n(\mathbf{v}_i,\mu_{\mathrm{U}})|\le\Delta n$, set $\mu_i=\mu_{\mathrm{U}}$ and halt;
   - if $\delta n(\mathbf{v}_i,\mu_{\mathrm{U}})>\Delta n$ do nothing;
   - if $\delta n(\mathbf{v}_i,\mu_{\mathrm{U}})<-\Delta n$, shift up the upper boundary, $\mu_{\mathrm{U}}\to\mu_{\mathrm{U}}+\Delta\mu$ and repeat this step;

5. At this point, the the lower boundary underestimates density, while the upper boundary overestimates it. Start iterations:
   a) Compute the midpoint,
   $$\mu_{\rm M} = \frac{\mu_{\rm L} + \mu_{\rm U}}{2} \, .$$
   b) Evaluate:
      - If $|\delta n(\mathbf{v}_i, \mu_{\rm M})| \leq \Delta n$, set $\mu_i = \mu_{\rm M}$ and halt;
      - If $\delta n(\mathbf{v}_i, \mu_{\rm M}) > \Delta n$, set $\mu_{\rm U} = \mu_{\rm M}$ and repeat from point a);
      - If $\delta n(\mathbf{v}_i, \mu_{\rm M}) < -\Delta n$, set $\mu_{\rm L} = \mu_{\rm M}$ and repeat from point a);

The bottleneck of the procedure is our ability to compute expectation values. This depends on how much we are able to simplify analytically the computations for each phase.

As will turn out in the derivation, chemical potential will be renormalised by interaction. This means that applying self-consistent HF methods to the EHM will produce an effective hamiltonian containing a shift to the chemical potential,

$$\mu \to \mu + F[\mathbf{v}] \equiv \tilde{\mu}[\mathbf{v}] \, ,$$

being F some function of the HFPs. The chemical potential is an example of "renormalised model parameter" (RMP).

Note that at step i of the iHF algorithm, our bisection procedure determines $\tilde{\mu}_i$, not μ_i. This consideration is relevant when dealing with ground state energy estimation. Let us abstract from the EHM and iHF, and consider a generic hamiltonian $\hat{H}$. To find the ground state $|\Psi\rangle$ means precisely to solve

$$\frac{\delta}{\delta \langle\Psi|} \langle\Psi|\hat{H}|\Psi\rangle = 0 \, .$$

We work with a fixed particles number N. Therefore, we are performing a constrained minimisation (i.e. taking the functional derivative) on the composite operator

$$\hat{H} - \mu(\hat{N} - N) \equiv \hat{K} + \mu N \, ,$$

being μ the Lagrange multiplier, $\hat{N}$ the number operator and $\hat{K} \equiv \hat{H} - \mu\hat{N}$. To find the approximate ground state $|\Psi[\mathbf{v}]\rangle$ means to find the HFPs vector $\mathbf{v}$ such that

$$|\Psi[\mathbf{v}]\rangle \quad : \quad \begin{cases} \dfrac{\delta}{\delta \langle\Psi|} \langle\Psi|\hat{K} + \mu N|\Psi\rangle \Big|_{|\Psi\rangle = |\Psi[\mathbf{v}]\rangle} = 0 \, , \\ \langle\Psi[\mathbf{v}]|\hat{N}|\Psi[\mathbf{v}]\rangle = N \, . \end{cases}$$

In the first line μN is derived to zero; the constraint is imposed in the second line. The key point, here, is that the HF method is structured in such a way that we compute EVs of $\hat{K}$, not $\hat{H}$, because we are working at a precise density. Therefore, energy is given by

$$\begin{aligned} \langle\hat{H}\rangle &= \langle\Psi[\mathbf{v}]|\hat{K}|\Psi[\mathbf{v}]\rangle + \mu N \\ &= \Omega[\mathbf{v}] + (\tilde{\mu}[\mathbf{v}] - F[\mathbf{v}])N \, , \end{aligned}$$

being Ω the expectation value (EV) of $\hat{K}$. To correctly compute energy, we need to add $+\mu N$; since we determine $\tilde{\mu}[\mathbf{v}] \neq \mu$, everything needs to be expressed in terms of it. Using this relation, when the algorithm has converged, we can estimate ground-state energy.

2.3.2 The convergence scheme

Let us discuss briefly the convergence criterion of our algorithm and the iteration mechanism. The iHF scheme we design is a sequential loop that runs up to a logical halt condition. In order to specify such condition, we need to define our notion of "convergence". Recall the definition of $\mathrm{A}(\mathbf{v})$, the non-linear map in HFPs space whose components are given by a set of self-consistency equations. We claim to have reached convergence when $\mathbf{v}$ comes "close enough" to a fixed point of the map A, as in Eq. (2.23). Explicitly, we stop whenever the following logic condition is satisfied for each component of $\mathbf{v} = (v_1, v_2, \cdots)$:

$$\forall v_j \quad : \quad |v_j - \mathrm{A}_j(\mathbf{v})| \leq \Delta v_j \, ,$$

being $\mathrm{A}(\mathbf{v})$ the map image of the HFP vector $\mathbf{v}$, and A_j its j-th component. The tolerance Δv_j is arbitrary. If just one component of $\mathbf{v}$ is mapped "too far" (with respect to Δv_j) iterations continue. In this text we deal in with quantities of order 10^{-1}, so convergence values of order $10^{-3} \div 10^{-4}$ will be in general sufficient.

At step i, we check the distance of the two vectors $\mathrm{A}(\mathbf{v}_i)$ and $\mathbf{v}_i$ and if possible, halt. If instead convergence is not reached, we repeat the cycle with a new initialiser $\mathbf{v}_{i+1}$. A very simple choice would be to set $\mathbf{v}_{i+1} = \mathbf{v}_i$. We do something slightly different:

$$\mathbf{v}_{i+1} \equiv g\mathrm{A}\left(\mathbf{v}_i\right) + (1-g)\mathbf{v}_i \, . \tag{2.25}$$

The "mixing parameter" g is a simple but useful knob to tune algorithmic evolution. Its usefulness comes from this observation: even if a fixed point exists, we don't know how easy it is to actually get there following A. Setting $g = 1$, the new initialiser is just the image of the previous initialiser. By moving "slower" ($g < 1$) with an accurate optimisation of the mixing, it is possible to reach convergence in a controlled way.

Let us give some concrete justification about the relevance of controlling g. When it comes to HF iterative schemes of this kind, it can sometimes happen for the algorithm to remain stuck in infinite loops. An example of this behaviour is given in Sec. 4.5.3. A typical situation is the following: during algorithm evolution we reach a couple of points $\mathbf{v}^{(a)}$, $\mathbf{v}^{(b)}$ such that

$$\cdots \xrightarrow{\mathrm{A},g} \mathbf{v}^{(a)} \xrightarrow{\mathrm{A},g} \mathbf{v}^{(b)} \xrightarrow{\mathrm{A},g} \mathbf{v}^{(a)} \xrightarrow{\mathrm{A},g} \mathbf{v}^{(b)} \xrightarrow{\mathrm{A},g} \cdots .$$

The notation "$\xrightarrow{\mathrm{A},g}$" indicates one algorithmic step at mixing parameter g. We are stuck forever in this loop. Now, to predict if couples like $(\mathbf{v}^{(a)}, \mathbf{v}^{(b)})$ exist, how they depend on g, where they are located in HFPs space... is impossible at this level. Thus there is no immediate way to avoid these loops *a priori*. One workaround we found to work in most cases, however, was to just make g smaller. We gained empirical evidence that to "slow down" in HFPs space makes numeric instabilities less frequent. The trade-off, however, is that if we move too slow, runtime could get very large. Note that inaccurate setup of the algorithm could lead to resources-intensive runs also in "ordinary" situations, when we are not stuck in a loop.

To keep the number of iterations under control, we also defined an upper bound p to the total number of steps. If after p steps we algorithm failed to converge, it halts. In that case, $\mathbf{v}$ is saved as a "`NaN`" (Not a number) vector and we move to another simulation. Introducing this simple rule, runtime is controlled and we gain an easy method to identify the regions of parameter space requiring more care – just read from the raw data where `NaN`s have been saved and repeat simulations fine-tuning the parameters. This basically means to lower g and increase p. These considerations end this technical introductory chapter on iHF methods.

Chapter 3

Phase transitions and spontaneous symmetry breaking in the extended Hubbard model

In this chapter, we discuss the symmetries of the extended Hubbard model, and classify the various phases discussed in this thesis based on the spontaneous symmetry breaking (SSB) mechanism. Therefore, we describe the explicit implementation of the HF method to investigate phase transitions hosted by the model.

3.1 Symmetries of the extended Hubbard model on the square lattice

In this section, we present the symmetries of the extended Hubbard model (EHM) introduced in Eq. (1.4)

$$\hat{H}_{\mathrm{EHM}} = \underbrace{-t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}}_{\hat{H}_t}\ \underbrace{+\,U\sum_{i\in\mathcal{L}}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}}_{\hat{H}_U}\ \underbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{n}_{i\sigma}\hat{n}_{j\sigma'}}_{\hat{H}_V}\,,$$

where $t, U, V > 0$ and $\langle ij\rangle$ is a shorthand notation to indicate nearest neighbours (NNs) on the lattice $\mathcal{L}$. The symmetries of the model can be divided in two groups: the ones related to the geometry of the square lattice, and those related to the algebraic properties of $\hat{H}_{\mathrm{EHM}}$.

3.1.1 Lattice symmetries

We study the EHM on the planar square lattice, with one orbital per site. We indicate with L_x the number of sites in direction x, and L_y in direction y. The hamiltonian of Eq. (1.4) respects three fundamental lattice symmetries:

1. Discrete translations symmetry in both the x and y directions. We refer to the groups of one-site translations as as $\mathrm{T}_1^{(x)}$, $\mathrm{T}_1^{(y)}$. It holds:

$$\mathrm{T}_n^{(\ell)} \supset \mathrm{T}_{kn}^{(\ell)} \quad \text{where} \quad k \in \mathbb{N}\,. \tag{3.1}$$

   Namely, a shift in direction ℓ by kn sites can be decomposed in k shift by n sites. Invariance by n-sites shifts, thus, implies invariance also for kn-sites shifts.

2. Discrete $\pi/2$ rotations symmetry, described by the four-fold cyclic point group C_4, containing rotations by angle $\pi/2, \pi, 3\pi/2$ and the identity. A similar property as for the translations holds for cyclic point groups,

$$\mathrm{C}_n \supset \mathrm{C}_{n/k} \quad \text{where} \quad k, n/k \in \mathbb{N}\,. \tag{3.2}$$

   Invariance for rotations of angle $2\pi/n$ implies invariance for rotations of angle $k \times 2\pi/n$.

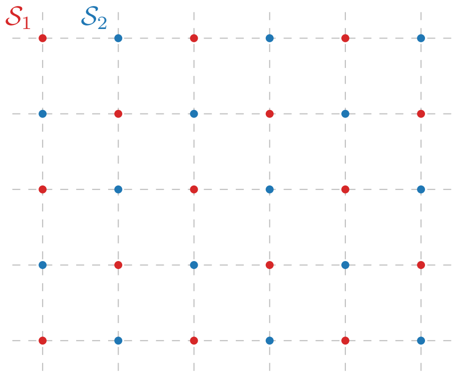


Figure 3.1 | Square lattice $\mathcal{L}$ bipartition in two sublattices $\mathcal{S}_1$ (red), $\mathcal{S}_2$ (blue).

3. Inversion symmetry. We define I_2 as the inversion group in two-dimensional space, containing the reflection operator and identity. Notice that

$$\mathrm{I}_2 \equiv \mathrm{C}_2\,, \tag{3.3}$$

implying that, in 2D space, a rotation by π around the origin and the reflection of the position of a point are the same operation.

The planar square lattice is bipartite: it can be divided in two square lattices, as represented in Fig. 3.1. Two sites are "connected" if linked by a non-zero hopping operator. Which is, if they are NNs. Note that odd-site translations leave the hamiltonian invariant but exchange the two sublattices, while even-sites translations don't. Specifically, let us consider a graph $\mathcal{L}$ and the set of its vertices $V(\mathcal{L}) = \{v_i\}$. Let $E(\mathcal{L})$ be the set of graph edges,

$$E(\mathcal{L}) = \{(v, w) \mid v, w \in V(\mathcal{L}) \wedge v,\, w \text{ are connected}\}\ .$$

The graph is said to be bipartite if two sub-graphs $\mathcal{S}_1$, $\mathcal{S}_2$ exist such that:

- The vertices of $\mathcal{L}$ split in the two sub-graphs,

$$V(\mathcal{L}) = V(\mathcal{S}_1) \cup V(\mathcal{S}_2)\,.$$

- Vertices of one sub-graph are linked only to vertices of the other sub-graph. In other words

$$v, w \in \mathcal{S}_i \quad \Longrightarrow \quad (v, w) \notin E(\mathcal{L})\,.$$

The two sublattices of the square lattice are coloured differently in Fig. 3.1.

3.1.2 Spin and charge symmetries

The conventional HM shows a global U(2) symmetry, extended to $\mathrm{SU}(2) \times \mathrm{SU}(2)/\mathbb{Z}_2 = \mathrm{SO}(4)$ symmetry at half-filling on a bipartite lattice [9], such as the square lattice. For the EHM, we limit to prove that we can identify a $\mathrm{U}(2) \subset \mathrm{SO}(4)$ global symmetry.

$\mathrm{U}^z(1)$ symmetry and charge conservation. The EHM is symmetric under the global gauge transformation

$$\hat{c}_{i\sigma} \to e^{-i\eta}\hat{c}_{i\sigma}\,, \quad \hat{c}^\dagger_{i\sigma} \to e^{i\eta}\hat{c}^\dagger_{i\sigma} \quad \text{where} \quad \eta \in \mathbb{R}\,. \tag{3.4}$$

To prove this, it suffices to note that any product of fermionic operators with an equal number of annihilations and creations is gauge-symmetric,

$$\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'} \quad \to \quad e^{i\eta}\hat{c}^\dagger_{i\sigma}e^{-i\eta}\hat{c}_{j\sigma'} = \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'}\,. \tag{3.5}$$

Hence the kinetic part $\hat{H}_t$ and the interactions $\hat{H}_U$, $\hat{H}_V$ are invariant. We will refer to this symmetry also as "gauge symmetry".

We observe that the on-site density is gauge invariant,

$$\hat{n}_{i\sigma} \to \hat{n}_{i\sigma} \,. \tag{3.6}$$

Since $\left[\hat{H}, \hat{N}\right] = 0$, it follows that the total number of particles (charge)

$$\hat{N} = \sum_{i\sigma} \hat{c}^\dagger_{i\sigma} \hat{c}_{i\sigma}$$

is a good quantum number.

Gauge symmetry is described by the transformation of Eq. (3.4), which is a pure phase shift by η. The gauge group is $\mathrm{U}^{\mathrm{c}}(1)$, with a superscript "c" indicating "charge".

Global spin symmetry Let the local spin operator

$$\hat{S}^\alpha_i \equiv \frac{1}{2} \hat{\Psi}^\dagger_i \tau^\alpha \hat{\Psi}_i \,, \tag{3.7}$$

where we defined the spinors $\hat{\Psi}_i$

$$\hat{\Psi}_i \equiv \begin{bmatrix} \hat{c}_{i\uparrow} \\ \hat{c}_{i\downarrow} \end{bmatrix} , \tag{3.8}$$

and the Pauli matrices

$$\tau^x \equiv \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \qquad \tau^y \equiv \begin{bmatrix} & -i \\ i & \end{bmatrix} \qquad \tau^z \equiv \begin{bmatrix} 1 & \\ & -1 \end{bmatrix} . \tag{3.9}$$

Everywhere in the following, we will represent zeros in matrices as blank entries. The spin operators of Eq. (3.7) obey an $\mathfrak{su}(2)$ algebra,

$$\left[\hat{S}^\alpha_j, \hat{S}^\beta_j\right] = i \sum_\gamma \epsilon_{\alpha\beta\gamma} \hat{S}^\gamma_j \,, \tag{3.10}$$

with $\epsilon_{\alpha\beta\gamma}$ the Levi-Civita tensor.

Proof 3.1 — Proof of Eq. (3.10): spin operators of Eq. (3.7) respect the $\mathfrak{su}(2)$ algebra.

To prove that $\mathfrak{su}(2)$ algebra is respected, we must compute all possible commutators. This involves many very similar calculations. For the sake of conciseness, we sketch the proof for $\alpha = x$, $\beta = y$:

$$\begin{aligned}
\left[\hat{S}^x_j, \hat{S}^y_j\right] &= \frac{1}{4} \left[\hat{\Psi}^\dagger_j \begin{bmatrix} & 1 \\ 1 & \end{bmatrix} \hat{\Psi}_j, \hat{\Psi}^\dagger_j \begin{bmatrix} & -i \\ i & \end{bmatrix} \hat{\Psi}_j\right] \\
&= \frac{1}{4} \left[\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow} + \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}, -i\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow} + i\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\right] \\
&= \frac{i}{4} \left[\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}, \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\right] - \frac{i}{4} \left[\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}, \hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}\right] \\
&= \frac{i}{2} \left[\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}, \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\right] \\
&= \frac{i}{2} \left(\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow} - \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}\right) \\
&= \frac{i}{2} \left(-\hat{c}^\dagger_{j\uparrow}\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\hat{c}_{j\uparrow} + \hat{c}^\dagger_{j\uparrow}\hat{c}_{j\uparrow} + \hat{c}^\dagger_{j\downarrow}\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\uparrow}\hat{c}_{j\downarrow} - \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\right) \\
&= \frac{i}{2} \left(\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\uparrow} - \hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\right) \\
&= i\hat{S}^z_j \,.
\end{aligned}$$

We used Fermi anticommutation rules, $\{\hat{c}_{i\sigma}, \hat{c}^\dagger_{j\sigma'}\} = \delta_{ij}\delta_{\sigma\sigma'}$. Identical derivations can be carried out for all other commutators, giving in the end the $\mathfrak{su}(2)$ spin algebra.

We define the on-site raising and lowering operators:

$$\hat{S}_j^+ \equiv \frac{\hat{S}_j^x + i\hat{S}_j^y}{2}\,, \qquad \hat{S}_j^- \equiv \frac{\hat{S}_j^x - i\hat{S}_j^y}{2}\,, \tag{3.11}$$

and the global spin operators

$$\hat{S}^\alpha \equiv \sum_{i\in\mathcal{L}} \hat{S}_i^\alpha\,, \qquad \left[\hat{S}^\alpha, \hat{S}^\beta\right] = i\sum_\gamma \epsilon_{\alpha\beta\gamma}\hat{S}^\gamma\,, \tag{3.12}$$

being $\mathcal{L}$ the lattice. Explicitly,

$$\hat{S}^z \equiv \frac{1}{2}\sum_{i\in\mathcal{L}} (\hat{n}_{i\uparrow} - \hat{n}_{i\downarrow})\,, \qquad \hat{S}^+ \equiv \frac{1}{2}\sum_{i\in\mathcal{L}} \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\downarrow}\,, \qquad \hat{S}^- \equiv \left(\hat{S}^+\right)^\dagger .$$

The EHM hamiltonian and the number operator commutate with the global spin operator. For the kinetic part $\hat{H}_t$ and the local repulsion $\hat{H}_U$ this is a well known result [9, 43],

$$\left[\hat{H}_t + \hat{H}_U, \hat{\mathbf{S}}\right] = 0\,.$$

Also $\hat{H}_V$ and $\hat{N}$ commutate with each component of the global spin given in Eq. (3.12),

$$\left[\hat{H}_V, \hat{\mathbf{S}}\right] = 0\,, \qquad \left[\hat{N}, \hat{\mathbf{S}}\right] = 0\,. \tag{3.13}$$

Proof 3.2 — Proof of Eq. (3.13): $\hat{H}_V$ and $\hat{N}$ are SU(2) symmetric.

The proof is organised through a sequence of results:

1. The non-local attraction can be written as:

$$\begin{aligned}\hat{H}_V &= V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'} \hat{n}_{i\sigma}\hat{n}_{j\sigma'} \\ &= V\sum_{\langle ij\rangle} \hat{\Psi}_i^\dagger\hat{\Psi}_i\hat{\Psi}_j^\dagger\hat{\Psi}_j\,,\end{aligned} \tag{3.14}$$

   having we used $\hat{\Psi}_i^\dagger\hat{\Psi}_i = \hat{c}_{i\uparrow}^\dagger\hat{c}_{i\uparrow} + \hat{c}_{i\downarrow}^\dagger\hat{c}_{i\downarrow}$.

2. Similarly, the number operator is given by:

$$\begin{aligned}\hat{N} &= \sum_{i\in\mathcal{L}}\sum_\sigma \hat{n}_{i\sigma} \\ &= \sum_{i\in\mathcal{L}} \hat{\Psi}_i^\dagger\hat{\Psi}_i\,.\end{aligned} \tag{3.15}$$

3. Let now $\hat{R}(\boldsymbol{\theta})$ be any global spin rotation,

$$\hat{R}(\boldsymbol{\theta}) \equiv \exp\left(-i\boldsymbol{\theta}\cdot\hat{\mathbf{S}}\right) \quad \text{where} \quad \boldsymbol{\theta} = (\theta_x, \theta_y, \theta_z)\,,$$

   with $\hat{\mathbf{S}}$ the global spin given in Eq. (3.12). Spinors transform as:

$$\hat{R}(\boldsymbol{\theta})\hat{\Psi}_i\hat{R}(\boldsymbol{\theta})^\dagger = \mathcal{R}(\boldsymbol{\theta})\hat{\Psi}_i\,,$$

   with $\mathcal{R}(\boldsymbol{\theta}) \in \mathrm{SU}(2)$ the unitary matrix associated to said rotation,

$$\mathcal{R}(\boldsymbol{\theta}) = \exp\left(-\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\right) \quad \text{where} \quad \boldsymbol{\tau} = (\tau^x, \tau^y, \tau^z)\,. \tag{3.16}$$

Let us prove this. Let $\hat{A} = i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}$ e $\hat{B} = \hat{\Psi}_i$. The following identity, a consequence of Baker-Campbell-Housdorff formula, holds:

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + \left[\hat{A}, \hat{B}\right] + \frac{1}{2!}\left[\hat{A}, \left[\hat{A}, \hat{B}\right]\right] + \frac{1}{3!}\left[\hat{A}, \left[\hat{A}, \left[\hat{A}, \hat{B}\right]\right]\right] + \cdots .$$

The first order of the expansion gives:

$$\begin{aligned}
[i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, \hat{\Psi}_i] &= i\sum_\alpha \theta_\alpha [\hat{S}^\alpha, \hat{\Psi}_i] \\
&= -\frac{i}{2}\sum_\alpha \theta_\alpha \tau^\alpha \hat{\Psi}_i \\
&= -\frac{i}{2}\boldsymbol{\theta} \cdot \boldsymbol{\tau}\hat{\Psi}_i .
\end{aligned}$$

We omitted some purely algebraic passages. The second order is given by

$$\begin{aligned}
\left[i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, \left[i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, \hat{\Psi}_i\right]\right] &= \left[i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, -\frac{i}{2}(\boldsymbol{\theta} \cdot \boldsymbol{\tau})\hat{\Psi}_i\right] \\
&= -\frac{i}{2}\boldsymbol{\theta} \cdot \boldsymbol{\tau}\left[i\boldsymbol{\theta} \cdot \hat{\mathbf{S}}, \Psi_i\right] \\
&= \frac{1}{4}(\boldsymbol{\theta} \cdot \boldsymbol{\tau})^2\hat{\Psi}_i ,
\end{aligned}$$

and so on. Collecting all terms, we obtain

$$\begin{aligned}
\hat{R}(\boldsymbol{\theta})\hat{\Psi}_i\hat{R}(\boldsymbol{\theta})^\dagger &= \hat{\Psi}_i - \frac{i}{2}\boldsymbol{\theta} \cdot \boldsymbol{\tau}\hat{\Psi}_i + \frac{1}{2! \times 4}(\boldsymbol{\theta} \cdot \boldsymbol{\tau})^2\hat{\Psi}_i + \cdots \\
&= \sum_{k=1}^{\infty}\frac{1}{k!}\left(-\frac{i}{2}\boldsymbol{\theta} \cdot \boldsymbol{\tau}\right)^k\hat{\Psi}_i \\
&= \mathcal{R}(\boldsymbol{\theta})\hat{\Psi}_i .
\end{aligned}$$

In the last line, we identified the series as the Taylor expansion of the exponential, recognising the expression for $\mathcal{R}(\boldsymbol{\theta})$ given in Eq. (3.16).

4. Then, the group acts on the product $\hat{\Psi}_i^\dagger\hat{\Psi}_i$ as

$$\begin{aligned}
\hat{R}\hat{\Psi}_i^\dagger\hat{\Psi}_i\hat{R}^\dagger &= \hat{R}\hat{\Psi}_i^\dagger\hat{R}^\dagger\hat{R}\hat{\Psi}_i\hat{R}^\dagger \\
&= \hat{\Psi}_i^\dagger\mathcal{R}^\dagger\mathcal{R}\hat{\Psi}_i \\
&= \hat{\Psi}_i^\dagger\hat{\Psi}_i
\end{aligned}$$

being $\mathcal{R} \in \mathrm{SU}(2)$. Recalling Eqs. (3.14), (3.15), we conclude that $\hat{H}_V$ and $\hat{N}$ are SU(2) symmetric, i.e. do not change under global spin rotations. This fact implies Eq. (3.13).

We observe that the above proof can be used to show that the on-site number operator is spin-invariant,

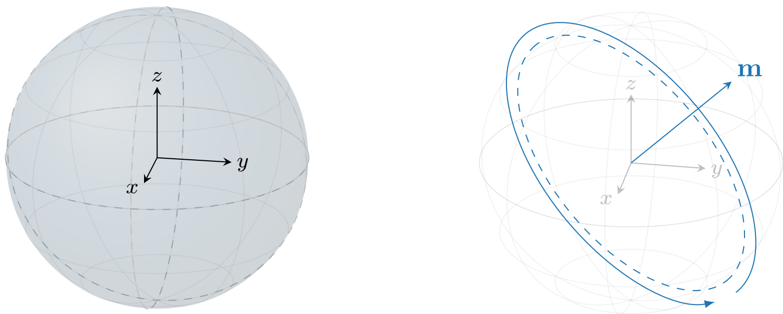
$$\hat{n}_{i\sigma} = \hat{\Psi}_i^\dagger\hat{\Psi}_i \quad \Longrightarrow \quad \left[\hat{n}_{i\sigma}, \hat{\mathbf{S}}\right] = 0 . \tag{3.17}$$

The EHM is symmetric under global spin rotations. Specifically, both $\hat{H}_U$ and $\hat{H}_V$ are symmetric also under local spin rotations, whereas the kinetic operator is symmetric only under global spin rotations.

The $\mathrm{U^c}(1)$ gauge symmetry and the SU(2) spin rotations symmetry can be combined,

$$\mathrm{U^c}(1) \otimes \mathrm{SU}(2) = \mathrm{U}(2) .$$

We will refer to U(2) symmetry to indicate full charge-spin conservation. Finally, note the group



a | SU(2) rotations domain.

b | $\mathrm{U}^z(1)$ rotations domain for a given vector $\mathbf{m}$.

Figure 3.2 | Graphic representation of the domain of SU(2) rotations (left figure) and $\mathrm{U}^z(1)$ rotations (right figure). For any given direction $\mathbf{m}$, the group of rotations around said direction is contained in the larger group SU(2).

of 3D rotations contains the subgroup of 2D rotations around a given axis z,

$$\mathrm{U}^z(1) \subset \mathrm{SU}(2)\,,$$

where the z superscript serves to distinguish from the gauge group. The z axis does not need to be identified with the "real" z axis, perpendicular with the lattice plane. Any given direction works. In Fig. 3.2 we show schematically how $\mathrm{U}^z(1)$ is a subgroup of SU(2) for a given direction $\mathbf{m}$.

3.1.3 Particle-hole symmetry at half-filling

At half-filling (equal number of electrons and lattice sites), the EHM on the square lattice exhibits particle-hole symmetry (PHS). The presence of this additional symmetry is a consequence of the fact that the square lattice is bipartite. The generic unitary particle-hole (PH) transformation is given by

$$\mathrm{P} \quad : \quad \hat{c}_{i\sigma} \quad \rightarrow \quad e^{i\eta_{i\sigma}}\hat{h}^{\dagger}_{i\sigma}\,, \tag{3.18}$$

being in general η_i a phase dependent on the fermionic quantum state $(i\sigma)$. Inserting the above transformation in the EHM, we get

$$\mathrm{P}\{\hat{H}\} = -t\sum_{\langle ij\rangle}\sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma})}\hat{h}_{i\sigma}\hat{h}^{\dagger}_{j\sigma} + U\sum_{i}\hat{h}_{i\uparrow}\hat{h}_{i\downarrow}\hat{h}^{\dagger}_{i\downarrow}\hat{h}^{\dagger}_{i\uparrow} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^{\dagger}_{j\sigma'}\hat{h}^{\dagger}_{i\sigma}\,.$$

For the two interactions $\hat{H}_U$, $\hat{H}_V$ the phase factors cancelled out. For a specific choice of the η_i phases,

$$\mathrm{P} \quad : \quad \begin{cases} \hat{c}_{i\sigma} \rightarrow \hat{h}^{\dagger}_{i\sigma} & \text{for} \quad i \in \mathcal{S}_1\,, \\ \hat{c}_{i\sigma} \rightarrow -\hat{h}^{\dagger}_{i\sigma} & \text{for} \quad i \in \mathcal{S}_2\,, \end{cases} \tag{3.19}$$

the above hamiltonian is mapped onto the EHM of (1.4),

$$\mathrm{P}\{\hat{H}\} = -t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{h}^{\dagger}_{i\sigma}\hat{h}_{j\sigma} + U\sum_{i\in\mathcal{L}}\hat{h}^{\dagger}_{i\uparrow}\hat{h}^{\dagger}_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}^{\dagger}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma}\,.$$

The equality $\mathrm{P}\{\hat{H}\} = \hat{H}$ expresses PHS, which holds as long as we work at half-filling.

Proof 3.3 — PHS at half filling, with η_i given by Eq. (3.19).

This proof is an extension of the argument carried out for the conventional HM [9]. For the hole operators, Fermi anticommutation rules hold regardlessly of the phase factor:

$$\begin{aligned}\{\hat{h}_{i\sigma}, \hat{h}^\dagger_{j\sigma'}\} &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})}\{\hat{c}^\dagger_{i\sigma}, \hat{c}_{j\sigma'}\} \\ &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})}\delta_{ij}\delta_{\sigma\sigma'} \\ &= \delta_{ij}\delta_{\sigma\sigma'}\,.\end{aligned}$$

The on-site density operator is given by:

$$\hat{h}_{i\sigma}\hat{h}^\dagger_{i\sigma} = \hat{n}_{i\sigma}\,,$$

being $\hat{n}_{i\sigma}$ the fermionic number operator on site i with spin σ. It follows, for quartic interactions

$$\begin{aligned}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma} &= -\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'} \\ &= \hat{h}_{i\sigma}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\ &= -\hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\ &= \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\ &= -\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'} \\ &= \hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} - (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) + 1 - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'}\,.\end{aligned} \tag{3.20}$$

The first term of the last passage is mirrored with respect to the starting term. In next paragraphs we derive the constraint of Eq. (3.19) and the requirement to work at half-filling.

Kinetic term. Consider the kinetic term,

$$\hat{H}_t = -t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\,.$$

We apply the PH transform of Eq. (3.18)

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma}e^{i(\eta_{i\sigma}+\eta_{j\sigma})}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma}\,.$$

Anti-commutating the two fermionic operators, and exchanging site labels $i \leftrightarrow j$, we get:

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma}e^{i(\eta_{i\sigma}+\eta_{j\sigma}+\pi)}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma}\,.$$

To obtain $\mathrm{P}\{\hat{H}_t\} \stackrel{!}{=} \hat{H}_t$, a necessary condition on the unitary operation P is

$$\eta_{i\sigma} + \eta_{j\sigma} \stackrel{!}{=} (2k-1)\pi \quad \text{with} \quad k \in \mathbb{Z}\,, \tag{3.21}$$

valid for NN sites. This requirement on the phases η_i gives Eq. (3.19).

Local repulsion. Substitute Eq. (3.20) in $\hat{H}_U$,

$$\begin{aligned}\mathrm{P}\{\hat{H}_U\} &= U\sum_{i\in\mathcal{L}}\hat{h}_{i\uparrow}\hat{h}_{i\downarrow}\hat{h}^\dagger_{i\downarrow}\hat{h}^\dagger_{i\uparrow} \\ &= U\sum_{i\in\mathcal{L}}\hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} - U\sum_{i\in\mathcal{L}}(\hat{n}_{i\uparrow}+\hat{n}_{i\downarrow}) + U\sum_{i\in\mathcal{L}}(1) \\ &= U\sum_{i}\hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} - U(\hat{N} - L_xL_y)\,.\end{aligned}$$

Symmetry	Operations
$\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$	One-site discrete translations separately in both directions
C_4	Four-fold point group (real $\pi/2$ angle rotations)
$\mathrm{U}^{\mathrm{c}}(1)$	Charge-conserving global phase shifts
$\mathrm{SU}(2)$	Three-dimensional rotations in global spin space

Table 3.1 | Model symmetries and relative group operations. Note that $\mathrm{U}^{\mathrm{c}}(1)$ represents the gauge symmetry linked to charge conservation.

At half filling, $\langle \hat{N} \rangle = L_x L_y$. The unitary map of Eq. (3.18), independently of the condition of Eq. (3.21), leaves the hamiltonian identical at half-filling.

Non-local attraction. We proceed identically for $\hat{H}_V$,

$$\begin{aligned}
\mathrm{P}\{\hat{H}_V\} &= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}^\dagger_{j\sigma'} \hat{h}^\dagger_{i\sigma} \\
&= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}^\dagger_{i\sigma} \hat{h}^\dagger_{j\sigma'} \hat{h}_{j\sigma'} \hat{h}_{i\sigma} + V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} (1) \\
&= -V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}^\dagger_{i\sigma} \hat{h}^\dagger_{j\sigma'} \hat{h}_{j\sigma'} \hat{h}_{i\sigma} + 2zV(\hat{N} - L_x L_y) \,,
\end{aligned}$$

with $z = 4$ the lattice coordination number. In the last line, we used that on the square lattice

$$\begin{aligned}
\sum_{\langle ij \rangle} (1) &= \sum_{i \in \mathcal{L}/2} z \\
&= \frac{L_x L_y}{2} \times z \,.
\end{aligned} \tag{3.22}$$

As before, the non-local attraction is particle-hole symmetric at half-filling. We conclude that the entire EHM exhibits PHS at half filling, if P is defined as in Eq. (3.19).

As for the conventional HM [9], this result can be generalised to any bipartite lattice. In turn, if a lattice is not bipartite, PHS is not present. We report in Tab. 3.1 all identified symmetries.

3.2 The HF method and symmetry breaking

In this thesis, we use the HF method to investigate the different many-body phases hosted by the EHM. We classify these phases in terms of the symmetry broken by spontaneous symmetry breaking (SSB) mechanism:

- *Normal phase*: no broken symmetry.
- *Commensurate antiferromagnetic phase*: simultaneous $\mathrm{SU}(2) \overset{\mathrm{SSB}}{\to} \mathrm{U}^z(1)$ and $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)} \overset{\mathrm{SSB}}{\to} \mathrm{T}_2^{(x)} \otimes \mathrm{T}_2^{(y)}$.
- *Nematic superconducting phase*: simultaneous $\mathrm{U}^c(1) \overset{\mathrm{SSB}}{\to} 0$ and $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$, namely superconductivity and breaking of planar rotational symmetry, observed for example in iron-based superconductors [37].

In the subsections that follow we discuss how to implement these SSBs in the context of the HF method. Specifically, as is described in Chap. 2, we investigate each of these phases by approximating the EHM hamiltonian

$$\hat{H} \overset{\mathrm{HF}}{\simeq} \hat{H}_{\mathrm{HF}} \,,$$

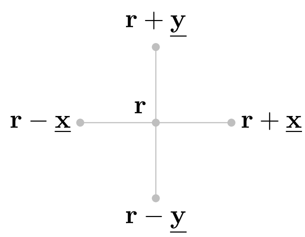


Figure 3.3 | Schematic representation of the four NNs of a given site i for a planar square lattice.

with an operator $\hat{H}_{\mathrm{HF}}$ that respects the symmetries of the given phase. We start with the general decomposition of the two interactions $\hat{H}_U$ and $\hat{H}_V$, given respectively in Eqs. (2.21) and (2.22), and discuss the symmetry properties of each sector.

In the following, we switch from the site index $i \in \mathcal{L}$ to the vector notation $\mathbf{r} \in \mathbb{N}^2$, indicating the site 2D position on the lattice as

$$i \quad \rightarrow \quad \mathbf{r} = (x, y)\,,$$

with $x, y \in \mathbb{N}$. The lattice versors are:

$$\underline{\mathbf{x}} \equiv (1, 0) \quad \text{and} \quad \underline{\mathbf{y}} \equiv (0, 1)\,,$$

as is shown in Fig. 3.3.

3.2.1 Local repulsion

The local repulsion $\hat{H}_U$ is given by:

$$\begin{aligned} \hat{H}_U &= U \sum_{\mathbf{r} \in \mathcal{L}} \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}\downarrow} \\ &= U \sum_{\mathbf{r} \in \mathcal{L}} \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}^\dagger_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow}\,. \end{aligned}$$

Applying Wick's theorem and using the denominations of Eq. (2.1) we have

$$\langle \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}^\dagger_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle = \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}_{\mathbf{r}\uparrow} \rangle \langle \hat{c}^\dagger_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\downarrow} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}_{\mathbf{r}\downarrow} \rangle \langle \hat{c}^\dagger_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}^\dagger_{\mathbf{r}\downarrow} \rangle \langle \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Cooper}}\,, \tag{3.23}$$

where the equality holds exactly as long as we perform the averaging operation on the subspace of gaussian states. We now discuss the above expectation values (EVs) separately, obtaining for each phase a set of "symmetry selection rules". We define a symmetry selection rule as a theoretical argument that allows us to assess if a given EV is zero, based on symmetry considerations.

Hartree term. The Hartree term of the above decomposition is a product of density EVs,

$$\langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle\,.$$

From Eq. (3.6), (3.17) we know that single-site number operators are invariant under the entire group U(2). Moreover, shifting or rotating the lattice leaves density operators unchanged. Thus no symmetry selection rule applies, and the above EVs are in principle non-zero.

Fock terms. For the Fock term, we notice that

$$\hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}_{\mathbf{r}\downarrow} = \hat{S}^+_{\mathbf{r}}\,,$$

being $S^+_{\mathbf{r}}$ the local raising operator. From $\mathfrak{su}(2)$ algebra we have:

$$\left[\hat{S}^+_{\mathbf{r}}, \hat{S}^\alpha\right] \neq 0 \quad \text{where} \quad \alpha \in \{x, y, z\}\,.$$

If the ground-state is invariant under global spin rotations, $\langle \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}_{\mathbf{r}\downarrow} \rangle$ must be zero. instead, gauge symmetry gives no symmetry selection rules, due to Eq. (3.5).

Term from $\hat{H}_U$	Normal phase	Antiferromagnet	Superconductor
Hartree	Allowed	Allowed	Allowed
Fock	Prohibited	Prohibited	Prohibited
Cooper	Prohibited	Prohibited	Allowed

Table 3.2 | Summary of symmetry selection rules for the Wick's decomposition of Eq. (3.23) of the local repulsion $\hat{H}_U$.

Cooper terms. The Cooper term can be non-zero only if charge is not conserved. In terms of the gauge symmetry, it suffices to apply the map of Eq. (3.4) to $\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}$,

$$\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow} \quad \to \quad e^{2i\eta}\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\,,$$

to conclude that this operator is in general not gauge invariant. It is instead SU(2)-symmetric.

Proof 3.4 — Local Cooper terms are SU(2)-symmetric.

We must compute the commutators of $\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}$ with $\hat{S}^z$, $\hat{S}^+$:

$$\begin{aligned}\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{S}^z\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}(\hat{n}_{\mathbf{r}'\uparrow}-\hat{n}_{\mathbf{r}'\downarrow})\right] \\ &= \frac{1}{2}\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\uparrow}-\hat{n}_{\mathbf{r}\downarrow}\right] \\ &= \frac{1}{2}\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}, \hat{n}_{\mathbf{r}\uparrow}\right]\hat{c}^\dagger_{\mathbf{r}\downarrow} - \frac{1}{2}\hat{c}^\dagger_{\mathbf{r}\uparrow}\left[\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\downarrow}\right].\end{aligned}$$

Since:

$$\begin{aligned}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right] &= \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma} - \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\sigma} \\ &= -\hat{c}^\dagger_{\mathbf{r}\sigma}\left(1-\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\right) \\ &= -\hat{c}^\dagger_{\mathbf{r}\sigma}\,,\end{aligned} \tag{3.24}$$

we have:

$$\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{S}^z\right] = -\frac{1}{2}\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow} + \frac{1}{2}\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow} = 0\,.$$

Hence, $\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}$ is $\mathrm{U}^z(1)$-symmetric.

Moreover, considering the raising operator:

$$\begin{aligned}\left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{S}^+\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}\hat{c}^\dagger_{\mathbf{r}'\uparrow}\hat{c}_{\mathbf{r}'\downarrow}\right] \\ &= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow}\right] \\ &= \left[\hat{c}^\dagger_{\mathbf{r}\uparrow}, \hat{c}^\dagger_{\mathbf{r}\uparrow}\right]\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow} - \left(\hat{c}^\dagger_{\mathbf{r}\uparrow}\right)^2\left[\hat{c}^\dagger_{\mathbf{r}\downarrow}, \hat{c}_{\mathbf{r}\downarrow}\right] \\ &= 0\,,\end{aligned}$$

where we used the facts that every operator commutates with itself, and a squared creation operator is null. These commutation relations, together, prove that Cooper EVs are legitimately non-zero for a SU(2)-preserving phase.

In Tab. 3.2 we collect the symmetry selection rules for the Hartree, Fock and Cooper terms of Eq. (3.23), based on the symmetry specifications given for each phase in above paragraphs. Note that Fock-like EVs are prohibited for all phases.

3.2.2 Non-local attraction

The non-local attraction $\hat{H}_V$ is given by:

$$\begin{aligned}\hat{H}_V &= -V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma,\sigma'} \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\sigma'} \\ &= V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma,\sigma'} \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}'\sigma'}\hat{c}_{\mathbf{r}'\sigma'}\hat{c}_{\mathbf{r}\sigma} \, .\end{aligned} \tag{3.25}$$

Let

$$\hat{H}_V^{\sigma\sigma'} \equiv -V \sum_{\langle \mathbf{r},\mathbf{r}'\rangle} \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}'\sigma'} \, .$$

The interaction decomposes in two "spin sectors"

$$\begin{aligned}\hat{H}_V &= \underbrace{\hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow}}_{\text{Same spin}} + \underbrace{\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}}_{\text{Opposite spin}} \\ &\equiv \hat{H}^{(\text{s.s.})} + \hat{H}^{(\text{o.s.})} \, ,\end{aligned} \tag{3.26}$$

where we further separated in s.s. and o.s. operators.

The operator $\hat{H}^{(\text{s.s.})}$ can be written as follows:

$$\begin{aligned}\hat{H}_V^{(\text{s.s.})} &\equiv \hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow} \\ &= -V \sum_{\mathbf{r}\in\mathcal{L}/2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \sum_{\sigma\in\{\uparrow,\downarrow\}} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma}\right) \, ,\end{aligned} \tag{3.27}$$

where the sum over $\mathcal{L}/2$, in last line, means that we are summing over just one sublattice (it does not matter which one). Applying Wick's theorem in the subspace of gaussian states, we get:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle \\ &= \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle\langle \hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle}_{\text{Cooper}} \, .\end{aligned} \tag{3.28}$$

where $\boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}$, using the notation of Fig. 3.3.

Proof 3.5 — Proof of Eq. (3.27).

Let $f(\mathbf{r}, \mathbf{r}')$ be a function of two variables on the square lattice. Summing over NNs, we get the identity

$$\sum_{\langle \mathbf{r}\mathbf{r}'\rangle} f(\mathbf{r},\mathbf{r}') = \sum_{\mathbf{r}\mathbf{r}'} f(\mathbf{r},\mathbf{r}') \left(\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right) \, ,$$

where we used the notation of Fig. 3.3. The above sum is identically written as:

$$\sum_{\langle \mathbf{r}\mathbf{r}'\rangle} f(\mathbf{r},\mathbf{r}') = \sum_{\mathbf{r}\in\mathcal{L}/2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left[f(\mathbf{r},\mathbf{r}+\boldsymbol{\delta}) + f(\mathbf{r},\mathbf{r}-\boldsymbol{\delta})\right] \, . \tag{3.29}$$

The symbol "$\mathcal{L}/2$" indicates one of the two sublattices $\mathcal{S}_1$, $\mathcal{S}_2$ represented in Fig. 3.1. Inserting Eq. (3.29) in Eq. (3.25), we obtain:

$$\hat{H}_V = -V \sum_{\mathbf{r}\in\mathcal{L}/2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \sum_{\sigma,\sigma'} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'}\right) \, . \tag{3.30}$$

By taking the s.s. sector of the above equation, namely $\sigma = \sigma'$, and performing Wick's contractions, we obtain Eq. (3.27).

The operator $\hat{H}^{(\text{o.s.})}$ can be written as follows:

$$\begin{aligned}\hat{H}_V^{(\text{o.s.})} &\equiv \hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow} \\ &= -V \sum_{\mathbf{r}\in\mathcal{L}} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow}\right) .\end{aligned} \tag{3.31}$$

From Wick's theorem:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \\ &= \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}} .\end{aligned} \tag{3.32}$$

Again, equality holds in the subspace of gaussian states.

Proof 3.6 — Proof of Eq. (3.31).

Consider Eq. (3.30), restricting to $\sigma =\uparrow$, $\sigma' =\downarrow$ and setting $\mathcal{L}/2 = \mathcal{S}_1$. We obtain

$$\hat{H}_V^{\uparrow\downarrow} = -V \sum_{\mathbf{r}\in\mathcal{S}_1} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow}\right) .$$

Note that spin $\uparrow$ operators act on the sublattice $\mathcal{S}_1$; spin $\downarrow$ operators act on the sublattice $\mathcal{S}_2$. If instead, from Eq. (3.30), we restrict to $\sigma =\downarrow$, $\sigma' =\uparrow$ and set $\mathcal{L}/2 = \mathcal{S}_2$, we get:

$$\hat{H}_V^{\downarrow\uparrow} = -V \sum_{\mathbf{r}\in\mathcal{S}_2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\downarrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\uparrow}\right) .$$

We observe that if $\mathbf{r} \in \mathcal{S}_2$, all its NNs belong to $\mathcal{S}_1$. Again, spin $\uparrow$ operators act on the sublattice $\mathcal{S}_1$; spin $\downarrow$ operators act on the sublattice $\mathcal{S}_2$.

Therefore, summing $\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}$, we obtain

$$\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow} = -V \sum_{\mathbf{r}\in\mathcal{L}} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow}\right) ,$$

which coincides with Eq. (3.31).

The separation of $\hat{H}_V$ in o.s. and s.s. sector is necessary in order to derive a set of symmetry selection rules for the Wick's contractions of Eqs. (3.28), (3.32). This is done in next paragraphs.

Hartree terms. As was for the local repulsion $\hat{H}_U$, the non-local Hartree terms of Eqs. (3.27), (3.31) are not subject to symmetry selection rules for any symmetry of Tab. 3.1. Thus they are allowed for all sectors.

Fock terms. The Fock EVs from the two sectors behave differently. Let us compute the following generic commutation relations:

$$\begin{aligned}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{S}^z\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}(\hat{n}_{\mathbf{r}'\uparrow} - \hat{n}_{\mathbf{r}'\downarrow})\right] \\ &= \frac{(-1)^{\delta_{\sigma=\downarrow}}}{2}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right]\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\hat{c}^\dagger_{\mathbf{r}\sigma}\left[\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}\right] \\ &= \left(-\frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\right)\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} ,\end{aligned} \tag{3.33}$$

where in last passage we used Eq. (3.24) and its conjugate version,

$$\left[\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma}\right] = \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} .$$

Sector	Term from $\hat{H}_V$	Normal phase	Antiferromagnet	Superconductor
o.s.	Hartree	Allowed	Allowed	Allowed
	Fock	Prohibited	Prohibited	Prohibited
	Cooper	Prohibited	Prohibited	Allowed
s.s.	Hartree	Allowed	Allowed	Allowed
	Fock	Allowed	Allowed	Allowed
	Cooper	Prohibited	Prohibited	Prohibited

Table 3.3 Summary of symmetry selection rules for the Wick's decomposition of Eqs. (3.32), (3.28) of the non-local attraction $\hat{H}_V$.

Now, if $\sigma = \sigma'$ the last passage of Eq. (3.33) is zero; otherwise is not. This proves that the o.s. Fock EVs are non-zero only for a phase that completely breaks SU(2); instead, in the s.s. sector we can have non-zero EVs for a phase that reduces SU(2) to $\mathrm{U}^z(1)$.

Cooper terms. Cooper terms are not gauge invariant. The proof is analogous to the one given for $\hat{H}_U$. Instead, the o.s. and s.s. spin sectors behave differently under global spin rotations. With an identical procedure as for Eq. (3.33), we get:

$$\begin{aligned}\left[\hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}^{\dagger}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{S}^z\right] &= \left[\hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}^{\dagger}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}(\hat{n}_{\mathbf{r}'\uparrow} - \hat{n}_{\mathbf{r}'\downarrow})\right] \\ &= \frac{(-1)^{\delta_{\sigma=\downarrow}}}{2}\left[\hat{c}^{\dagger}_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right]\hat{c}^{\dagger}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\hat{c}^{\dagger}_{\mathbf{r}\sigma}\left[\hat{c}^{\dagger}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}\right] \\ &= \left(-\frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} - \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\right)\hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}^{\dagger}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}\,. \end{aligned} \tag{3.34}$$

We have a "mirrored" situation with respect to the Fock terms. For the Cooper channel, the s.s. EVs are zero if $\mathrm{U}^z(1)$ is unbroken, while the o.s. sector are not.

In Tab. 3.3 we summarise these considerations, listing a set of symmetry selection rules about the EVs of Eqs. (3.32), (3.28), dividing in the two sectors "o.s." and "s.s.". By identifying these symmetry selection rules, we set grounds for the detailed derivation of each chapter and heavily simplify application of HF model to the EHM.

3.3 The EHM in reciprocal space

The EHM is symmetric under discrete lattice translations. We will investigate phases that do not (completely) break translational invariance. Therefore, It is convenient to move to reciprocal space, by Fourier-transforming the hamiltonian. We will use the following convention:

$$\hat{c}_{\mathbf{r}\sigma} \overset{\mathcal{F}}{=} \frac{1}{\sqrt{L_x L_y}}\sum_{\mathbf{k}\in\mathrm{BZ}} e^{-i\mathbf{k}\cdot\mathbf{r}}\hat{c}_{\mathbf{k}\sigma}\,. \tag{3.35}$$

Throughout the text, $\overset{\mathcal{F}}{=}$ will indicate a passage where a Fourier transform is used. "BZ" is shorthand notation for the Brillouin Zone.

It is useful to recall that the following equation for Kronecker's delta holds:

$$\frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}} e^{i\mathbf{k}\cdot\mathbf{r}} = \delta_{\mathbf{k}=\mathbf{0}}\,. \tag{3.36}$$

On one sublattice:

$$\frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}/2} e^{i\mathbf{k}\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}=\mathbf{0}}\,. \tag{3.37}$$

The notation $\mathcal{L}/2$ is shorthand to indicate either $\mathcal{S}_1$ or $\mathcal{S}_2$.

Kinetic part. The kinetic operators transforms as follows:

$$\hat{H}_t \stackrel{\mathcal{F}}{=} \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \,, \tag{3.38}$$

where we defined the bare bands:

$$\epsilon_{\mathbf{k}} \equiv -2t\left(\cos k_x + \cos k_y\right) \,. \tag{3.39}$$

Proof 3.7 — Proof of Eq. (3.38): $\mathcal{F}$-transform of $\hat{H}_t$.

Consider the kinetic operator $\hat{H}_t$:

$$\hat{H}_t = -t \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \left[\hat{c}^\dagger_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}'\sigma} + \mathrm{h.c.}\right] \,.$$

Applying the convention of Eq. (3.35), we get:

$$\begin{aligned}
\hat{H}_t &\stackrel{\mathcal{F}}{=} -t \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \left(\frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\mathrm{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}} \hat{c}^\dagger_{\mathbf{k}\sigma}\right) \left(\frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}'\in\mathrm{BZ}} e^{-i\mathbf{k}'\cdot\mathbf{r}'} \hat{c}_{\mathbf{k}'\sigma}\right) + \mathrm{h.c.} \\
&= -2t \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}'\sigma} \times \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}/2} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \times \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} e^{i\mathbf{k}'\cdot\boldsymbol{\delta}} \,.
\end{aligned}$$

We used Eq. (3.29) and recognised that the h.c. term gives an identical contribution. We recognise in the central term the relation of Eq. (3.37). Hence,

$$\begin{aligned}
\hat{H}_t &\stackrel{\mathcal{F}}{=} -2t \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \left[\frac{e^{ik_x} + e^{-ik_x}}{2} + \frac{e^{ik_y} + e^{-ik_y}}{2}\right] \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \,,
\end{aligned}$$

which proves Eq. (3.38).

Local repulsion. The local repulsion transforms as follows:

$$\hat{H}_U \stackrel{\mathcal{F}}{=} \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \,. \tag{3.40}$$

Sums over the variables $\mathbf{K}, \mathbf{k}, \mathbf{k}'$ must be intended as over the BZ.

Proof 3.8 — Proof of Eq. (3.40): $\mathcal{F}$-transform of $\hat{H}_U$.

Using the convention of Eq. (3.35), we get for $\hat{H}_U$:

$$\begin{aligned}
\hat{H}_U &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}^\dagger_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \\
&\stackrel{\mathcal{F}}{=} U \sum_{\mathbf{r}\in\mathcal{L}} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i\mathbf{k}_1\cdot\mathbf{r}} \hat{c}^\dagger_{\mathbf{k}_1\uparrow} e^{i\mathbf{k}_2\cdot\mathbf{r}} \hat{c}^\dagger_{\mathbf{k}_2\downarrow} e^{-i\mathbf{k}_3\cdot\mathbf{r}} \hat{c}_{\mathbf{k}_3\downarrow} e^{-i\mathbf{k}_4\cdot\mathbf{r}} \hat{c}_{\mathbf{k}_4\uparrow} \\
&= \frac{U}{L_x L_y} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} \hat{c}^\dagger_{\mathbf{k}_1\uparrow} \hat{c}^\dagger_{\mathbf{k}_2\downarrow} \hat{c}_{\mathbf{k}_3\downarrow} \hat{c}_{\mathbf{k}_4\uparrow} \times \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \\
&= \frac{U}{L_x L_y} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} \hat{c}^\dagger_{\mathbf{k}_1\uparrow} \hat{c}^\dagger_{\mathbf{k}_2\downarrow} \hat{c}_{\mathbf{k}_3\downarrow} \hat{c}_{\mathbf{k}_4\uparrow} \delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4} \,,
\end{aligned}$$

where we used Eq. (3.36). Let $\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3 + \mathbf{k}_4 = \mathbf{K}$, and define $\mathbf{k}$, $\mathbf{k}'$ such that

$$\mathbf{k}_1 \equiv \mathbf{K} + \mathbf{k}\,, \qquad \mathbf{k}_2 \equiv \mathbf{K} - \mathbf{k}\,, \qquad \mathbf{k}_3 \equiv \mathbf{K} - \mathbf{k}'\,, \qquad \mathbf{k}_4 \equiv \mathbf{K} + \mathbf{k}'\,. \tag{3.41}$$

Substituting, we get the expression of Eq. (3.40).

Non-local attraction. In o.s. sector, the non-local attraction transforms as:

$$\hat{H}_V^{(\text{o.s.})} \overset{\mathcal{F}}{=} -\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos\left(\delta k_x\right) + \cos\left(\delta k_y\right)\right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow}\,, \tag{3.42}$$

where $\delta k_\ell \equiv k_\ell - k'_\ell$. In the s.s. sector:

$$\begin{aligned} \hat{H}_V^{(\text{s.s.})} \overset{\mathcal{F}}{=} -\frac{V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} &\left[\cos\left(\delta k_x\right) + \cos\left(\delta k_y\right)\right] \\ &\times \left[\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\uparrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\uparrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} + \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\downarrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\downarrow}\right]\,. \end{aligned} \tag{3.43}$$

Sums over the variables $\mathbf{K}, \mathbf{k}, \mathbf{k}'$ must be intended as over the BZ.

Proof 3.9 — Proof of Eqs. (3.42), (3.43): $\mathcal{F}$-transform of $\hat{H}_V$.

Let

$$h_{\mathbf{r}\sigma\sigma'} \equiv -V \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'}\right)\,, \tag{3.44}$$

which implies

$$\hat{H}_V = \sum_{\mathbf{r}\in\mathcal{L}/2} \sum_{\sigma,\sigma'} h_{\mathbf{r}\sigma\sigma'}\,. \tag{3.45}$$

The product of densities on neighbouring sites transforms as:

$$\begin{aligned} \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'} &= \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\sigma} \\ &\overset{\mathcal{F}}{=} \frac{1}{(L_xL_y)^2} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} e^{\pm i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}} \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}\,. \end{aligned} \tag{3.46}$$

Then, inserting Eq. (3.46) in Eq. (3.44), we obtain

$$\begin{aligned} h_{\mathbf{r}\sigma\sigma'} &\overset{\mathcal{F}}{=} -\frac{V}{(L_xL_y)^2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \\ &\qquad\qquad \times \left(e^{i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}} + e^{-i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}\right) \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma} \\ &= -\frac{2V}{(L_xL_y)^2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \cos\left[(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}\right] \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}\,. \end{aligned}$$

The full non-local attraction $\hat{H}_V$ is given, as in Eq. (3.45), by summing over all sites of $\mathcal{L}/2$ (i.e. one sublattice). Applying Eq. (3.37) gives back momentum conservation

$$\frac{1}{L_xL_y} \sum_{\mathbf{r}\in\mathcal{L}/2} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}\,,$$

where we used Eq. (3.37). We define $\mathbf{k}$, $\mathbf{k}'$ and $\mathbf{K}$ as we did fo the local repulsion in Eq. (3.41), and

$$\delta\mathbf{k} \equiv \mathbf{k} - \mathbf{k}'\,.$$

Hence:

$$\begin{aligned}\hat{H}_V &= \sum_{\mathbf{r}\in\mathcal{L}/2}\sum_{\sigma,\sigma'} h_{\mathbf{r}\sigma\sigma'} \\ &\overset{\mathcal{F}}{=} -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\cos\left(\delta\mathbf{k}\cdot\boldsymbol{\delta}\right)\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \\ &= -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\,. \end{aligned} \tag{3.47}$$

Finally, we can separate in same spin and opposite spin channels. This gives:

$$\begin{aligned}\hat{H}_V \overset{\mathcal{F}}{=} &\overbrace{-\frac{V}{L_xL_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}^{\text{s.s.}} \\ &\underbrace{-\frac{V}{L_xL_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}_{\text{o.s.}}\,.\end{aligned}$$

The s.s. part confirms Eq. (3.43). The o.s. part can be simplified: since

$$\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} = \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}$$

due to fermionic anti-commutation rules, and since the composite transform

$$\mathbf{k}\to-\mathbf{k} \quad \text{and} \quad \mathbf{k}'\to-\mathbf{k}'$$

leaves the form factor $\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)$ invariant, we reduce the spin sum to two times the $\sigma=\uparrow$ part. This confirms Eq. (3.42)

In reciprocal space, following the general decomposition presented in Eq. (C.9) for a quartic operator $\hat{c}^{\dagger}_{\mathbf{k}_1}\hat{c}^{\dagger}_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4}$, we distinguish contractions as follows:

- Hartree contractions:
$$\hat{c}^{\dagger}_{\mathbf{k}_1}\hat{c}^{\dagger}_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4} \quad \text{and} \quad \hat{c}^{\dagger}_{\mathbf{k}_1}\hat{c}^{\dagger}_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4}\,.$$
- Fock contractions:
$$\hat{c}^{\dagger}_{\mathbf{k}_1}\hat{c}^{\dagger}_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4} \quad \text{and} \quad \hat{c}^{\dagger}_{\mathbf{k}_1}\hat{c}^{\dagger}_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4}\,.$$
- Cooper contractions:
$$\hat{c}^{\dagger}_{\mathbf{k}_1}\hat{c}^{\dagger}_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4} \quad \text{and} \quad \hat{c}^{\dagger}_{\mathbf{k}_1}\hat{c}^{\dagger}_{\mathbf{k}_2}\hat{c}_{\mathbf{k}_3}\hat{c}_{\mathbf{k}_4}\,.$$

with appropriate sign factors.

3.4 Square lattice spatial symmetry channels

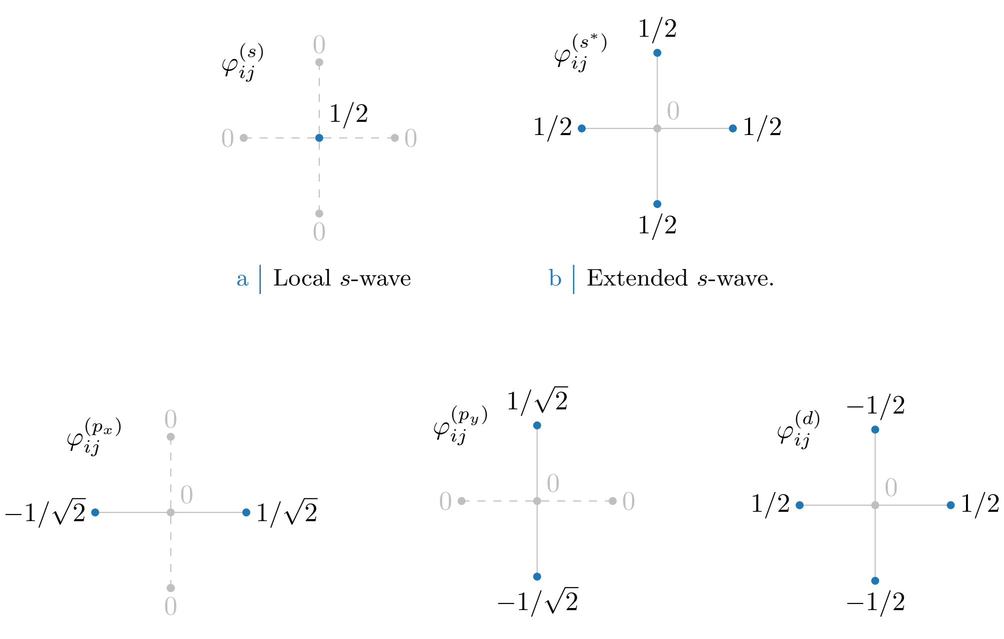
In this section, we discuss the decomposition in different spatial symmetry channels for expectation values defined on the square lattice. We define the set of functions $\varphi^{(\gamma)}_{\mathbf{r}\mathbf{r}'}$, with $\gamma\in\{s,s^*,p_x,p_y,d\}$, defined in Tab. 3.4 and plotted in Fig. 3.4. We will often refer to them as "harmonics". By construction, these functions obey an orthonormality condition:

$$\frac{1}{L_xL_y}\sum_{\mathbf{r}\mathbf{r}'}\varphi^{(\gamma)}_{\mathbf{r}\mathbf{r}'}\varphi^{(\gamma')*}_{\mathbf{r}\mathbf{r}'} = \delta_{\gamma\gamma'}\,, \tag{3.48}$$

which is proven by direct computation.

Channel	Form factor	Graph
s-wave	$\varphi^{(s)}_{\mathbf{rr}'} = \delta_{\mathbf{rr}'}$	Fig. 3.4a
s^*-wave	$\varphi^{(s^*)}_{\mathbf{rr}'} = \frac{1}{2}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 3.4b
p_x-wave	$\varphi^{(p_x)}_{\mathbf{rr}'} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}}\right]$	Fig. 3.4c
p_y-wave	$\varphi^{(p_y)}_{\mathbf{rr}'} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 3.4d
$d_{x^2-y^2}$-wave	$\varphi^{(d)}_{\mathbf{rr}'} = \frac{1}{2}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 3.4e

Table 3.4 | The harmonics $\varphi^{(\gamma)}_{\mathbf{rr}'}$. The notation of Fig. 3.3 is used. In the last column, we indicate a graphic representation given in the subfigures Fig. 3.4.



a | Local s-wave

b | Extended s-wave.

c | p_x-wave.

d | p_y-wave.

e | $d_{x^2-y^2}$-wave.

Figure 3.4 | The harmonics listed in Tab. 3.4. In figures five sites are represented: the hub site i and its four NNs. Solid lines connect the hub sites to a rim site where the harmonic is non-zero. Dashed lines connect to a rim site where the harmonic is zero.

Decomposition in symmetry channels. Consider a generic function of two variables $f(\mathbf{r}, \mathbf{r}')$. Suppose that, for a given centre $\mathbf{r}$, the function is non-zero only on the site itself ($\mathbf{r}' = \mathbf{r}$) and its NNs $\mathbf{r}' = \mathbf{r} \pm \boldsymbol{\delta}$, with $\boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}$. In other words,

$$f(\mathbf{r}, \mathbf{r}') \neq 0 \quad \Longleftrightarrow \quad |\mathbf{r} - \mathbf{r}'| = 0, 1\,, \tag{3.49}$$

in units of the lattice step. Using the harmonics $\varphi^{(\gamma)}_{\mathbf{rr}'}$, we get the following decomposition:

$$f(\mathbf{r}, \mathbf{r}') = \sum_{\gamma} f^{(\gamma)} \varphi^{(\gamma)}_{\mathbf{rr}'}\,, \tag{3.50}$$

where we defined:

$$f^{(\gamma)} = \frac{1}{L_x L_y} \sum_{\mathbf{r},\mathbf{r}'} f(\mathbf{r}, \mathbf{r}') \varphi^{(\gamma)}_{\mathbf{rr}'}\,. \tag{3.51}$$

Eq. (3.50) will be referred to as "decomposition in symmetry channels". The sum over γ is extended to the five symmetries of Tab. 3.4.

Proof 3.10 — Decomposition in symmetry channels of Eq. (3.50).

Consider a function $f(\mathbf{r},\mathbf{r}')$ defined as in Eq. (3.49) and the harmonics of Tab. 3.4. By direct computation, the following identity holds

$$\begin{aligned} f(\mathbf{r},\mathbf{r}+\underline{\mathbf{x}}) &= \sum_{\mathbf{r}'\in\mathcal{L}} f(\mathbf{r},\mathbf{r}')\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} \\ &= \sum_{\mathbf{r}'\in\mathcal{L}} f(\mathbf{r},\mathbf{r}')\left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} + \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} + \sqrt{2}\varphi^{(p_x)}_{\mathbf{r}\mathbf{r}'}\right) . \end{aligned}$$

Similarly, also the following relations hold

$$\begin{aligned} f(\mathbf{r},\mathbf{r}) &= \sum_{\mathbf{r}'\in\mathcal{L}} f(\mathbf{r},\mathbf{r}')\varphi^{(s)}_{\mathbf{r}\mathbf{r}'} , \\ f(\mathbf{r},\mathbf{r}-\underline{\mathbf{x}}) &= \sum_{\mathbf{r}'\in\mathcal{L}} f(\mathbf{r},\mathbf{r}')\left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} + \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} - \sqrt{2}\varphi^{(p_x)}_{\mathbf{r}\mathbf{r}'}\right) , \\ f(\mathbf{r},\mathbf{r}+\underline{\mathbf{y}}) &= \sum_{\mathbf{r}'\in\mathcal{L}} f(\mathbf{r},\mathbf{r}')\left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} - \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} + \sqrt{2}\varphi^{(p_y)}_{\mathbf{r}\mathbf{r}'}\right) , \\ f(\mathbf{r},\mathbf{r}-\underline{\mathbf{y}}) &= \sum_{\mathbf{r}'\in\mathcal{L}} f(\mathbf{r},\mathbf{r}')\left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} - \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} - \sqrt{2}\varphi^{(p_y)}_{\mathbf{r}\mathbf{r}'}\right) . \end{aligned}$$

Hence, by defining

$$\begin{aligned} f^{(s)} &\equiv f(\mathbf{r},\mathbf{r}) , \\ f^{(s^*)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{x}}) + f(\mathbf{r},\mathbf{r}-\underline{\mathbf{x}}) + f(\mathbf{r},\mathbf{r}+\underline{\mathbf{y}}) + f(\mathbf{r},\mathbf{r}-\underline{\mathbf{y}}) , \\ f^{(p_x)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{x}}) - f(\mathbf{r},\mathbf{r}-\underline{\mathbf{x}}) , \\ f^{(p_y)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{y}}) - f(\mathbf{r},\mathbf{r}-\underline{\mathbf{y}}) , \\ f^{(d)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{x}}) + f(\mathbf{r},\mathbf{r}-\underline{\mathbf{x}}) - f(\mathbf{r},\mathbf{r}+\underline{\mathbf{y}}) - f(\mathbf{r},\mathbf{r}-\underline{\mathbf{y}}) . \end{aligned}$$

we get the decomposition:

$$f(\mathbf{r},\mathbf{r}') = \sum_{\gamma} f^{(\gamma)}\varphi^{(\gamma)}_{\mathbf{r}\mathbf{r}'} .$$

The function f depends only on the difference of its two spatial variables. In other words, all the $f^{(\gamma)}$ components we defined above are independent of $\mathbf{r}$. Hence we can redefine them as in Eq. (3.51), averaging over $\mathbf{r}$. The decomposition obtained above still holds. With this redefinition of the $f^{(\gamma)}$ components, we recognise Eq. (3.50).

We decomposed the original function in its harmonics. Moving to reciprocal space, we get the $\mathcal{F}$-transformed functions of Tab. 3.5, also satisfying the orthonormality condition:

$$\frac{1}{L_x L_y}\sum_{\mathbf{k}} \varphi^{(\gamma)}_{\mathbf{k}}\varphi^{(\gamma')*}_{\mathbf{k}} = \delta_{\gamma\gamma'} .$$

The functions of Tab. 3.5 are plotted in the BZ in Fig. 3.5.

A useful identity. Using the harmonics functions, it is useful to prove the following identity:

$$\cos(\delta k_x) + \cos(\delta k_y) = \frac{1}{2}\sum_{\gamma\neq s} \varphi^{(\gamma)}_{\mathbf{k}}\varphi^{(\gamma)*}_{\mathbf{k}'} \tag{3.52}$$

$$= \frac{1}{2}\sum_{\gamma\neq s} \varphi^{(\gamma)*}_{\mathbf{k}}\varphi^{(\gamma)}_{\mathbf{k}'} . \tag{3.53}$$

Structure	Structure factor	Graph
s-wave	$\varphi_{\mathbf{k}}^{(s)} = 1$	Fig. 3.4a
Extended s-wave	$\varphi_{\mathbf{k}}^{(s^*)} = \cos k_x + \cos k_y$	Fig. 3.4b
p_x-wave	$\varphi_{\mathbf{k}}^{(p_x)} = i\sqrt{2}\sin k_x$	Fig. 3.4c
p_y-wave	$\varphi_{\mathbf{k}}^{(p_y)} = i\sqrt{2}\sin k_y$	Fig. 3.4d
$d_{x^2-y^2}$-wave	$\varphi_{\mathbf{k}}^{(d)} = \cos k_x - \cos k_y$	Fig. 3.4e

Table 3.5 | $\mathcal{F}$-transforms of the harmonics of Tab. 3.4. These functions are plotted in Fig. 3.5.

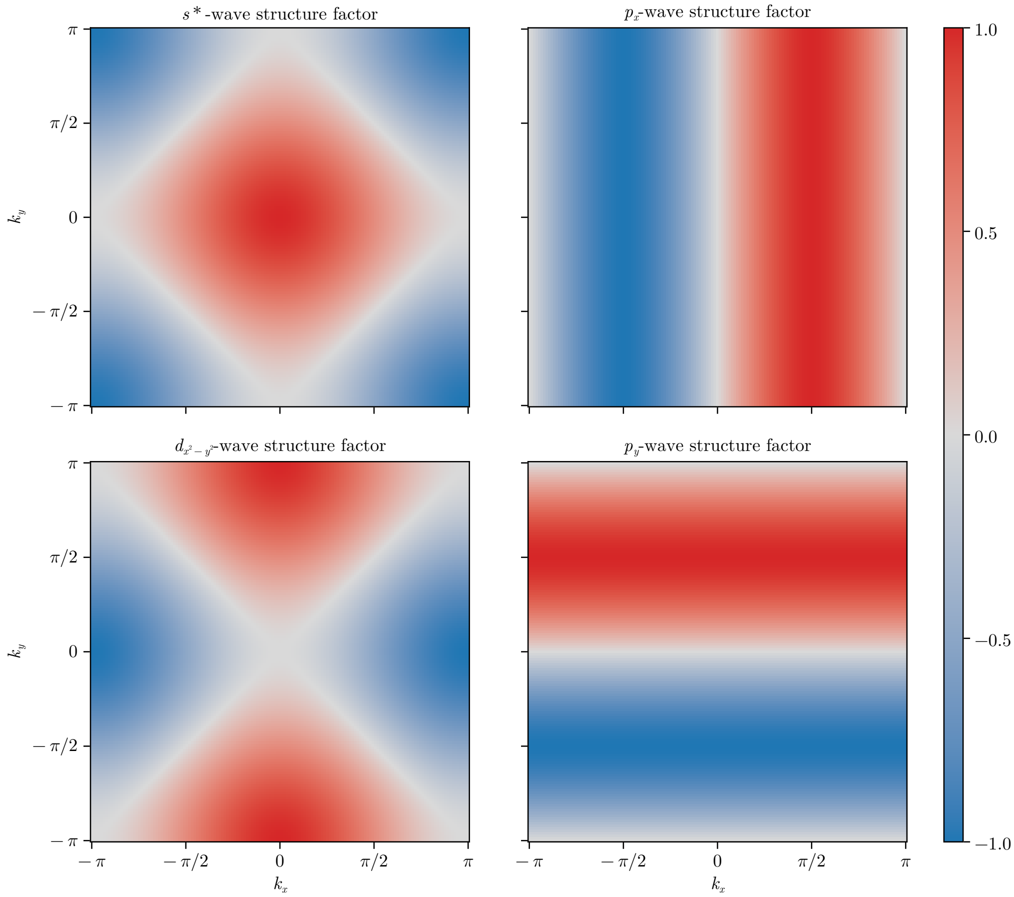


Figure 3.5 | Representation of the various structure factors listed in Tab. 3.5 on the BZ. For the s^* and $d_{x^2-y^2}$-wave factors, the real part is plotted; for the p_ℓ factors, the imaginary part is plotted.

Note that in both rows above the sum is intended for $\gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$, excluding s.

Proof 3.11 — Proof of Eq. (3.52).

Let $k_\ell \in \mathbb{R}$, with $\ell \in \{x, y\}$. We define:

$$c_\ell \equiv \cos k_\ell\,, \qquad s_\ell \equiv \sin k_\ell\,, \qquad c'_\ell \equiv \cos k'_\ell\,, \qquad s'_\ell \equiv \sin k'_\ell\,.$$

Let $\delta k_\ell \equiv k_\ell - k'_\ell$. Using basic trigonometric identities we get:

$$\cos(\delta k_x) + \cos(\delta k_y) = \underbrace{\left(c_x c'_x + c_y c'_y\right)}_{\text{Symmetric}} + \underbrace{\left(s_x s'_x + s_y s'_y\right)}_{\text{Anti-symmetric}} . \tag{3.54}$$

All four terms are invariant under simultaneous inversion of both momenta, $(\mathbf{k}, \mathbf{k}') \to (-\mathbf{k}, -\mathbf{k}')$. Instead, if we invert just one momentum of the pair, the first two are symmetric for both arguments $\mathbf{k}$, $\mathbf{k}'$; the second two are anti-symmetric. Decoupling the symmetric part, we obtain

$$c_x c'_x + c_y c'_y = \frac{1}{2}(c_x + c_y)(c'_x + c'_y) + \frac{1}{2}(c_x - c_y)(c'_x - c'_y) ,$$

which finally gives:

$$\begin{aligned}
\cos(\delta k_x) + \cos(\delta k_y) = {} & \frac{1}{2}(c_x + c_y)(c'_x + c'_y) && (s^*\text{-wave}) \\
& + s_x s'_x && (p_x\text{-wave}) \\
& + s_y s'_y && (p_y\text{-wave}) \\
& + \frac{1}{2}(c_x - c_y)(c'_x - c'_y) , && (d_{x^2-y^2}\text{-wave})
\end{aligned}$$

which is written in compact form as in Eq. (3.52).

Chapter 4

Mott localisation in the normal phase

In this chapter, we study the normal phase, namely the phase characterised by the absence of any spontaneous symmetry breaking mechanism.

We show that the non-local pairing induces a renormalisation of the hopping amplitude, leading to a Mott localisation effect. The chapter is organised as follows. In Sec. 4.1 we discuss the symmetries of the normal phase. In Secs. 4.2, 4.3 we derive the HF approximation to the EHM in the normal phase. In Sec. 4.4 we analyse the self-consistency equations and derive constraints on the self-consistent parameters identified by the Hartree-Fock method. In Sec. 4.5 we present the result of numeric simulations.

4.1 The normal phase and its symmetries

In the normal phase no symmetry is broken, as summarised in Tab. 3.1. Therefore, the HF approximation to the hamiltonian must exhibit the full $\mathrm{U}(2) \otimes \mathrm{C}_4 \otimes \mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$ symmetry discussed in Sec. 3.1.

We anticipate here that we will actually allow for $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$, namely, breaking of planar rotational symmetry. As will result from our discussion, the normal phase does not break C_4 even if we allow it to. Nevertheless, we carry out this general derivation because breaking of planar rotational symmetry will be observed in Chap. 6 in the context of superconductivity, and the computations carried out for the normal phase will be re-used.

Translational invariance implies the following result:

$$\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}'\sigma'} \rangle = \delta_{\mathbf{k}=\mathbf{k}'} \delta_{\sigma=\sigma'} \langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \rangle \,. \tag{4.1}$$

Proof 4.1 — Proof of Eq. (4.1).

Consider the generic EV on the left side of Eq. (4.1)

$$\left\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}'\sigma'} \right\rangle ,$$

for two momenta $\mathbf{k}$, $\mathbf{k}'$. Moving to real space through an $\mathcal{F}$-transform, we get

$$\left\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}'\sigma'} \right\rangle \overset{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} \sum_{\mathbf{r}'\in\mathcal{L}} e^{-i\left(\mathbf{k}\cdot\mathbf{r} - \mathbf{k}'\cdot\mathbf{r}'\right)} \left\langle \hat{c}^\dagger_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}'\sigma'} \right\rangle ,$$

where we used the convention of Eq. (3.35). Let now $\Delta\mathbf{r} \equiv \mathbf{r}' - \mathbf{r}$. Translational invariance implies that the function

$$g(\Delta\mathbf{r}, \sigma, \sigma') = \left\langle \hat{c}^\dagger_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}+\Delta\mathbf{r}\sigma'} \right\rangle$$

Operator	Sector	Channel	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
	s.s.	Hartree	μ shift by nzV
		Fock	Bands renormalisation and resymmetrization

Table 4.1 Wick's contractions for each interaction and channel in the HF framework, and their net effect in the normal phase.

is independent of $\mathbf{r}$. Thus we have:

$$\begin{aligned}\langle \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma'}\rangle &\overset{\mathcal{F}}{=} \sum_{\Delta\mathbf{r}\in\mathcal{L}} e^{-i\mathbf{k}'\cdot\Delta\mathbf{r}} g(\Delta\mathbf{r},\sigma,\sigma')\frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}} e^{-i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \\ &= \sum_{\Delta\mathbf{r}\in\mathcal{L}} e^{-i\mathbf{k}'\cdot\Delta\mathbf{r}} g(\Delta\mathbf{r},\sigma,\sigma')\delta_{\mathbf{k}=\mathbf{k}'}\,,\end{aligned}$$

where we used the identity of Eq. (3.36). Then the EV of Eq. (4.1) vanishes for $\mathbf{k}\neq\mathbf{k}'$.

Recall now Eq. (3.33). Upon substituting $\boldsymbol{\delta}\to\Delta\mathbf{r}$, we obtain that $g(\Delta\mathbf{r},\sigma,\overline{\sigma})=0$ for a phase that is SU(2)-symmetric. Then, the EV of Eq. (4.1) is non-zero only if $\sigma=\sigma'$. This proves Eq. (4.1).

Translational invariance and spin rotational symmetry imply that the site occupation is independent of site and spin indices,

$$\langle \hat{n}_{i\sigma}\rangle = n\,, \tag{4.2}$$

with $n\in[0,1]$.

We define the single-state occupation operator in reciprocal space:

$$\hat{n}_{\mathbf{k}\sigma}\equiv \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}$$

and its EV

$$u_{\mathbf{k}\sigma}\equiv\langle\hat{n}_{\mathbf{k}\sigma}\rangle\,,\qquad u_{\mathbf{k}\sigma}=u^*_{\mathbf{k}\sigma}\,. \tag{4.3}$$

The relation on the right follows from the fact that $\hat{n}_{\mathbf{k}\sigma}$ is hermitian by construction. Note that $\hat{n}_{\mathbf{k}\sigma}$ is not the Fourier-transform of $\hat{n}_{i\sigma}$. Finally, using the functions $\varphi^{(\gamma)}_{\mathbf{k}}$ of Tab. 3.5, we define the harmonics

$$u^{(\gamma)}_\sigma \equiv \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}}\langle\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\rangle\,. \tag{4.4}$$

These harmonics are the HFPs of the normal phase, and Eq. (4.4) is their self-consistency equation.

4.2 Application of Hartree-Fock methods to the normal phase

Applying the HF methods to the interactions $\hat{H}_U$ and $\hat{H}_V$, from Tabs. 3.2 and 3.3, we know that in the normal phase the allowed contractions are:

$\hat{H}_U$	Hartree contractions,
$\hat{H}_V$ (o.s. sector)	Hartree contractions,
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions.

As we will see in detail and is summarised in Tab. 4.1, Hartree contractions give a chemical potential shift, while the Fock contractions renormalise the bare bands.

4.2.1 Effects of the Hartree terms: chemical potential renormalisation

In this section we show that the Hartree channels of $\hat{H}_U$ and $\hat{H}_V$ reduce to multiples of the number operator. Applying HF methods, we get the decomposition of Eqs. (2.21), (2.22). Specifically, the Hartree channels are given by

$$\hat{H}_U \overset{\text{HF}}{\simeq} U \sum_i \left(\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}\langle \hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\rangle + \langle \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}\rangle \hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow} - \langle \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}\rangle\langle \hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\rangle \right) ,$$

$$\hat{H}_V \overset{\text{HF}}{\simeq} -V \sum_{\langle ij\rangle}\sum_{\sigma,\sigma'} \left(\hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'}\rangle + \langle \hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\rangle \hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'} - \langle \hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\rangle\langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'}\rangle \right) + (\text{Fock channel}) .$$

Summing the two terms, we obtain

$$\hat{H}_U + \hat{H}_V \overset{\text{HF}}{\simeq} -n(2zV - U)\hat{N} - \left(E^{(\text{N})}_{\text{H},U} + E^{(\text{N})}_{\text{H},V} \right) + (\text{Fock channel}) , \tag{4.5}$$

where we introduces the "Hartree contraction energy", namely a shift to energy coming from the scalar terms of Eqs. (2.21), (2.22):

$$E^{(\text{N})}_{\text{H},U} = -U \sum_{i\in\mathcal{L}} \langle \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}\rangle\langle \hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\rangle , \tag{4.6}$$

$$E^{(\text{N})}_{\text{H},V} = V \sum_{\langle ij\rangle}\sum_{\sigma,\sigma'} \langle \hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\rangle\langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'}\rangle . \tag{4.7}$$

From Eq. (4.5), we see that the chemical potential is shifted as

$$\tilde{\mu} \equiv \mu + n(2zV - U) , \tag{4.8}$$

as is reported in Tab. 4.1. This result is valid for all the other phases discussed in this thesis.

Proof 4.2 — Proof of Eq. (4.5): HF approximation of $\hat{H}_U + \hat{H}_V$ in Hartree channel.

We analyse $\hat{H}_U$ and $\hat{H}_V$ separately.

Local repulsion. In the normal phase, the local repulsion $\hat{H}_U$ decomposes in the Hartree channel as is specified in Eq. (2.21),

$$\hat{H}_U \overset{\text{HF}}{\simeq} U \sum_i \left(\langle \hat{n}_{i\uparrow}\rangle \hat{n}_{i\downarrow} + \hat{n}_{i\uparrow}\langle \hat{n}_{i\downarrow}\rangle - \langle \hat{n}_{i\downarrow}\rangle\langle \hat{n}_{i\uparrow}\rangle \right) .$$

Using Eq. (4.2), we obtain:

$$\hat{H}_U \overset{\text{HF}}{\simeq} nU \sum_i (\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}) - E^{(\text{N})}_{\text{H},U} \tag{4.9}$$

$$= nU \times \hat{N} - E^{(\text{N})}_{\text{H},U} . \tag{4.10}$$

Non-local attraction. The non-local attraction is divided in two sectors, as in Eq. (3.26). From Eq. (2.22), its Hartree-like terms are:

$$\hat{H}_V \overset{\text{HF}}{\simeq} \overbrace{-V \sum_{\langle ij\rangle}\sum_{\sigma} \left(\langle \hat{n}_{i\sigma}\rangle \hat{n}_{j\sigma} + \hat{n}_{i\sigma}\langle \hat{n}_{j\sigma}\rangle - \langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\sigma}\rangle \right)}^{\text{s.s.}} \underbrace{-V \sum_{\langle ij\rangle}\sum_{\sigma} \left(\langle \hat{n}_{i\sigma}\rangle \hat{n}_{j\bar{\sigma}} + \hat{n}_{i\sigma}\langle \hat{n}_{j\bar{\sigma}}\rangle - \langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\bar{\sigma}}\rangle \right)}_{\text{o.s.}} + (\text{Fock channel}) . \tag{4.11}$$

Using Eq. (4.2), we get

$$\hat{H}_V \overset{\text{HF}}{\simeq} \underbrace{-nV\sum_{\langle ij\rangle}\sum_{\sigma}(\hat{n}_{j\sigma}+\hat{n}_{i\sigma})}_{\text{s.s.}}\underbrace{-nV\sum_{\langle ij\rangle}\sum_{\sigma}(\hat{n}_{j\bar{\sigma}}+\hat{n}_{i\sigma})}_{\text{o.s.}}-E^{(\text{N})}_{\text{H},V}+(\text{Fock channel})\,, \tag{4.12}$$

Both sums give a number operator. Note that we sum over NNs: each site is connected to $z = 4$ sites. Including this combinatorial factor, we have

$$\hat{H}_V \simeq -2nzV\times\hat{N}-E^{(\text{N})}_{\text{H},V}+(\text{Fock channel})\,. \tag{4.13}$$

Summing Eqs. (4.10) and (4.13), we obtain Eq. (4.5).

The contraction energies of Eqs. (4.6), (4.7) read

$$E^{(\text{N})}_{\text{H},U} = -n^2U \tag{4.14}$$
$$E^{(\text{N})}_{\text{H},V} = L_xL_y\times 2zn^2V\,. \tag{4.15}$$

where $z = 4$ is the lattice coordination number.

Proof 4.3 — Proof of Eqs. (4.14), (4.15): Hartree contraction energies.

Using Eq. (4.2) in Eq. (4.6), we obtain

$$\begin{aligned}E^{(\text{N})}_{\text{H},U} &= -U\sum_{\mathbf{r}\in\mathcal{L}} n\\ &= -n^2U\,,\end{aligned}$$

which proves Eq. (4.14). Doing the same in Eq. (4.7), we get

$$\begin{aligned}E^{(\text{N})}_{\text{H},V} &= V\sum_{\langle ij\rangle}\sum_{\sigma}\left(\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\sigma}\rangle+\langle\hat{n}_{i\sigma}\rangle\langle\hat{n}_{j\bar{\sigma}}\rangle\right)\\ &= 2V\sum_{\langle ij\rangle}\sum_{\sigma} n^2\\ &= L_xL_y\times 2zn^2V\,,\end{aligned}$$

where in last line we used Eq. (3.22). We obtained Eq. (4.15).

4.2.2 Effects of the Fock terms: bare bands renormalisation

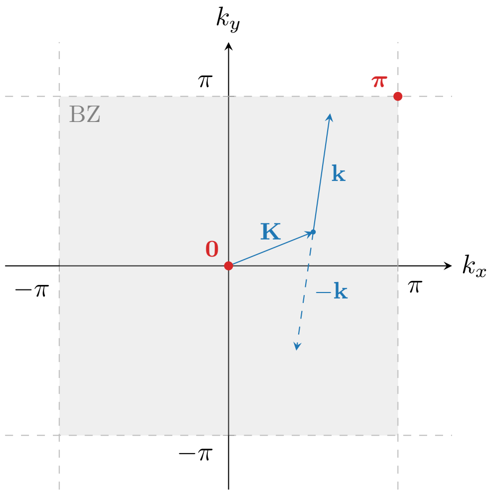
From the Wick's decomposition of $\hat{H}_V$ (see Eq. (2.22)), the only Fock term allowed in the normal phase comes from the same-spin sector. See Tab. 3.3 for a summary. Explicitly,

$$\hat{H}_V \overset{\text{HF}}{\simeq} (\text{Hartree channel})+V\sum_{\langle ij\rangle}\sum_{\sigma}\left(\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right)\,. \tag{4.16}$$

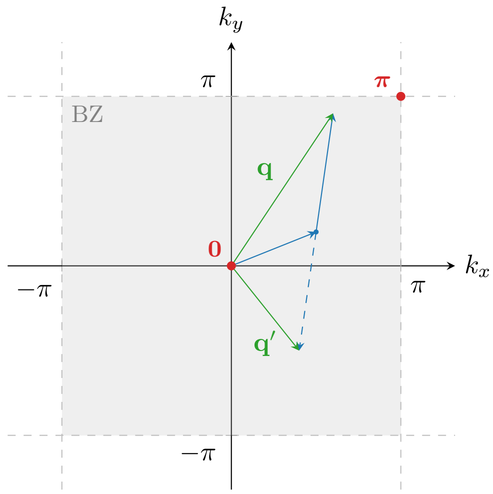
We notice that the *plus* sign in front of the displayed term comes from the *minus* sign in Fock term of Eq. (2.1), applied to the interaction $-V$.

Using the Fourier transform of $\hat{H}_V$ given in Eq. (3.43), we obtain

$$\begin{aligned}\hat{H}_V \overset{\text{HF}}{\simeq}\ &(\text{Hartree channel})\\ &+\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left[\cos(\delta k_x)+\cos(\delta k_y)\right]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}-E^{(\text{N})}_{\text{F},V}\,.\end{aligned} \tag{4.17}$$



a | Generic vectors.



b | Change of variables.

Figure 4.1 | Representation of the variable change employed to pass from Eq. (4.20) to (4.21). In Fig. 4.1a generic vectors are considered, cycling over all values of $\mathbf{K}, \mathbf{k} \in \mathrm{BZ}$. In Fig. 4.1b is depicted the variables change to the new vectors $\mathbf{q}, \mathbf{q}'$.

Here, the 2 prefactor comes from the fact that the first two operators in square brackets in Eq. (4.16) are equal in reciprocal space. The contraction energy shift is given by

$$E^{(\mathrm{N})}_{\mathrm{F},V} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} [\cos(\delta k_x) + \cos(\delta k_y)] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \langle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle \,. \tag{4.18}$$

By recalling the definition of $u_{\mathbf{k}\sigma} = \langle \hat{n}_{\mathbf{k}\sigma} \rangle$, the non local attraction is approximated, in Fock channel, by:

$$\hat{H}_V \overset{\mathrm{HF}}{\simeq} (\text{Hartree channel}) + V \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \left(u^{(s^*)}_{\sigma} \varphi^{(s^*)*}_{\mathbf{k}} + u^{(d)}_{\sigma} \varphi^{(d)*}_{\mathbf{k}} \right) \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} - E^{(\mathrm{N})}_{\mathrm{F},V} \,, \tag{4.19}$$

where $u^{(\gamma)}_{\sigma}$ are the harmonics of the occupation $u_{\mathbf{k}\sigma}$, defined in Eq. (4.4).

Proof 4.4 — Proof of Eq. (4.19): HF approximation of $\hat{H}_V$ in Fock channel.

Consider Eq. (4.17). Applying Eq. (4.1), it becomes:

$$\begin{aligned} \hat{H}_V \overset{\mathrm{HF}}{\simeq} & (\text{Hartree channel}) \\ & + \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} (\cos 2k_x + \cos 2k_y) \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} - E^{(\mathrm{N})}_{\mathrm{F},V} \,. \end{aligned} \tag{4.20}$$

Let now $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$, $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. A representation of this variables change is given in Fig. 4.1. Being (for $\ell = x, y$)

$$\begin{aligned} \delta q_\ell &\equiv q_\ell - q'_\ell \\ &= (K_\ell + k_\ell) - (K_\ell - k_\ell) \\ &= 2k_\ell \,, \end{aligned}$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. 3.4b
$d_{x^2-y^2}$-wave	$u^{(d)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. 3.4e

Table 4.2 | Symmetry resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

we reduce Eq. (4.17) to

$$\hat{H}_V \overset{\mathrm{HF}}{\simeq} (\text{Hartree channel}) + \frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} [\cos(\delta q_x) + \cos(\delta q_y)] \underbrace{\langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle}_{u_{\mathbf{q}\sigma}} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F},V} \,, \quad (4.21)$$

where we recognised $u_{\mathbf{q}\sigma}$, as was defined in Eq. (4.3). Using Eq. (3.52), we obtain

$$\hat{H}_V \overset{\mathrm{HF}}{\simeq} (\text{Hartree channel}) + \sum_{\gamma,\sigma} \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} u_{\mathbf{q}\sigma} \right) \times V \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F},V} \,, \quad (4.22)$$

where we used the functions defined in Tab. 3.5. We recognise the harmonics $u^{(\gamma)}_\sigma$ from Eq. (4.4),

$$\hat{H}_V \overset{\mathrm{HF}}{\simeq} (\text{Hartree channel}) + V \sum_{\mathbf{q}'\in\mathrm{BZ}} \sum_{\gamma,\sigma} u^{(\gamma)}_\sigma \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F},V} \,, \quad (4.23)$$

which is almost Eq. (4.19).

To complete the proof, we recognise that in the normal phase inversion symmetry is not broken, thus

$$u_{\mathbf{q}\sigma} = u_{-\mathbf{q}\sigma} \,.$$

This follows from the fact that in the normal phase the EV of the operator $\hat{n}_{\mathbf{k}\sigma}$ must be inversion-symmetric. It follows that antisymmetric harmonics are zero,

$$u^{(p_\ell)}_\sigma = 0 \quad \text{for} \quad \ell \in \{x, y\} \,. \quad (4.24)$$

Using this observation in Eq. (4.23), we obtain Eq. (4.19).

Bringing together Eqs. (4.5) and (4.19), we obtain that the quadratic HF approximation to the EHM, for the normal phase, depends on the renormalised chemical potential $\tilde{\mu}$, and the occupation harmonics $u^{(s^*)}$ and $u^{(d)}$. The chemical potential $\tilde{\mu}$ is fixed by density and determined during computations. To find the ground-state, we must determine self-consistently the two HFPs $u^{(s^*)}_\sigma$, $u^{(d)}_\sigma$ through the self-consistency Eq. (4.4), which is reported symmetry-resolved in Tab. 4.2.

The operator of Eq. (4.19) has the same algebraic structure of the kinetic operator $\hat{H}_t$ in reciprocal space, given in Eq. (3.38). We can collapse the two into one effective kinetic operator with renormalised bands. Therefore,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u^{(s^*)}_\sigma V(\cos k_x + \cos k_y) + u^{(d)}_\sigma V(\cos k_x - \cos k_y) \,. \quad (4.25)$$

In Fock channel, we have:

$$\hat{H}_t + \hat{H}_V \overset{\mathrm{HF}}{\simeq} (\text{Hartree channel}) + \underbrace{\sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \tilde{\epsilon}_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}}_{\hat{H}_{\tilde{t}}} - E^{(\mathrm{N})}_{\mathrm{F},V} \,, \quad (4.26)$$

where we defined the effective kinetic operator, $\hat{H}_{\tilde{t}}$.

The renormalised bands of Eq. (4.25) are a sum of s^* and d-wave components. Instead, bare bands as defined Eq. (3.39) are purely s^*-symmetric functions,

$$\epsilon_{\mathbf{k}\sigma} = -2t(\cos k_x + \cos k_y) = -4t\varphi^{(s^*)}_{\mathbf{k}} \,. \tag{4.27}$$

We define the symmetry-resolved hoppings,

$$\tilde{t}^{(s^*)}_{\sigma} \equiv t - \frac{u^{(s^*)}_{\sigma} V}{2} \quad \text{and} \quad \tilde{t}^{(d)}_{\sigma} \equiv -\frac{u^{(d)}_{\sigma} V}{2} \,, \tag{4.28}$$

so that the bands renormalisation of Eq. (4.25) can be written as

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv -4\tilde{t}^{(s^*)}_{\sigma}\varphi^{(s^*)}_{\mathbf{k}} - 4\tilde{t}^{(d)}_{\sigma}\varphi^{(d)}_{\mathbf{k}} \,. \tag{4.29}$$

We notice the formal similarity with Eq. (4.27), showing how bands renormalisation reduces to a renormalisation of the hopping parameter t,

$$t \ (s^*\text{-wave}) \quad \rightarrow \quad \tilde{t}^{(s^*)}_{\sigma} \ (s^*\text{-wave}) \text{ and } \tilde{t}^{(d)}_{\sigma} \ (d\text{-wave}) \,.$$

As we will discuss in Sec. 4.3.1, the presence of $\tilde{t}^{(d)} \neq 0$ implies a nematic state that breaks C_4.

We conclude this section by proving that the Fock contraction energy is given by:

$$E^{(\mathrm{N})}_{\mathrm{F},V} = L_x L_y \times \frac{V}{2} \sum_{\sigma} \left(\left|u^{(s^*)}_{\sigma}\right|^2 + \left|u^{(d)}_{\sigma}\right|^2 \right) \,. \tag{4.30}$$

Later, using spin degeneracy, we will omit the sum on σ and the $1/2$ prefactor.

Proof 4.5 — Proof of Eq. (4.30): non-local Fock contraction energy.

Recall Eq. (4.18). Using Eq. (4.1), it reduces to:

$$E^{(\mathrm{N})}_{\mathrm{F},V} = \frac{V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos\left(\delta q_x\right) + \cos\left(\delta q_y\right)\right] \langle \hat{c}^{\dagger}_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}\sigma}\rangle \langle \hat{c}^{\dagger}_{\mathbf{q}'\sigma}\hat{c}_{\mathbf{q}'\sigma}\rangle \,.$$

Inserting Eq. (4.3), we get:

$$\begin{aligned} E^{(\mathrm{N})}_{\mathrm{F},V} &= L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}\in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} u_{\mathbf{q}\sigma} \right) \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in \mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} u^{*}_{\mathbf{q}'\sigma} \right) \\ &= L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left|u^{(\gamma)}_{\sigma}\right|^2 , \end{aligned}$$

which, using the fact that $u^{(p_\ell)}_{\sigma} = 0$, proves Eq. (4.30).

4.3 The HF hamiltonian and its properties

Collecting all the various terms, the HF approximation to the EHM hamiltonian in the normal phase is given by

$$\hat{H} - \mu\hat{N} \overset{\mathrm{HF}}{\simeq} \sum_{\mathbf{k}\in \mathrm{BZ}} \sum_{\sigma} \tilde{\epsilon}_{\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} - \tilde{\mu}\hat{N} + E^{(\mathrm{N})}_{\mathrm{c}} \,, \tag{4.31}$$

where in first line we included the all contraction energies from Eqs. (4.14), (4.15) and (4.30) in $E^{(\mathrm{N})}_{\mathrm{c}}$,

$$E^{(\mathrm{N})}_{\mathrm{c}} \equiv -\left(E^{(\mathrm{N})}_{\mathrm{H},U} + E^{(\mathrm{N})}_{\mathrm{H},V} + E^{(\mathrm{N})}_{\mathrm{F},V} \right) \,. \tag{4.32}$$

All dependence on the HFPs is inside the renormalised bands of Eq. (4.25) and the contributions to contraction energy.

Apart from energy shifts, this hamiltonian describes a perfectly degenerate Fermi gas whose bare bands are $\tilde{\epsilon}_{\mathbf{k}\sigma}$ as defined in Eq. (4.25), and whose chemical potential is $\tilde{\mu}$ as defined in Eq. (4.8). The latter is fixed self-consistently by setting the density; the former must be determined using self-consistency equations. The thermal state of the approximated hamiltonian obeys Fermi-Dirac statistics,

$$\langle \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}'\sigma'} \rangle = \delta_{\mathbf{k}=\mathbf{k}'} \delta_{\sigma=\sigma'} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \,, \tag{4.33}$$

being f the Fermi-Dirac distribution,

$$f(\epsilon; \beta, \mu) \equiv \frac{1}{e^{\beta(\epsilon-\mu)} + 1} \,.$$

In next subsections, we discuss in detail the renormalisation of the hopping parameter in real space, with a focus on the meaning of the d-wave component, and derive the free energy and entropy densities for the ground-state.

4.3.1 Nematicity: d-wave bands and anisotropic hopping

In Eq. (4.26) we defined $\hat{H}_{\tilde{t}}$ in reciprocal space. In real space, $\hat{H}_{\tilde{t}}$ takes the form

$$\begin{aligned} \hat{H}_{\tilde{t}} &= \hat{H}_t + V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} + \text{h.c.} \right] \\ &= -\sum_{\langle ij \rangle} \sum_{\sigma} \left[\tilde{t}_{ij\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] \,. \end{aligned}$$

where we introduce the renormalised hopping $\tilde{t}_{ij\sigma}$,

$$\tilde{t}_{ij\sigma} \equiv \begin{cases} \frac{1}{2} \left(t - V \langle \hat{c}^{\dagger}_{j\sigma} \hat{c}_{i\sigma} \rangle \right) & \text{if } i, j \text{ are NNs,} \\ 0 & \text{otherwise.} \end{cases}$$

Due to translational invariance, the renormalised hopping $\tilde{t}_{ij\sigma}$ depends only on the difference $i-j$. Therefore we can use the decomposition of Eq. (3.50),

$$\tilde{t}_{ij\sigma} = \sum_{\gamma} \tilde{t}^{(\gamma)}_{\sigma} \varphi^{(\gamma)}_{ij} \,.$$

The harmonics, given by Eq. (3.51), are identified with $\tilde{t}^{(s^*)}_{\sigma}$, $\tilde{t}^{(d)}_{\sigma}$ already introduced in Eq. (4.28). From the same argument we know that $\tilde{t}^{(p_x)}_{\sigma} = \tilde{t}^{(p_y)}_{\sigma} = 0$. Then, the renormalised hopping is explicitly given by

$$\begin{aligned} \tilde{t}_{\mathbf{r}\mathbf{r}'\sigma} = \frac{\tilde{t}^{(s^*)}_{\sigma}}{2} & \left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}} \right] \\ & + \frac{\tilde{t}^{(d)}_{\sigma}}{2} \left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}} \right] \,. \end{aligned}$$

Let us define the direction-resolved renormalised hopping,

$$\tilde{t}^{(x)}_{\sigma} \equiv \frac{\tilde{t}^{(s^*)}_{\sigma} + \tilde{t}^{(d)}_{\sigma}}{2} \,, \tag{4.34}$$

$$\tilde{t}^{(y)}_{\sigma} \equiv \frac{\tilde{t}^{(s^*)}_{\sigma} - \tilde{t}^{(d)}_{\sigma}}{2} \,. \tag{4.35}$$

This shows that, as anticipated, a finite $\tilde{t}^{(d)} \neq 0$ implies a nematic order characterised by the breaking of planar rotational symmetry, highlighted by the fact that kinetic operators in directions x and y differ. We can rewrite $\hat{H}_{\tilde{t}}$ as:

$$\hat{H}_{\tilde{t}} = -\sum_{\mathbf{r}\in\mathcal{L}} \sum_{\sigma} \left[\tilde{t}^{(x)}_{\sigma} \left(\hat{c}^{\dagger}_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}+\underline{\mathbf{x}}\sigma} + \hat{c}^{\dagger}_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}-\underline{\mathbf{x}}\sigma} \right) + \tilde{t}^{(y)}_{\sigma} \left(\hat{c}^{\dagger}_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}+\underline{\mathbf{y}}\sigma} + \hat{c}^{\dagger}_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}-\underline{\mathbf{y}}\sigma} \right) \right] \,.$$

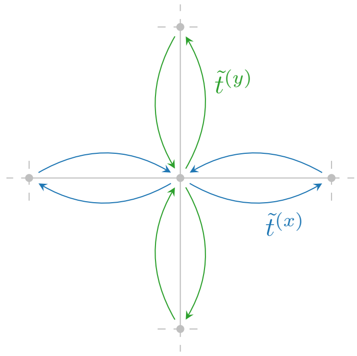


Figure 4.2 | Graphic rendition of anisotropic hopping, with a s^*-wave component (symmetric by $\pi/2$ rotations) and a d-wave component (antisymmetric).

A schematics of anisotropic hopping is given in Fig. 4.2: for each bond linking two sites, the hopping parameter can differ, as the link direction is vertical or horizontal. We notice, however, that even in the presence of a non-zero d-wave hopping the density remains uniform.

Proof 4.6 — A d-wave hopping is compatible with uniform density.

Consider the on-site occupation operator $\hat{n}_{\mathbf{r}\sigma} = \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}$. Applying a Fourier transform as defined in Eq. (3.35), we obtain

$$\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma} \overset{\mathcal{F}}{=} \frac{1}{L_x L_y}\sum_{\mathbf{k},\mathbf{k}'} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma}\,.$$

We compute the expectation value of the above operator and apply Eq. (4.33),

$$\begin{aligned}\langle\hat{n}_{\mathbf{r}\sigma}\rangle &= \frac{1}{L_x L_y}\sum_{\mathbf{k},\mathbf{k}'} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}}\langle\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma}\rangle \\ &= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\,. \end{aligned} \tag{4.36}$$

The right-hand side of last line is independent of $\mathbf{r}$, therefore also $\langle\hat{n}_{\mathbf{r}\sigma}\rangle$ is.

Note that we never used the actual symmetry of the renormalised bands $\tilde{\epsilon}_{\mathbf{k}}$, but only the fact that the thermal state obeys the Fermi-Dirac distribution. A perfectly degenerate Fermi gas is always translational invariant and thus cannot exhibit non-uniform charge density.

From now on, we will omit the spin index σ from both the renormalised bands and the renormalised hopping,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \to \tilde{\epsilon}_{\mathbf{k}} \qquad \forall\alpha \quad : \quad \tilde{t}^{(\alpha)}_\sigma \to \tilde{t}^{(\alpha)}\,.$$

The normal phase does not break spin rotational invariance. Therefore the renormalised bands $\tilde{\epsilon}_{\mathbf{k}}$ must be independent of the spin index. Inverting Eq. (4.28), we can express $\tilde{t}^{(x)}$ and $\tilde{t}^{(y)}$ in terms of the HFPs $u^{(s^*)}$ and $u^{(d)}$

$$\begin{aligned}\tilde{t}^{(x)} &\equiv t - \frac{u^{(s^*)}+u^{(d)}}{2}V\,,\\ \tilde{t}^{(y)} &\equiv t - \frac{u^{(s^*)}-u^{(d)}}{2}V\,.\end{aligned}$$

Inverting these relations we obtain:

$$u^{(s^*)} = \frac{\left(t-\tilde{t}^{(x)}\right)+\left(t-\tilde{t}^{(y)}\right)}{V}\,, \tag{4.37}$$

$$u^{(d)} = \frac{\tilde{t}^{(x)}-\tilde{t}^{(y)}}{V}\,. \tag{4.38}$$

It follows that if $u^{(d)} > 0$, the effective hopping parameter is larger in direction x with respect to direction y. If $u^{(d)} < 0$, roles are inverted.

4.3.2 The entropy density of the normal ground-state

In this section, we compute the density of entropy of the ground state in the normal phase. Entropy density is given by

$$s^{(\mathrm{N})} = \frac{S^{(\mathrm{N})}}{L_x L_y} \,,$$

being $S^{(\mathrm{N})}$ the total entropy.

The HF hamiltonian of Eq. (4.31) describes a non-interacting Fermi gas. The entropy density of a free Fermi gas in its ground state [4] is

$$s^{(\mathrm{N})} = -\frac{2k_B}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \left[(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}))\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right] \,. \tag{4.39}$$

The 2 prefactor was included to account for spin degeneracy. The above equation can be written in another way:

$$s^{(\mathrm{N})} = -\frac{2k_B}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \left[\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right] \,. \tag{4.40}$$

Proof 4.7 — Proof of Eq. (4.40): entropy density for the normal phase.

The following identity holds,

$$\begin{aligned} \ln\left(1 - \frac{1}{e^x + 1}\right) &= \ln\left(\frac{e^x}{e^x+1}\right) \\ &= x + \ln\left(\frac{1}{e^x+1}\right) \,. \end{aligned} \tag{4.41}$$

Substituting $x = \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu})$and using Eq. (4.41) in Eq. (4.39) we obtain

$$\begin{aligned} s^{(\mathrm{N})} = -\frac{2k_B}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \Big[&\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \\ &- \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\ln f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\Big] \,. \end{aligned}$$

The second and the fourth terms cancel out. The remaining parts give Eq. (4.40).

4.3.3 The free energy density of the normal ground-state

In this section we compute the free energy density of the ground state, for the normal phase. The density of free energy is given by

$$f^{(\mathrm{N})} \equiv \frac{F^{(\mathrm{N})}}{L_x L_y} \,, \tag{4.42}$$

being $F^{(\mathrm{N})}$ the total free energy,

$$F^{(\mathrm{N})} = E^{(\mathrm{N})}_{\mathrm{HF}} - \frac{S^{(\mathrm{N})}}{k_{\mathrm{B}}\beta} \,. \tag{4.43}$$

Here $E^{(\mathrm{N})}_{\mathrm{HF}}$ is the ground-state energy for the HF hamiltonian. In the normal phase, we compute $f^{(\mathrm{N})}$ using the following formula:

$$f^{(\mathrm{N})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in \mathrm{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})) - V\left(\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right) + \tilde{\mu} n \,. \tag{4.44}$$

Proof 4.8 — Proof of Eq. (4.44): free energy density for the normal phase.

Using Eqs. (4.42), (4.43), we have

$$f^{(\mathrm{N})} = \frac{E^{(\mathrm{N})}_{\mathrm{HF}}}{L_x L_y} - \frac{s^{(\mathrm{N})}}{k_{\mathrm{B}}\beta}\,,$$

with $s^{(\mathrm{N})}$ the entropy density computed in Eq. (4.40). Then, $f^{(\mathrm{N})}$ is obtained by summing the following terms:

1. The free particles ground state energy density,

$$e^{(\mathrm{N})}_{\mathrm{g}} = \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})\,.$$

   The prefactor 2 accounts for spin degeneracy.

2. The contraction energy density,

$$\begin{aligned} e^{(\mathrm{N})}_{\mathrm{c}} &\equiv \frac{E^{(\mathrm{N})}_{\mathrm{c}}}{L_x L_y} \\ &= -n^2 U + V\left(2zn^2 - \left|u^{(s^*)}\right|^2 - \left|u^{(d)}\right|^2\right)\,, \end{aligned}$$

   where we used Eq. (4.32).

3. The entropic contribution to free energy

$$\begin{aligned} e^{(\mathrm{N})}_{s} &\equiv -\frac{s^{(\mathrm{N})}}{k_{\mathrm{B}}\beta} \\ &= \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln\left(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})\right) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})\,. \end{aligned}$$

   We used Eq. (4.40).

4. The chemical potential correction

$$e^{(\mathrm{N})}_{n} = \mu n\,.$$

   We explained at the end of Sec. 2.3.1 why this correction is needed. Note here that we are using the "physical" value of μ, not its RMP counterpart $\tilde{\mu}$.

Thus, composing everything

$$\begin{aligned} f^{(\mathrm{N})} &\equiv e^{(\mathrm{N})}_{\mathrm{g}} + e^{(\mathrm{N})}_{\mathrm{c}} + e^{(\mathrm{N})}_{s} + e^{(\mathrm{N})}_{n} \\ &= \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln\left(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})\right) - V\left(\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right) + \left[\mu + n(2zV - U)\right] n\,. \end{aligned}$$

Using Eq. (4.8), we recognise

$$\mu + n(2zV - U) = \tilde{\mu}\,.$$

Hence Eq. (4.44) is proven.

4.4 Constraints on the self-consistent solution

In this section, we collect some constraints on the HFPs $u^{(s^*)}$, $u^{(d)}$, that follow from a formal analysis of the self-consistency problem for the normal phase. It is possible to prove that:

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0\,, \tag{4.45}$$

for any choice of $U, V > 0$ and doping δ. The proof is reported in App. F. Eq. (4.45) implies that in the normal phase there is no nematic effect, and the renormalised hopping parameter $\tilde{t}_{ij}$

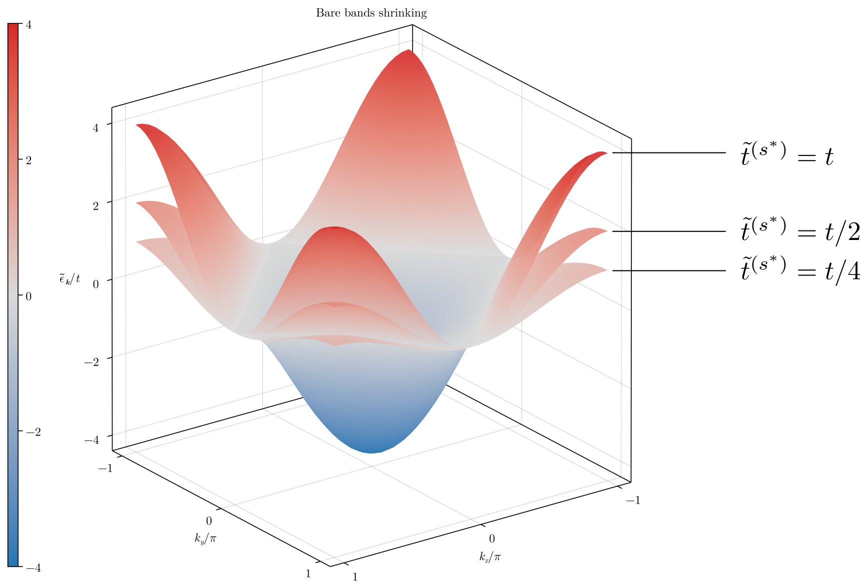


Figure 4.3 Graphic representation of the shrinking of renormalised band $\tilde{\epsilon}_{\mathbf{k}} = -4t\varphi^{(s^*)}_{\mathbf{k}}$, as the hopping parameter is decreased by renormalisation.

is isotropic with an amplitude smaller than t,

$$0 < \tilde{t}^{(s^*)} < t\,.$$

Here we used Eq. (4.28). Then, bands renormalisation

$$\epsilon_{\mathbf{k}} \quad \rightarrow \quad \tilde{\epsilon}_{\mathbf{k}}\,,$$

is actually a flattening. We denote this effect as "bands shrinking", and show a schematics in Fig. 4.3. The flattening of electronic bands is a signature of a Mott transition, as we discussed in Chap. 1.

From Eq. (4.28) we notice that, if the converged value of $u^{(s^*)}$ approaches its boundaries, we have

$$\begin{aligned} u^{(s^*)} &\rightarrow 0 &\qquad \Longrightarrow \qquad & \tilde{t}^{(s^*)} \rightarrow t\,,\\ u^{(s^*)} &\rightarrow \frac{2t}{V} &\qquad \Longrightarrow \qquad & \tilde{t}^{(s^*)} \rightarrow 0\,. \end{aligned}$$

In the second line, we see that the upper boundary can be interpreted as a critical value for Mott transition. Hence we define

$$u_{\mathrm{c}}^{(s^*)}(V) \equiv \frac{2t}{V}\,. \tag{4.46}$$

If $u^{(s^*)} \rightarrow u_{\mathrm{c}}^{(s^*)}(V)$ (from below), bands are completely flattened.

We conclude this section by proving some analytic properties of $u^{(s^*)}$. Its self-consistency equation, from Tab. 4.2, reads

$$u^{(s^*)}(\beta, \tilde{\mu}) = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})\,, \tag{4.47}$$

where we explicitly highlighted the dependence of $u^{(s^*)}$ on the chemical potential and the temperature. We also notice that the renormalised bands, inside the Fermi-Dirac factor on the right-hand side, depend on $u^{(s^*)}$. We now prove that in the low-temperature limit the above equation implies that $u^{(s^*)}$ is a monotonically descendent function of doping,

$$\frac{\partial}{\partial \delta} u^{(s^*)}(\beta, \tilde{\mu}) \leq 0\,, \tag{4.48}$$

with the equality being verified when bands are completely flattened. The fact that $u^{(s^*)}$ cannot increase as we increase δ implies that the bands shrinking effect must be suppressed as we move away from half-filling.

Proof 4.9 — Proof of Eq. (4.48): $u^{(s^*)}$ decreases with doping at low temperature.

Whatever the shape of the renormalised bands, the chemical potential cannot decrease with doping,

$$\frac{\partial\tilde{\mu}}{\partial\delta} \geq 0\,,$$

the equality holding only when adding electrons the Fermi level does not rise – namely, when the band is flat. Moreover,

$$\frac{\partial}{\partial\delta}u^{(s^*)}(\beta,\tilde{\mu}) = \frac{\partial\tilde{\mu}}{\partial\delta}\frac{\partial}{\partial\tilde{\mu}}u^{(s^*)}(\beta,\tilde{\mu})\,.$$

Then, if we shows that $\partial u^{(s^*)}/\partial\tilde{\mu} < 0$, Eq. (4.48) is proven.

Upon taking the derivative of Eq. (4.47), we obtain

$$\begin{aligned}
\frac{\partial}{\partial\tilde{\mu}}u^{(s^*)}(\beta,\tilde{\mu}) &= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x + \cos k_y)\frac{\partial}{\partial\tilde{\mu}}f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \\
&= -\frac{\beta}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x + \cos k_y)\frac{e^{\beta(\tilde{\epsilon}_{\mathbf{k}}-\tilde{\mu})}}{\left[e^{\beta(\tilde{\epsilon}_{\mathbf{k}}-\tilde{\mu})}+1\right]^2} < 0
\end{aligned}$$

which ends the proof.

This is the first expected behaviour of $u^{(s^*)}$: if there exists a region of parametric space where the shrinking effect is not strong enough to flatten the band, a rise in the doping level must lower $u^{(s^*)}$. Such a region must exist: at least all the segment $(V = 0, \delta)$ is included, since in that region of parametric space the HF hamiltonian reduces to the that of the free Fermi gas.

We conclude this section with the analytic computation of $u^{(s^*)}$ at half filling and unitary filling, both at low temperature:

$$\lim_{\beta\to\infty} u^{(s^*)}[\beta, 0] = \left(\frac{2}{\pi}\right)^2 \approx 0.405\,, \tag{4.49}$$

$$\lim_{\beta\to\infty} u^{(s^*)}\left[\beta, 2\tilde{t}^{(s^*)}\right] = 0\,, \tag{4.50}$$

where, at unitary filling, we set the chemical potential equal to the maximum energy reached by a band of amplitude $4\tilde{t}^{(s^*)}$. All the properties hereby discussed are independent of the sign of δ. Namely, all we derived is symmetric as we add or remove electrons.

In order to prove Eq. (4.49), it is necessary to introduce the Magnetic Brillouin Zone (MBZ):

$$\mathrm{MBZ} \equiv \left\{\mathbf{k}\in\mathrm{BZ} \mid \varphi^{(s^*)}_{\mathbf{k}} \geq 0\right\}\,. \tag{4.51}$$

In Chap. 5 the reason for such a name will be clarified. The MBZ coincides with the set of points (k_x, k_y) that satisfy the relation

$$|k_x| + |k_y| \leq \pi\,.$$

We draw the MBZ in Fig. 4.4. The renormalised bands satisfy

$$\tilde{\epsilon}_{\mathbf{k}} = -4\tilde{t}^{(s^*)}\varphi^{(s^*)}_{\mathbf{k}}\,.$$

Hence, $\tilde{\epsilon}_{\mathbf{k}} < 0$ inside the MBZ and $\tilde{\epsilon}_{\mathbf{k}} > 0$ outside. This is also visible in the top-left panel of Fig. 3.5. Along the boundary ∂MBZ we have $\tilde{\epsilon}_{\mathbf{k}} = 0$.

Proof 4.10 — Proof of Eqs. (4.49), (4.50).

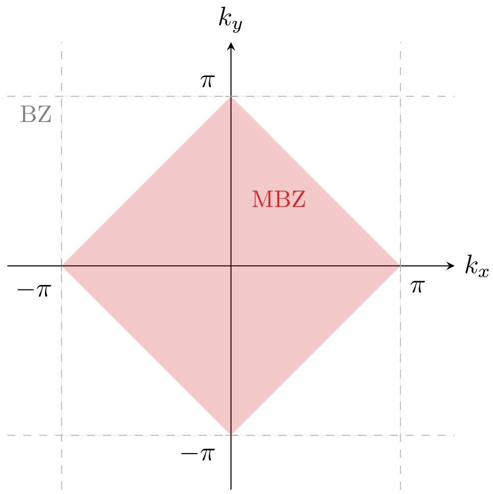


Figure 4.4 | Representation of the MBZ. The dashed lines represent the boundaries of the BZ.

The two proofs are the following:

1. At half-filling, the Fermi surface coincides with the boundary of the MBZ. Hence, being the Fermi-Dirac distribution a step-function at low temperatures, Eq. (4.47) reduces to

$$\lim_{\beta\to\infty} u^{(s^*)}(\beta,0) \simeq \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}(\cos k_x+\cos k_y)\,.$$

In thermodynamic limit, performing integration on the MBZ shown in Fig. 4.4, we obtain

$$\begin{aligned}
\lim_{\beta\to\infty} u^{(s^*)}(\beta,0) &\simeq \frac{1}{(2\pi)^2}\int_{-\pi}^{\pi}\mathrm{d}k_y\int_{-\pi+|k_y|}^{\pi-|k_y|}\mathrm{d}k_x\,\cos k_x + \frac{1}{(2\pi)^2}\int_{-\pi}^{\pi}\mathrm{d}k_x\int_{-\pi+|k_x|}^{\pi-|k_x|}\mathrm{d}k_y\,\cos k_y\\
&= \frac{1}{2\pi^2}\int_{-\pi}^{0}\mathrm{d}k_y\int_{-\pi-k_y}^{\pi+k_y}\mathrm{d}k_x\,\cos k_x + \frac{1}{2\pi^2}\int_{0}^{\pi}\mathrm{d}k_y\int_{-\pi+k_y}^{\pi-k_y}\mathrm{d}k_x\,\cos k_x\\
&= \frac{1}{\pi^2}\int_{-\pi}^{0}\mathrm{d}k_y\,\sin(\pi+k_y) + \frac{1}{\pi^2}\int_{0}^{\pi}\mathrm{d}k_y\,\sin(\pi-k_y)\\
&= \left(\frac{2}{\pi}\right)^2,
\end{aligned}$$

which proves Eq. (4.49).

2. At unitary filling we have

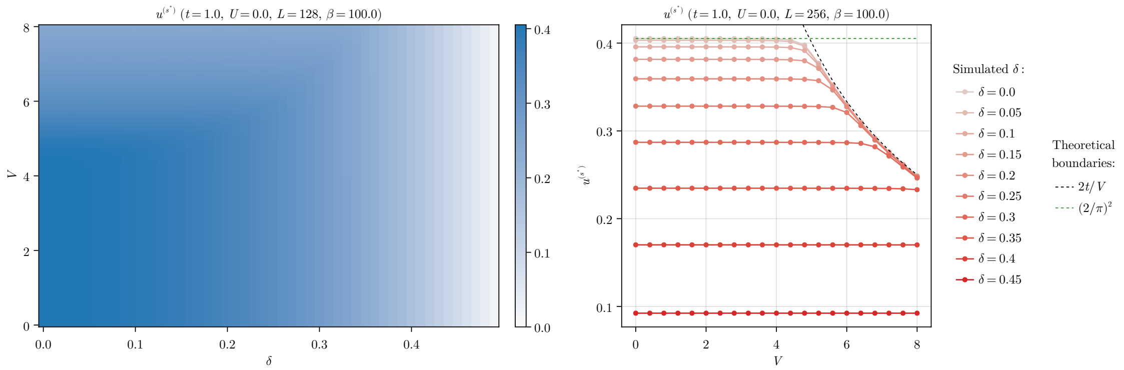
$$\lim_{\beta\to\infty} u^{(s^*)}\left(\beta,2\tilde{t}^{(s^*)}\right) = \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x+\cos k_y) = 0\,,$$

which vanishes due to the structure factor periodicity by $\boldsymbol{\pi}$. This proves Eq. (4.50).

4.5 Numeric results for the normal phase

In this section we present and discuss the numeric results we obtained for the normal phase. In the end we present a concrete scenario where careful tuning of the algorithm is necessary to ensure convergence. As is discussed above, in the normal phase the HFPs vector (defined in Sec. 2.3.2) has just one component:

$$\mathbf{v} = \left[u^{(s^*)}\right]\,.$$



a | Heatmap of $u^{(s^*)}$ in δV plane.

b | V-dependency of $u^{(s^*)}$ varying δ.

Figure 4.5 | Numerical results for $u^{(s^*)}$. On the left: colour plot in the (δ, V) plane. On the right: line plot at doubled lattice size, varying δ. The theoretical value for $u^{(s^*)}$ computed in Eqs. (4.49) at half-filling is plotted as a green dashed line.

We have two RMPs,

$$\tilde{t}_{ij} \equiv \left(t - \frac{u^{(s^*)}V}{2}\right)\varphi_{ij}^{(s^*)}\,,$$
$$\tilde{\mu} \equiv \mu + n(2zV - U)\,,$$

the second one being only dependent on the model parameters and not on HFPs. Finally, the phase free energy is given by Eq. (4.44). The normal phase is algebraically independent of U. Therefore, we present the results varying the other parameters, keeping everywhere $U = 0$. We ran simulations in the (V, δ) plane, with the setup:

$$U = 0\,, \qquad \Delta n = 10^{-2}\,, \qquad \mathbf{\Delta v} = \left[u^{(s^*)} : 5\times 10^{-4}\right]\,, \qquad \beta = 100.0\,.$$

Here $\mathbf{\Delta v}$ is the tolerance vector, defined as in Sec. 2.3.2. We have set $\beta = 100.0$ as an (arbitrary) approximation of zero temperature. We measure temperature in adimensional units,

$$\frac{\beta}{\beta_0} \quad\to\quad \beta$$

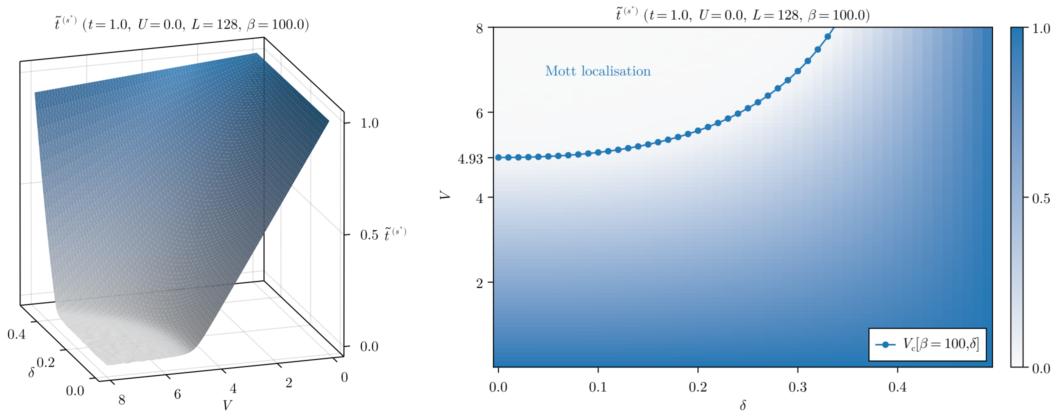
being $\beta_0 = 1/(1\mathrm{K}\times k_\mathrm{B})$.

The trend of Eq. (4.48) is confirmed in Fig. 4.5, where we show $u^{(s^*)}$ as a function of V and varying δ, with $\beta = 100$. We see that $u^{(s^*)}$ decreases as we increase δ while keeping V fixed. In the colour plot of Fig. 4.5a we observe that in the top-left corner (low-δ and high-V limit) there is a brighter region in which, increasing δ while keeping V fixed, the converged value of $u^{(s^*)}$ is approximately constant. This effect is more visible in Fig. 4.5b, where we doubled L and ran simulations with a smaller δ resolution. At small values of V, the convergence value of $u^{(s^*)}$ is approximately independent of V. Note that, as was expected from the discussion of Sec. 4.4, the half-filling convergence value of $u^{(s^*)}$ in the V-independent region is approximately given by Eq. (4.49), which we report in Fig. 4.5b as an horizontal dashed green line. We did not find significant discrepancies in the converged data for $L = 128$ and $L = 256$.

For larger values of V, $u^{(s^*)}$ starts to decrease monotonically, consistently with the constraints of Eq. (4.45). We sketch the upper boundary $u_\mathrm{c}^{(s^*)}(V) = 2t/V$, defined in Eq. (4.46), in Fig. 4.5b as a black dashed line. We observe that when the red lines (numerical results) reach the black line (theoretical upper boundary), we have

$$u^{(s^*)} \lesssim \frac{2t}{V} \qquad\Longrightarrow\qquad \tilde{t}^{(s^*)} \approx 0$$

Interestingly, this would correspond to a complete flattening of the renormalised bands.



a | Numerical results for $\tilde{t}^{(s^*)}$

b | Critical Mott boundary.

Figure 4.6 | Plots of $t^{(s^*)}$ and the Mott-boundary V_c, computed as in Eq. (4.52).

4.5.1 Mott localisation

In Sec. 1.1.1 we introduced Mott insulation, and argued that the flattening of bands is a signature of a Mott transition. Interestingly, our results suggest that Mott localisation is induced by the non-local attraction $-V$, instead of the usual Mott localisation induced by the local repulsion $+U$.

In Fig. 4.5b,we see that the converged value of $u^{(s^*)}$ is approximately horizontal up to the boundary $u_c^{(s^*)} = 2t/V$, represented as a dashed black line. Therefore, in the region where it is constant, we have

$$u^{(s^*)}[\beta, \tilde{\mu}(\delta)]\big|_V = u^{(s^*)}[\beta, \tilde{\mu}(\delta)]\big|_{V=0} .$$

We define the critical Mott boundary as:

$$V_c[\beta, \delta] \equiv \frac{2t}{u^{(s^*)}[\beta, \tilde{\mu}(\delta)]\big|_{V=0}} . \tag{4.52}$$

When V becomes larger than this critical value, due to the constraint of Eq. (4.45), $u^{(s^*)}$ is forced to decrease while $\tilde{t}^{(s^*)}$ drops to 0. In Fig. 4.6a the converged values of $\tilde{t}^{(s^*)}$, extracted from the same data of Fig. 4.5a, are plotted in parametric space.

The localisation effect is particularly effective at half-filling, where the kinetic energy vanishes for $V/t \approx 5$. This value needs to be compared with the critical value of the local repulsion U_c for Mott localisation. In DMFT, $U_c \simeq 3W$, with W half the bare bandwidth [11]. In Fig. 4.6b we report, superimposed, the convergence values of the RMP $\tilde{t}^{(s^*)}$ and $V_c[\beta = 100, \delta]$ given by Eq. (4.52). Our prediction of the critical value for V approximates reasonably the localisation region.

To extract the value of $V_c[\beta, \delta]$ is, in general, a non-trivial analytic problem due to the insurgence of elliptic integrals and the smearing effects of temperature. At low temperature, however, we can use Eq. (4.49) and obtain the asymptotic critical boundary at half-filling

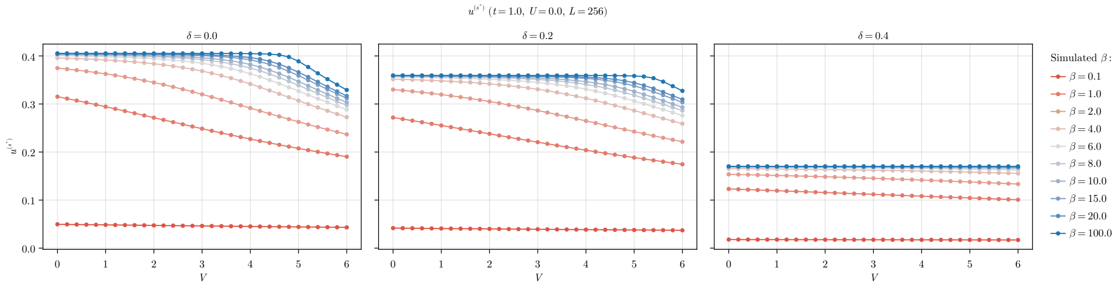
$$V_c[\beta \to \infty, 0] = \frac{2t}{(2/\pi)^2} \approx 4.93t , \tag{4.53}$$

which we report in Fig. 4.6b as a tick of the vertical axis.

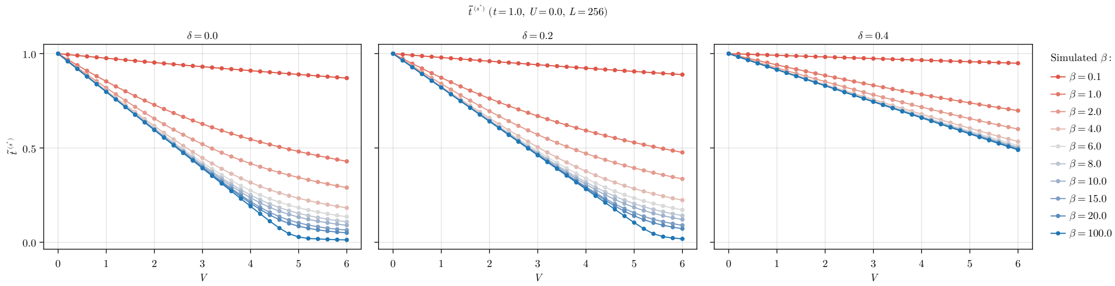
Increasing temperature, Mott localisation is suppressed. In Fig. 4.7a we report results for the HFP $u^{(s^*)}$ computed at decreasingly lower temperature and three example doping levels,

$$1.0 \leq \beta \leq 100.0 , \qquad \delta = 0.0, 0.2, 0.4 .$$

Increasing temperature, the constant $u^{(s^*)}$ region progressively disappears in favour of a monotonically decreasing behaviour. In Fig. 4.7b the same dataset is plotted with focus on the RMP $\tilde{t}^{(s^*)}$. Temperature increase disrupts localisation effects, at least at small values of V. However, the bands shrinking effect is still relevant up to $\beta = 1.0$, with the hopping parameter approximately halving at $V \approx 5 \div 6$.



a | Temperature dependence of $u^{(s^*)}$ at various dopings.



b | Temperature dependence of $\tilde{t}^{(s^*)}$ at various dopings.

Figure 4.7 | Comparative plots of the HFP $u^{(s^*)}$ (top panel) and RMP $\tilde{t}^{(s^*)}$ (bottom panel) in null-doped (left plots), "medium-doped" ($\delta = 0.2$, central plots) and "heavily-doped" ($\delta = 0.4$, right plots) contexts. Increasing temperature progressively destroys Mott localisation, reducing the bands shrinking effect.

4.5.2 Solution free energy and entropy densities

In order to compare phases, we need an estimation of free energy of each. The plots of Fig. 4.8 report our findings on the normal phase free energy, computed based on Eq. (4.44). The free energy is intended in unities of t. For a non-flat band, doping increases energy because the band fills up and electrons populate higher energy levels.

Looking at the data of Fig. 4.8a at fixed δ, free energy density remains approximately constant up to the Mott boundary. This is better appreciated in Fig. 4.8b. Indeed, as long as a band with non-zero amplitude is present, the HFP correction of Eq. (4.44) compensates the free energy increase due to shrinking of bands,

$$
\begin{aligned}
-V\left|u^{(s^*)}\right|^2 &= -\frac{V u^{(s^*)*}}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} (\epsilon_{\mathbf{k}} - \tilde{\epsilon}_{\mathbf{k}})\, f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})\,,
\end{aligned}
$$

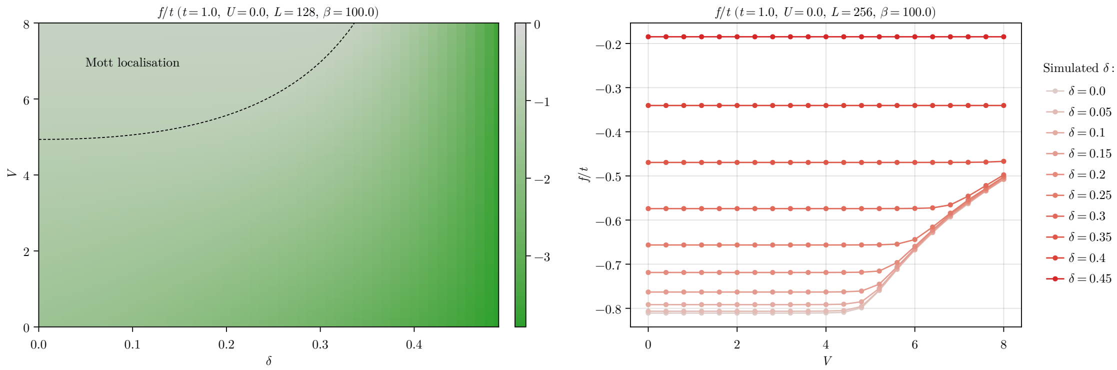
having we used in the second passage

$$
\tilde{\epsilon}_{\mathbf{k}} = \epsilon_{\mathbf{k}} + u^{(s^*)*} V(\cos k_x + \cos k_y)\,.
$$

Recalling the first point of the proof given in Sec. 4.3.3, to add the HFP correction to free energy is the same as removing the *tilde* sign from single-state energies, thus using the free model ($V = 0$) free energy density expression.

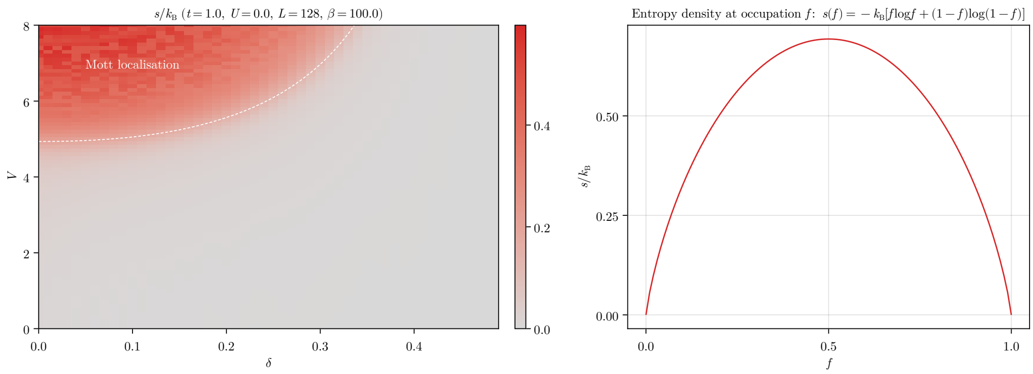
Mott localisation is energetically expensive. This is visible in both panels of Fig. 4.8. The V-dependent energy increase is an entropic effect: indeed, for a flat band, the Fermi-Dirac distribution equals

$$
f(0; \beta, \tilde{\mu}(\delta)) = \frac{1}{2}
$$



a | Heatmap of $f^{(\mathrm{N})}$ in δV plane. b | V-dependency of $f^{(\mathrm{N})}$ varying δ.

Figure 4.8 | Free energy density of the normal phase. On the left-side panel, the dashed black line has the same meaning as the dotted line of Fig. 4.6b, indicating the Mott transition.



a | Heatmap of $s^{(\mathrm{N})}$ in δV plane. b | Theoretical single-state entropy at occupation f.

Figure 4.9 | Entropy density of the normal phase (left panel) and theoretical contribution to entropy for a state with occupation f. On the left-side panel, the dashed white line has the same meaning as the dotted line of Fig. 4.6b, indicating the Mott transition.

at any temperature. The expression for the contribution to entropy of a single particle state with occupation f [4] is given by

$$s(f) = -k_{\mathrm{B}} \left[f \log f + (1-f) \log(1-f) \right] ,$$

and is represented in Fig. 4.9b. The maximum is located at $f = 1/2$, then a flat band realises the maximally entropic condition. The Mott insulating region can be seen as a degeneration of the Fermi surface (FS) to the entire BZ, a process that maximises entropy.

All of this holds, as all the rest, at low temperature: by lowering β, finite-temperature effects become more and more relevant and the statements above need to be corrected.

4.5.3 Convergence problems and mixing fine-tuning

In this subsection we describe some of the typical convergence problems arising in HF iterative schemes anticipated in Sec. 2.3.2. Bands flattening is a source of algorithmic instability: numerics can lead to values of $u^{(s^*)}$, during iterations, slightly out of bounds with respect to Eq. (4.45). Even if the numeric effect is small, the consequence can be vast. To heal this effect and extract the actual convergence values, it is necessary to optimise the mixing parameter $g \in [0, 1]$.

We defined g in Sec. 2.3.2, and operatively in Eq. (2.25). This parameter controls deterministically how, from an initialiser (the HFPs vector we plug in the algorithm) and an output (the HFPs vector we get back from the algorithm) the new initialiser for the following step is generated. We

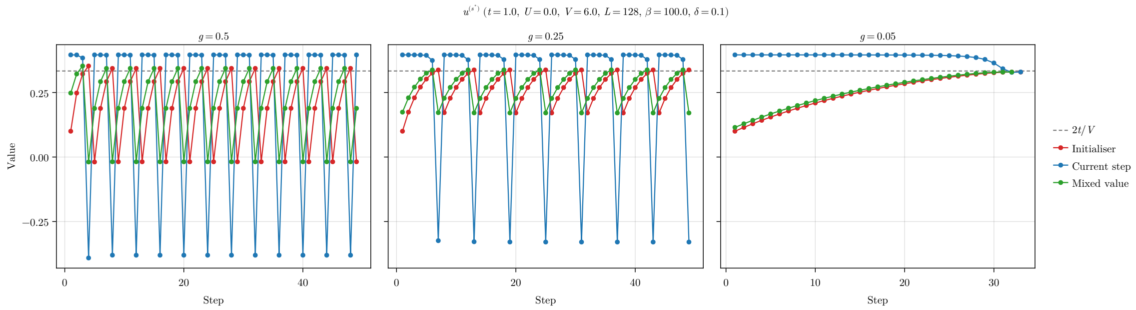


Figure 4.10 | Comparison of three example tracks for the single-valued HFPs vector of the normal phase, simulated at a “Mott-deep” point.

show in Fig. 4.10 the evolutions at different mixing parameters of a simulation at

$$V = 6.0\,, \qquad \delta = 0.1\,, \qquad \beta = 100.0\,.$$

This point is inside the Mott region, as can be checked in Fig. 4.6b. The HFP $u^{(s^*)}$ needs to converge at $2t/V = t/3$ in order to give complete localisation. The initialiser

$$\text{Step 1} \quad : \quad u^{(s^*)} = 0.1\,,$$

was arbitrarily chosen. In each panel of Fig. 4.10, the red track represents the value of $u^{(s^*)}$ we plug in the self-consistency equations, the blue the result of the self-consistent equation, the green one the mixed value of the two based on different values of g. The gray dashed line is the theoretical convergence target.

In the first two panels, there is an evident infinite loop occurring whenever the green line crosses the threshold $2t/V$. At those steps, the green point is out of bounds with respect to Eq. (4.45). When this happens, the successive red point gives a band with inverted sign, a feature that “breaks” the self-consistency equation and generates a blue point “far away”.

To overcome this problem it is necessary to move “slowly”. As is shown in the right panel, the choice of a small enough mixing parameter ensures convergence. In the track, each blue dot is above the target and would produce a band with inverted sign. Mixing blue and red dots with a 95% unbalance on the red ones, we ensure that the green track approaches the target from below.

This kind of problems can turn out to be pretty common for iHF schemes, and to understand how to knob the algorithmic parameters without sacrificing accurateness and precision, as well as avoiding computationally costly and frustrating infinite loops, was an essential part when designing this algorithm.

Chapter 5

Antiferromagnetic phase

In this chapter, we study antiferromagnetic (AF) ordering in the extended Hubbard model, limiting ourselves to commensurate antiferromagnetism at half-filling. This phase is well known and studied on the square Hubbard lattice. Here we show that the insertion of the non-local attraction $\hat{H}_V$ leads to an enhancement of the AF instability, as a result of two cooperative effects. First, the direct amplification of the AF gap, and, second, the shrinking of electronic bandwidth that lead to the aforementioned Mott localisation.

This chapter is organised as follows. In Sec. 5.1, we present the commensurate AF phase and discuss its symmetries. In Sec. 5.2, we discuss the HF approximation in the AF phase for the conventional HM. In Sec. 5.3, we generalise to the EHM, showing how the HF hamiltonian obtained for the HM can be renormalised. In Sec. 5.4, we prove that Mott localisation can enhance long-range AF ordering. Finally, in Sec. 5.5, we present the results of numeric simulations.

5.1 The AF phase and its symmetries

Long-range AF ordering is defined by the insurgence of a staggered pattern of spin density with a given periodicity, with zero net magnetisation. In Fig. 5.1, we show three (of many) possible AF orderings, with the blue/red arrows indicating an abundance of spin-up/down electrons on-site. The left panel of Fig. 5.1 shows an alternating structure with two-sites periodicity in both directions. We denote this configuration as “commensurate”. This structure is periodic by shifts of an amount $(1, 1)$ in lattice units. Therefore, in reciprocal space this phase has a non-zero component at wavevector

$$\boldsymbol{\pi} = (\pi, \pi)\,. \tag{5.1}$$

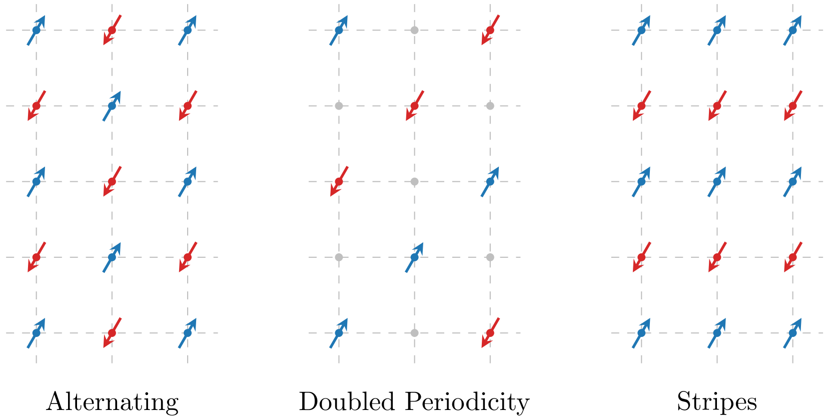


Figure 5.1 Three possible AF structures. Dots and arrows indicate spin abundance on a given site, while empty sites are intended as spin-balanced. Within this text we deal with the lowest-periodicity commensurate phase, the antiferromagnet on left panel.

which we denote, from now on, the "AF vector". The central panel of Fig. 5.1 presents another staggered configuration with doubled periodicity, where sites with a given polarisation are intermediated by sites where spin populations are balanced (gray dots). The right panel shows a possible "striped" antiferromagnet, where AF ordering is established along one particular direction. In our example, in along direction y we have AF ordering with periodicity of 2 sites, while along x we have ferromagnetic ordering. Striped antiferromagnetism is studied in HM physics [29] as a possible long-range ordering energetically competing with anisotropic superconductivity. Nematic antiferromagnets, establishing long-range magnetic ordering along one particular direction, are interesting due to competition and interplay against superconducting ordering at finite doping [38]. Geometrically non-trivial magnetic ordering are at the core of SDW formation in copper oxides at finite doping [39, 40].

In this thesis, we focus on the commensurate configuration on the left panel of Fig. 5.1, which is known to be favourite at half-filling [3]. As we anticipated in Sec. 3.2 and summarised in Tab. 3.1, the AF phase is described by the breaking of global spin rotation and translational symmetries:

$$\mathrm{SU}(2) \overset{\mathrm{SSB}}{\to} \mathrm{U}^z(1)\,,$$
$$\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)} \overset{\mathrm{SSB}}{\to} \mathrm{T}_2^{(x)} \otimes \mathrm{T}_2^{(y)}\,.$$

As we did for the normal phase, we also allow for

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2\,.$$

We will show that the commensurate AF phase actually leaves C_4 unbroken.

Commensurate antiferromagnetism is specified by the Ansatz:

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m \quad \text{and} \quad n, m \in [0,1]\,. \tag{5.2}$$

where $\mathrm{r} = (x, y) \in \mathcal{L}$. Eq. (5.2) represents the staggered distribution of spin populations shown in the left panel of Fig. 5.1. At site $\mathbf{r}$ the two spin populations average to n and differ by $2m$. Note that $\langle \hat{n}_{\mathbf{r}\sigma} \rangle \leq 1$ due to Pauli exclusion principle.

5.1.1 Symmetries of the commensurate AF Ansatz

The Ansatz of Eq. (5.2) breaks translational invariance in each spin sector, but preserves $\mathrm{U}^z(1)$ and $\mathrm{U}^c(1)$ symmetries. Eq. (5.2) is formulated equivalently as

$$\langle \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \rangle = (-1)^{x+y} 2m\,. \tag{5.3}$$

Antiferromagnetism establishes when $m \neq 0$.

Proof 5.1 — The Ansatz of Eq. (5.2) is $\mathrm{U}^z(1)$-symmetric.

We use the spinors $\hat{\Psi}_{\mathbf{r}}$ defined in Eq. (3.8)

$$\hat{\Psi}_{\mathbf{r}} = \begin{bmatrix} \hat{c}_{\mathbf{r}\uparrow} \\ \hat{c}_{\mathbf{r}\downarrow} \end{bmatrix}\,,$$

and the local spin operators of Eq. (3.7),

$$\hat{S}^+_{\mathbf{r}} = \frac{\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow}}{2}\,, \qquad \hat{S}^-_{\mathbf{r}} = \frac{\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}}{2}\,, \qquad \hat{S}^z_{\mathbf{r}} = \frac{\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow} + \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}}{2}\,.$$

Consider Eq. (3.24) and its conjugate,

$$\left[\hat{c}^\dagger_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right] = -\hat{c}^\dagger_{\mathbf{r}\sigma} \quad \text{and} \quad \left[\hat{c}_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right] = \hat{c}_{\mathbf{r}\sigma}\,.$$

The operators $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$ and $\hat{S}^{\pm}_{\mathbf{r}}$ do not commute. Indeed,

$$\begin{aligned}
\left[\hat{S}^{+}_{\mathbf{r}}, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right] &= \frac{1}{2}\left[\hat{c}^{\dagger}_{\mathbf{r}\uparrow}, \hat{n}_{\mathbf{r}\uparrow}\right]\hat{c}_{\mathbf{r}\downarrow} - \frac{1}{2}\hat{c}^{\dagger}_{\mathbf{r}\uparrow}[\hat{c}_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\downarrow}] \\
&= -\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow} \,, \\
\left[\hat{S}^{-}_{\mathbf{r}}, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right] &= \frac{1}{2}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}[\hat{c}_{\mathbf{r}\uparrow}, \hat{n}_{\mathbf{r}\uparrow}] - \frac{1}{2}\left[\hat{c}^{\dagger}_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\downarrow}\right]\hat{c}_{\mathbf{r}\uparrow} \\
&= \hat{c}^{\dagger}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow} \,.
\end{aligned}$$

Instead, $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$ and $\hat{S}^{z}_{\mathbf{r}}$ commute, because all on-site number operators commute among themselves. Then, in order to have $m \neq 0$ in Eq. (5.3), SU(2) must be broken while $\mathrm{U}^{z}(1)$ does not. Therefore the commensurate AF phase breaks global spin rotational invariance, reducing it to the smaller symmetry of rotations around the magnetisation axis.

Proof 5.2 — The Ansatz of Eq. (5.2) is $\mathrm{U}^{\mathrm{c}}(1)$-symmetric.

From Eq. (3.5) we know that on-site density operators are gauge-invariant separately in both spin sectors,

$$\hat{n}_{\mathbf{r}\sigma} \quad \to \quad e^{i(\eta-\eta)}\hat{n}_{\mathbf{r}\sigma} = \hat{n}_{\mathbf{r}\sigma} \,.$$

Then no selection rule applies to $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$: its EV can be non-zero without breaking $\mathrm{U}^{z}(1)$.

Eq. (5.2) contains two physical observables: the average density n and the staggered magnetisation m. These two quantities obey the self-consistency equations

$$n = \frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}} \frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle + \langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2} \,, \tag{5.4}$$

$$m = \frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}} (-1)^{x+y}\frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle - \langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2} \,, \tag{5.5}$$

where the expectation values on the right-hand side depend on n and m. Since we work at fixed density, Eq. (5.4) is used to determine self-consistently the chemical potential.

Proof 5.3 — Proof of Eqs. (5.4) and (5.5).

The on-site averaged density is given by:

$$\forall\mathbf{r}\in\mathcal{L} \quad : \quad n \equiv \frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle + \langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2} \,, \tag{5.6}$$

while on-site magnetisation is, inverting Eq. (5.3),

$$\forall\mathbf{r} = (x,y)\in\mathcal{L} \quad : \quad m \equiv (-1)^{x+y}\frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle - \langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2} \,. \tag{5.7}$$

By construction, both n and m are translationally invariant quantities. Hence

$$\begin{aligned}
n &= \frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}} \frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle + \langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2} \,, \\
m &= \frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}} (-1)^{x+y}\frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle - \langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2} \,,
\end{aligned}$$

which ends the proof.

5.1.2 The AF Ansatz in reciprocal space

Translational symmetry is not completely broken in the AF phase; rather, is reduced. Therefore, it is useful to move to reciprocal space and transform Eq. (5.2). We define

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\,, \tag{5.8}$$
$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}\sigma}\,, \tag{5.9}$$

and their expectation values (EVs),

$$u_{\mathbf{k}\sigma} \equiv \langle \hat{n}_{\mathbf{k}\sigma}\rangle\,, \tag{5.10}$$
$$v_{\mathbf{k}\sigma} \equiv i\langle \hat{\psi}_{\mathbf{k}\sigma}\rangle\,. \tag{5.11}$$

In the definition of $v_{\mathbf{k}\sigma}$ we included a i prefactor that simplifies analytic calculations. Using the functions of Tab. 3.5, we define the harmonics

$$u^{(\gamma)}_{\sigma} \equiv \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} u_{\mathbf{k}\sigma}\,, \tag{5.12}$$
$$v^{(\gamma)}_{\sigma} \equiv \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} v_{\mathbf{k}\sigma}\,. \tag{5.13}$$

As will be discussed, these harmonics are HFPs in the AF phase and must be determined self-consistently. Eqs. (5.12) and (5.13) are their self-consistency equations.

A finite staggered magnetisation ($m \neq 0$) implies, at least for a subset of momenta $\{\mathbf{k}\}$,

$$m \neq 0 \quad \Longrightarrow \quad \exists \mathbf{k} \in \mathrm{BZ} \quad : \quad \left|\langle\hat{\psi}_{\mathbf{k}\sigma}\rangle\right| > 0\,. \tag{5.14}$$

Recalling Eq. (5.9), we see that removing one electron at $(\mathbf{k}, \sigma)$ from the ground-state and creating another at $(\mathbf{k}+\boldsymbol{\pi}, \sigma)$ produces a state with a finite overlap.

Proof 5.4 — If $m \neq 0$, for some $\mathbf{k}$ it must hold $\langle\hat{\psi}_{\mathbf{k}\sigma}\rangle \neq 0$.

We assume $m \neq 0$. From Eq. (5.2) we get

$$\begin{aligned}
(-1)^{\delta_{\sigma=\downarrow}} m &= \overbrace{\frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}}(-1)^{x+y} n}^{=0} + \frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}}(-1)^{\delta_{\sigma=\downarrow}} m \\
&= \frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}}(-1)^{x+y}\left(n + (-1)^{x+y+\delta_{\sigma=\downarrow}} m\right) \\
&= \frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}}(-1)^{x+y}\langle\hat{n}_{\mathbf{r}\sigma}\rangle\,,
\end{aligned} \tag{5.15}$$

where we used $1 = \left((-1)^{x+y}\right)^2$. $\mathcal{F}$-transforming the operator and its sign factor inside the sum in last line, we get

$$\begin{aligned}
\sum_{\mathbf{r}\in\mathcal{L}}(-1)^{x+y}\hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{r}\in\mathcal{L}}(-1)^{x+y}\hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma} \\
&\overset{\mathcal{F}}{=} \sum_{\mathbf{r}\in\mathcal{L}} e^{i\boldsymbol{\pi}\cdot\mathbf{r}} \frac{1}{\sqrt{L_x L_y}}\sum_{\mathbf{k}\in\mathrm{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}}\hat{c}^{\dagger}_{\mathbf{k}\sigma} \frac{1}{\sqrt{L_x L_y}}\sum_{\mathbf{k}'\in\mathrm{BZ}} e^{-i\mathbf{k}'\cdot\mathbf{r}}\hat{c}_{\mathbf{k}'\sigma} \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\mathbf{k}'\in\mathrm{BZ}} \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma}\frac{1}{L_x L_y}\sum_{\mathbf{r}\in\mathcal{L}} e^{-i[\mathbf{k}'-(\mathbf{k}+\boldsymbol{\pi})]\cdot\mathbf{r}}\,.
\end{aligned}$$

Using Eq. (3.36), last line is reduced to

$$\sum_{\mathbf{r}\in\mathcal{L}}(-1)^{x+y}\hat{n}_{\mathbf{r}\sigma} = \sum_{\mathbf{k}\in\mathrm{BZ}} \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \tag{5.16}$$

$$= \sum_{\mathbf{k}\in\mathrm{BZ}} \hat{\psi}_{\mathbf{k}\sigma}\,. \tag{5.17}$$

We insert the last line in Eq. (5.15),

$$(-1)^{\delta_{\sigma=\downarrow}} m = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \langle\hat{\psi}_{\mathbf{k}\sigma}\rangle\,. \tag{5.18}$$

Since $m \neq 0$, at least for some momentum $\mathbf{k}$ it must hold $\langle\hat{\psi}_{\mathbf{k}\sigma}\rangle \neq 0$.

The self-consistency equation for m reads, in reciprocal space,

$$m = \frac{1}{2L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \langle\hat{\psi}_{\mathbf{k}\uparrow} - \hat{\psi}_{\mathbf{k}\downarrow}\rangle\,. \tag{5.19}$$

This is the transformed version of Eq. (5.5).

Proof 5.5 — Proof of Eq. (5.19): $\mathcal{F}$-transform of Eq. (5.5).

From Eq. (5.18), we have

$$m = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \langle\hat{\psi}_{\mathbf{k}\uparrow}\rangle \tag{5.20}$$

$$= -\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \langle\hat{\psi}_{\mathbf{k}\downarrow}\rangle\,. \tag{5.21}$$

Summing the two lines, we obtain Eq. (5.19). From Eq. (5.13), we notice that $v^{(s)}_{\sigma} = i(-1)^{\delta_{\sigma=\downarrow}}m$.

The operator $\hat{\psi}_{\mathbf{k}\sigma}$ satisfies the relation

$$\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}}\hat{\psi}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma} = -\varphi^{(\gamma)}_{\mathbf{k}}\hat{\psi}_{\mathbf{k}\sigma}\,, \tag{5.22}$$

where we used the harmonics $\varphi^{(\gamma)}_{\mathbf{k}}$ defined in Tab. 3.5.

Proof 5.6 — Proof of Eq. (5.22).

The following identity holds

$$\begin{aligned}\hat{\psi}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma} &= \left(\hat{c}^{\dagger}_{\mathbf{k}+2\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right)^{\dagger}\\ &= \hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}\sigma}\\ &= \hat{\psi}_{\mathbf{k}\sigma}\,.\end{aligned} \tag{5.23}$$

Moreover, being combinations of sines and cosines, the functions $\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}}$ are antisymmetric under $\boldsymbol{\pi}$ shifts,

$$\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}} = -\varphi^{(\gamma)}_{\mathbf{k}}\,. \tag{5.24}$$

Combining Eq. (5.23) and (5.24), we obtain Eq. (5.22).

5.2 The AF phase in the conventional HM

In this section, we set $V = 0$ and describe antiferromagnetism in the conventional HM, $\hat{H}_t + \hat{H}_U$. We start by deriving the analytic expression of the HF hamiltonian for the AF phase, and then diagonalise it.

5.2.1 Derivation of the HF hamiltonian

The two-body repulsive interaction is

$$\hat{H}_U = U \sum_{\mathbf{r}\in\mathcal{L}} \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\,.$$

We know from Tab. 3.2 that the only contribution to the HF hamiltonian comes from Hartree channel, while Fock and Cooper channel are identically zero in the AF phase.

Local Hartree terms. In Eq. (2.21) we decomposed $\hat{H}_U$ through Wick's theorem. The decoupling in the Hartree channel reads

$$\hat{H}_U \overset{\mathrm{HF}}{\simeq} U \sum_{\mathbf{r}\in\mathcal{L}} \left[\hat{n}_{\mathbf{r}\uparrow}\langle\hat{n}_{\mathbf{r}\downarrow}\rangle + \langle\hat{n}_{\mathbf{r}\uparrow}\rangle\hat{n}_{\mathbf{r}\downarrow} - \langle\hat{n}_{\mathbf{r}\uparrow}\rangle\langle\hat{n}_{\mathbf{r}\downarrow}\rangle\right]\,. \tag{5.25}$$

We define the MBZ as in Eq. (4.51), as the set of points in reciprocal space that satisfy the inequality $|k_x|+|k_y|\le\pi$. We sketched it in Fig. 4.4. Applying the Ansatz of Eq. (5.2), we obtain the HF approximation

$$\hat{H}_U \overset{\mathrm{HF}}{\simeq} -\sum_{\mathbf{k}\in\mathrm{MBZ}}\sum_{\sigma}\left(\Delta_{\mathbf{k}\sigma}\hat{\psi}_{\mathbf{k}\sigma} + \Delta^*_{\mathbf{k}\sigma}\hat{\psi}^\dagger_{\mathbf{k}\sigma}\right) + nU\hat{N} - E^{(\mathrm{AF})}_{\mathrm{H},U}\,, \tag{5.26}$$

where we defined the gap function

$$\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU\,, \tag{5.27}$$

ans the energy shift

$$E^{(\mathrm{AF})}_{\mathrm{H},U} \equiv U\sum_{\mathbf{r}\in\mathcal{L}}\langle\hat{n}_{\mathbf{r}\uparrow}\rangle\langle\hat{n}_{\mathbf{r}\downarrow}\rangle\,. \tag{5.28}$$

We observe, in Eq. (5.26), the presence of a number operator, meaning that chemical potential is shifted in the Hartree channel by

$$\mu\to\mu-nU\,. \tag{5.29}$$

We report these results in the summary of Tab. 5.1, provided in next section.

Proof 5.7 — Proof of Eq. (5.26): HF approximation of $\hat{H}_U$ in Hartree channel.

Consider Eq. (5.25). Using Eqs. (5.4), (5.5) we can re-write

$$\begin{aligned}\hat{n}_{\mathbf{r}\uparrow}\langle\hat{n}_{\mathbf{r}\downarrow}\rangle + \langle\hat{n}_{\mathbf{r}\uparrow}\rangle\hat{n}_{\mathbf{r}\downarrow} &= \frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle+\langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2}\left(\hat{n}_{\mathbf{r}\uparrow}+\hat{n}_{\mathbf{r}\downarrow}\right) - \frac{\langle\hat{n}_{\mathbf{r}\uparrow}\rangle-\langle\hat{n}_{\mathbf{r}\downarrow}\rangle}{2}\left(\hat{n}_{\mathbf{r}\uparrow}-\hat{n}_{\mathbf{r}\downarrow}\right)\\ &= n\left(\hat{n}_{\mathbf{r}\uparrow}+\hat{n}_{\mathbf{r}\downarrow}\right) - m\left(\hat{n}_{\mathbf{r}\uparrow}-\hat{n}_{\mathbf{r}\downarrow}\right)\,.\end{aligned}$$

Inserting this decomposition and the expression for $E^{(\mathrm{AF})}_{\mathrm{H},U}$ given in Eq. (5.28) into Eq. (5.25), we obtain

$$\hat{H}_U \overset{\mathrm{HF}}{\simeq} nU\sum_{\mathbf{r}\in\mathcal{L}}\left(\hat{n}_{\mathbf{r}\uparrow}+\hat{n}_{\mathbf{r}\downarrow}\right) - mU\sum_{\mathbf{r}\in\mathcal{L}}(-1)^{x+y}\left(\hat{n}_{\mathbf{r}\uparrow}-\hat{n}_{\mathbf{r}\downarrow}\right) - E^{(\mathrm{AF})}_{\mathrm{H},U}\,. \tag{5.30}$$

The first term gives a number operator,

$$nU\sum_{\mathbf{r}\in\mathcal{L}}\left(\hat{n}_{\mathbf{r}\uparrow}+\hat{n}_{\mathbf{r}\downarrow}\right) = nU\times\hat{N}\,.$$

Using Eq. (5.16), the second term of Eq. (5.30) becomes

$$mU\sum_{\mathbf{r}\in\mathcal{L}}(-1)^{x+y}\left(\hat{n}_{\mathbf{r}\uparrow}-\hat{n}_{\mathbf{r}\downarrow}\right) = mU\sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\sigma}(-1)^{\delta_{\sigma=\downarrow}}\hat{\psi}_{\mathbf{k}\sigma}\,.$$

Recall the definition of the MBZ given in Eq. (4.51). The full BZ can be seen as the sum of two

MBZs shifted by $\boldsymbol{\pi}$. Then, from Eq. (5.23), we get

$$\begin{aligned} mU\sum_{\mathbf{r}\in\mathcal{L}}(-1)^{x+y}\left(\hat{n}_{\mathbf{r}\uparrow}-\hat{n}_{\mathbf{r}\downarrow}\right) &= mU\sum_{\mathbf{k}\in\mathrm{MBZ}}\sum_{\sigma}(-1)^{\delta_{\sigma=\downarrow}}\left(\hat{\psi}_{\mathbf{k}\sigma}+\hat{\psi}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right) \\ &= mU\sum_{\mathbf{k}\in\mathrm{MBZ}}\sum_{\sigma}(-1)^{\delta_{\sigma=\downarrow}}\left(\hat{\psi}_{\mathbf{k}\sigma}+\hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\right) . \end{aligned}$$

Composing everything, Eq. (5.30) becomes

$$\hat{H}_U \stackrel{\mathrm{HF}}{\simeq} -mU\sum_{\mathbf{k}\in\mathrm{MBZ}}\sum_{\sigma}(-1)^{\delta_{\sigma=\downarrow}}\left(\hat{\psi}_{\mathbf{k}\sigma}+\hat{\psi}^{\dagger}_{\mathbf{k}\sigma}\right)+nU\hat{N}-E^{(\mathrm{AF})}_{\mathrm{H},U} .$$

By defining the gap function $\Delta_{\mathbf{k}\sigma}$ as in Eq. (5.27), we obtain the HF approximation of Eq. (5.26).

The contraction energy, defined in Eq. (5.28), is given by

$$E^{(\mathrm{AF})}_{\mathrm{H},U} = L_xL_y\times U\left(n^2-m^2\right) . \tag{5.31}$$

Compared with that for the normal phase given in Eq. (4.14), we observe the presence of an additional term dependent on m^2.

Proof 5.8 — Proof of Eq. (5.31): local Hartree contraction energy.

Inserting Eq. (5.2) in Eq. (5.28), we obtain

$$\begin{aligned} E^{(\mathrm{AF})}_{\mathrm{H},U} &\equiv U\sum_{\mathbf{r}\in\mathcal{L}}\langle\hat{n}_{\mathbf{r}\uparrow}\rangle\langle\hat{n}_{\mathbf{r}\downarrow}\rangle \\ &= U\sum_{\mathbf{r}\in\mathcal{L}}\left(n+(-1)^{x+y}m\right)\left(n-(-1)^{x+y}m\right) . \end{aligned}$$

Expanding the product above we get:

$$\begin{aligned} \left(n+(-1)^{x+y}m\right)\left(n-(-1)^{x+y}m\right) &= n^2+\left((-1)^{x+y}-(-1)^{x+y}\right)nm-\left((-1)^{x+y}\right)^2m^2 \\ &= n^2-m^2 . \end{aligned}$$

The sum over $\mathcal{L}$ gives L_xL_y identical terms. Hence we obtain Eq. (5.31).

5.2.2 Diagonalisation of the HF hamiltonian

In this section, we diagonalise the HF hamiltonian identified by Eqs. (5.26) and (5.31). The procedure hereby sketched will be reused when inserting the non-local attraction, and when analysing the SC phase in Chap. 6.

Matrix representation. For the AF phase, the HF hamiltonian can be written as

$$\hat{H}_t+\hat{H}_U \stackrel{\mathrm{HF}}{\simeq} \sum_{\mathbf{k}\in\mathrm{MBZ}}\sum_{\sigma}\hat{\Psi}^{\dagger}_{\mathbf{k}\sigma}h_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma}+nU\hat{N}-E^{(\mathrm{AF})}_{\mathrm{H},U} , \tag{5.32}$$

where we defined the 2×2 matrix:

$$h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta^*_{\mathbf{k}\sigma} \\ -\Delta_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix} , \tag{5.33}$$

and the spinors

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix} . \tag{5.34}$$

Proof 5.9 — Proof of Eq. (5.32).

Recall Eq. (3.38): the kinetic sector is represented in reciprocal space by

$$\begin{aligned}\hat{H}_t &= \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\sigma}\epsilon_{\mathbf{k}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\\ &= \sum_{\mathbf{k}\in\mathrm{MBZ}}\sum_{\sigma}\epsilon_{\mathbf{k}}\left(\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}-\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right),\end{aligned}$$

where we used antiperiodicity by $\boldsymbol{\pi}$ of the bare bands $\epsilon_{\mathbf{k}}$ defined in Eq. (3.39),

$$\epsilon_{\mathbf{k}} = -\epsilon_{\mathbf{k}+\boldsymbol{\pi}}\,.$$

The kinetic operator can be written in terms of the spinors $\hat{\Psi}_{\mathbf{k}\sigma}$ defined in Eq. (5.34),

$$\hat{H}_t = \sum_{\mathbf{k}\in\mathrm{MBZ}}\sum_{\sigma}\hat{\Psi}^\dagger_{\mathbf{k}\sigma}\begin{bmatrix}\epsilon_{\mathbf{k}\sigma} & \\ & -\epsilon_{\mathbf{k}\sigma}\end{bmatrix}\hat{\Psi}_{\mathbf{k}\sigma}\,.$$

The local repulsion $\hat{H}_U$ of Eq. (5.26) can also be written in terms of these spinors:

$$\hat{H}_U = \sum_{\mathbf{k}\in\mathrm{MBZ}}\sum_{\sigma}\hat{\Psi}^\dagger_{\mathbf{k}\sigma}\begin{bmatrix} & -\Delta^*_{\mathbf{k}\sigma}\\ -\Delta_{\mathbf{k}\sigma} & \end{bmatrix}\hat{\Psi}_{\mathbf{k}\sigma} + nU\hat{N} - E^{(\mathrm{AF})}_{\mathrm{H},U}\,.$$

Summing $\hat{H}_t + \hat{H}_U$ and defining $h_{\mathbf{k}\sigma}$ as in Eq. (5.33), we obtain the matricial representation of Eq. (5.32).

Pseudospin representation. The matrix $h_{\mathbf{k}\sigma}$ can be written in term of the Pauli matrices τ^α of Eq. (3.9),

$$h_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma}\tau^z - \Delta_{\mathbf{k}\sigma}\tau^x\,.$$

Let us define the pseudospin operators $\hat{\mathbf{s}}_{\mathbf{k}\sigma}$ as

$$\hat{s}^\alpha_{\mathbf{k}\sigma} \equiv \hat{\Psi}^\dagger_{\mathbf{k}\sigma}\tau^\alpha\hat{\Psi}_{\mathbf{k}\sigma} \quad\text{with}\quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} = \begin{bmatrix}\hat{s}^x_{\mathbf{k}\sigma}\\ \hat{s}^y_{\mathbf{k}\sigma}\\ \hat{s}^z_{\mathbf{k}\sigma}\end{bmatrix}\,. \tag{5.35}$$

These operators satisfy a $\mathfrak{su}(2)$ algebra up to a constant C,

$$\left[\hat{s}^\alpha_{\mathbf{k}\sigma}, \hat{s}^\beta_{\mathbf{k}\sigma}\right] = C\times i\sum_{\gamma}\epsilon_{\alpha\beta\gamma}\hat{s}^\gamma_{\mathbf{k}\sigma}\,. \tag{5.36}$$

The proof of Eq. (5.36) is identical to the one presented in Sec. 3.1.2. Let us express the pseudospin components and identity in terms of the operators of Eqs. (5.8), (5.9):

$$\hat{s}^x_{\mathbf{k}\sigma} = \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma}\,, \tag{5.37}$$

$$\hat{s}^y_{\mathbf{k}\sigma} = i\left(\hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}^\dagger_{\mathbf{k}\sigma}\right)\,, \tag{5.38}$$

$$\hat{s}^z_{\mathbf{k}\sigma} = \hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\,, \tag{5.39}$$

$$\begin{aligned}\hat{\mathbb{I}}_{\mathbf{k}\sigma} &\equiv \hat{\Psi}^\dagger_{\mathbf{k}\sigma}\mathbb{I}_{2\times2}\hat{\Psi}_{\mathbf{k}\sigma}\\ &= \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\,.\end{aligned} \tag{5.40}$$

We define the pseudofield $\mathbf{b}_{\mathbf{k}\sigma}$:

$$\mathbf{b}_{\mathbf{k}\sigma} \equiv \begin{bmatrix}-\Delta_{\mathbf{k}\sigma}\\ 0\\ \epsilon_{\mathbf{k}\sigma}\end{bmatrix}\,.$$

In pseudospin representation, the hamiltonian of Eq. (5.32) can be expressed as

$$\hat{H}_t + \hat{H}_U \overset{\mathrm{HF}}{\simeq} \sum_{\mathbf{k}\in\mathrm{MBZ}}\sum_{\sigma}\mathbf{b}_{\mathbf{k}\sigma}\cdot\hat{\mathbf{s}}_{\mathbf{k}\sigma} + nU\hat{N} - E^{(\mathrm{AF})}_{\mathrm{H},U}\,, \tag{5.41}$$

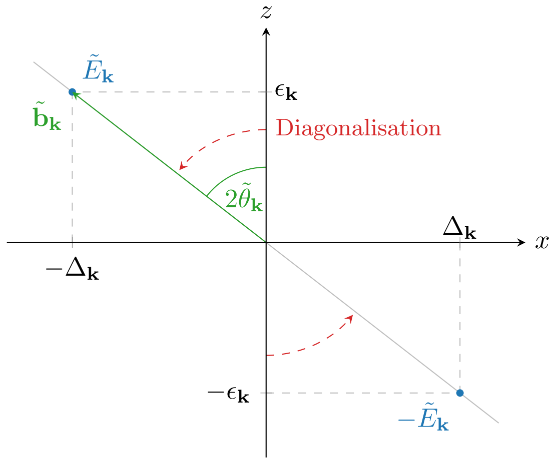


Figure 5.2 | Sketch of the diagonalisation procedure for the HM antiferromagnet. To diagonalise the HF hamiltonian, we align the z axis with the pseudo-field.

describing a set of (pseudo)spins subject to local (pseudo)magnetic fields. The field $\mathbf{b}_{\mathbf{k}\sigma}$ is orthogonal to the y plane, and is tilted in the z plane by an angle $2\theta_{\mathbf{k}\sigma}$, where

$$\cos 2\theta_{\mathbf{k}\sigma} = \frac{\epsilon_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}}\,, \qquad \sin 2\theta_{\mathbf{k}\sigma} = \frac{\Delta_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}}\,, \qquad E_{\mathbf{k}\sigma} \equiv \sqrt{\epsilon_{\mathbf{k}\sigma}^2 + |\Delta_{\mathbf{k}\sigma}|^2}\,.$$

A sketch of these angles and the pseudofield is given in Fig. 5.2.

We observe that we could have redefined the pseudospin operators of Eq. (5.35) dividing them by an appropriate constant, in order to satisfy exactly the $\mathfrak{su}(2)$ algebra, and at the same time multiply the pseudo-magnetic field by the same constant. The product $\mathbf{b}_{\mathbf{k}\sigma} \cdot \hat{\mathbf{s}}_{\mathbf{k}\sigma}$ would be unchanged, and the diagonalisation matrix $W_{\mathbf{k}\sigma}$ would be the same. Because of this, the constant C in Eq. (5.36) is irrelevant.

Diagonalisation. In order to diagonalise the HF hamiltonian, Eq. (5.41), we perform a unitary operation on the $\hat{\Psi}_{\mathbf{k}\sigma}$ spinors which reduces the 2×2 matrix $h_{\mathbf{k}\sigma}$ of Eq. (5.33) to a diagonal form. This means to diagonalise each $h_{\mathbf{k}\sigma}$. This is obtained through a rotation around the y axis:

$$d_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} W_{\mathbf{k}\sigma}^\dagger \quad \text{with} \quad d_{\mathbf{k}\sigma} \equiv \begin{bmatrix} E_{\mathbf{k}\sigma} & \\ & -E_{\mathbf{k}\sigma} \end{bmatrix}\,. \tag{5.42}$$

This kind of procedure is referred to as "Bogliubov rotation". The diagonal values of $d_{\mathbf{k}\sigma}$ are the "Bogoliubov bands" $\pm E_{\mathbf{k}\sigma}$, with

$$E_{\mathbf{k}\sigma} = \sqrt{\epsilon_{\mathbf{k}\sigma}^2 + |\Delta_{\mathbf{k}\sigma}|^2}\,. \tag{5.43}$$

In Eq. (5.42), the matrix $W_{\mathbf{k}\sigma}$ is unitary, and gives a counter-clockwise rotation by an angle $2\theta_{\mathbf{k}\sigma}$ around the y axis:

$$\begin{aligned} W_{\mathbf{k}\sigma} &\equiv e^{-i\theta_{\mathbf{k}\sigma}\tau^y} \\ &= \cos\theta_{\mathbf{k}\sigma} - i\sin\theta_{\mathbf{k}\sigma}\tau^y \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}\sigma} & -\sin\theta_{\mathbf{k}\sigma} \\ \sin\theta_{\mathbf{k}\sigma} & \cos\theta_{\mathbf{k}\sigma} \end{bmatrix}\,. \end{aligned} \tag{5.44}$$

In Fig. 5.2 we sketch the pseudo-magnetic field in the xz plane, and the diagonalisation procedure. $W_{\mathbf{k}\sigma}$ is a rotation by an angle $2\theta_{\mathbf{k}\sigma}$ around the y axis, and in the end aligns the original z axis with the pseudofield axis. We notice that $E_{\mathbf{k}\sigma}$ is actually spin independent: only the gap depends on σ through a sign factor, which is irrelevant in Eq. (5.43).

Applying the rotation matrix to the spinors we obtain the "tilted" spinors,

$$\hat{\Gamma}_{\mathbf{k}\sigma} \equiv W_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma} \quad \text{with} \quad \hat{\Gamma}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} \\ \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \end{bmatrix}\,, \tag{5.45}$$

where we defined a new set of operators $\hat{\gamma}^{(\pm)}_{\mathbf{k}\sigma}$. Being the transformation $W_{\mathbf{k}\sigma}$ unitary, these operators obey Fermi anticommutation rules (see App. B for details),

$$\left\{\hat{\gamma}^{(p)}_{\alpha}, \hat{\gamma}^{(q)}_{\beta}\right\} = \delta_{pq}\delta_{\alpha\beta}\,.$$

Bringing all together, we obtain the non-interacting hamiltonian:

$$\begin{aligned}
\hat{H}_t + \hat{H}_U &\overset{\text{HF}}{\simeq} \sum_{\mathbf{k}\in\text{MBZ}}\sum_{\sigma}\hat{\Psi}^{\dagger}_{\mathbf{k}\sigma}h_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma} + (\cdots) \\
&= \sum_{\mathbf{k}\in\text{MBZ}}\sum_{\sigma}\hat{\Psi}^{\dagger}_{\mathbf{k}\sigma}W^{\dagger}_{\mathbf{k}\sigma}W_{\mathbf{k}\sigma}h_{\mathbf{k}\sigma}W^{\dagger}_{\mathbf{k}\sigma}W_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma} + (\cdots) \\
&= \sum_{\mathbf{k}\in\text{MBZ}}\sum_{\sigma}\hat{\Gamma}^{\dagger}_{\mathbf{k}\sigma}d_{\mathbf{k}\sigma}\hat{\Gamma}_{\mathbf{k}\sigma} + (\cdots) \\
&= \sum_{\mathbf{k}\in\text{MBZ}}\sum_{\sigma}E_{\mathbf{k}\sigma}\left(\hat{\gamma}^{(+)\dagger}_{\mathbf{k}\sigma}\hat{\gamma}^{(+)}_{\mathbf{k}\sigma} - \hat{\gamma}^{(-)\dagger}_{\mathbf{k}\sigma}\hat{\gamma}^{(-)}_{\mathbf{k}\sigma}\right) + (\cdots)\,,
\end{aligned} \tag{5.46}$$

where we intend $(\cdots) = nU\hat{N} - E^{(\text{AF})}_{\text{H},U}$. The final non-interacting hamiltonian describes a free Fermi gas of fermions described by the operators $\hat{\gamma}^{(\pm)}_{\mathbf{k}\sigma}$, with energies $\pm E_{\mathbf{k}\sigma}$.

Pseudospin EVs. The expectation value of the original pseudospin operators in the various directions can be expressed in terms of the tilted pseudospins $\hat{\Gamma}^{\dagger}_{\mathbf{k}\sigma}\tau^{\alpha}\hat{\Gamma}_{\mathbf{k}\sigma}$,

$$\begin{aligned}
\langle\hat{\Psi}^{\dagger}_{\mathbf{k}\sigma}\tau^{x}\hat{\Psi}_{\mathbf{k}\sigma}\rangle &= -\sin 2\theta_{\mathbf{k}\sigma}\langle\hat{\Gamma}^{\dagger}_{\mathbf{k}\sigma}\tau^{z}\hat{\Gamma}_{\mathbf{k}\sigma}\rangle\,, \\
\langle\hat{\Psi}^{\dagger}_{\mathbf{k}\sigma}\tau^{y}\hat{\Psi}_{\mathbf{k}\sigma}\rangle &= 0\,, \\
\langle\hat{\Psi}^{\dagger}_{\mathbf{k}\sigma}\tau^{z}\hat{\Psi}_{\mathbf{k}\sigma}\rangle &= \cos 2\theta_{\mathbf{k}\sigma}\langle\hat{\Gamma}^{\dagger}_{\mathbf{k}\sigma}\tau^{z}\hat{\Gamma}_{\mathbf{k}\sigma}\rangle\,,
\end{aligned}$$

where we used the angular relations of Fig. 5.2. Moving to the tilted frame, the z component of the pseudo-spin reduces to the thermal occupation of "up" and "down" states

$$\langle\hat{\Gamma}^{\dagger}_{\mathbf{k}\sigma}\tau^{z}\hat{\Gamma}_{\mathbf{k}\sigma}\rangle = f\left(E_{\mathbf{k}\sigma};\beta,\tilde{\mu}\right) - f\left(-E_{\mathbf{k}\sigma};\beta,\tilde{\mu}\right)\,. \tag{5.47}$$

Let us define the thermal factors:

$$\text{Th}^{(\pm)}_{\mathbf{k}\sigma}[\beta,\mu] \equiv f\left(-E_{\mathbf{k}\sigma};\beta,\mu\right) \pm f\left(E_{\mathbf{k}\sigma};\beta,\mu\right)\,, \tag{5.48}$$

which will often appear in computations. Being $E_{\mathbf{k}\sigma}$ spin independent, so are the thermal factors. We have:

$$\langle\hat{s}^{x}_{\mathbf{k}\sigma}\rangle = \frac{\Delta_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}}\text{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu]\,, \tag{5.49}$$

$$\langle\hat{s}^{y}_{\mathbf{k}\sigma}\rangle = 0\,, \tag{5.50}$$

$$\langle\hat{s}^{z}_{\mathbf{k}\sigma}\rangle = -\frac{\epsilon_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}}\text{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu]\,. \tag{5.51}$$

Magnetisation self-consistency equation. We gave the self-consistency equation for the magnetisation m in Eq. (5.19). By further elaboration, it can be written equivalently as

$$m = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\text{MBZ}}\frac{\Delta_{\mathbf{k}\uparrow}}{E_{\mathbf{k}\uparrow}}\text{Th}^{(-)}_{\mathbf{k}\uparrow}[\beta,\mu]\,. \tag{5.52}$$

This equation is the "operational" version of the self-consistency equation for the HFP m.

Proof 5.10 — Proof of Eq. (5.52): self-consistency equation for m.

Consider Eq. (5.19). Restricting the sum to the MBZ and using Eq. (5.23), we get

$$m = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left\langle\left(\hat{\psi}_{\mathbf{k}\uparrow}+\hat{\psi}^\dagger_{\mathbf{k}\uparrow}\right)-\left(\hat{\psi}_{\mathbf{k}\downarrow}+\hat{\psi}^\dagger_{\mathbf{k}\downarrow}\right)\right\rangle .$$

From Eq. (5.37), we recognise a pseudospin EV in direction x. From Eq. (5.49)

$$\begin{aligned} m &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left(\langle\hat{s}^x_{\mathbf{k}\uparrow}\rangle-\langle\hat{s}^x_{\mathbf{k}\downarrow}\rangle\right) \\ &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left(\frac{\mathrm{Re}\{\Delta_{\mathbf{k}\uparrow}\}}{E_{\mathbf{k}\uparrow}}\mathrm{Th}^{(-)}_{\mathbf{k}\uparrow}[\beta,\mu]-\frac{\mathrm{Re}\{\Delta_{\mathbf{k}\downarrow}\}}{E_{\mathbf{k}\downarrow}}\mathrm{Th}^{(-)}_{\mathbf{k}\downarrow}[\beta,\mu]\right) \\ &= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\Delta_{\mathbf{k}\uparrow}}{E_{\mathbf{k}\uparrow}}\mathrm{Th}^{(-)}_{\mathbf{k}\uparrow}[\beta,\mu]\,, \end{aligned}$$

which proves Eq. (5.52). In last line we used $\Delta_{\mathbf{k}\uparrow}=-\Delta_{\mathbf{k}\downarrow}$, which follows from Eq. (5.27), and the fact that $E_{\mathbf{k}\sigma}$ and $\mathrm{Th}^{(-)}_{\mathbf{k}\sigma}[\beta,\mu]$ are spin actually spin-independent.

Density. The averaged density obeys the equations

$$n = \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\mathrm{Th}^{(+)}_{\mathbf{k}\uparrow}[\beta,\mu] \tag{5.53}$$

$$= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\mathrm{Th}^{(+)}_{\mathbf{k}\uparrow}[\beta,\mu]\,. \tag{5.54}$$

The two are equivalent, but the second one is handier to implement in numeric computations.

Proof 5.11 — Proof of Eq. (5.53): self-consistency equation for n.

The $\mathcal{F}$-transform of the on-site density operator satisfies

$$\begin{aligned} \sum_{\mathbf{r}\in\mathcal{L}}\hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{r}\in\mathcal{L}}\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma} \\ &\stackrel{\mathcal{F}}{=} \sum_{\mathbf{r}\in\mathcal{L}}\frac{1}{\sqrt{L_xL_y}}\sum_{\mathbf{k}\in\mathrm{BZ}}e^{i\mathbf{k}\cdot\mathbf{r}}\hat{c}^\dagger_{\mathbf{k}\sigma}\frac{1}{\sqrt{L_xL_y}}\sum_{\mathbf{k}'\in\mathrm{BZ}}e^{-i\mathbf{k}'\cdot\mathbf{r}}\hat{c}_{\mathbf{k}'\sigma}\,, \end{aligned} \tag{5.55}$$

where we used Eq. (3.35). By further elaboration, we obtain

$$\begin{aligned} \sum_{\mathbf{r}\in\mathcal{L}}\hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\mathbf{k}'\in\mathrm{BZ}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma}\frac{1}{L_xL_y}\sum_{\mathbf{r}\in\mathcal{L}}e^{-i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{r}} \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\mathbf{k}'\in\mathrm{BZ}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma}\delta_{\mathbf{k}=\mathbf{k}'}\,, \end{aligned}$$

where we used Eq. (3.36). Therefore,

$$\begin{aligned} \sum_{\mathbf{r}\in\mathcal{L}}\hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{k}\in\mathrm{BZ}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}}\hat{n}_{\mathbf{k}\sigma}\,. \end{aligned} \tag{5.56}$$

Hence, applying Eq. (5.2),

$$\begin{aligned} n &= \frac{1}{2}\sum_{\sigma}\left(\frac{1}{L_xL_y}\sum_{\mathbf{r}\in\mathcal{L}} n + \overbrace{\frac{1}{L_xL_y}\sum_{\mathbf{r}\in\mathcal{L}}(-1)^{x+y}m}^{=0}\right) \\ &= \frac{1}{2L_xL_y}\sum_{\sigma}\sum_{\mathbf{r}\in\mathcal{L}}\langle\hat{n}_{\mathbf{r}\sigma}\rangle \\ &= \frac{1}{2L_xL_y}\sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{BZ}}\langle\hat{n}_{\mathbf{k}\sigma}\rangle\,. \end{aligned} \tag{5.57}$$

Restricting the sum of Eq. (5.57) to the MBZ, we get

$$\begin{aligned} n &= \frac{1}{2L_xL_y}\sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{MBZ}}\langle\hat{n}_{\mathbf{k}\sigma}+\hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle \\ &= \frac{1}{2L_xL_y}\sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left\langle\hat{\Psi}^{\dagger}_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma}\right\rangle\,, \end{aligned} \tag{5.58}$$

where we used the spinors of Eq. (5.34). The mapping $\hat{\Psi}\to\hat{\Gamma}$ is unitary, so

$$\hat{\Psi}^{\dagger}_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma} = \hat{\Gamma}^{\dagger}_{\mathbf{k}\sigma}\hat{\Gamma}_{\mathbf{k}\sigma}\,.$$

Therefore, Eq. (5.58) reduces to

$$\begin{aligned} n &= \frac{1}{2L_xL_y}\sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left\langle\hat{\Gamma}^{\dagger}_{\mathbf{k}\sigma}\hat{\Gamma}_{\mathbf{k}\sigma}\right\rangle \\ &= \frac{1}{2L_xL_y}\sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left(\left\langle\hat{\gamma}^{(+)\dagger}_{\mathbf{k}\sigma}\hat{\gamma}^{(+)}_{\mathbf{k}\sigma}\right\rangle+\left\langle\hat{\gamma}^{(-)\dagger}_{\mathbf{k}\sigma}\hat{\gamma}^{(-)}_{\mathbf{k}\sigma}\right\rangle\right) \\ &= \frac{1}{2L_xL_y}\sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{MBZ}}\mathrm{Th}^{(+)}_{\mathbf{k}\sigma}[\beta,\mu]\,. \end{aligned}$$

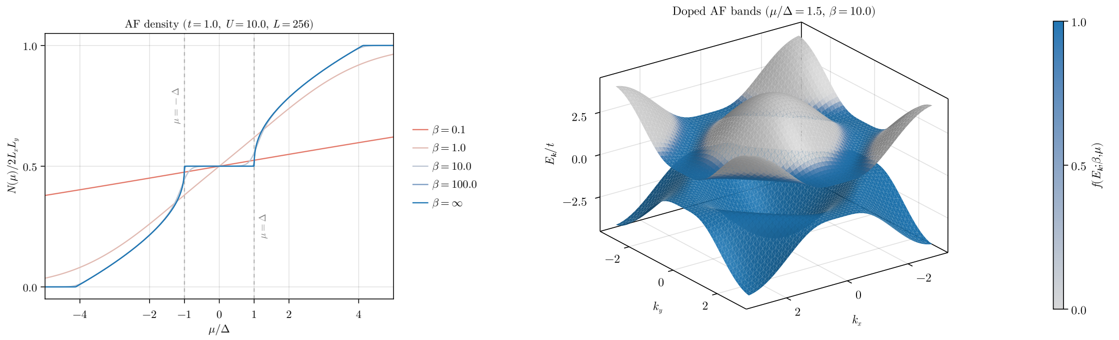
In last line we used Eq. (5.48). The thermal factor is spin-independent, hence we can set $\sigma=\uparrow$ and remove the sum over spin. This proves Eq. (5.53).

In Fig. 5.3a we sketch density as a function of the chemical potential, computed using Eq. (5.54) at $U/t = 10$ and $L = 256$. As we increase β, we observe that a sort of plateau emerges at $\mu\in[-\Delta,\Delta]$, becoming sharper as we decrease temperature. For an insulating phase, any infinitesimal shift to μ does not increase n. Indeed, in that case the chemical potential lies within a gap and no states can be filled by incrementing a little the chemical potential. On the contrary, for a metallic phase n changes as we vary infinitesimally μ. In Fig. 5.3a, we observe that the plateau is located at $n = 1/2$, i.e. half-filling, while for $|\mu| > \Delta$ the density varies with μ. Therefore, the commensurate AF phase is insulating only at half-filling.

In Fig. 5.3b we sketch the bands $\pm E_{\mathbf{k}\sigma}$ at $\mu/\Delta = 1.5$. The intense colour indicates an almost filled single-particle state, while the lighter gray colour an almost empty single-particle state. The band are gapped at zero energy, while μ crosses the upper band. This can be explained algebraically as follows. Consider the gran canonical hamiltonian $\hat{H}-\mu\hat{N}$, and use Eq. (5.40). The resulting operator is diagonalised by the same matrix given in Eq. (5.44), being

$$-\mu\hat{N} = \sum_{\mathbf{k}\in\mathrm{MBZ}}\begin{bmatrix}-\mu & \\ & -\mu\end{bmatrix} \quad\Longrightarrow\quad W_{\mathbf{k}\sigma}\left[h_{\mathbf{k}\sigma}-\mu\mathbb{I}_{2\times 2}\right]W^{\dagger}_{\mathbf{k}\sigma} = d_{\mathbf{k}\sigma}-\mu\mathbb{I}_{2\times 2}\,, \tag{5.59}$$

thus for non-null chemical potential (i.e. finite doping) the eigenvalues are just shifted, $\pm E_{\mathbf{k}}-\mu$. This means that the chemical potential for the new fermions, described by the operators $\hat{\gamma}_{\mathbf{k}\sigma}$, is the same we had for the electrons. From Fig. 5.3a we know that at finite doping the chemical potential becomes larger than the gap, therefore μ crosses the upper/lower band for electron/hole doping. The resulting phase is metallic.



a | Electronic density.   b | Example doped bands.

Figure 5.3 | Left panel: electronic density in the AF phase as a function of the chemical potential, computed through Eq. (5.54), at various temperatures. Right panel: example doped bands, with $\mu/\Delta = 1.5$ set arbitrarily and temperature set to $\beta = 10$ to enhance smearing effects around the FS. Single-state filling is proportional to the colour intensity.

Instability of the doped commensurate antiferromagnet. As we said at the beginning of this chapter, there exist several ways to realise an Hubbard antiferromagnet. For the conventional HM, it is known [27, 28] that the commensurate AF phase is unstable at any finite doping. This is discussed briefly in App. D. The key takeaway is that insulating antiferromagnets are stable with respect to spin fluctuations, while metallic antiferromagnets are not.

In fact, the mere existence of a self-consistent solution does not guarantee stability: indeed, when in Sec. 2.2.2 we derived the variational solution, we only required it to be a stationary point of the HF energy. There is no need for the solution to be a minimum. In particular, the commensurate AF phase is a saddle point of HF energy and it can be shown that any finite doping disrupts commensurate ordering towards uncommensurate phases [27].

5.3 Generalisation to the EHM: the HF hamiltonian and the variational parameters

In this section, we generalise the discussion carried out for the conventional HM, including the non-local attraction $\hat{H}_V$. We obtain an HF hamiltonian that is formally identical to the one of Eq. (5.41), apart from additional energy shift and a renormalisation of the various quantities,

$$\mu \to \tilde{\mu}\,, \qquad \epsilon_{\mathbf{k}\sigma} \to \tilde{\epsilon}_{\mathbf{k}\sigma}\,, \qquad \Delta_{\mathbf{k}\sigma} \to \tilde{\Delta}_{\mathbf{k}\sigma}\,.$$

We report all details of the analytic derivation in App. E.

The role of each contraction channel. We recall now Tab. 3.3. The contractions that, through Wick's theorem, possibly give non-zero contributions are, for $\hat{H}_V$,

$$\begin{aligned} &\hat{H}_V \text{ (o.s. sector)} && \text{Hartree contractions,}\\ &\hat{H}_V \text{ (s.s. sector)} && \text{Hartree and Fock contractions.} \end{aligned}$$

Tab. 5.1 summarises the roles of the various contraction channels (including those from the local repulsion).

As we discuss thoroughly in App. E, $\hat{H}_V$ contributes in three ways: by renormalising the chemical potential,

$$\tilde{\mu} = \mu + n(2zV - U)\,, \tag{5.60}$$

the bare bands,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} + V \sum_{\gamma \in \{s^*, d\}} u_\sigma^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*}\,, \tag{5.61}$$

Operator	Sector	Channel	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$, real part of the AF gap
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
	s.s.	Hartree	μ shift by nzV
		Fock	Bands renormalisation and resymmetrization, imaginary part of the AF gap

Table 5.1 | Wick's contractions for each interaction and channel in the HF framework, and their net effect in the AF phase.

and the AF gap

$$\tilde{\Delta}_{\mathbf{k}\sigma} \equiv \Delta_{\mathbf{k}\sigma} - iV \sum_{\gamma \neq s} v_\sigma^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} , \tag{5.62}$$

where we used the functions $\varphi_{\mathbf{k}}^{(\gamma)}$ defined in Tab. 3.5, and the harmonics of Eqs. (5.12), (5.13).

The HF hamiltonian in the AF phase. By following a procedure very similar to Sec. 5.2, we obtain the renormalised AF hamiltonian

$$\hat{H} - \mu\hat{N} \overset{\mathrm{HF}}{\simeq} \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \tilde{\epsilon}_{\mathbf{k}\sigma} \left(\hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right)$$
$$- \sum_{\sigma} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left(\tilde{\Delta}_{\mathbf{k}\sigma} \hat{\psi}_{\mathbf{k}\sigma} + \tilde{\Delta}^*_{\mathbf{k}\sigma} \hat{\psi}^\dagger_{\mathbf{k}\sigma}\right) - \tilde{\mu}\hat{N} + E_{\mathrm{c}}^{(\mathrm{AF})} , \tag{5.63}$$

where we defined the contraction energy,

$$\frac{E_{\mathrm{c}}^{(\mathrm{AF})}}{L_x L_y} \equiv U(m^2 - n^2) + V \left[2zn^2 - \frac{1}{2} \sum_{\gamma \neq s} \sum_{\sigma} \left(\left|u_\sigma^{(\gamma)}\right|^2 + \left|v_\sigma^{(\gamma)}\right|^2 \right) \right] . \tag{5.64}$$

The operator of Eq. (5.63) can be further elaborated to obtain the Bogoliubov representation of the HF hamiltonian, which describes a free Fermi gas of fermions,

$$\hat{H} - \mu\hat{N} \overset{\mathrm{HF}}{\simeq} \sum_{\mathbf{k}\in\mathrm{MBZ}} \sum_{\sigma} \tilde{E}_{\mathbf{k}\sigma} \left(\hat{\gamma}^{(+)\dagger}_{\mathbf{k}\sigma} \hat{\gamma}^{(+)}_{\mathbf{k}\sigma} - \hat{\gamma}^{(-)\dagger}_{\mathbf{k}\sigma} \hat{\gamma}^{(-)}_{\mathbf{k}\sigma} \right) - \tilde{\mu}\hat{N} + E_{\mathrm{c}}^{(\mathrm{AF})} , \tag{5.65}$$

where the $\{\hat{\gamma}^{(\pm)}_{\mathbf{k}\sigma}\}$ operators are linked to electronic operators $\{\hat{c}_{\mathbf{k}\sigma}, \hat{c}^\dagger_{\mathbf{k}\sigma}\}$ by a unitary mapping, which is the renormalised version of the rotation of Eq. (5.44). We defined the renormalised Bogoliubov bands,

$$\tilde{E}_{\mathbf{k}\sigma} = \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + |\tilde{\Delta}^2_{\mathbf{k}\sigma}|} . \tag{5.66}$$

All observable quantities are independent of spin index, and therefore from now on we will omit the subscript σ. The analytic derivation of all these results is reported in App. E, where we also report the analytic expressions for the entropy density $s^{(\mathrm{AF})}$, and the free energy density $f^{(\mathrm{AF})}$, respectively in Eqs. (E.58) and (E.61).

The HFPs of the AF phase. In the conventional HM, magnetisation m is the only HFP and obeys the self-consistency Eq. (5.52), which holds identically in this context upon renormalising all involved quantities. A similar statement holds for density n, for which Eq. (5.53) can be extended through a renormalisation procedure. We report the renormalised version of these equations, respectively, in Eqs. (E.19), (E.21).

The other variational parameters of the AF phase are "hidden" inside the HF hamiltonian of Eq. (5.63). Indeed, recalling the definitions of the renormalised bands, Eq. (5.61), and the renormalised gap, Eq. (5.62), we see that the harmonics

$$u^{(s^*)}, u^{(d)} ,$$

Phase	**HFPs**
sAF	$\{m, u^{(s^*)}, u^{(d)}, v^{(s^*)}, v^{(d)}\}$
aAF	$\{m, u^{(s^*)}, u^{(d)}, -iv^{(p_x)}, -iv^{(p_y)}\}$

Table 5.2 | Set of HFP for the two possible realisations of the AF phase, respectively with a symmetric and an antisymmetric imaginary part of the gap.

and

$$v^{(s^*)}, v^{(p_y)}, v^{(p_x)}, v^{(d)},$$

are the remaining HFPs. These harmonics obey, respectively, the self-consistency Eqs. (5.12), (5.13), which can be reformulated by expressing the EVs $\langle \hat{n}_{\mathbf{k}\sigma} \rangle$ and $\langle \hat{\psi}_{\mathbf{k}\sigma} \rangle$ in terms of EVs of components of pseudospin operators, similarly to what we did for the magnetisation m in Eq. (5.52). The explicit version of these self-consistency equations is reported in Tabs. E.1, E.2.

The harmonics $u_\sigma^{(\gamma)}$ have the identical role we described for the normal phase, in Sec. 4.3.1. Indeed, the renormalisation of bare bands of Eq. (5.61) coincides exactly with that discussed for the normal phase, and reported in Eq. (4.25). Therefore, Eq. (4.28) hold identically,

$$\tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2} \quad \text{and} \quad \tilde{t}^{(d)} \equiv -\frac{u^{(d)}V}{2}, \tag{5.67}$$

and so do Eqs. (4.34), (4.35),

$$\tilde{t}^{(x)} \equiv \frac{\tilde{t}^{(s^*)} + \tilde{t}^{(d)}}{2}, \tag{5.68}$$

$$\tilde{t}^{(y)} \equiv \frac{\tilde{t}^{(s^*)} - \tilde{t}^{(d)}}{2}. \tag{5.69}$$

If $u^{(s^*)} \neq 0$, the hopping amplitude is isotropically renormalised. If $u^{(d)} \neq 0$, planar rotational symmetry C_4 is broken and renormalised hopping is different in directions x and y.

It can be shown (see Sec. E.1.2) that

$$v^{(s^*)}, v^{(d)} \in \mathbb{R}, \qquad v^{(p_x)}, v^{(p_y)} \in i\mathbb{R}. \tag{5.70}$$

Recalling the definition of gap function of Eq. (5.62), and the functions $\varphi_{\mathbf{k}}^{(\gamma)}$ of Tab. 3.5, we see that the correction to the gap function in Eq. (5.62) is purely imaginary,

$$\tilde{\Delta}_{\mathbf{k}} - \Delta_{\mathbf{k}} = -iV \sum_{\gamma \neq s} v^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} \in i\mathbb{R}.$$

Therefore, the non-local attraction $\hat{H}_V$ renormalises the AF gap by adding an imaginary part, as we anticipated in Tab. 5.1. We denote this effect as "gap complexification".

The four harmonics $v^{(\gamma)}$ cannot be all non-zero simultaneously. In particular, the imaginary correction to the gap function can be either inversion-symmetric or inversion antisymmetric (see Sec. E.2.1 for details). Therefore, we can divide the HFPs in two sets, which we report in Tab. 5.2, based on the symmetry properties of the harmonics $v^{(\gamma)}$. Specifically, we can identify two possible "flavours" of commensurate antiferromagnetism, respectively "sAF" (symmetric antiferromagnet) and "aAF" (antisymmetric antiferromagnet).

Self-consistency constraints. In Tab. 5.2 we collected the two sets HFPs for the two flavours of the AF phase. Each set contains five parameters. The scenario appears quite complicated, but it is possible to show that two constraints hold:

- The bands renormalisation is actually a shrinking, and there is no nematic effect,

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0. \tag{5.71}$$

Looking at Eqs. (5.68), (5.69), the second relation implies that C_4 is unbroken, identically to the normal phase. Moreover, looking at Eqs. (5.67), the first relation implies that the renormalised hopping is purely isotropic (s^*-wave), and

$$0 < \tilde{t}^{(s^*)} < t\,.$$

- The imaginary correction to the AF gap is identically zero, i.e.

$$V \geq 0 \quad \Longrightarrow \quad \forall\gamma \in \{s^*, p_x, p_y, d\} \quad : \quad v^{(\gamma)} = 0\,. \tag{5.72}$$

  Therefore, in both rows of Tab. 5.2, only two HFPs remain: m and $u^{(s^*)}$. As a consequence, there is no need to distinguish two AF phases,

$$\mathrm{sAF} = \mathrm{aAF}\,.$$

The proofs of Eqs. (5.71) and (5.72) are given in App. F. A consequence of Eq. (5.72) is that the AF gap coincides with the one given by Eq. (5.27),

$$\tilde{\Delta}_{\mathbf{k}} = \Delta_{\mathbf{k}} = mU\,. \tag{5.73}$$

5.4 Mott localisation-enhanced magnetisation

From the analysis of antiferromagnetism in the Hubbard model, we know that the local repulsion $\hat{H}_U$ induces magnetisation. In this section, we show that the bands shrinking induced by $\hat{H}_V$ enhances it at half-filling.

The AF phase is described by two HFPs: $u^{(s^*)}$ and m. Moreover, we know from the constraint of Eq. (5.71) that the hopping parameter is reduced in amplitude,

$$t \quad \rightarrow \quad 0 < \tilde{t}^{(s^*)} < t\,.$$

Bands are effectively shrunk by the non-local attraction, as happened for the normal phase. Consider the Bogoliubov bands $\tilde{E}_{\mathbf{k}}$ given in Eq. (5.66),

$$\tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + |mU|^2}\,,$$

where we inserted Eq. (5.73). If $mU \gg \tilde{t}^{(s^*)}$, these are approximated by

$$\begin{aligned}\tilde{E}_{\mathbf{k}} &= |mU|\sqrt{1 + \left(\frac{\tilde{\epsilon}_{\mathbf{k}}}{mU}\right)^2} \\ &\simeq mU\left[1 + \frac{1}{2}\left|\frac{\tilde{\epsilon}_{\mathbf{k}}}{mU}\right|^2 + \mathcal{O}\left(r^4\right)\right]\,,\end{aligned} \tag{5.74}$$

where we defined the ratio

$$r \equiv \frac{\tilde{t}^{(s^*)}}{mU}\,, \tag{5.75}$$

being $\tilde{\epsilon}_{\mathbf{k}} = \mathcal{O}\left(\tilde{t}^{(s^*)}\right)$.

Let us limit to the half-filling condition, $\tilde{\mu} = 0$. It can be shown (see Sec. E.1) that the self-consistency Eq. (5.52) for the magnetisation m can be reformulated, with renormalised quantities, as

$$m = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\,. \tag{5.76}$$

By inserting Eq. (5.74) in Eq. (5.76), we obtain

$$\begin{aligned}m &= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{mU}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, 0] \\ &\simeq \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \left[1 - \frac{1}{2}\left|\frac{\tilde{\epsilon}_{\mathbf{k}}}{mU}\right|^2 + \mathcal{O}\left(r^4\right)\right] \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\,.\end{aligned} \tag{5.77}$$

Let $m \neq 0$. From Eq. (E.63) we know that $\tilde{E}_{\mathbf{k}} \neq 0$. The hyperbolic tangent is, at low temperature, approximated by,

$$\lim_{\beta\to\infty} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) = 1 \,.$$

Therefore, Eq. (5.77) becomes

$$m \simeq \frac{1}{2} - \underbrace{\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \left|\frac{\tilde{\epsilon}_{\mathbf{k}}}{mU}\right|^2}_{\mathcal{O}(r^2)} + \mathcal{O}\left(r^4\right) \,. \tag{5.78}$$

The second order correction can be made small in two ways:

1. By increasing U. From Eq. (5.75), we see that r decreases as U grows. This is the "conventional" way of magnetising the Hubbard lattice, by taking advantage of local repulsion which induces direct exchange (see App. A).

2. By shrinking bands. From Eq. (5.75), we see that as the renormalised bandwidth goes to zero, so does r. This is the "unconventional" way of magnetising the Hubbard lattice, where localisation is not necessarily due to a strong on-site repulsion, rather a low electronic mobility on the lattice.

Hence, as we wanted to prove, a strong Mott flattening can induce commensurate antiferromagnetism.

5.5 Numeric results for the commensurate AF phase

In this section, we present numeric results. For the AF phase, the HFPs vector has two components:

$$\mathbf{v} = \begin{bmatrix} m \\ u^{(s^*)} \end{bmatrix} \quad \text{with} \quad \mathbf{v} = \mathbb{R}^2 \,.$$

The RMPs are, as for the normal phase:

$$\tilde{t}_{ij} \equiv \left(t - \frac{u^{(s^*)}V}{2}\right)\varphi_{ij}^{(s^*)} \,,$$
$$\tilde{\mu} \equiv \mu + n(2zV - U) \,.$$

We simulate in the ranges:

$$0 \leq U \leq 8 \,, \qquad 0 \leq V \leq 8 \,,$$

and limit to $\delta = 0$, since we restrict to the commensurate AF phase which is stable only at half-filling (see Sec. 5.2.2, last paragraph). Tolerances on density and HFPs have been set to

$$\Delta n = 10^{-2} \,, \qquad \Delta\mathbf{v} = \begin{bmatrix} m\colon 5\times 10^{-4} \\ u^{(s^*)}\colon 5\times 10^{-4} \end{bmatrix} \,.$$

We will show heatmaps in the (U, V) plane. For each heatmap, we will superimpose six arbitrarily chosen contour lines, in order to highlight how isovalued domains are shaped. These contour lines will be coloured in gray scale. We limit to low temperature,

$$\beta = 100.0 \,,$$

where we are using absolute units (see Sec. 4.5).

5.5.1 Mott localisation in the AF phase

In this subsection, we present results for the HFP $u^{(s^*)}$ and the RMP $\tilde{t}^{(s^*)}$.

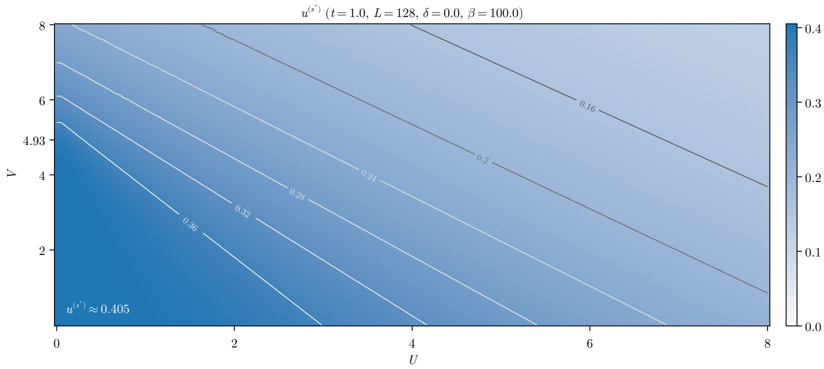
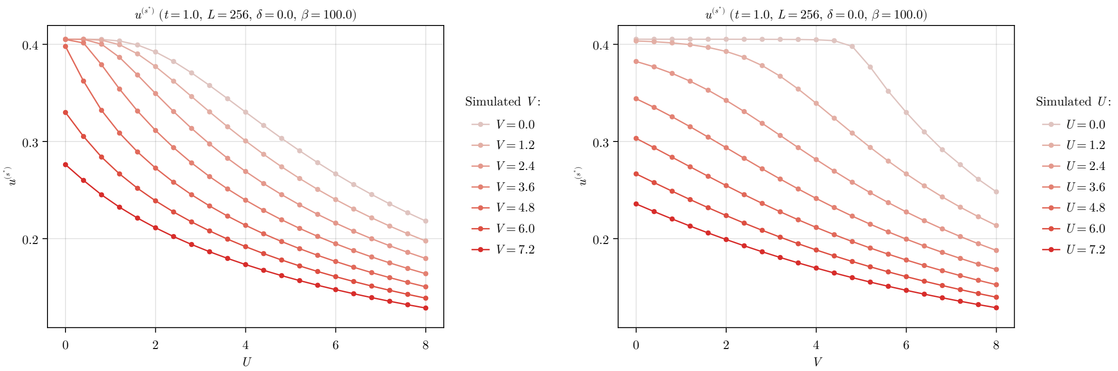


Figure 5.4 | Numerical results for the HFP $u^{(s^*)}$. The gray curves indicate isovalued lines.



a | U-dependency of $u^{(s^*)}$ varying V.

b | V-dependency of $u^{(s^*)}$ varying U.

Figure 5.5 | Numerical results for the HFP $u^{(s^*)}$ as we vary U (left) and V (right), keeping the other variable fixed.

Numerical results for $u^{(s^*)}$. In Fig. 5.4, we show the converged values of the HFP $u^{(s^*)}$ in the (U, V) plane at $t = 1.0$. We see that $u^{(s^*)}$ is maximised in the bottom-left corner (low-U, low-V limit). As we see from the colour plot, $u^{(s^*)}$ exhibits a sort of triangle-shaped plateau in this region, where it reaches the value

$$u^{(s^*)} \lesssim 0.405\,.$$

The above converged value coincides with that computed at half-filling for the normal phase, in Eq. (4.49). This is expected at $U = 0$, where magnetisation is necessarily null and we reduce to the normal phase. The approximate plateau in the bottom-left corner of Fig. 5.4 extends to $U/t \lesssim 2$, and indicates the region of parameters space where we expect magnetisation to be low enough to retrieve, approximately, the normal phase. We indicated the tick $V = 4.93t$ on the vertical axis, where $U = 0$. We know from Eq. (4.53) that, at half-filling, it is the critical value for the Mott transition. Indeed, we see that the plateau of $u^{(s^*)}$ has the point

$$(U/t, V/t) = (0, 4.93)$$

as one of its "corners".

In Fig. 5.4, we see that isovalued domains are, specifically, straight lines. Moreover, we see that as we move to the top-right corner (high-U, high-V) the isovalued lines become approximately parallel and $u^{(s^*)}$ decreases steadily.

The fact that $u^{(s^*)}$ is a decreasing function of both U and V is confirmed by looking at Fig. 5.5. In the left panel (Fig. 5.5a), we see that $u^{(s^*)}$ decreases as a function of U. For small V (brighter colour), the curves show a pleateau for $U \to 0$, which disappears upon increasing V. An analogous behaviour is observed in the right panel (Fig. 5.5b), by varying V at fixed values of U. We notice that, since $m = 0$ for $U = 0$, the $U = 0$ curve of Fig. 5.5b coincides with the one at $\delta = 0$ for the normal phase, reported in Fig. 4.5b.

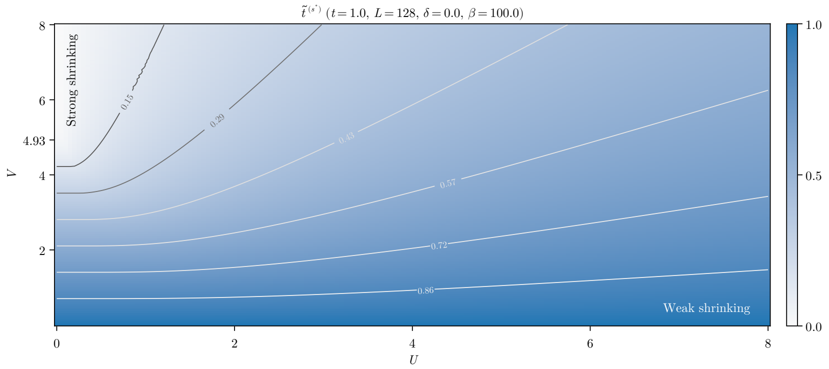
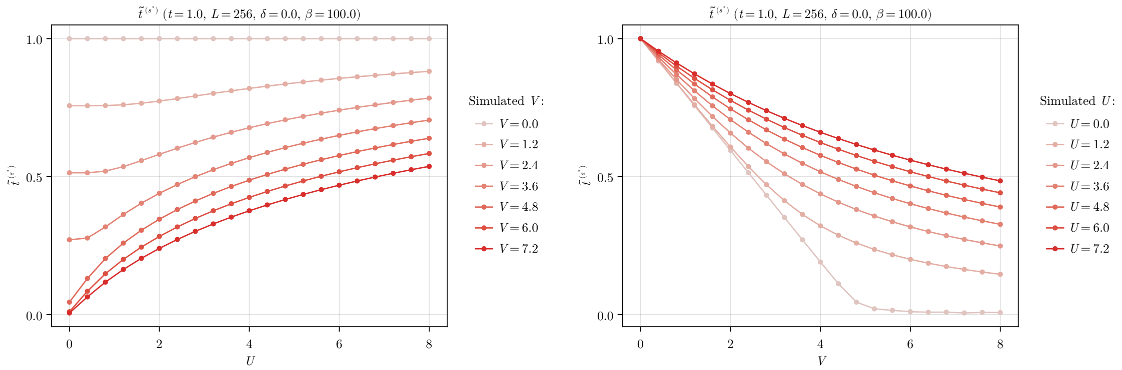


Figure 5.6 | Numerical results for the RMP $\tilde{t}^{(s^*)}$. The gray curves indicate isovalued lines.



a | U-dependency of $t^{(s^*)}$ varying V.

b | V-dependency of $t^{(s^*)}$ varying U.

Figure 5.7 | Numerical results for the RMP $\tilde{t}^{(s^*)}$ as we vary U (left) and V (right), keeping the other variable fixed.

Hopping renormalisation. In Fig. 5.6, we show the the corresponding values for the renormalised hopping amplitude, defined as

$$\tilde{t}^{(s^*)} = t - \frac{u^{(s^*)}V}{2} \,.$$

As is clear from the definition, isovalued curves of $\tilde{t}^{(s^*)}$ and $u^{(s^*)}$ do not coincide. Specifically, we see that the top-left (low-U, high-V) and bottom-right (high-U, low-V) corners differ significantly (while, for $u^{(s^*)}$, they were similar enough). In the first case, $\tilde{t}^{(s^*)}$ is minimised; in the second, is maximised. The two corners correspond, respectively, to an “strong shrinking” and a “weak shrinking” of bare bands. The region of strong shrinking, namely the brighter region of Fig. 5.6, extends from the critical value $(U/t, V/t) = (0, 4.93)$ to small values of the local repulsion, approximately $U \lesssim 1$. While isovalued lines in the weak shrinking region are approximately parallel to the horizontal axis, moving towards the strong shrinking region they bend progressively.

Apart from the vertical axis, $U = 0$, the converged value of $\tilde{t}^{(s^*)}$ never vanishes (within numerical tolerance). This is better appreciated in Fig. 5.7, where we show data for $\tilde{t}^{(s^*)}$ as we vary U and V. In the right panel (Fig. 5.7b) we see that increasing V while U is fixed makes $\tilde{t}^{(s^*)}$ decrease monotonically, but at any $U \neq 0$ we do not reach in our simulation range a condition of total flattening. In the left panel (Fig. 5.7a), we see that as we increase U, the renormalised hopping amplitude gets larger.

In Fig. 5.8 we show a comparative plot of the converged values for $\tilde{t}^{(s^*)}$ for the normal phase (red surface) and the AF phase (blue surface). Being the normal phase independent of U, the data for the normal phase are a concatenation at various values of the non-local repulsion of the $\delta = 0.0$ data shown in Fig. 4.6. We see that the blue surface reduces to zero only at $U = 0$, whereas the red surface is consistently null at $V \gtrsim 4.93t$. As we mentioned before, in the AF phase we

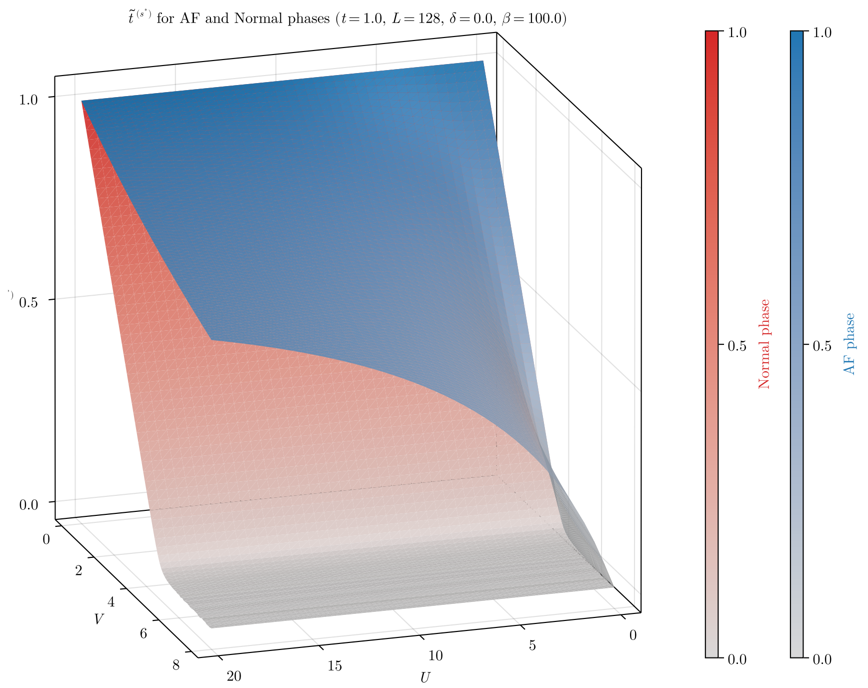


Figure 5.8 Comparative plot of the converged values for $\tilde{t}^{(s^*)}$ for the normal phase and the AF phase.

do not observe a complete flattening of bare bands. On the contrary, the presence of U reduces significantly the bands shrinking effect.

For the AF phase, we conclude that the non-local attraction $\hat{H}_V$ induces the flattening of bare bands, while the local repulsion $\hat{H}_U$ does the opposite. When we discussed Mott localisation in the context of the normal phase (see Sec. 4.5.1), we argued that bands flattening was induced by V, while U was irrelevant. Here we see that in the AF phase, the local repulsion opposes Mott localisation. This is somewhat the opposite of what we would expect, based on the phenomenology of Mott localisation, which is in general enhanced by the repulsion of two electrons on the same site (see Sec. 1.1.1).

The above observation need to be addressed in the context of AF ordering, where magnetisation m is a macroscopic observable. Recall the discussion of Sec. 5.4: we claimed that the flattening of bare bands could induce magnetisation. Indeed, if by increasing U we are "by force" enhancing the AF gap (even at low magnetisation), by increasing V we are reducing the bare bandwidth. This suggests that we can obtain the same macroscopic magnetisation even if the underlying physics is much different. Indeed, as we observe from Fig. 5.7, it is only by increasing the non-local attraction that the actual hopping amplitude of electrons in reduced. From the above considerations, we observe that either increasing U or V we should expect approximately the same magnetic ordering.

5.5.2 Mott-induced enhancement of magnetisation

In Fig. 5.9 we show the converged results for m as a function of U and V. In the bottom-left corner (low-U, low-V), a bright triangle-shaped region is present in this area, where magnetisation is low. While we cannot assess precisely if this low-magnetisation region has the same shape of the plateau observed in Fig. 5.4, we see that they share the same top-left corner at $(U/t, V/t) = (0, 4.93)$. Indeed, consider a path starting from the low-m region and moving vertically, namely increasing V while keeping U fixed. If U is small enough, the transition of m to approximately zero to its saturation value happens as soon as we reach the strong shrinking region highlighted in Fig. 5.6.

In Fig. 5.9, we see that isomagnetic curves are actually straight lines, and become approximately parallel as we move towards the top-right corner of the plot. This is similar to what we found for the HFP $u^{(s^*)}$. The fact that isovalued curves are linearly shaped suggests that possibly some

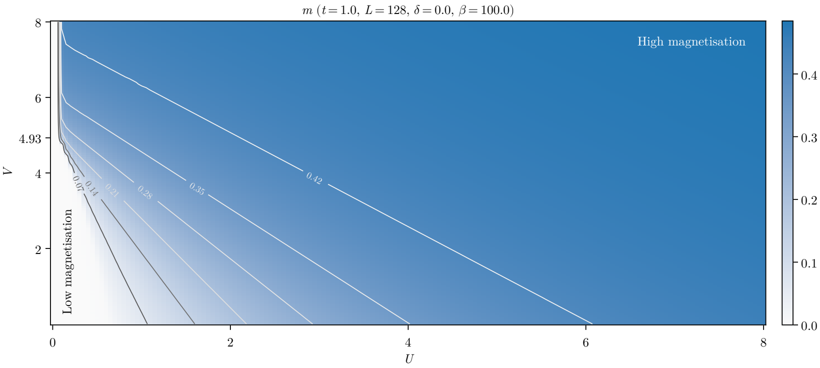
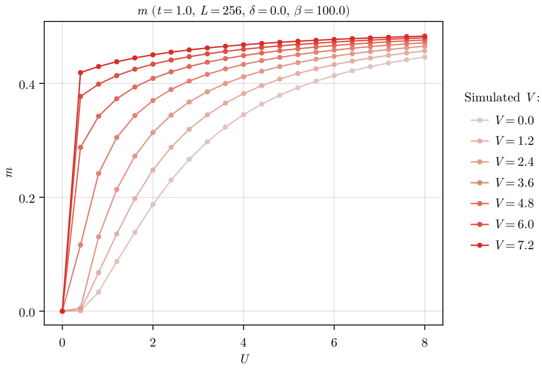
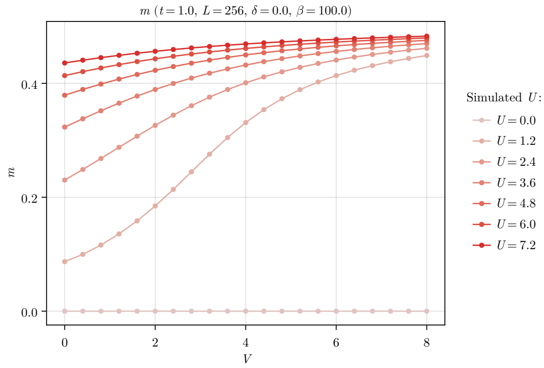


Figure 5.9 | Numerical results for the HFP m. The gray curves indicate isomagnetic lines.



a | U-dependency of m varying V.



b | V-dependency of m varying U.

Figure 5.10 | Numerical results for the HFP m as we vary U (left) and V (right), keeping the other variable fixed.

kind of critical ratio among the interactions exists, such that the net sum of local repulsion and Mott physics leads to the same macroscopic magnetic ordering. This relation should be, roughly, something like

$$\frac{V}{V_0} = \frac{U_0 - U}{U_0},$$

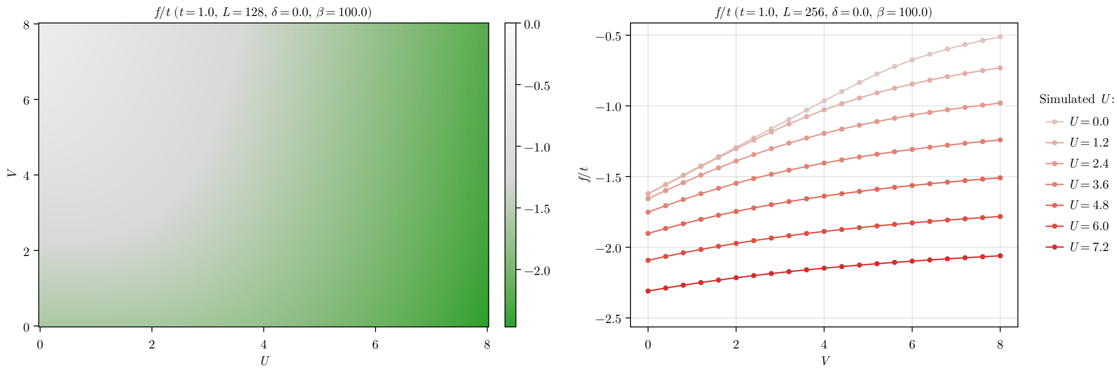
where V_0 is the intersection of the isomagnetic line with the vertical axis ($U = 0$) and U_0 with the horizontal axis ($V = 0$).

In Fig. 5.10 we show the converged data for m as we vary, alternatively, U (left panel) and V (right panel). As we already commented, magnetisation is induced by both interactions, but the general shape of the curves is slightly different. In Fig. 5.10a we see that m saturates rapidly, but as we lower U all curves go to zero. In Fig. 5.10b, instead, lowering V the curves reach a finite intersection with the vertical axis. The local repulsion is necessary in order to establish AF ordering, while the non-local attraction acts – at fixed U – as a sort of magnetisation boost.

5.5.3 Solution free energy density

In this section, we show the data for the free energy density, computed at half-filling using Eq. (E.62). In Fig. 5.11 the free energy density is plotted. In the right panel, Fig. 5.11b, we see that free energy is minimised in the bottom-right corner (low-V and high-U). In the top-left corner of Fig. 5.11a free energy grows, eventually reaching zero in the strong shrinking region.

It is then evident how, even if the macroscopic magnetic properties are the same along one isomagnetic line of those plotted in Fig. 5.9, and thus the realised state exhibits an identical value of m, much different is the energetic content of the state itself. Increasing U while keeping V fixed, we amplify the gap. Hence, Bogoliubov bands $\pm\tilde{E}_{\mathbf{k}}$ are shifted away, namely the lower band is pushed towards more negative energies and the upper band towards more positive energies. At



a | Heatmap of f in δV plane.

b | V-dependency of f varying δ.

Figure 5.11 | Free energy density of the AF phase.

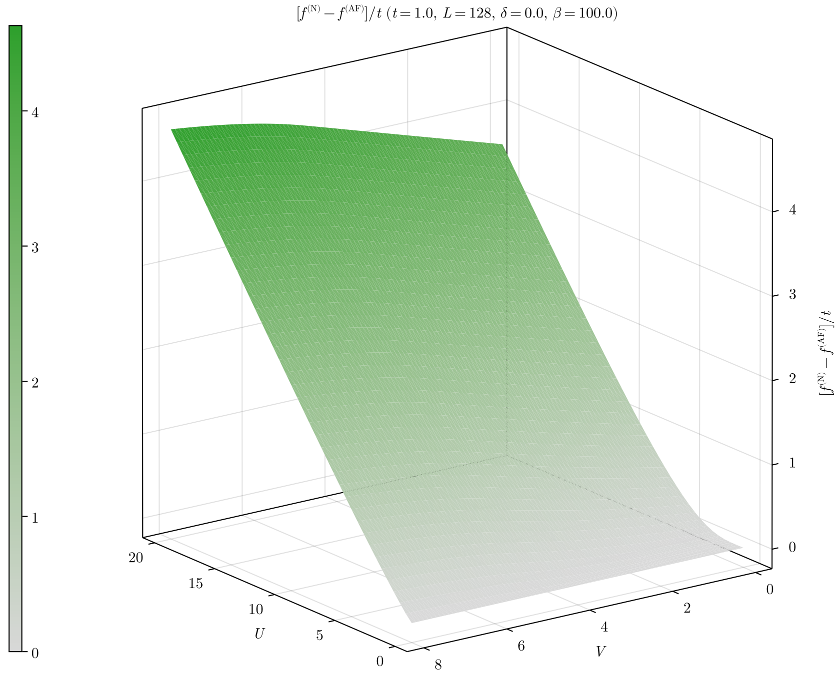


Figure 5.12 | Difference of free energies for the normal phase minus the AF phase.

half-filling and low temperature, the lower band is approximately completely occupied while the upper band is approximately completely empty. Hence, quasiparticles lower their energy due to the amplification of the gap. On the other hand, increasing V while keeping U fixed does the opposite. As bands get flattened, their distance lowers and quasiparticles increase their energies (namely, the lower band gets closer to zero). Hence, even if we observe isomagnetic domains, this does not mean that the content in energy of the HF ground state is identical.

The commensurate AF phase is always energetically favoured at half-filling, with respect to the normal phase. In Fig. 5.12 we plot the energy difference

$$f^{(\mathrm{N})} - f^{(\mathrm{AF})} .$$

The data for the normal phase, similarly to Fig. 5.8, are a concatenation of those of Fig. 4.8 at $\delta = 0.0$. We see that, on the entire $(U/t, V/t)$ region of our simulations, the free energy difference is positive. Hence the AF phase is consistently more stable with respect to the normal phase, except for the segment at $U = 0$, where the two coincide. Looking at Fig. 5.12, we see that the energy difference is enhanced as we increase V at fixed U.

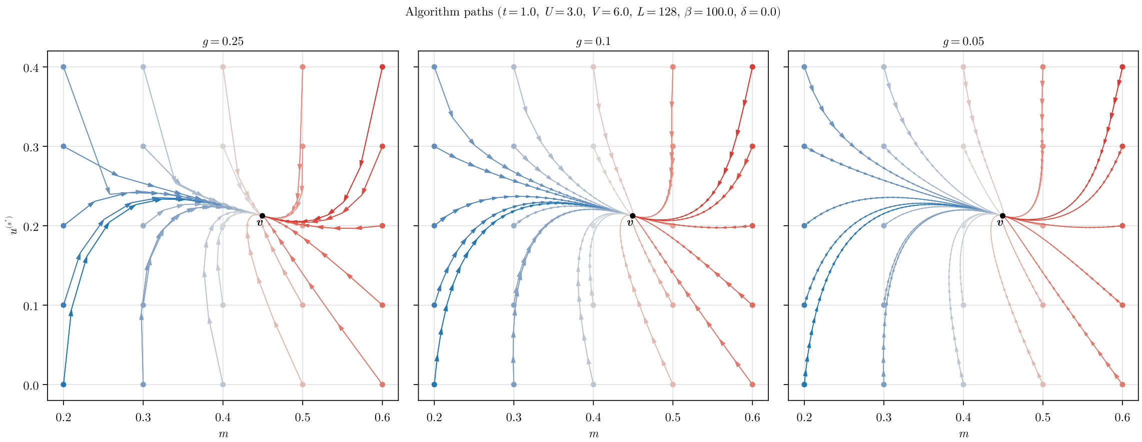


Figure 5.13 Quiver plot of algorithmic evolution in HFPs space at fixed model parametrisation. The black dot $\mathbf{v}$ represents the nonlinear map fixed point, while the coloured paths the sequences of HFPs coordinates simulated via iterative HF procedure, starting from different HFPs initialisers (coloured dots).

At low temperature and half filling, it is not worth plotting entropy density. Half-filled commensurate antiferromagnets are insulators, with two gapped Bogoliubov bands. At low temperature, electronic distribution fills completely the lower band and empties the upper one. Because of this, insulating gapped systems inherently are low entropic and our results are compatible with zero in the entire region of parametric space we explored.

5.5.4 Fixed-point interpretation of HF solution

We end this chapter, by presenting a final remark regarding the numerical structure of the algorithm. As we anticipated in Sec. 2.3.2, our convergence scheme can be seen as an effective search for the singular fixed point of a nonlinear map. A physical solution is represented as a HFP vector $\mathbf{v}$ such that the algorithm is a local identity,

$$\mathrm{A}(\mathbf{v}) = \mathbf{v}\,.$$

To predict, for a given initialiser $\mathbf{v}_0$, the path of the algorithm in HFPs space is in general non-trivial due to the intrinsic nonlinearity of the problem. An example of such nonlinearity is given in the derivation of Secs. F.2.2, F.2.3.

In Fig. 5.13 we sketch this concept in a practical situation. In figure are plotted various algorithmic paths through AF HFPs space, with operational setup at

$$U = 3\,, \qquad V = 6\,, \qquad L = 128\,, \qquad \beta = 100\,, \qquad \delta = 0\,.$$

At this point, the self-consistent HF “fixed-point” solution is given by

$$m \approx 0.449\,, \qquad u^{(s^*)} \approx 0.212\,,$$

indicated in panels as the black dot. The three panels show different map evolutions varying the mixing parameter within three choices, $g \in \{0.25, 0.1, 0.05\}$. We chose various initialisers in the set:

$$\mathbf{v}_0 = \begin{bmatrix} m_0 \\ u_0^{(s^*)} \end{bmatrix} \quad : \quad m_0 \in \{0.2,\ 0.3,\ ...,\ 0.6\} \quad \text{and} \quad u_0^{(s^*)} \in \{0.0,\ 0.1,\ ...,\ 0.4\}\,.$$

In figures, different colours indicate different evolutions, while the tip size indicates the magnitude of the algorithmic step in passing from one iteration to the following.

All paths are able to converge to $\mathbf{v}$, and the quantity of steps necessary to reach the fixed point, a sort of attractor, increases as we lower g. To select the correct g turns out to be the real algorithm fine-tuning knob: as we saw for the normal phase, some model parametrisation could lead to unavoidable algorithmic instabilities. The problem is that we do not know beforehand where the instabilities will occur, and how. The correct g is the compromise between keeping low the number of steps (larger mixing) and moving slow enough to not diverge when encountering unstable points (smaller mixing). By recursively fine-tuning g repeating non-converged simulation with more care, we are essentially making the nonlinear map “softer” and curvier, as is in the right panel of Fig. 5.13. It’s not always resource convenient to do so, so for most simulations we stick to a rougher map (like the left panel one) by setting $g = 0.5$.

Chapter 6

Superconducting phase

In this chapter, we study the translationally invariant superconducting (SC) phase. We observe various effects: bare bands shrinking, insurgence of mixed-symmetry superconductivity and nematicity.

This chapter is organised as follows. In Sec. 6.1, we present the translationally invariant superconducting (SC) phase and its symmetries. In Sec. 6.2, we report the HF approximation to the EHM hamiltonian in the normal phase, the self-consistency equations and two constraints for the various HFPs. In Sec. 6.3, we present the results of numeric simulations.

The analytical computations for the SC phase are almost identical to the ones carried out for the AF phase. For the sake of conciseness, in this section we limit to reporting the essential result, while the entire derivation is moved to App. G.

6.1 The SC phase and its symmetries

The translationally invariant SC phase is described by the SSB

$$\mathrm{U}^z(1) \overset{\mathrm{SSB}}{\to} 0\,,$$

as we discussed in Sec. 3.2. In the SC phase, we have

$$\langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \rangle \neq 0\,,$$

because charge is not conserved. We discussed this fact in Sec. 3.2, when we proved that EVs of products of two creation operators are zero when charge is conserved. The above inequality has a precise meaning: adding two electrons to the ground state produces a state with a finite overlap with the ground state itself. In BCS theory, this is interpreted with the appearance of pairs of electrons bound together by an attractive interaction, i.e. Cooper pairs [4, 14].

Cooper pairs are composite quantum objects, with spatial and spin structure. Being couples of electrons, they obey antisymmetry rules under internal exchange, and their total spin follows the composition rules of spin algebras

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1\,,$$

which means that two $\mathfrak{su}(2)$ algebras are composed into a singlet and a triplet. The first is antisymmetric under spin exchange; the second is symmetric. Hence, we can distinguish two Cooper pairing channels:

1. Singlet, with total spin equal to 0. In this channel, the Cooper pairs must exhibit spatial inversion symmetry.

2. Triplet, with total spin equal to 1. In this channel, the Cooper pairs must exhibit spatial inversion anti-symmetry.

This concept is summarised in Tab. 6.1. We can choose to restrict to just one of these pairing channels. This provides specific symmetry selection rules on the SC gap function harmonics. In this thesis, we restrict to the singlet channel.

Spatial structure of Cooper pairs	Pairing channel
Inversion symmetric	Singlet pairing
Inversion anti-symmetric	Triplet pairing

Table 6.1 Relation between the spatial inversion (intended as $(x, y) \to (-x, -y)$) symmetry properties and the pairing channel for Cooper pairs.

As we did previously, we also allow for breaking of rotational symmetry,

$$\mathrm{C}_4 \stackrel{\mathrm{SSB}}{\to} \mathrm{C}_2 \,.$$

We will observe the breaking of planar rotational symmetry in the SC phase.

6.2 The HF hamiltonian and the variational parameters

In this section, we summarise the computations carried out in App. G and report the final form of the HF hamiltonian. We define the operators

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \,, \tag{6.1}$$

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}_{-\mathbf{k}\downarrow}\hat{c}_{\mathbf{k}\uparrow} \,, \tag{6.2}$$

and their expectation values (EVs),

$$u_{\mathbf{k}\sigma} \equiv \langle \hat{n}_{\mathbf{k}\sigma} \rangle \,, \tag{6.3}$$

$$w_{\mathbf{k}} \equiv \langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \,. \tag{6.4}$$

The breaking of charge conservation, namely $\mathrm{U}^z(1)$ symmetry, allows $w_{\mathbf{k}}$ to become non-zero. Instead, for a phase which preserves $\mathrm{U}^z(1)$, $w_{\mathbf{k}}$ is identically zero.

Using the functions of Tab. 3.5, we define the harmonics

$$u^{(\gamma)}_\sigma \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} u_{\mathbf{k}\sigma} \,, \tag{6.5}$$

$$w^{(\gamma)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} w_{\mathbf{k}} \,. \tag{6.6}$$

As was for the AF phase, these harmonics are HFPs in the SC phase and must be determined self-consistently through Eqs. (6.5) and (6.6).

The role of each contraction channel. Collecting the symmetry selection rules reported in Tabs. 3.2, 3.3, we see that the only non-zero contributions to the HF hamiltonian come from the following channels:

$\hat{H}_U$	Hartree and Cooper contractions,
$\hat{H}_V$ (o.s. sector)	Hartree and Cooper contractions,
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions.

Based on the discussion of Sec. 3.2.2, we divided the contributions of $\hat{H}_V$ in s.s. and o.s. channels. In Tab. 6.2 we report the role of each contraction channel.

The HF hamiltonian in the SC phase. In order to proceed, it is necessary to define the renormalised chemical potential,

$$\tilde{\mu} = \mu + n(2zV - U) \,, \tag{6.7}$$

Operator	**Sector**	**Channel**	**Net effect**
$\hat{H}_U$		Hartree	μ shift by $-nU$
		Cooper	Isotropic pairing in singlet channel
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
		Cooper	Anisotropic pairing in singlet and triplet channels
	s.s.	Hartree	μ shift by nzV
		Fock	Band renormalisation and resymmetrisation

Table 6.2 | Wick's contractions for each interaction and channel in the HF framework, and their net effect in the SC phase.

the renormalised shifted bands,

$$\tilde{\xi}_{\mathbf{k}} = \epsilon_{\mathbf{k}} + V \sum_{\gamma \in \{s^*, d\}} u^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} - \tilde{\mu}\,, \tag{6.8}$$

and the full SC gap

$$\tilde{\Delta}_{\mathbf{k}} = -U w^{(s)} + V \sum_{\gamma \neq s} w^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*}\,. \tag{6.9}$$

If the gap function is zero, $\tilde{\Delta}_{\mathbf{k}} = 0$, the SC phase reduces to the normal phase. These quantities are "renormalised" when compared to their respective values for the SC phase in the conventional HM.

Proceeding analogously to what we did for the AF phase, we obtain the HF approximation in the SC phase

$$\hat{H} - \mu \hat{N} \overset{\text{HF}}{\simeq} \sum_{\mathbf{k} \in \text{BZ}} \tilde{\xi}_{\mathbf{k}} \left(\hat{c}_{\mathbf{k}\uparrow}^{\dagger} \hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{-\mathbf{k}\downarrow}^{\dagger} \right) - \sum_{\mathbf{k} \in \text{BZ}} \left(\tilde{\Delta}_{\mathbf{k}} \hat{\phi}_{\mathbf{k}} + \tilde{\Delta}_{\mathbf{k}}^{*} \hat{\phi}_{\mathbf{k}}^{\dagger} \right) + E_{\text{a}}^{(\text{SC})} + E_{\text{c}}^{(\text{SC})}\,, \tag{6.10}$$

where we defined the anticommutation energy,

$$E_{\text{a}}^{(\text{SC})} \equiv \sum_{\mathbf{k} \in \text{BZ}} \tilde{\xi}_{\mathbf{k}}\,. \tag{6.11}$$

and the total contraction energy,

$$\frac{E_{\text{c}}^{(\text{SC})}}{L_x L_y} = -U \left(n^2 + \left| w^{(s)} \right|^2 \right) + V \left(2zn^2 - \sum_{\gamma} \left| u^{(\gamma)} \right|^2 + \sum_{\gamma} \left| w^{(\gamma)} \right|^2 \right)\,. \tag{6.12}$$

By further elaboration, we obtain the Bogoliubov representation of the HF hamiltonian, which describes a free Fermi gas of fermions,

$$\hat{H} - \tilde{\mu} \hat{N} \overset{\text{HF}}{\simeq} \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \tilde{E}_{\mathbf{k}} \left(\hat{\gamma}_{\mathbf{k}}^{(+)\dagger} \hat{\gamma}_{\mathbf{k}}^{(+)} - \hat{\gamma}_{\mathbf{k}}^{(-)\dagger} \hat{\gamma}_{\mathbf{k}}^{(-)} \right) + E_{\text{a}}^{(\text{SC})} + E_{\text{c}}^{(\text{SC})}\,, \tag{6.13}$$

where the $\{\hat{\gamma}_{\mathbf{k}\sigma}^{(\pm)}\}$ operators are linked to electronic operators $\{\hat{c}_{\mathbf{k}\sigma}, \hat{c}_{\mathbf{k}\sigma}^{\dagger}\}$ by a unitary mapping. We defined the Bogoliubov bands,

$$\tilde{E}_{\mathbf{k}} \equiv \sqrt{\tilde{\xi}_{\mathbf{k}}^2 + \left| \tilde{\Delta}_{\mathbf{k}} \right|^2}\,. \tag{6.14}$$

The analytic derivation of all these results is reported in App. G, where we also report the analytic expressions for the entropy density $s^{(\text{SC})}$, and the free energy density $f^{(\text{SC})}$, respectively in Eqs. (G.81) and (G.84).

Phase	HFPs
Singlet SC	$\{u^{(s^*)}, u^{(d)}, w^{(s)}, w^{(s^*)}, w^{(d)}\}$
Triplet SC	$\{u^{(s^*)}, u^{(d)}, w^{(p_x)}, w^{(p_y)}\}$

Table 6.3 | Set of HFP for the two possible realisations of the SC phase, respectively with a symmetric and an antisymmetric SC gap.

The HFPs of the SC phase. In the SC phase, we have two possible channels of Cooper pairing: singlet and triplet. We recall Tab. 6.1. The singlet pairing channel is described by a symmetric gap function $\tilde{\Delta}_{\mathbf{k}}$, while for the triplet channel the gap function is antisymmetric. This implies that, for a superconductor that exhibits translational invariance as well as C_2 symmetry, we have two sets of HFPs, reported in Tab. 6.3.

As was for the normal and AF phase, and is reported in Eq. (6.8), the occupation harmonics $u^{(\gamma)}$ are involved in the renormalisation of the bare bands. We report their self-consistency equations in explicit form, reformulating Eq. (6.5), in Tab. G.1. Instead, the HFPs $w^{(\gamma)}$ are components of the gap function, as we see in Eq. (6.9). Indeed,

$$\tilde{\Delta}^{(s)} = -Uw^{(s)}\,, \qquad \tilde{\Delta}^{(\gamma)} = Vw^{(\gamma)} \qquad \text{for } \gamma \neq s.$$

We report the explicit self-consistency equations of the harmonics $w^{(\gamma)}$, reformulating Eq. (6.6), in Tab. G.2.

In this thesis, we focus on the singlet channel, therefore the first rule of HFPs in Tab. 6.3. It is possible to prove that:

- Superconductivity is impossible on the conventional HM at HF level. This translates in the constraint:
$$V = 0 \quad \Longrightarrow \quad w^{(s)} = 0\,, \tag{6.15}$$
which implies that the SC gap is identically zero. This can be seen by plugging in Eq. (6.9) both $V = 0$ and $w^{(s)} = 0$. Eq. (6.15) is proven in Sec. G.1.4
- At $V \neq 0$, we can have $w^{(s)} \neq 0$. The harmonics of $w_{\mathbf{k}}$ obey the relation
$$\sum_{\gamma \neq s} \left|w^{(\gamma)}\right|^2 \geq M^{(\mathrm{SC})}\,, \tag{6.16}$$
where we defined:
$$M^{(\mathrm{SC})} \equiv \left|w^{(s)}\right|^2 \frac{U}{V}\,.$$
In other words, $w^{(s)}$ cannot be non-zero without at least one of the other two singlet harmonics – namely, s^*-wave and d-wave – being non-zero. Eq. (6.16) is proven in Sec. G.3.4.

Hopping renormalisation and nematicity. We conclude this summary by recalling the renormalisation of the hopping parameter in the context of the normal phase. Being the SC phase translationally invariant, and since the band renormalisation of Eq. (6.8) is formally identical to the one given in Eq. (4.25), the discussion of Sec. 4.3.1 holds identically, upon substituting in Eq. (4.36)

$$f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \to u_{\mathbf{k}}\,.$$

In particular, being

$$\tilde{t}^{(s^*)} \equiv t - \frac{u^{(s^*)}V}{2} \quad \text{and} \quad \tilde{t}^{(d)} \equiv -\frac{u^{(d)}V}{2}\,, \tag{6.17}$$

Eqs. (4.34), (4.35) hold,

$$\tilde{t}^{(x)} \equiv \frac{\tilde{t}^{(s^*)} + \tilde{t}^{(d)}}{2}\,, \tag{6.18}$$

$$\tilde{t}^{(y)} \equiv \frac{\tilde{t}^{(s^*)} - \tilde{t}^{(d)}}{2}\,. \tag{6.19}$$

If $u^{(s^*)} \neq 0$, the hopping amplitude is isotropically renormalised. If $u^{(d)} \neq 0$, planar rotational symmetry C_4 is broken and renormalised hopping is different in directions x and y.

6.3 Numeric results for the the singlet SC phase

In this section, we present the numeric results obtained from the iHF procedure. We limited simulations to the the singlet channel, leaving the triplet channel for future works. The singlet SC phase is described by five HFPs

$$\mathbf{v} = \begin{bmatrix} u^{(s^*)} \\ u^{(d)} \\ w^{(s)} \\ w^{(s^*)} \\ w^{(d)} \end{bmatrix} \quad \text{with} \quad \mathbf{v} = \mathbb{R}^5 \,,$$

leading to three RMPs

$$\begin{aligned} \tilde{t}^{(s^*)}_{ij} &\equiv \left(t - \frac{u^{(s^*)}V}{2} \right) \varphi^{(s^*)}_{ij} \,, \\ \tilde{t}^{(d)}_{ij} &\equiv -\frac{u^{(d)}V}{2} \varphi^{(d)}_{ij} \,, \\ \tilde{\mu} &\equiv \mu + n(2zV - U) \,. \end{aligned}$$

We notice that, in contrast to the AF and normal phases, the renormalised hopping has a finite d-wave component. We ran simulations at $\beta = 100.0$ (in absolute units, see Sec. 4.5) within the boundaries:

$$0 \leq U \leq 12 \,, \qquad 0 \leq V \leq 8 \,, \qquad 0 \leq \delta \leq 0.45 \,.$$

6.3.1 Multi-symmetry singlet superconductivity

In the singlet channel, the most general gap function is a combination of s, s^* and d components, and the resulting gap function $\tilde{\Delta}_{\mathbf{k}}$ emerges from a competition among these channels. Let us discuss the gap components of the self-consistent solution for the three symmetry channels separately.

s-wave gap. Consider the isotropic harmonic of the gap function, namely the s-wave component. In Fig. 6.1, we show the data for $w^{(s)}$, which determines the isotropic gap through

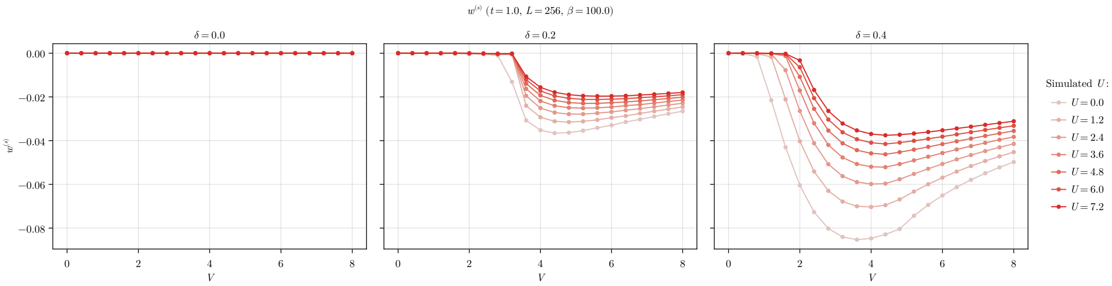
$$\tilde{\Delta}^{(s)} = -Uw^{(s)} \,.$$

We now recall the constraint of Eq. (6.15). In the conventional HM, at HF level, isotropic superconductivity is impossible. As we move to the EHM, namely setting $V > 0$, the SC gap can acquire a s-wave component, as long as the constraint of Eq. (6.16) is satisfied: $w^{(s)}$ can be non-zero if at least one of $w^{(s^*)}$, $w^{(d)}$ is non-zero. This is a necessary but not sufficient condition.

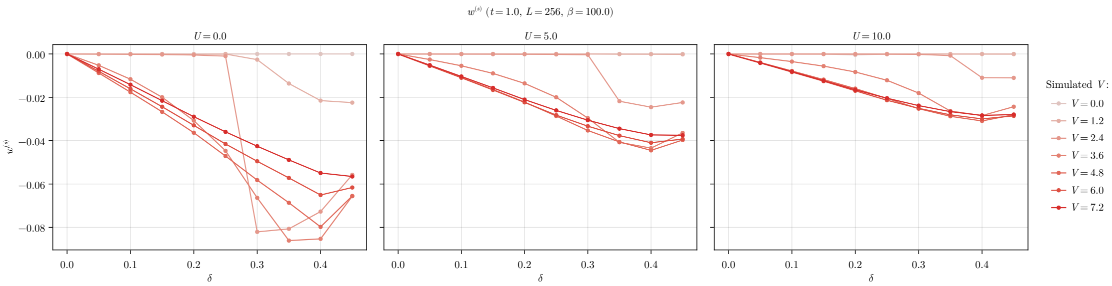
In Fig. 6.1a we see that $w^{(s)}$ is null at $\delta = 0$ (left panel), namely

$$\tilde{\Delta}^{(s)}\big|_{\delta=0} = 0 \,, \tag{6.20}$$

within the range of parameters we explored. Increasing doping (central and right panels), $w^{(s)}$ is null up to a critical value $V_c^{(s)}$, which depends also on U. For $V \geq V_c^{(s)}$, $w^{(s)}$ is negative and the shape of the curve depends on U. In particular, the amplitude $\left|w^{(s)}\right|$ exhibits a local maximum. We observe that doping lowers the critical value $V_c^{(s)}$, with $V_c^{(s)}/t \approx 1$, at almost unitary filling ($\delta = 0.4$). In Fig. 6.1b, we report the dependence of $w^{(s)}$ on doping. As anticipated, $w^{(s)} = 0$ at half-filling. Varying U/t and V/t we see that $w^{(s)} = 0$ reaches some negative minimum, whose absolute value lowers as U grows.

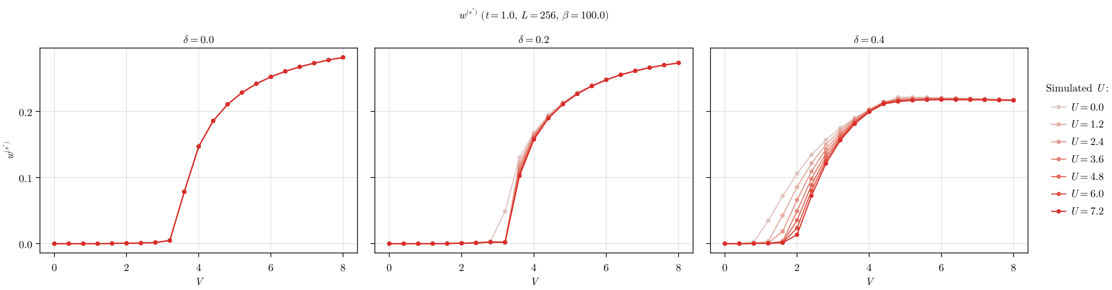


a | Dependency of $w^{(s)}$ on V at various Us, compared at three values of δ.

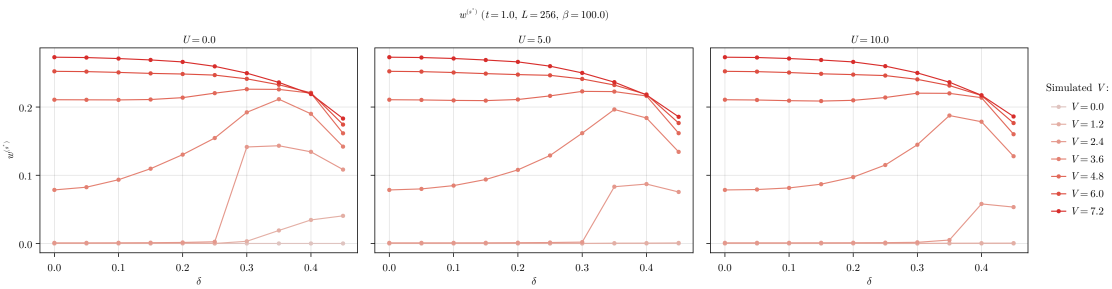


b | Dependency of $w^{(s)}$ on δ at various Vs, compared at three values of U.

Figure 6.1 | Dependency of the s-wave HFP $w^{(s)}$ on V at fixed U and δ (top panel, compared figures) and on δ at fixed V and U (bottom panel, compared figures).

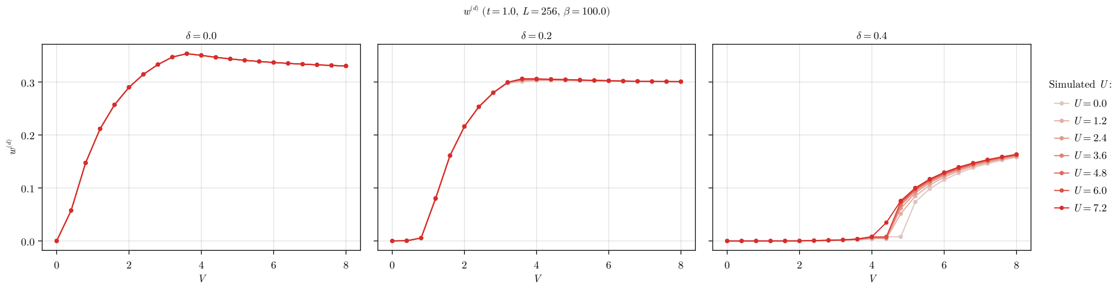


a | Dependency of $w^{(s^*)}$ on V at various Us, compared at three values of δ.

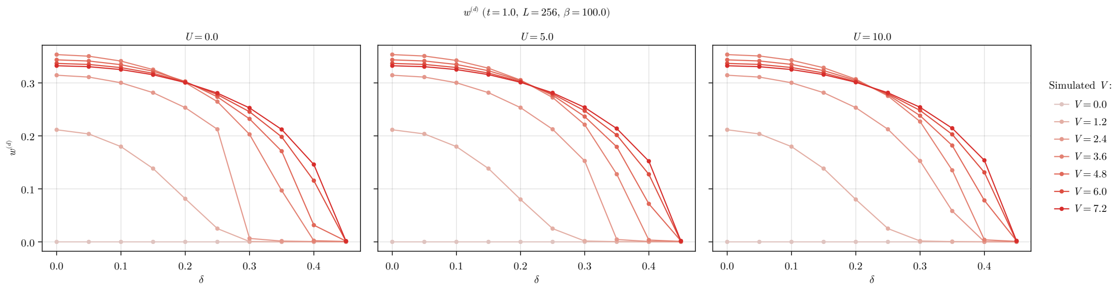


b | Dependency of $w^{(s^*)}$ on δ at various Vs, compared at three values of U.

Figure 6.2 | Dependency of the s^*-wave HFP $w^{(s^*)}$ on V at fixed U and δ (top panel, compared figures) and on δ at fixed V and U (bottom panel, compared figures).



a | Dependency of $w^{(d)}$ on V at various Us, compared at three values of δ.



b | Dependency of $w^{(d)}$ on δ at various Vs, compared at three values of U.

Figure 6.3 | Dependency of the d-wave HFP $w^{(d)}$ on V at fixed U and δ (top panel, compared figures) and on δ at fixed V and U (bottom panel, compared figures).

s^*-wave gap. In Fig. 6.2, we show the results for the non-local isotropic HFP $w^{(s^*)}$. The resulting component of the full gap reads

$$\tilde{\Delta}^{(s^*)} = V w^{(s^*)} .$$

Consider first Fig. 6.2a. At null and finite fillings, $w^{(s^*)}$ is null up to a critical value $V_c^{(s^*)}$. Its position depends on δ and weakly on U/t. In particular, the half-filling curve (left panel) is independent of U/t. The local isotropic gap harmonic $\tilde{\Delta}^{(s)}$ was always null at half filling, see Eq. (6.20). In contrast, its non-local counterpart $\tilde{\Delta}^{(s^*)}$ is null up to the critical value delimiting the phase transition. Therefore, thanks to the non-local pairing, the SC gap at half-filling can develop a purely non-local isotropic component.

In Fig. 6.2b, we see that the dependence on doping of $w^{(s^*)}$ is peculiar. At low doping ($\delta \lesssim 0.3$), $w^{(s^*)}$ is small for small values of V (left panel). At higher doping (central and right panels) we see that a local maximum emerges, indicating that $w^{(s^*)}$ converges to higher values at smaller values of V.

d-wave gap. In Fig. 6.3, we report our findings for the HFP $w^{(d)}$, which is related to the d-wave gap,

$$\tilde{\Delta}^{(d)} = V w^{(d)} .$$

From the top panel (Fig. 6.3a), we see that $w^{(d)}$ depends weakly on U. At half-filling (left panel), the d-wave gap is zero only at $V = 0$ for any value of U. Differently from the other components, $w^{(d)}$ grows to finite values already at infinitesimal V/t. Recalling the results of Figs. 6.1 and 6.2, we note that when the strength of the non-local attraction grows enough, the s^*-wave part sets in, meaning that, in this region, the SC gap has a mixed symmetry. At finite filling (central and right panels), the d-wave gap emerges for $V > V_c^{(d)}$.

In the bottom row (Fig. 6.3b), we show the dependence of $w^{(d)}$ on doping at different values of U. For any value of V/t, the HFP $w^{(d)}$ is a monotonically decreasing function of δ, and the critical value $V_c^{(d)}$ increases with doping. Because of this, we see that the lighter curves of Fig. 6.3b, corresponding to lower values of V, reach zero quite rapidly for $\delta \gtrsim 0.3$. The curves for larger values of V, instead, reach zero at almost unitary filling. In all three panels of Fig. 6.3b we see

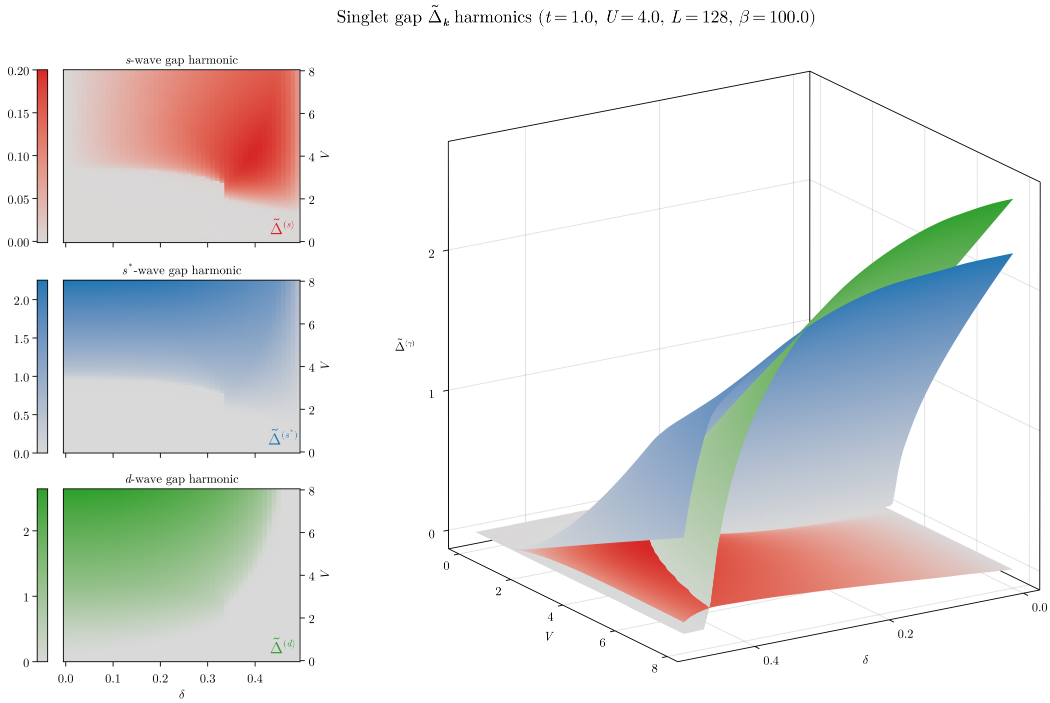


Figure 6.4 Plot of the singlet gap harmonics $\tilde{\Delta}^{(\gamma)}$, with $\gamma = s, s^*, d$ at $U/t = 4$. The panels on the left show the three harmonics separately as heatmaps, while the surface plot on the right shows them comparatively.

that at doping $\delta \approx 0.2$ various lines cross at $w^{(d)} \approx 0.3$. Finally, at null doping, there exists a finite value of V that maximises $w^{(d)}$, see Fig. 6.3a.

Competition among the channels. To understand the competition among channels, we compare the values of the harmonics $\tilde{\Delta}^{(\gamma)}$ for $\gamma = s, s^*, d$. In Fig. 6.4, we plot the different components as a function of δ and V for $U/t = 4$. Data and plots for other values of U are openly accessible at our repository.

The left column shows three heatmaps, one for each harmonic in singlet SC phase. The same data are shown, superimposed, in the surface plot on the right. The harmonics $\tilde{\Delta}^{(\gamma)}$ cross if V/t gets large enough, with the d-wave component reaching the largest values in the parametric region we investigated, followed closely by the s^*-wave harmonic. Isotropic harmonics (s and s^*) are significantly non-zero in the same region, as we see in the first two rows of the left column. The two channel significantly differ in amplitude: the maximum value reached by the local gap harmonic $\tilde{\Delta}^{(s)}$ is approximately one order of magnitude smaller than the maximum of the non-local harmonic $\tilde{\Delta}^{(s^*)}$. Moreover, we see that at $\delta = 0.0$ the local isotropic harmonic $\tilde{\Delta}^{(s)}$ is completely suppressed, as we expressed in Eq. (6.20).

All the harmonics share a sharp boundary around $\delta \approx 0.34$, $V/t \approx 2.5$, visible as a vertical segment in the three heatmaps on the left column of Fig. 6.4. The isotropic harmonics suddenly change from zero to large values moving from left to right, while the d-wave part does the opposite. With our coarse-grained simulations, we were not able to verify if this behaviour underlies some deeper structure.

Summarising all results, we can distinguish four phases: the normal state and three SC regions (pure d-wave, pure $s \oplus s^*$-wave, mixed). In Fig. 6.5, we show the phase diagram in the plane (δ, V) for $U/t = 0$ (left), $U/t = 4$ (centre) and $U/t = 12$ (right). The boundaries were computed by setting a threshold $\varepsilon = 10^{-2}$, with the following rule:

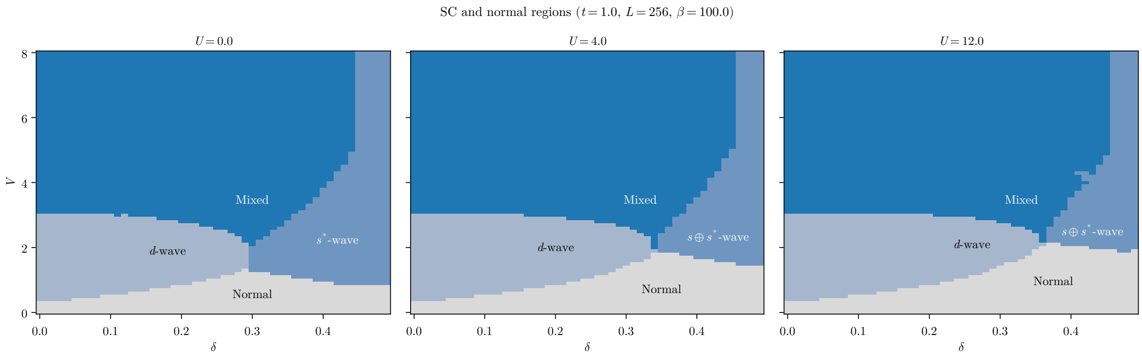


Figure 6.5 Depiction of the normal state region and the three singlet SC regions, at $U/t = 0, 4, 12$ from left to right. Lighter blue indicates the pure d-wave SC phase, blue the isotropic $s \oplus s^*$ phase, deeper blue the mixed region. In text is described the convention we used to identify the boundaries.

$$
\begin{array}{rccccc}
\text{Normal} & \left|\tilde{\Delta}^{(s)}\right| < \varepsilon & \wedge & \left|\tilde{\Delta}^{(s^*)}\right| < \varepsilon & \wedge & \left|\tilde{\Delta}^{(d)}\right| < \varepsilon \quad , \\
d\text{-wave} & \left|\tilde{\Delta}^{(s)}\right| < \varepsilon & \wedge & \left|\tilde{\Delta}^{(s^*)}\right| < \varepsilon & \wedge & \left|\tilde{\Delta}^{(d)}\right| \geq \varepsilon \quad , \\
s \oplus s^*\text{-wave} & \left[\, \left|\tilde{\Delta}^{(s)}\right| \geq \varepsilon \right. & \vee & \left. \left|\tilde{\Delta}^{(s^*)}\right| \geq \varepsilon \,\right] & \wedge & \left|\tilde{\Delta}^{(d)}\right| < \varepsilon \quad , \\
\text{Mixed} & \left[\, \left|\tilde{\Delta}^{(s)}\right| \geq \varepsilon \right. & \vee & \left. \left|\tilde{\Delta}^{(s^*)}\right| \geq \varepsilon \,\right] & \wedge & \left|\tilde{\Delta}^{(d)}\right| \geq \varepsilon \quad ,
\end{array}
$$

and by considering the s and s^* components as part of the same isotropic channel.

Comparing the three panels of Fig. 6.5, we see that increasing the value of the local repulsion U the shapes of the four regions change. In particular, the d-wave region expands to the right (larger δs); the normal-$s \oplus s^*$ SC boundary moves up (larger Vs). Note that on the left panel the isotropic region is labeled as purely s^*-wave: this is because if $U = 0$, the s-wave gap harmonic is null, even if $w^{(s)}$ converges to a non-zero value. The growth of the local repulsion U has two effects:

1. turning on the s-wave gap harmonic at finite doping (see Fig. 6.1);
2. expanding the d-wave region and restricting the isotropic and mixed region.

From the last point we understand that increasing U we allow for the gap function to have a finite s-wave component, but this acts as a sort of internal competition within the isotropic region. Indeed, from the constraint of Eq. (6.16), we know that $w^{(s)} \neq 0$ only if $w^{(\gamma)} \neq 0$, with $\gamma \neq s$. In the $s \oplus s^*$ region the constraint reads:

$$\left|w^{(s^*)}\right|^2 \geq \left|w^{(s)}\right|^2 \frac{U}{V} , \qquad \text{(Isotropic region)}$$

Suppose we work at fixed δ and $V > 0$ in the isotropic region at $U = 0$. There is no constraint whatsoever on $w^{(s^*)}$. Also, $w^{(s)}$ is free to converge at some finite value (we saw this in the left panel of Fig. 6.1b): the gap still has no s-wave component. Suppose now we switch on $U > 0$. Now, in order for $w^{(s)}$ to reach the same value as before, $w^{(s^*)}$ has to respect the constraint posed by the above equation. As U increases, the right-hand side gets bigger and $w^{(s^*)}$ needs to converge to larger and larger values to sustain $w^{(s)}$, otherwise to drop to zero. This is why increasing U the isotropic boundary moves up: $w^{(s^*)}$ isn't “free” anymore and must respect the constraint of Eq. (6.16).

In the three panels of Fig. 6.5, the mixing region is almost independent of U. At low doping, it is bounded from below by an horizontal line at $V/t \approx 3$. Increasing doping at fixed U, we see that the lower boundary reaches a cusp-like minimum around $\delta \approx 0.3 \div 0.4$ and then grows. This cusp coincides with the aforementioned sharp vertical segment, visible in Fig. 6.4. If we work at fixed doping and U, and choose δ such that we are aligned with the horizontal position of the cusp on the (δ, V) plane, the critical value of V/t for transitioning to a mixed SC state is minimised. At high values of V/t the mixing region extends to almost unitary doping, being limited on the right by a vertical line at $\delta \gtrsim 0.4$. We remind that we are discerning the SC phases by applying an arbitrary threshold ε to each gap component: we imagine that by using a more refined technique and by increasing the resolution of the simulation the boundary would appear less straight. Apart from a vertical stripe at almost unitary filling, at high V/t the dominant singlet SC phase is described by a mix of d-wave and $s \oplus s^*$-wave component. Recalling the surface plot of Fig. 6.4, we see that at $V/t \gg 1$ the s-wave component gets weaker with respect to the other two. In this limit, if $U \ll V$, the s-wave component is negligible, giving an approximate $d \oplus s^*$ mixed phase.

The mixing region is of particular interest also due its connection with the planar rotational SSB. Indeed, as we show in next section, the mixed-symmetry region coincides with the portion of the (δ, V) plane where planar rotational symmetry is broken,

$$\text{Symmetry mixing} \quad \Longrightarrow \quad \mathrm{C}_4 \overset{\text{SSB}}{\to} \mathrm{C}_2\,.$$

6.3.2 Rotational SSB and nematic superconductivity

As we discussed in Sec. 4.3.1, and at the end of Sec. 6.2, nematicity is established whenever $\tilde{t}^{(d)}$ becomes non-zero. If the renormalised hopping parameter has a non-zero d-wave component, this means that bare hopping is effectively boosted in one direction and damped in the other. Indeed, from Eqs. (6.18), (6.19), we know that

$$\tilde{t}^{(d)} = \frac{\tilde{t}^{(y)} - \tilde{t}^{(x)}}{2}\,.$$

If $\tilde{t}^{(d)} > 0$ the renormalised hopping is larger in direction y, if $\tilde{t}^{(d)} < 0$ instead it's larger in direction x. In terms of the Bogoliubov energies, this means that the Bogoliubov bands are not rotationally invariant,

$$\tilde{E}_{\mathbf{k}} \neq \tilde{E}_{\mathcal{R}\mathbf{k}}\,,$$

being $\mathcal{R}$ a $\pi/2$ rotation.

For the normal and AF phases we were able to prove that $u^{(d)} = 0$ for any choice of model parameters. For the SC phase, instead, $u^{(d)}$ converges to non-zero values. In Fig. 6.6, we show the quantity

$$\tilde{t}^{(d)} = -\frac{V u^{(d)}}{2}\,,$$

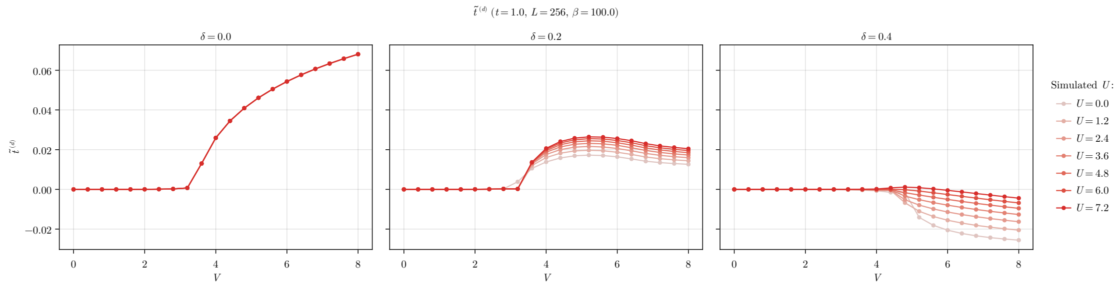
as a function of U, V and δ. In the top panel (Fig. 6.6a), we see that a critical value of V exists over which nematicity is established. Moreover, the $\delta = 0.0$ case (left panel) is independent of U. In all cases, the convergence values for $\tilde{t}^{(d)}$ are two order of magnitudes smaller than $t = 1.0$, meaning that nematicity is a sharp but small effect. Note that at $\delta = 0.2$ (central panel) $\tilde{t}^{(d)} \geq 0$, while at $\delta = 0.4$ (right panel) $\tilde{t}^{(d)} \leq 0$. Increasing the doping we change the sign of $\tilde{t}^{(d)}$.

This sign change is most evident in the first two figures of the bottom panel (Fig. 6.6b), where we vary the doping level while keeping fixed the local repulsion U. The behaviour of $\tilde{t}^{(d)}$ is similar at low doping in the three cases, but changes significantly at larger doping. Indeed, at $U/t = 0$ (left figure) and $U/t = 5$ (central figure) at some point $\tilde{t}^{(d)}$ changes sign. At $U/t = 10$ (right figure), all our data points are positive (possibly the sign flip occurs at higher Vs). Moreover, as U increases, the amplitude of the oscillation of $\tilde{t}^{(d)}$ as a function of δ reduces. We conclude that both the local repulsion and electronic doping disrupt nematicity, which is instead most effective for a purely NNs attractive model.

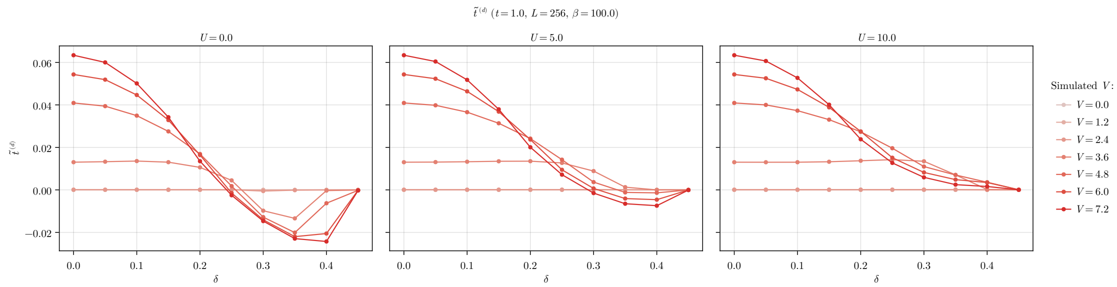
Nematic effects are directly related to symmetry mixing in the SC order parameter. In Fig. 6.7, we plot the convergence values for $\tilde{t}^{(d)}$ in the (δ, V) plane for $U/t = 0, 4, 12$. The used dataset is the same of Fig. 6.5. The superimposed black lines are the mixing boundaries, the set of extremal points of the deeper blue regions of Fig. 6.5. Despite a bit of roughness related to the gridded nature of our data, it is evident that these boundaries encircle the regions where $\tilde{t}^{(d)} \neq 0$, clearly highlighting a connection between symmetry mixing and breaking of planar rotational symmetry.

The mixing region coincides with the nematic region. As we said at the beginning of this section, nematicity can be established in two ways, based on the direction along which hopping is boosted. Hence it must hold

$$\mathrm{M} \simeq \mathrm{N}_x \cup \mathrm{N}_y$$



a | Dependency of $\tilde{t}^{(d)}$ on V at various Us, compared at three values of δ.



b | Dependency of $\tilde{t}^{(d)}$ on δ at various Vs, compared at three values of U.

Figure 6.6 Dependency of the d-wave renormalised hopping $t^{(d)}$ varying the doping δ and the local repulsion U. In the top panel we compare three doping levels $\delta = 0.0,\ 0.2,\ 0.4$ varying $0 \leq U \leq 8$. In the bottom panel we compare three repulsion values $U = 0.0,\ 5.0,\ 10.0$ varying $0 \leq \delta \leq 0.45$.

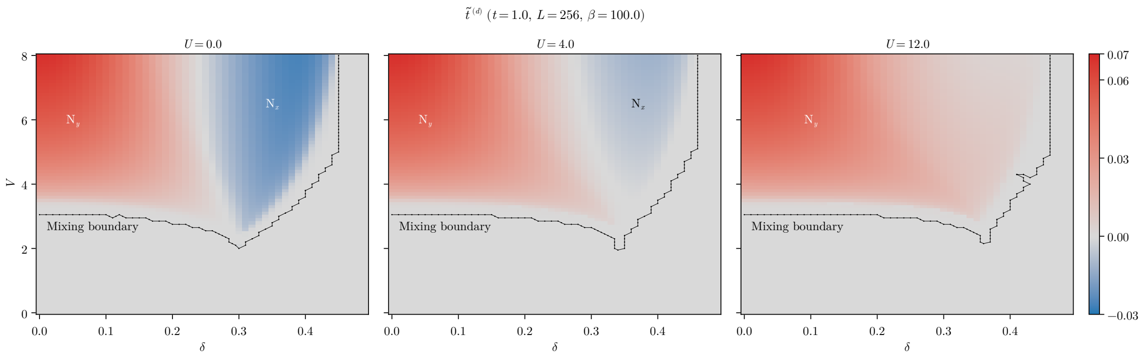


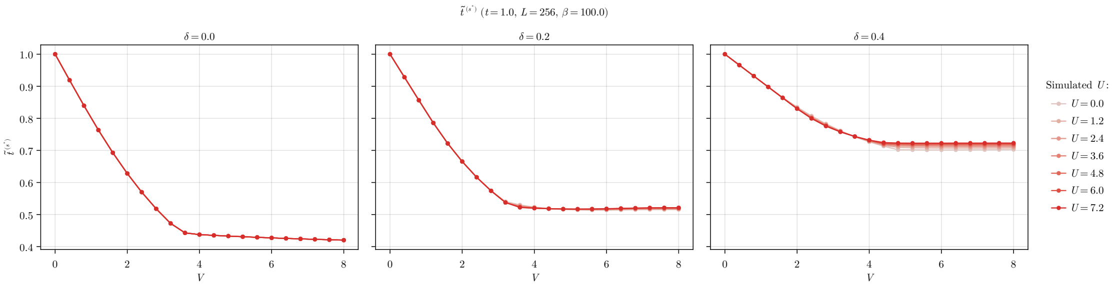
Figure 6.7 Heatmap of the anisotropic component of renormalised hopping, $\tilde{t}^{(d)}$, at $U/t = 0, 4, 12$ from left to right. The dataset is the same of Fig. 6.5, and the mixing boundary is the set of points separating the mixed region from the rest of the phases.

with the following convention on notation:

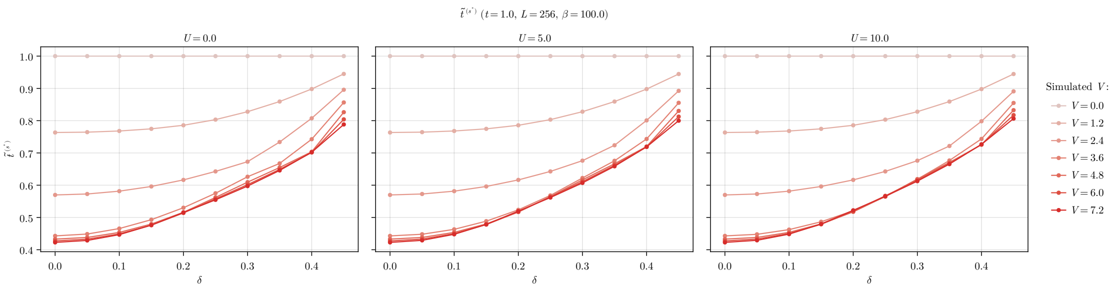
$$\begin{aligned} \mathrm{M} &= \text{Region of } (\delta, V) \text{ plane where mixing occurs,} \\ \mathrm{N}_x &= \text{Region of } (\delta, V) \text{ plane where } \tilde{t}^{(d)} < 0, \\ \mathrm{N}_y &= \text{Region of } (\delta, V) \text{ plane where } \tilde{t}^{(d)} > 0. \end{aligned}$$

The notation N_ℓ has the following meaning: the index ℓ indicates the direction along which renormalised hopping is larger. We indicate in Fig. 6.7 these regions for $U/t = 0, 4, 12$.

Note that the map were colour-corrected in order to enhance the visibility of the blue region. The intensity of the colouring there suggests that the amplitude is comparable with the red region, but is not. Indeed, we set up the colormap such that gray is aligned with $\tilde{t}^{(d)} = 0.0$, and intensity of colour grows asimmetrically in the negative and positive direction.



a | Dependency of $\tilde{t}^{(s^*)}$ on V at various Us, compared at three values of δ.



b | Dependency of $\tilde{t}^{(s^*)}$ on δ at various Vs, compared at three values of U.

Figure 6.8 | Dependency of the s^*-wave renormalised hopping $t^{(s^*)}$ varying the doping δ and the local repulsion U. In the top panel we compare three doping levels $\delta = 0.0,\ 0.2,\ 0.4$ varying $0 \leq U \leq 8$. In the bottom panel we compare three repulsion values $U = 0.0,\ 5.0,\ 10.0$ varying $0 \leq \delta \leq 0.45$.

The three panels of Fig. 6.7 differ crucially in the fact that the data for $U/t = 12$ (right) are all positive, while those for $U/t = 0, 4$ (left,centre) are negative in a portion of space. In other words, as we increase U/t, N_y expands and N_x shrinks. Moving from $U/t = 0$ (left) to $U/t = 4$ (centre) we see that the amplitude of $\tilde{t}^{(d)}$ decreases in N_x and not in N_y. We already observed this in Fig. 6.6b. Moreover, for $U/t = 12$ (right) N_x is apparently absent and N_y takes up the entirety of the mixing region.

Breaking of planar rotational symmetry and the insurgence of a d-wave harmonic for the renormalised hopping is certainly the most peculiar effect for the singlet SC phase. It adds to the isotropic bands shrinking, which is, the s^*-wave renormalisation of the hopping parameter. We discuss it in next section.

6.3.3 Isotropic bands shrinking

In this section, we discuss the behaviour of the HFP $u^{(s^*)}$, responsible of the pairing-induced Mott localisation effect through the isotropic renormalisation of the hopping parameter,

$$\tilde{t}^{(s^*)} = t - \frac{V u^{(s^*)}}{2} .$$

In Fig. 6.8, we plot $\tilde{t}^{(s^*)}$ as a function of U, V and δ. In the top panel (Fig. 6.8a), we show the dependence of $t^{(s^*)}$ on V. The three curves, obtained at three doping levels, are approximately independent of U. The common feature is that $\tilde{t}^{(s^*)}$ decreases linearly with V up to some critical value, above which it converges to a constant value which increases as a function of the doping. As was for the AF and normal phases, electronic doping suppresses the isotropic bands shrinking effect. At half-filling (left figure) we see that bands shrinking is most effective, reducing the hopping parameter to less than half its $V = 0$ value. In the bottom panel (Fig. 6.8b) we show $t^{(s^*)}$ as a function of δ, varying V and U. As we already noted, the dependence on U is very weak and the three figures are a lot alike. As we already mentioned, $t^{(s^*)}$ increases with doping.

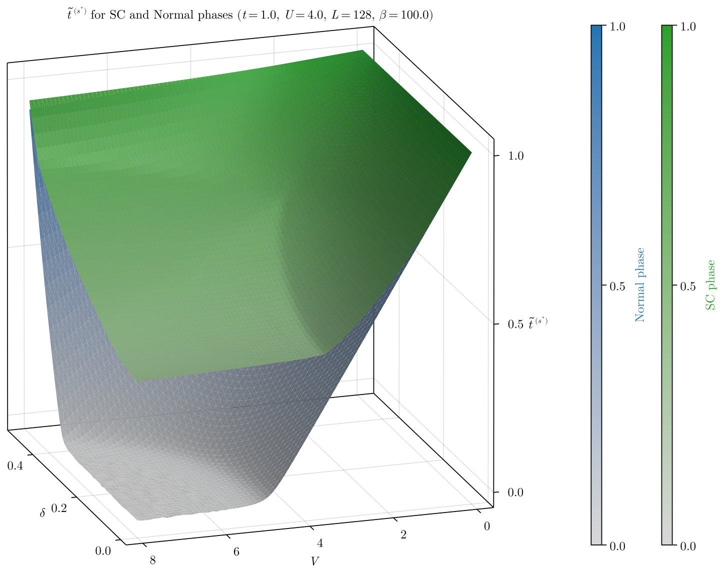


Figure 6.9 | Comparison of the isotropic part of the renormalised hopping $\tilde{t}^{(s^*)}$ for the SC (green) and normal (blue) phases on the (δ, V) plane. The data from the normal phase are the same of Fig. 4.6.

In Fig. 6.9, we compare $t^{(s^*)}$ for the normal phase (blue surface) and the SC phase (green surface). The two surfaces share a general feature: $t^{(s^*)}$ is approximately unitary at low V/t and almost unitary doping, and decreases significantly in the low-δ and high-V/t limit. In this region, for the normal phase we observe the Mott-localisation plateau of Fig. 4.6, discussed in Sec. 4.5.1, where $t^{(s^*)} \approx 0$. In the SC phase, looking at the same region, $t^{(s^*)}$ is minimised, but, in contrast to the normal phase, we do not observe the complete Mott localisation effect.

Recalling the plots of Fig. 6.5, the low-δ and high-V/t limit (the corner where $t^{(s^*)}$ is minimised) falls deep within the mixed region. Recalling also Fig. 6.4, in this corner the d-wave and s^*-wave harmonics of the gap function are large and comparable. Finally, we know from Fig. 6.7 (central panel) that here nematic effects are small, but significantly non-zero. In particular, considering that

$$\lim_{\delta \to 0} \lim_{V/t \gg 1} \frac{\tilde{t}^{(s^*)}}{t} \approx 0.4 \quad \text{and} \quad \lim_{\delta \to 0} \lim_{V/t \gg 1} \frac{\tilde{t}^{(d)}}{t} \approx 0.06\,,$$

we see that the two components differ by one order of magnitude.

With this consideration, we end the presentation of HFPs and RMPs and move to the last section, where we discuss the results regarding free energy and entropy densities.

6.3.4 Solution free energy and entropy densities

For the SC phase, free energy density is given by Eq. (G.84), while entropy density is given by Eq. (G.79). In Fig. 6.10 we plot the heatmaps for f and s at $U/t = 4$ in the (δ, V) plane. Superimposed, we show the boundaries between the four regions identified in Fig. 6.5. Consider first the free energy, on the left: our data suggest that free energy is maximised in the d-wave phase, followed by the mixed phase. At high doping, the $s \oplus s^*$-wave phase is low-energetic. Overall, f varies smoothly across phase boundaries, and we do not identify the black lines with some isoenergetic line separating phases.

This is not the case for entropy density. Indeed, the entropic region on the right panel of Fig. 6.10 coincides with the normal phase. The three SC regions are low-entropic. This is explained by the following argument:

- In the normal phase, the Bogoliubov band reduces to $\tilde{E}_{\mathbf{k}} \to \tilde{\xi}_{\mathbf{k}}$. The chemical potential crosses the bare band and a Fermi surface exists. Hence, at finite temperature, the Fermi-Dirac occupation of single-particle states varies continuously from almost unitary (deep in

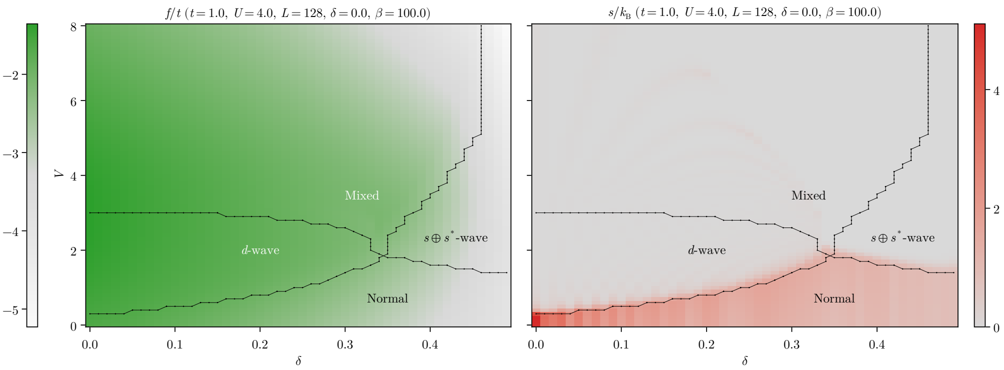
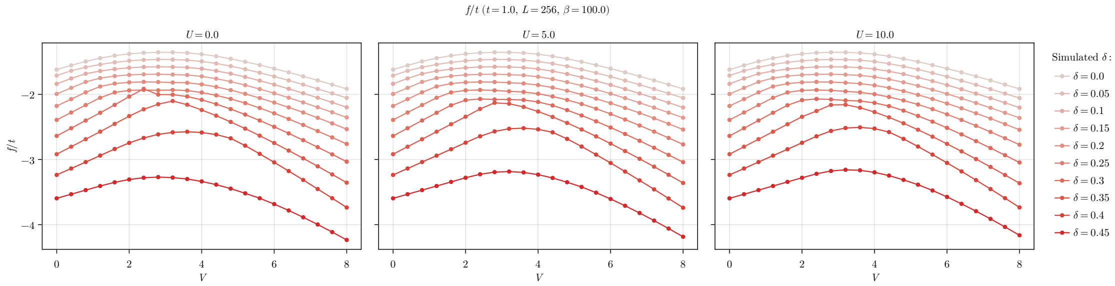
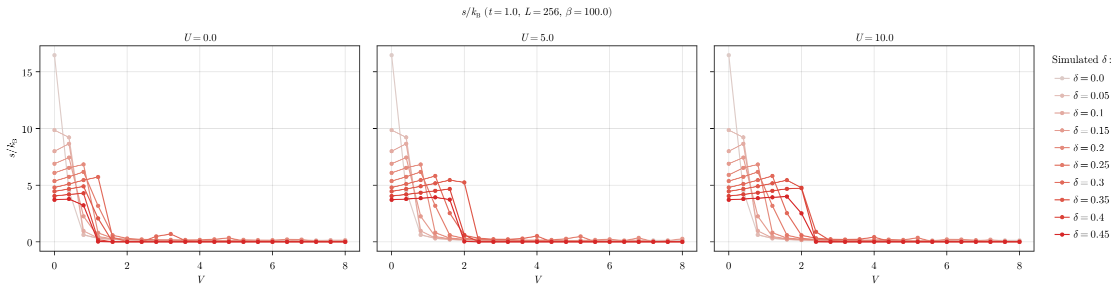


Figure 6.10 | Heatmaps of the free energy and entropy densities at $U/t = 4$. The black dotted lines represent the internal boundaries of the singlet SC phase, extracted from Fig. 6.5.



a | Dependency of f on V at various δs, compared at three values of U.



b | Dependency of s on V at various δs, compared at three values of U.

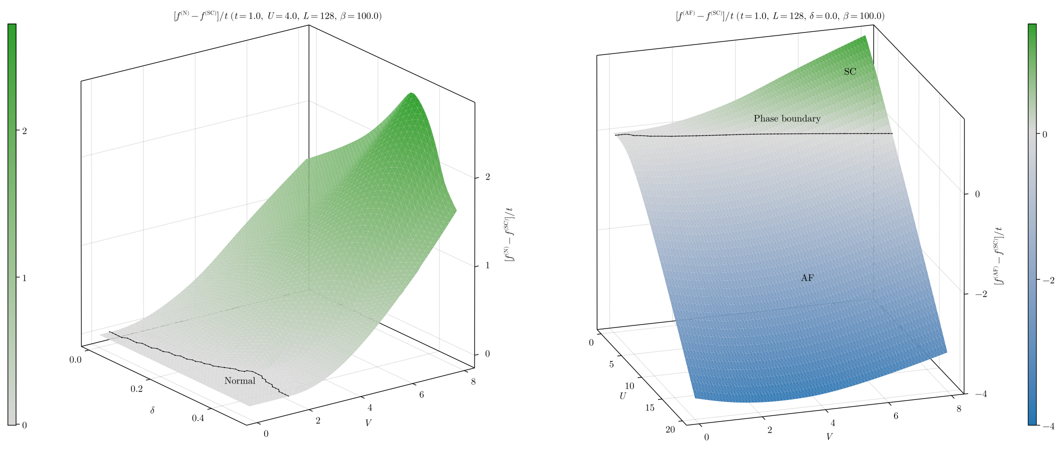
Figure 6.11 | Dependency of free energy (top) and entropy (bottom) densities on V varying the doping δ and the local repulsion U.

the Fermi sea) to almost zero (far outside). All those states in the middle, around the Fermi level, are highly entropic: we have shown this in Fig. 4.9b. It is the intrinsic metallic nature of the normal phase that makes it entropic.

- In the SC phase, we open a gap at the chemical potential, not at zero energy (as instead was in the AF phase). Computing entropy as in Sec. G.3.2, we apply the Fermi-Dirac distribution on the two bands $\pm\tilde{E}_{\mathbf{k}}$. As the gap opens, the lower one fills (occupation ≈ 1), while the upper empties (occupation ≈ 0). As we lower temperature, the above statement gets exact. Recalling Fig. 4.9b, we see that entropy is minimised at occupation equal to 0 or 1. Hence, the SC phase is low-entropic.

This is the reason behind the sharp separation of the normal region from the SC regions in the right panel of Fig. 6.10.

In Fig. 6.11 we show f and s as functions of V, varying δ and comparing for three values of U. On both rows, moving from left to right we do not observe significant differences: f and s depend weakly on U. We already observed the presence of a maximum of the free energy at ratios V/t



a | Energy difference with normal phase.   b | Energy difference with AF phase.

Figure 6.12 | Surface plots of the energies among the three phases. For the normal phase, we use that fact that $f^{(\mathrm{SC})}$ depends weakly on U, while $f^{(\mathrm{N})}$ does not depend on it at all, to limit to the case $U/t = 4$. For the AF phase, we limit to the half-filling condition and show data in the (U, V) plane.

of order unity. In Fig. 6.11a we see this maximum around $V/t \lesssim 4$. Comparing with the data of Fig. 6.10, we can imagine that this maximum could be located at at the boundary between d-wave and mixed regions.

In the bottom panel, entropy density (Fig. 6.11b) shows the aforementioned sharp separation between normal and SC regions, and is maximised at half-filling and $V = 0$. That is the case of the conventional HM. We already argued that superconductivity is impossible at HF level, while commensurate antiferromagnetism is favoured at null doping with respect to the normal phase.

To conclude, we compare the free energy of the SC phase with that of the normal phase and the AF phase. We said that f depends rather weakly on U for the singlet SC phase, so we carry out the comparison with normal phase at $U/t = 4$ in the (δ, V) plane. In Fig. 6.12a we plot the quantity

$$f^{(\mathrm{N})} - f^{(\mathrm{SC})} ,$$

and observe that it is greater or equal to zero over the entire explored region. The energy difference is null only at $V/t \lesssim 2$, where the SC gap is null. We plot, superimposed, the boundaries of the normal region extracted from Fig. 6.5. In the rest of the (δ, V) plane the SC phase is energetically favoured with respect to the normal phase.

In Fig. 6.12b we show the energy difference between AF and SC phases,

$$f^{(\mathrm{AF})} - f^{(\mathrm{SC})} .$$

We colour-corrected the surface plot to align gray with zero. Colour separates positive and negative regions: the SC phase is dominant where $f^{(\mathrm{AF})} > f^{(\mathrm{SC})}$ (green region), in the rest of the plot the AF phase dominates (blue region). We indicate the contour level at $f = 0$ as a black dotted line, obtained as the set of data points whose energy difference is closest to zero. At half-filling, commensurate antiferromagnetism is almost everywhere energetically favoured. Singlet superconductivity prevails in the $U/t \ll 1$, $V/t \gg 1$ limit. The fact that the phase boundary has somewhat a linear shape suggests the fact that possibly a critical ratio exists,

$$\left(\frac{V}{U}\right)_{\mathrm{c}} = k , \qquad \text{(Critical ratio)}$$

delimiting a phase transition from antiferromagnetic to SC ordering at half-filling.

Chapter 7

Conclusions

In this thesis, we studied the extended Hubbard model introduced in Eq. (1.4) by the means of the Hartree-Fock method, both analytically and numerically. We analysed three phases: normal, commensurate antiferromagnetic and nematic superconducting (in singlet channel). The aim of this project was to investigate the insurgence of Mott localisation related to the adding of a non-local attraction to the conventional Hubbard model.

Normal phase. For the normal phase (see Chap. 4), we demonstrated that, at HF level, the non-local attraction acts a source of Mott localisation. In the entire parametric space the non-local attraction produces an effective shrinking of bands. Indeed, increasing the value of V as we vary δ, we identified a "Mott region" (see Fig. 4.6), where the bare bands are completely flattened,

$$t \to \tilde{t}^{(s^*)} \approx 0\,.$$

We also allowed, theoretically, for the breaking of planar rotational symmetry and concluded that in the normal phase we observe no nematic effects. We gave a formal proof of these effects in App. F.

Commensurate antiferromagnetic phase. In the commensurate AF phase (see Chap. 5), we limited to the half-filling condition ($\delta = 0$), because at finite filling the phase is unstable. We have shown that the non-local attraction behaves similarly to the normal phase, inducing a shrinking of bare bands which, in turn, enhances the antiferromagnetic ordering already induced by the local repulsion. Specifically, we observed that the hopping amplitude t is reduced, similarly to the AF phase, but equals zero only when $U = 0$ – which is, where the AF and normal phases coincide. This was shown in Fig. 5.8.

While the local repulsion $\hat{H}_U$ induces, by itself, a staggered magnetisation (see Sec. 5.2), we have shown that the shrinking of bands, originated by $\hat{H}_V$, enhances magnetisation . We identified the presence of isomagnetic lines (see Fig. 5.9), namely straight lines in the (U, V) plane where macroscopic magnetisation is constant. The existence of these lines implies that, in the EHM, commensurate antiferromagnetism is a result of two cooperative effects, namely Mott localisation and virtual hopping (see App. A). Finally, we have shown that, at half-filling, the commensurate AF phase is energetically favoured with respect to the normal phase (see Fig. 5.12).

Nematic superconducting phase. We studied translationally invariant superconductivity in both the singlet and triplet channels, and limited our simulation to the first one (see Chap. 6). We collected numeric evidence of the presence of Mott localisation also in this phase, induced by the non-local attraction and manifest in the shrinking of electronic bands.

We studied the competition among two symmetries of the superconducting gap, namely d-wave (which is observed in copper-oxides based compounds) and $s \oplus s^*$-wave. In particular, we have identified four regions in the (δ, V) plane as we vary U: the normal region, the d-wave region, the $s \oplus s^*$-wave region and the mixed region (see Fig. 6.5). The latter is particularly interesting: we were able to claim, based on numerical evidence, that the mixing of symmetries in the singlet channel is connected to the insurgence of nematic effects (see Fig. 6.7), namely the breaking of symmetry of $\tilde{t}$ in directions x and y. In this phase, thus, the non-local attraction plays two roles: Mott localisation, and insurgence of nematicity.

Finally, we compared the free energies of the normal, AF and SC phases (Fig. 6.12). We already understood that the AF phase is favoured, at half-filling, with respect to the normal phase. We have shown that the normal phase is also always unfavoured with respect to the SC phase, at finite filling. Instead, the ratio U/V regulates the competition between SC and AF phases, as is visible in Fig. 6.12, right panel. We identified a phase boundary between the two, and understood that in the $U \gg V$ limit the AF phase is energetically favoured, at $\delta = 0$.

Further development and outlook. In this thesis we developed a general framework for the analysis of the EHM, or a similar model, using HF methods. This framework relies upon three main steps: identifying the symmetry of the phase, approximate the EHM hamiltonian accordingly, identify the variational parameters and the self-consistency equation for each. In principle, then, several other phases could be explored, for instance:

- *Uncommensurate antiferromagnetism*, namely insurgence of a staggered magnetisation with a periodicity larger than 2 sites;
- *Nematic antiferromagnetism*, namely the insurgence of a staggered magnetisation in just one direction, while in the other direction we observe ferromagnetic ordering;
- *Triplet (translationally invariant) superconductivity.* We already carried out the analytic computations for this one in App. G, and our data suggest that, in our solution, it is identically null;
- *Non-uniform superconductivity*, namely a kind of superconducting ordering where translational invariance is broken. It can be seen as a sort of "mixing" of the AF and SC phases here presented;
- And many others!

Moreover, one could further extend the model, adding next-nearest neighbours hopping, anisotropic hopping; we could explore the region $V > 0$, mimicking a sort of extension of the Coulomb repulsion to NN sites.

Another possibility is to adopt more advanced numerical techniques. For instance, we could use DMRG (density matrix renormalisation group) to simulate the EHM on a quasi-one dimensional system, such as a chain. This type of analysis was already carried out by Padma et al. [22], and could be extended by searching evidence of Mott localisation. As another example, we could implement a combination of DMFT (dynamical mean field theory, [11]) and Hartree-Fock methods to treat separately, respectively, the local repulsion and the non-local attraction. In this framework, we could search for the signatures of Mott physics induced by the local repulsion, while in this thesis all effects related to Mott localisation were induced by the non-local attraction.

Overall, recalling the purposes of this project we presented in Chap. 1, we conclude that the insertion of a non-local attraction is, at HF level, evidently related both to Mott localisation and the insurgence of nematic effects. More advanced techniques could help to further investigate this relation.

Appendix A

Superexchange and virtual hopping in Hubbard lattices

Superexchange is a key mechanism in the formation of AF ordering in Hubbard lattices. It is an extension of the direct exchange effect, that we describe in Sec. A.2, that favours antiferromagnetic ordering and takes advantage of the chemical structure of cuprates. The mechanism becomes clear enough if we start with a 2-sites Hubbard toy model.

A.1 Singlet pairing in Hubbard lattices

Consider the 2 sites Hubbard model of Eq. (1.2),

$$\hat{H} = -t\left(\hat{c}^{\dagger}_{1\uparrow}\hat{c}_{2\uparrow} + \hat{c}^{\dagger}_{1\downarrow}\hat{c}_{2\downarrow} + \text{h.c.}\right) + U\left(\hat{n}_{1\uparrow}\hat{n}_{1\downarrow} + \hat{n}_{2\uparrow}\hat{n}_{2\downarrow}\right),$$

with $i = 1, 2$ the site index. For an half-filled system (i.e. with two electrons) the Hilbert space is six-dimensional. We use the notation $|n_{1\uparrow}n_{1\downarrow}n_{2\uparrow}n_{2\downarrow}\rangle$ to indicate the six states of the computational basis:

$$\begin{aligned} |\psi_1\rangle &\equiv |1010\rangle\ , \qquad |\psi_3\rangle \equiv |1001\rangle\ , \qquad |\psi_5\rangle \equiv |0011\rangle\ , \\ |\psi_2\rangle &\equiv |1100\rangle\ , \qquad |\psi_4\rangle \equiv |0110\rangle\ , \qquad |\psi_6\rangle \equiv |0101\rangle\ . \end{aligned} \tag{A.1}$$

We give a pictorial representation of these states in Fig. A.1. We ordered the states in such a way that the two states with total spin different from zero, $|\psi_1\rangle$ and $|\psi_6\rangle$, are at the extrema of the set.

The matrix elements of the hamiltonian can be computed using this basis,

$$H_{ij} = \langle\psi_i|\hat{H}|\psi_j\rangle \qquad \Longrightarrow \qquad H = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & U & -t & -t & 0 & 0 \\ 0 & -t & 0 & 0 & -t & 0 \\ 0 & -t & 0 & 0 & -t & 0 \\ 0 & 0 & -t & -t & U & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \tag{A.2}$$

In the above matrix, rows and columns are ordered following the states indices of Eq. (A.1). The states $|\psi_1\rangle$ (both spins up) and $|\psi_6\rangle$ (both spins down) are zero-energy eigenstates. Indeed, they do not interact via Coulomb repulsion (are on different sites) and neither through NNs hopping, due to Pauli exclusion principle.

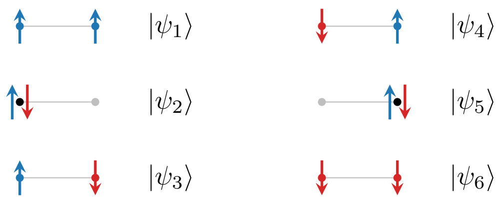


Figure A.1 | Computational basis of Eq. (A.1) for the 2-sites Hubbard model.

Structure	**Eigenstate**	**Energy**
Spin-1/2 singlet	$\frac{\lvert\psi_3\rangle - \lvert\psi_4\rangle}{\sqrt{2}}$	$E^- = \frac{U}{2} - \sqrt{\frac{U^2}{4} + 4t^2}$
Spin-1/2 triplet	$\lvert\psi_1\rangle\,, \frac{\lvert\psi_3\rangle + \lvert\psi_4\rangle}{\sqrt{2}}, \lvert\psi_6\rangle$	0
	$\frac{\lvert\psi_2\rangle - \lvert\psi_5\rangle}{\sqrt{2}}$	U
	$\frac{\lvert\psi_2\rangle + \lvert\psi_5\rangle}{\sqrt{2}}$	$E^+ = \frac{U}{2} + \sqrt{\frac{U^2}{4} + 4t^2}$

Table A.1 | List of exact eigenstates and relative energies for the 2-sites half-filled Hubbard model.

All states with null total spin projection, namely $\{|\psi_i\rangle\}$ for $2 \leq i \leq 5$, give a null matrix elements when computed with the aforementioned states, but non-zero matrix elements among themselves. Note that this set includes both the states with one doubly occupied site and one empty site ($|\psi_2\rangle$, $|\psi_5\rangle$) and those with two singly occupied sites ($|\psi_3\rangle$, $|\psi_4\rangle$). These states are connected by the central 4×4 matrix in Eq. (A.2). The latter is diagonalised by the means of a change of basis M,

$$M \begin{bmatrix} U & -t & -t & \\ -t & & & -t \\ -t & & & -t \\ & -t & -t & U \end{bmatrix} M^\dagger = \begin{bmatrix} E^- & & & \\ & 0 & & \\ & & U & \\ & & & E^+ \end{bmatrix} .$$

where blank slots indicate zeros. In Tab. A.1 we report the eigenvectors and relative eigenvalues obtained from diagonalisation.

The ground-state is the singlet state,

$$\frac{|\psi_3\rangle - |\psi_4\rangle}{\sqrt{2}} = \frac{|1010\rangle - |0101\rangle}{\sqrt{2}} = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}} ,$$

whose energy is

$$E^- = \frac{U}{2} - \sqrt{\frac{U^2}{4} + 4t^2} \simeq -\frac{4t^2}{U} .$$

The latter approximation is true if $U \gg t$ (strong coupling limit). The singlet state pairs with a spatially-symmetric (node-less) wavefunction, while triplet states with an antisymmetric wavefunction (which, instead, has a node). The entire triplet (second row of Tab. A.1) has zero energy. Excited states are anti-symmetrised and symmetrised version of the polarised states $|\psi_1\rangle$ and $|\psi_6\rangle$.

A.2 Direct exchange and virtual hopping

The fact that we limited to two sites allowed for an exact diagonalisation of the toy hamiltonian of Eq. (1.2). When moving to lattices, the above procedure cannot be extended easily. By using perturbation theory, instead, we can obtain the same result and easily extend to larger lattices.

Nevertheless, let us obtain the same result of Sec. A.1 by the means of perturbation theory. Consider Eq. (1.2) separating two parts,

$$\hat{H} = \hat{H}_0 + \hat{H}_1 ,$$

where

$$\hat{H}_0 \equiv U \left(\hat{n}_{1\uparrow}\hat{n}_{1\downarrow} + \hat{n}_{2\uparrow}\hat{n}_{2\downarrow}\right) ,$$
$$\hat{H}_1 \equiv -t \left(\hat{c}^\dagger_{1\uparrow}\hat{c}_{2\uparrow} + \hat{c}^\dagger_{1\downarrow}\hat{c}_{2\downarrow} + \text{h.c.}\right) .$$

Figure A.2 | Schematisation of a second order virtual hopping. Note that this is a virtual process induced by the kinetic perturbation, not an actual physical process.

At zeroth order in perturbation theory, thus considering just the Coulomb repulsion $\hat{H}_0$, the Hilbert space separates in two manifolds

$$\begin{aligned} \mathrm{Span}(|\psi_1\rangle\,, |\psi_3\rangle\,, |\psi_4\rangle\,, |\psi_6\rangle) \quad &\text{with} \quad E = 0\,, \\ \mathrm{Span}(|\psi_2\rangle\,, |\psi_5\rangle) \quad &\text{with} \quad E = U\,. \end{aligned}$$

We limit to the ground manifold, thus "excluding from the Hilbert space" the states with double occupation $|\psi_2\rangle\,, |\psi_5\rangle$.

First order corrections to energy are null:

$$\forall i, j \quad : \quad \langle \psi_i | \hat{H}_1 | \psi_j \rangle = 0\,,$$

because the hopping operator $\hat{H}_1$ produces a doubly occupied site when applied to $|\psi_3\rangle\,, |\psi_4\rangle$ and zero when applied to $|\psi_1\rangle\,, |\psi_6\rangle$ due to Pauli exclusion principle.

We need to move to second order corrections. Let us change basis, and define the singlet state

$$|s\rangle \equiv \frac{|\psi_3\rangle - |\psi_4\rangle}{\sqrt{2}}\,,$$

and the triplet states

$$|t_+\rangle \equiv |\psi_1\rangle\,, \qquad |t_0\rangle \equiv \frac{|\psi_3\rangle + |\psi_4\rangle}{\sqrt{2}}\,, \qquad |t_-\rangle \equiv |\psi_6\rangle\,.$$

Let also $E^{(0)}_\psi$ be the zeroth-order energy associated to state $|\psi\rangle$. Second order correction to the singlet state energy is given by

$$\begin{aligned} E^{(2)}_s &= \sum_{\psi \neq s} \frac{\langle s|\hat{H}_1|\psi\rangle\ \langle\psi|\hat{H}_1|s\rangle}{E^{(0)}_s - E^{(0)}_\psi} \\ &= \frac{2t^2}{E^{(0)}_s - E^{(0)}_{\psi_2}} + \frac{2t^2}{E^{(0)}_s - E^{(0)}_{\psi_5}} = -\frac{4t^2}{U}\,. \end{aligned}$$

The above result can be obtained easily by direct computation. Note that the + sign in the middle is obtained as a combination of the internal − sign in the definition of the singlet state, and another − sign arising from Fermi anticommutation rules.

Instead, for the central triplet state

$$\begin{aligned} E^{(2)}_{t_0} &= \sum_{\psi \neq t_0} \frac{\langle t_0|\hat{H}_1|\psi\rangle\ \langle\psi|\hat{H}_1|t_0\rangle}{E^{(0)}_{t_0} - E^{(0)}_\psi} \\ &= \frac{2t^2}{E^{(0)}_{t_0} - E^{(0)}_{\psi_2}} - \frac{2t^2}{E^{(0)}_{t_0} - E^{(0)}_{\psi_5}} = 0\,. \end{aligned}$$

This time, the central − sign is present precisely because of the internal sign of the state $|t_0\rangle$. Finally,

$$E^{(2)}_{t_\pm} = \sum_{\psi \neq t_0} \frac{\langle t_\pm|\hat{H}_1|\psi\rangle\ \langle\psi|\hat{H}_1|t_\pm\rangle}{E^{(0)}_{t_\pm} - E^{(0)}_\psi} = 0\,,$$

again, because of Pauli exclusion principle.

What makes $|s\rangle$ and $|t_0\rangle$ fundamentally different? The singlet state takes advantage of the second-order virtual process where one spin hops to the neighbouring site and returns back. This process is schematised in Fig. A.2, and is referred to as “virtual hopping”. But each one of the two electrons can do this. The antisymmetry properties of many-body electronic wavefunctions imply, in the end, that these two virtual processes interfere constructively. For the triplet state $|t_0\rangle$, the two processes are perfectly plausible but interfere destructively. The coupling of electronic spin on neighbouring atoms is known as “direct exchange”.

This perturbative scheme scales up to larger lattices. It can be shown analytically that the low-energy sector of the strong-coupled Hubbard hamiltonian of Eq. (1.1) is approximated by an effective Heisenberg hamiltonian with antiferromagnetic spin-spin coupling [42].

A.3 Superexchange in cuprates

The scheme of direct exchange explains the insurgence of antiferromagnetism in the Hubbard model at strong coupling, $U \gg t$. The physics of cuprates is slightly more complicated.

Indeed, in the copper oxides active layers, the Cu^{2+} ions are surrounded by oxygens in a closed shell configuration. Hence, copper $d_{x^2-y^2}$-wave orbitals (where conduction electrons reside) do not overlap. Instead, they overlap with p-wave orbitals of the neighbouring oxygens. This enables a mechanism known as “superexchange”: even if the copper spins cannot interact directly, they use the intermediate p-orbitals as a sort of “bridge” and establish an direct exchange mechanism with the intermediate spins. The resulting interaction, as is “seen” by the copper electrons, is again an effective antiferromagnetic coupling [46].

Appendix B

Gaussian states

In this appendix, we discuss the properties of the many-body states we use to approximate the real ground-state of the system within HF method, introduced in Chap. 2. We introduce Slater determinants to grasp the general idea behind HF variational method, and then extend to a larger set of states.

B.1 Slater determinants

A many-body wavefunction describing a collective electronic state needs to be antisymmetric under interchange of coordinates and spins $(\mathbf{r}_i, \sigma_i)$, $(\mathbf{r}_j, \sigma_j)$ of any couple of electrons. The simplest way to build a many-body wavefunction satisfying this rule is "to anti-symmetrise" a set of orthonormal N one-body spin-orbitals, being N the total number of electrons and D the total number of spin-orbitals. Let $\{\psi_i\}$ be said set,

$$\frac{1}{2\mathcal{V}} \sum_{\sigma \in \{\uparrow,\downarrow\}} \int \mathrm{d}\mathbf{r}\ \psi_i(\mathbf{r}, \sigma)\psi_j(\mathbf{r}, \sigma) = \delta_{ij}\ ,$$

being $\mathcal{V}$ the total volume.

Let S_N be the $N!$ elements symmetric group of permutations of N objects, and $\mathrm{P} \in \mathrm{S}_N$ a specific permutation of a given N-uple,

$$\mathrm{P} \quad : \quad (x_1, x_2, \cdots, x_N) \to \left(x_{\mathrm{P}(1)}, x_{\mathrm{P}(2)}, \cdots, x_{\mathrm{P}(N)}\right)\ .$$

Let F_P be the operator that implements the permutation P on the N variables of a function f,

$$F_\mathrm{P}\left[f(x_1, x_2, \cdots, x_N)\right] = f\left(x_{\mathrm{P}(1)}, x_{\mathrm{P}(2)}, \cdots, x_{\mathrm{P}(N)}\right)\ .$$

We define the anti-symmetrisation operator:

$$\mathcal{A}[f] \equiv \frac{1}{\sqrt{N!}} \sum_{\mathrm{P}\in\mathrm{S}_N} (-1)^{\sigma(\mathrm{P})} F_\mathrm{P}[f]\,,$$

being $\sigma(\mathrm{P})$ the "parity factor" of the permutation $\mathrm{P} \in \mathcal{S}_N$. The factor $\sigma(\mathrm{P})$ counts the number of exchanges needed to obtain P: this way, an even number of exchanges gives a total sign $+1$, an odd number gives -1. By construction, the function $\mathcal{A}[f]$ is antisymmetric under exchange of a pair of coordinates.

Consider now the many-body wavefunction

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2; \cdots; \mathbf{r}_N, \sigma_N) = \mathcal{A}\left[\prod_{i=1}^{N} \psi_i(\mathbf{r}_i, \sigma_i)\right]\ .$$

This class of wavefunctions, obtained as an antisymmetric combinations of product states, are representable as so-called Slater determinants,

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2; \cdots; \mathbf{r}_N, \sigma_N) = \frac{1}{\sqrt{N!}} \det \begin{bmatrix} \psi_1(\mathbf{r}_1, \sigma_1) & \psi_1(\mathbf{r}_2, \sigma_2) & \cdots & \psi_1(\mathbf{r}_N, \sigma_N) \\ \psi_2(\mathbf{r}_1, \sigma_1) & \psi_2(\mathbf{r}_2, \sigma_2) & \cdots & \psi_2(\mathbf{r}_N, \sigma_N) \\ \vdots & \vdots & & \vdots \\ \psi_N(\mathbf{r}_1, \sigma_1) & \psi_N(\mathbf{r}_2, \sigma_2) & \cdots & \psi_N(\mathbf{r}_N, \sigma_N) \end{bmatrix}\ .$$

This representation includes automatically Pauli exclusion principle: determinants of matrices with two equal rows or two equal columns are null. Thus, all spin-orbitals must differ, as well as no single spin-orbital ψ_i can be doubly occupied (which is, all spin-coordinate couples must be different).

Any normalised combinations of Slater determinants gives an antisymmetric wavefunction, although not necessarily representable as a single Slater determinant. In HF theory, we search for the best possible choice of orthonormal one-body spin orbitals $\{\psi_1, \psi_2, \cdots, \psi_N\}$ to approximate the true ground-state of the model through a single Slater determinant. The space of possible "Slater-like" many-body wavefunctions is the parametric space where we perform the variational constrained search.

The EHM is formulated in second-quantisation. It is then useful to represent Slater-like states in this language. Let us map all the spin-orbitals from the discussion above to pairs of annihilation-creation fermionic operators,

$$\psi_i \quad \rightarrow \quad \hat{c}_i, \hat{c}_i^\dagger \,.$$

Any Slater state can be expressed in second-quantisation as:

$$|\Psi\rangle = \prod_{i=1}^{N} \hat{c}_i^\dagger |\Omega\rangle \,, \tag{B.1}$$

being $|\Omega\rangle$ the vacuum. Antisymmetry is guaranteed by Fermi anticommutation rules.

B.2 Wick's theorem applied to Slater determinants

The concept of vacuum state needs to be interpreted in relation to an appropriate choice of second-quantisation annihilation and creation operators. For a given set of fermion operators, the vacuum is that state mapped to zero by any annihilation operator. As we will see, to correctly define the vacuum state is crucial in order to apply Wick's theorem and develop an HF approximation of EHM. Before moving to generalisation, let us discuss a playground setup.

Consider some unspecified model whose ground state at null temperature is given in Eq. (B.1) for a given set of electronic operators $\hat{c}_i$, $\hat{c}_i^\dagger$. Let $\hat{c}_i$ with $N < i < D$ identify all spin-orbitals not already included in the definition of $|\Psi\rangle$ of Eq. (B.1) (thus, non populated). Then, it holds:

$$\begin{aligned} 0 \leq i \leq N \qquad & \hat{c}_i^\dagger |\Psi\rangle = 0 \,, \\ i > N \qquad & \hat{c}_i |\Psi\rangle = 0 \,. \end{aligned}$$

Let:

$$\hat{a}_i \equiv \begin{cases} \hat{c}_i^\dagger & \text{for } 0 \leq i \leq N, \\ \hat{c}_i & \text{for } i > N. \end{cases} \tag{B.2}$$

These new operators satisfy a Fermi algebra $\{\hat{a}_i, \hat{a}_j^\dagger\} = \delta_{ij}$ and annihilate the "vacuum" state $|\Psi\rangle$. In other words, any Slater state like Eq. (B.1) can be seen as a vacuum state for a suitably chosen set of fermionic operators.

This property is handy for evaluation of expectation values (EV). Let $\hat{O}^{(\ell)}$ be a linear combination of $\hat{c}_i$ or $\hat{c}_i^\dagger$,

$$\hat{O}^{(\ell)} \equiv \sum_{i=1}^{D} \left(u_i^{(\ell)} \hat{c}_i + v_i^{(\ell)} \hat{c}_i^\dagger \right) \quad \text{where} \quad u_i^{(\ell)}, v_i^{(\ell)} \in \mathbb{C} \,. \tag{B.3}$$

Consider the product operator:

$$\hat{\mathcal{O}} \equiv \hat{O}^{(1)} \hat{O}^{(2)} \cdots \hat{O}^{(2k)} \quad \text{where} \quad k \in \mathbb{N} \,.$$

Inserting the above decomposition, we get a large sum of many terms, each containing some sequence of annihilations and creations. Using linearity we can concentrate on one specific term of these, derive results and then recompose them for the original operator. Consider one of these terms,

$$\hat{\mathcal{O}} = \cdots + (\text{coefficient}) \times \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} + \cdots \,,$$

with $\hat{C}_i$ some fermionic operator, either $\hat{c}_j$ or $\hat{c}_j^\dagger$ for some j. Now, Wick's theorem states that

$$\hat{C}_1\hat{C}_2\cdots\hat{C}_{2k} = \mathrm{N}\left\{\hat{C}_1\hat{C}_2\cdots\hat{C}_{2k}\right\} + \sum_{(\alpha\beta)}(-1)^{\zeta_{\alpha\beta}}D_{\alpha\beta}\mathrm{N}\left\{\prod_{i\neq\alpha,\beta}\hat{C}_i\right\}$$
$$+ \sum_{(\alpha\beta)(\gamma\delta)}(-1)^{\zeta_{\alpha\beta}}D_{\alpha\beta}(-1)^{\zeta_{\gamma\delta}}D_{\gamma\delta}\mathrm{N}\left\{\prod_{i\neq\alpha,\beta,\gamma,\delta}\hat{C}_i\right\} + \cdots\,, \quad \text{(B.4)}$$

where $\mathrm{N}\{\cdots\}$ indicates normal ordering, $D_{\alpha\beta}$ is the contraction of the pair $\hat{C}_\alpha\hat{C}_\beta$ and $\zeta_{\alpha\beta}$ is a symmetry factor included to take care of the number of fermionic algebra. In this context contractions are just expectation values over $|\Psi\rangle$,

$$D_{\alpha\beta} \equiv \langle\Psi|\hat{C}_\alpha\hat{C}_\beta|\Psi\rangle\ .$$

Sums are performed over pairs of indices, implicitly taken to be different from each other. To take averages of this last operator over $|\Psi\rangle$ is a not-so-hard task, given the simple expression of the state in terms of electronic annihilations and creations. However it can be made simpler: let us change basis and express all operators in terms of the operators defined in Eq. (B.2),

$$\hat{C}_i \quad\rightarrow\quad \hat{A}_i\,,$$

being $\hat{A}_i$ either $\hat{a}_i$ or $\hat{a}_i^\dagger$. The Wick's decomposition of Eq. (B.4) holds identically, upon making substitutions. Now, taking a zero temperature average on $|\Psi\rangle$ (the vacuum with respect to the new operators) all normal-ordered terms disappear,

$$\langle\Psi|\mathrm{N}\{\hat{A}_i\hat{A}_j\cdots\}|\Psi\rangle = 0\,,$$

thus in the decomposition above only fully contracted terms survive. Going back to $\hat{c}$ operators, we can re-express fully contracted terms in the original basis. Thus:

$$\left\langle\hat{C}_1\hat{C}_2\cdots\hat{C}_{2k}\right\rangle = \sum(\text{total sign factor})\times(\text{product of complete contractions})\,,$$

where sum is intended, implicitly, over all possible ways of choosing k couples of indices from a starting set of $2k$ elements.

Taking another step back, due to linearity we can apply this result to all terms of $\mathcal{O}$ and conclude that the EV of $\mathcal{O}$ decomposes as a sum of 2-points EVs. This can be seen as the defining property of gaussian states. In this procedure, the only key property of Slater states we used was the fact that they are vacuum to some fermionic basis. Finite-temperature extension of Wick's theorem as well as more complicated change of basis are possible, but the concept remains the same: on a Slater state, which is essentially a non-interacting state for some set of fermions, Wick's theorem is particularly well-behaved.

From the point above we can conclude that:

$$\langle\hat{c}_\alpha^\dagger\hat{c}_\beta^\dagger\hat{c}_\gamma\hat{c}_\delta\rangle = \langle\hat{c}_\alpha^\dagger\hat{c}_\delta\rangle\langle\hat{c}_\beta^\dagger\hat{c}_\gamma\rangle - \langle\hat{c}_\alpha^\dagger\hat{c}_\beta\rangle\langle\hat{c}_\gamma^\dagger\hat{c}_\delta\rangle\,,$$

being the average intended on a Slater state. In this context, only contractions with an equal number of creation and annihilation operators contribute.

B.3 Generalisation to Bogoliubov-Valatin states

In HF theory we look for a good approximation of the true many-body wavefunction constraining search to a class of states. Introducing Slater determinants was useful to grasp the concrete idea behind the method. However, in order to treat some physical phenomena we need to extend the variational procedure to a other sets of states. For example, Bardeen-Cooper-Shrieffer (BCS) theory of superconductivity gives a many-body ground-state which is not representable as a Slater determinant of "standard" electronic operators. In this section, we extend the idea introduced above.

Consider the Nambu spinor

$$\hat{\Phi} \equiv \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix} ,$$

being $\hat{\mathbf{c}}$ the vector of all annihilation operators, ordered, and $\hat{\mathbf{c}}^\dagger$ of all creation operators. Let us assume the vectors have D entries. Consider an unitary map M, such that

$$\hat{\Gamma} \equiv M\hat{\Phi} \quad \text{being} \quad \hat{\Gamma} \equiv \begin{bmatrix} \hat{\boldsymbol{\gamma}} \\ \hat{\boldsymbol{\gamma}}^\dagger \end{bmatrix} .$$

The new Nambu vectors $\hat{\boldsymbol{\gamma}}$ and $\hat{\boldsymbol{\gamma}}^\dagger$ contain our new, generalised fermionic operators in second quantisation. Let:

$$M \equiv \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix} .$$

If M is unitary, it follows:

$$\begin{aligned} \mathbb{I}_{2D\times 2D} &= MM^\dagger \\ &= \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix} \begin{bmatrix} U^\dagger & U \\ V^\dagger & V \end{bmatrix} \\ &= \begin{bmatrix} UU^\dagger + VV^\dagger & U^2 + V^2 \\ \left(U^\dagger\right)^2 + \left(V^\dagger\right)^2 & U^\dagger U + V^\dagger V \end{bmatrix} , \end{aligned}$$

therefore,

$$UU^\dagger + VV^\dagger = \mathbb{I}_{D\times D} , \tag{B.5}$$

$$U^2 + V^2 = 0 , \tag{B.6}$$

where in last line 0 is intended as a $D \times D$ zero matrix.

The new operators in $\hat{\Gamma}$ satisfy Fermi anticommutation rules. In fact, writing the mapping explicitly for any of the new operators and some index $i \in [1, D]$,

$$\hat{\gamma}_i \equiv \sum_{j=1}^{D} U_{ij}\hat{c}_j + \sum_{j=1}^{D} V_{ij}\hat{c}_j^\dagger \quad \text{and} \quad \hat{\gamma}_i^\dagger \equiv \sum_{j=1}^{D} U_{ji}^*\hat{c}_j^\dagger + \sum_{j=1}^{D} V_{ji}^*\hat{c}_j , \tag{B.7}$$

and substituting in the anti-commutator

$$\begin{aligned} \{\hat{\gamma}_i, \hat{\gamma}_j^\dagger\} &= \sum_{i'=1}^{D} U_{ii'} \sum_{j'=1}^{D} U_{j'j}^* \{\hat{c}_{i'}, \hat{c}_{j'}^\dagger\} + \sum_{i'=1}^{D} V_{ii'} \sum_{j'=1}^{D} V_{j'j}^* \{\hat{c}_{i'}^\dagger, \hat{c}_{j'}\} \\ &= \sum_{i'=1}^{D} U_{ii'} \sum_{j'=1}^{D} U_{j'j}^* \delta_{i'j'} + \sum_{i'=1}^{D} V_{ii'} \sum_{j'=1}^{D} V_{j'j}^* \delta_{i'j'} \\ &= \left[UU^\dagger + VV^\dagger\right]_{ij} \\ &= \delta_{ij} , \end{aligned}$$

being M unitary. In last line we used Eq. (B.5).

Assume now $|\Psi\rangle$ to be the vacuum state for the new operators,

$$\forall i \quad : \quad \hat{\gamma}_i \left|\Psi\right\rangle = 0 .$$

Suppose we want to evaluate, with identical meaning of the notation as in previous section, something like

$$\left\langle \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} \right\rangle .$$

We can apply the result obtained above. Indeed, the inverse version of the maps of Eq. (B.7), linking old and new operators, read

$$\hat{c}_i \equiv \sum_{j=1}^{D} U_{ij}\hat{\gamma}_j + \sum_{j=1}^{D} U_{ji}^*\hat{\gamma}_j^\dagger \quad \text{and} \quad \hat{c}_i^\dagger \equiv \sum_{j=1}^{D} V_{ji}^*\hat{\gamma}_j^\dagger + \sum_{j=1}^{D} V_{ij}\hat{\gamma}_j . \tag{B.8}$$

In Sec. B.2 we argued that, when averaging a product of operators like the one of Eq. (B.3), only 2-points EVs contribute. This time the idea is the same, substituting $\mathcal{O}^{(\ell)} \to \hat{C}_\ell$ and $\hat{c}_i, \hat{c}_i^\dagger \to \hat{\gamma}_i, \hat{\gamma}_i^\dagger$. Thus, also here we can decompose the product operator, apply Wick's theorem, discard all non-fully-contracted terms and recompose the results. All we used was just the unitary mapping between $\hat{\Phi}$ and $\hat{\Gamma}$, the fact that $\gamma_i, \hat{\gamma}_i^\dagger$ obey Fermi anticommutation rules and "see" $|\Psi\rangle$ as a vacuum state.

As in Sec. B.2, we conclude that:

$$\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \rangle = \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\delta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\gamma \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \langle \hat{c}^\dagger_\gamma \hat{c}_\delta \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \langle \hat{c}_\gamma \hat{c}_\delta \rangle}_{\text{Cooper}} \, . \tag{B.9}$$

The subscripts indicate the name we give to these terms throughout the text. Note that, differently from "pure Slater states" EVs with two annihilations or two creations do not vanish, in general.

All of this holds when averaging is performed on a state, $|\Psi\rangle$, which is vacuum for a set of operators linked to fermionic Nambu spinor $\hat{\Phi}$ by some unitary operation. This set of states is the domain of our variational constrained search. This kind of unitary mapping is often referred to as Bogoliubov-Valatin transformation, and the method hereby presented as Hartree-Fock-Bogoliubov. For simplicity, in this thesis we stick to the name "HF method" referring to this discussion. This argument has been carried out at zero temperature, but can be extended to finite temperature.

Appendix C

Connection with mean-field theories

In this appendix, we bridge the gap between HF methods and mean-field theory (MFT). In general, in MFT approaches we substitute the two-body (or more) interactions of the hamiltonian with a single-body interaction with a mean field, determined locally via self-consistency equations. This is, for example, the Weiß mean-field solution to the quantum Ising model. The HF method of Chap. 2 does not work like that. In the context of Hubbard models, the analogous of Weiß MFT would be dynamical mean-field theory (DMFT) [11]. However, there are some points of connection. Let us sketch first the general scheme behind MFT.

C.1 Mean-field theory

Suppose we have some operator $\hat{A} = \hat{B}\hat{C}$. Let

$$\delta\hat{O} \equiv \hat{O} - \langle\hat{O}\rangle \quad \text{with} \quad \hat{O} \in \{\hat{A}, \hat{B}, \hat{C}\}\,.$$

Then we have:

$$\begin{aligned}
\hat{A} &= \left(\langle\hat{B}\rangle + \delta\hat{B}\right)\left(\langle\hat{C}\rangle + \delta\hat{C}\right) \\
&= \langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle\delta\hat{C} + \delta\hat{B}\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C} \\
&= \langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle\left(\hat{C} - \langle\hat{C}\rangle\right) + \left(\hat{B} - \langle\hat{B}\rangle\right)\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C} \\
&= -\langle\hat{B}\rangle\langle\hat{C}\rangle + \langle\hat{B}\rangle\hat{C} + \hat{B}\langle\hat{C}\rangle + \delta\hat{B}\delta\hat{C}\,.
\end{aligned}$$

Up to now, this relation is exact. MFT relies on the idea that the last term, of order $\mathcal{O}(\delta^2)$, can be neglected. This amounts to say that the operator fluctuations about its average are "small". This gives the approximation:

$$\hat{A} \simeq \langle\hat{B}\rangle\hat{C} + \hat{B}\langle\hat{C}\rangle - \langle\hat{B}\rangle\langle\hat{C}\rangle\,. \tag{C.1}$$

Within the assumption that, independently, $\hat{B}$ and $\hat{C}$ fluctuations are "small", we are able to simplify $\hat{A}$ with a sum of three terms: the product of each operator with the averaged value of the other, minus the product of the averages.

In the context of HF methods, we are in an slightly different situation. We are not making assumptions on the fluctuations of the quartic sectors of the hamiltonian. We are restricting the space of wavefunctions and minimising energy there.

C.2 Hartee-Fock methods

Let us restart from the interacting hamiltonian of Eq. (2.2). In this thesis we deal with the EHM, thus (limitedly to this section) let us restrict to the two-body generic quartic interaction:

$$\hat{V} = \sum_{\alpha,\beta} V_{\alpha\beta}\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\hat{c}_\beta\hat{c}_\alpha\,,$$

with $\forall\alpha,\beta$: $V_{\alpha\beta} = V^*_{\beta\alpha}$. In terms of the general hamiltonian of Eq. (2.2), we have:

$$B_{\alpha\beta\gamma\delta} = 2V_{\alpha\beta}\delta_{\alpha\delta}\delta_{\beta\gamma}\,. \tag{C.2}$$

We work with HF states. Thus Eq. (2.1) holds exactly (adapted for two indices instead of four)

$$\alpha \neq \beta \quad : \quad \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\beta \hat{c}_\alpha \rangle = \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \langle \hat{c}_\beta \hat{c}_\alpha \rangle}_{\text{Cooper}}\,.$$

This gives:

$$\begin{aligned}
\langle \hat{V} \rangle &= \sum_{\alpha,\beta} V_{\alpha\beta} \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\beta \hat{c}_\alpha \rangle \\
&= \sum_{\alpha,\beta} V_{\alpha\beta} \Big(\langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle - \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle + \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \langle \hat{c}_\beta \hat{c}_\alpha \rangle \Big)\,,
\end{aligned} \tag{C.3}$$

from which we have the equations:

$$\frac{\partial \langle \hat{V} \rangle}{\partial \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle} = \delta_{\alpha\beta} \left[\sum_\gamma V_{\alpha\gamma} \langle \hat{c}^\dagger_\gamma \hat{c}_\gamma \rangle \right] - V_{\alpha\beta} \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle\,, \tag{C.4}$$

$$\frac{\partial \langle \hat{V} \rangle}{\partial \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle} = V_{\alpha\beta} \langle \hat{c}_\beta \hat{c}_\alpha \rangle\,. \tag{C.5}$$

Now, our aim is to approximate $\hat{V}$ as:

$$\hat{V} \simeq \sum_{\alpha\beta} \left[x_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta + y_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta + y^*_{\beta\alpha} \hat{c}_\alpha \hat{c}_\beta \right] + (\text{some constant})\,, \tag{C.6}$$

where $x_{\alpha\beta} = x^*_{\beta\alpha}$ to ensure hermiticity. Note that we are actively ignoring terms like $\hat{c}\hat{c}^\dagger$. Anti-commuting these, we get sums of operators like $\hat{c}^\dagger\hat{c}$ and constants. The above expression is the most general. Take its EV:

$$\langle \hat{V} \rangle \simeq \sum_{\alpha\beta} \left[x_{\alpha\beta} \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle + y_{\alpha\beta} \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle + y^*_{\beta\alpha} \langle \hat{c}_\alpha \hat{c}_\beta \rangle \right] + (\text{some constant})\,.$$

Deriving the equation above, we obtain:

$$x_{\alpha\beta} = \frac{\partial \langle \hat{V} \rangle}{\partial \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle} \quad \text{and} \quad y_{\alpha\beta} = \frac{\partial \langle \hat{V} \rangle}{\partial \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle}\,.$$

Applying Eqs. (C.4) and (C.5), we get the identities:

$$x_{\alpha\beta} = \delta_{\alpha\beta} \left[\sum_\gamma V_{\alpha\gamma} \langle \hat{c}^\dagger_\gamma \hat{c}_\gamma \rangle \right] - V_{\alpha\beta} \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle\,, \tag{C.7}$$

$$y_{\alpha\beta} = V_{\alpha\beta} \langle \hat{c}_\beta \hat{c}_\alpha \rangle\,. \tag{C.8}$$

These equations are a rephrasing of Eqs. (2.14), (2.15) for the fields $h^{pq}_{\alpha\beta}$, setting $A_{\alpha\beta} = 0$. This can be checked by using Eq. (C.2).

Now: consider the special case where $\forall\alpha$: $V_{\alpha\alpha} = 0$. This is the case for the EHM. Then the above equations imply:

$$\begin{aligned}
\hat{V} &= \sum_{\alpha\neq\beta} V_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\beta \hat{c}_\alpha \\
&\simeq \sum_{\alpha\neq\beta} V_{\alpha\beta} \Big(\overbrace{\hat{c}^\dagger_\alpha \hat{c}_\alpha \langle \hat{c}^\dagger_\beta \hat{c}_\beta \rangle + \langle \hat{c}^\dagger_\alpha \hat{c}_\alpha \rangle \hat{c}^\dagger_\beta \hat{c}_\beta}^{\text{Hartree}} \overbrace{- \hat{c}^\dagger_\alpha \hat{c}_\beta \langle \hat{c}^\dagger_\beta \hat{c}_\alpha \rangle - \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \hat{c}^\dagger_\beta \hat{c}_\alpha}^{\text{Fock}} \\
&\qquad + \underbrace{\hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \langle \hat{c}_\beta \hat{c}_\alpha \rangle + \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \hat{c}_\beta \hat{c}_\alpha}_{\text{Cooper}} \Big) + (\text{some constant})\,.
\end{aligned}$$

To determine the constant, we take the EV of the above equation. The result should equal Eq. (C.3). Take e.g. the Hartree sector:

$$\big\langle \hat{c}_\alpha^\dagger \hat{c}_\alpha \big\rangle \big\langle \hat{c}_\beta^\dagger \hat{c}_\beta \big\rangle + \big\langle \hat{c}_\alpha^\dagger \hat{c}_\alpha \big\rangle \big\langle \hat{c}_\beta^\dagger \hat{c}_\beta \big\rangle + C = \big\langle \hat{c}_\alpha^\dagger \hat{c}_\alpha \big\rangle \big\langle \hat{c}_\beta^\dagger \hat{c}_\beta \big\rangle \qquad \Longrightarrow \qquad C = -\big\langle \hat{c}_\alpha^\dagger \hat{c}_\alpha \big\rangle \big\langle \hat{c}_\beta^\dagger \hat{c}_\beta \big\rangle\,,$$

with C a constant. A similar implication holds for the Fock and the Cooper sectors. Then:

$$\begin{aligned}
\hat{V} &= \sum_{\alpha\neq\beta} V_{\alpha\beta} \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\beta \hat{c}_\alpha && \text{(C.9)}\\
&\overset{\text{HF}}{\simeq} \sum_{\alpha\neq\beta} V_{\alpha\beta} \Big(\hat{c}_\alpha^\dagger \hat{c}_\alpha \big\langle \hat{c}_\beta^\dagger \hat{c}_\beta \big\rangle + \big\langle \hat{c}_\alpha^\dagger \hat{c}_\alpha \big\rangle \hat{c}_\beta^\dagger \hat{c}_\beta - \big\langle \hat{c}_\alpha^\dagger \hat{c}_\alpha \big\rangle \big\langle \hat{c}_\beta^\dagger \hat{c}_\beta \big\rangle && \text{(Hartree)}\\
&\qquad - \hat{c}_\alpha^\dagger \hat{c}_\beta \big\langle \hat{c}_\beta^\dagger \hat{c}_\alpha \big\rangle - \big\langle \hat{c}_\alpha^\dagger \hat{c}_\beta \big\rangle \hat{c}_\beta^\dagger \hat{c}_\alpha + \big\langle \hat{c}_\alpha^\dagger \hat{c}_\beta \big\rangle \big\langle \hat{c}_\beta^\dagger \hat{c}_\alpha \big\rangle && \text{(Fock)}\\
&\qquad + \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \big\langle \hat{c}_\beta \hat{c}_\alpha \big\rangle + \big\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \big\rangle \hat{c}_\beta \hat{c}_\alpha - \big\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \big\rangle \big\langle \hat{c}_\beta \hat{c}_\alpha \big\rangle \Big)\,. && \text{(Cooper)}
\end{aligned}$$

Passing from the first row to the following ones, we decomposed the original quartic operator to a sum of three decomposition much similar to Eq. (C.1). This is the main connection with MFT. In HF theory, we do not write the quartic operator as a product of two quadratic operators and decompose it as we would do in MFT. Instead, a very similar decomposition (summed over the three sectors) is given by the fact that EVs are computed over HF states.

Appendix D

Antiferromagnetism in the HM

In this appendix we give some details about the antiferromagnetic phase in the conventional Hubbard model,

$$\hat{H} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \left[\hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \quad \text{where} \quad t, U > 0 \,, \tag{D.1}$$

approximated by HF methods. We use the results discussed in Sec. 5.2.

D.1 The AF ground-state for the conventional HM

Let the Nambu spinors:

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix} .$$

Let also the gap function:

$$\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU \,.$$

We know from Eq. (5.41) that the approximated hamiltonian is:

$$\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \mathbf{b}_{\mathbf{k}\sigma} \cdot \hat{\mathbf{s}}_{\mathbf{k}\sigma} + nU\hat{N} - E^{(\text{AF})}_{\text{H}/U} \,,$$

where:

$$\mathbf{b}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} -\Delta_{\mathbf{k}\sigma} \\ 0 \\ \epsilon_{\mathbf{k}\sigma} \end{bmatrix} \qquad \hat{s}^{\alpha}_{\mathbf{k}\sigma} \equiv \hat{\Psi}^{\dagger}_{\mathbf{k}\sigma} \tau^{\alpha} \hat{\Psi}_{\mathbf{k}\sigma} \qquad \hat{\mathbf{s}}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{s}^{x}_{\mathbf{k}\sigma} \\ \hat{s}^{y}_{\mathbf{k}\sigma} \\ \hat{s}^{z}_{\mathbf{k}\sigma} \end{bmatrix} .$$

Through the unitary matrix $W_{\mathbf{k}\sigma}$ given in Eq. (5.44), the hamiltonian is mapped to a gas of non-interacting fermions collected in the $\hat{\Gamma}$ spinors of Eq. (5.45),

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{\gamma}^{(+)}_{\mathbf{k}\sigma} \\ \hat{\gamma}^{(-)}_{\mathbf{k}\sigma} \end{bmatrix} .$$

The final non-interacting hamiltonian is:

$$\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} E_{\mathbf{k}\sigma} \left(\hat{\gamma}^{(+)\dagger}_{\mathbf{k}\sigma} \hat{\gamma}^{(+)}_{\mathbf{k}\sigma} - \hat{\gamma}^{(-)\dagger}_{\mathbf{k}\sigma} \hat{\gamma}^{(-)}_{\mathbf{k}\sigma} \right) + nU\hat{N} - E^{(\text{AF})}_{\text{H}/U} \,,$$

with $E_{\mathbf{k}\sigma}$ the Bogoliubov bands of Eq. (5.43). In next section we give the explicit form of the many-body ground-state at zero temperature and half-filling. From now on, limitedly to this appendix, we use the simplified notation:

$$\Delta_{\mathbf{k}\uparrow} \quad \rightarrow \quad \Delta \,,$$

with $\Delta_{\mathbf{k}\uparrow} = -\Delta_{\mathbf{k}\downarrow}$, and $\Delta = mU$.

D.2 Explicit AF ground state at half-filling and $\beta \to \infty$

Let $|\Uparrow_{\mathbf{k}\sigma}\rangle$ and $|\Downarrow_{\mathbf{k}\sigma}\rangle$ be respectively the "up" and "down" $\mathbf{k}$-th pseudo-spin states. Let $|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle$ be the "down" state in the tilted frame. The ground state is given by

$$|\Psi\rangle \equiv \bigotimes_{\sigma} \bigotimes_{\mathbf{k}\in\mathrm{MBZ}} |\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle \,. \tag{D.2}$$

Let us rotate the "down" states,

$$\begin{aligned}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle &= e^{-i\theta_{\mathbf{k}\sigma}\hat{s}^y_{\mathbf{k}\sigma}} |\Downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos\theta_{\mathbf{k}\sigma} |\Downarrow_{\mathbf{k}\sigma}\rangle - i\sin\theta_{\mathbf{k}\sigma}\hat{s}^y_{\mathbf{k}\sigma} |\Downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos\theta_{\mathbf{k}\sigma} |\Downarrow_{\mathbf{k}\sigma}\rangle - \sin\theta_{\mathbf{k}\sigma}(\hat{s}^+_{\mathbf{k}\sigma} - \hat{s}^-_{\mathbf{k}\sigma}) |\Downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos\theta_{\mathbf{k}\sigma} |\Downarrow_{\mathbf{k}\sigma}\rangle - \sin\theta_{\mathbf{k}\sigma} |\Uparrow_{\mathbf{k}\sigma}\rangle \,,\end{aligned} \tag{D.3}$$

having we used

$$\hat{s}^\pm_{\mathbf{k}\sigma} \equiv \frac{\hat{s}^x_{\mathbf{k}\sigma} \pm i\hat{s}^y_{\mathbf{k}\sigma}}{2} \,.$$

From Eqs. (5.40) and (5.39),

$$\frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}^z_{\mathbf{k}\sigma}}{2} = \hat{n}_{\mathbf{k}\sigma} \,, \qquad \frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}^z_{\mathbf{k}\sigma}}{2} = \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \,.$$

Evaluating the above operators on the "up" and "down" states, we get

$$\begin{aligned}&\frac{1}{2}\langle\Uparrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}^z_{\mathbf{k}\sigma}|\Uparrow_{\mathbf{k}\sigma}\rangle = 1 \,, \qquad &&\frac{1}{2}\langle\Uparrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}^z_{\mathbf{k}\sigma}|\Uparrow_{\mathbf{k}\sigma}\rangle = 0 \,, \\ &\frac{1}{2}\langle\Downarrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}^z_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle = 0 \,, \qquad &&\frac{1}{2}\langle\Downarrow_{\mathbf{k}\sigma}|\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}^z_{\mathbf{k}\sigma}|\Downarrow_{\mathbf{k}\sigma}\rangle = 1 \,.\end{aligned}$$

Thus, up to a phase

$$|\Uparrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}^\dagger_{\mathbf{k}\sigma} |\Omega_{\mathbf{k}\sigma}\rangle \,, \qquad |\Downarrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma} |\Omega_{\mathbf{k}\sigma}\rangle \,,$$

being $|\Omega_{\mathbf{k}\sigma}\rangle$ the $(\mathbf{k},\sigma)$ Hilbert subspace vacuum.

It follows that composing Eqs. (D.2) and (D.3) the ground state $|\Psi\rangle$ can be written as:

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k}\in\mathrm{MBZ}} \left[\sin\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}\sigma} + e^{2i\zeta_{\mathbf{k}\sigma}}\cos\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right] |\Omega_{\mathbf{k}\sigma}\rangle \,, \tag{D.4}$$

having we absorbed in the relative phase $e^{2i\zeta_{\mathbf{k}\sigma}}$ the two states phases, as well as the original sign difference of Eq. (D.3). Now, inserting the explicit form of the ground state from equation above:

$$\begin{aligned}\langle\overline{\Downarrow}_{\mathbf{k}\sigma}|\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle &= \left(e^{2i\zeta_{\mathbf{k}\sigma}} + e^{-2i\zeta_{\mathbf{k}\sigma}}\right)\sin\theta_{\mathbf{k}\sigma}\cos\theta_{\mathbf{k}\sigma} \\ &= \cos 2\zeta_{\mathbf{k}\sigma}\sin 2\theta_{\mathbf{k}\sigma} \,.\end{aligned}$$

From Eq. (5.37)

$$\begin{aligned}\langle\overline{\Downarrow}_{\mathbf{k}\sigma}|\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}^\dagger_{\mathbf{k}\sigma}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle &= \langle\overline{\Downarrow}_{\mathbf{k}\sigma}|\hat{s}^x_{\mathbf{k}\sigma}|\overline{\Downarrow}_{\mathbf{k}\sigma}\rangle \\ &= \sin 2\theta_{\mathbf{k}\sigma} \,,\end{aligned}$$

since the pseudofield has no y component. This implies necessarily

$$\cos 2\zeta_{\mathbf{k}\sigma} \quad \Longrightarrow \quad \zeta_{\mathbf{k}\sigma} = k\pi \quad \text{with} \quad k\in\mathbb{Z} \,.$$

Using Eq. (D.4), we conclude

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k}\in\mathrm{MBZ}} \left[\sin\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}\sigma} + \cos\theta_{\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\right] |\Omega_{\mathbf{k}\sigma}\rangle \,. \tag{D.5}$$

For low-temperature physics, we can expect the ground-state density matrix to be highly dominated by this BCS-like zero-temperature ground-state, which highlights the essential features of the system.

Finally, using Eq. (5.45), we check that correctly the ground-state describes a perfect degenerate Fermi gas described by operators $\hat{\gamma}^{(-)}_{\mathbf{k}\sigma}$. Indeed,

$$\begin{aligned}\hat{\gamma}^{(-)}_{\mathbf{k}\sigma} &= \Big[W_{\mathbf{k}\sigma}\hat{\Psi}_{\mathbf{k}\sigma}\Big]_2 \\ &= \sin\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} + \cos\theta_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\end{aligned}$$

Therefore, Eq. (D.5) reduces to

$$|\Psi\rangle = \bigotimes_{\sigma}\bigotimes_{\mathbf{k}\in\mathrm{MBZ}} \gamma^{(-)\dagger}_{\mathbf{k}\sigma}|\Omega_{\mathbf{k}\sigma}\rangle\,. \tag{D.6}$$

The AF phase can be thought of as a long-range coordination between fermionic states related by a $\boldsymbol{\pi}$ shift in reciprocal space, producing a new free gas where all interactions have been "condensed" in these AF pairs.

D.3 Instability of the commensurate AF phase

The commensurate AF phase is unstable at any finite doping. This result is discussed in [27, 28], and here we recall just the main passages. Consider the local spin operator defined in Eq. (3.7),

$$\hat{S}^\alpha_j = \frac{1}{2}\hat{\Psi}^\dagger_j\tau^\alpha\hat{\Psi}_j\,,\qquad \hat{S}^+_j \equiv \frac{\hat{S}^x_j + i\hat{S}^y_j}{2}\,,\qquad \hat{S}^-_j \equiv \frac{\hat{S}^x_j - i\hat{S}^y_j}{2}\,.$$

Let us take a continuum limit, $L_x\to\infty$ and $L_y\to\infty$, while the lattice spacing goes to zero.

Continuous SSBs are connected to the presence of a stable gapless Golstone mode. Such a Goldstone mode is essentially a sharp peak at zero frequency for some dressed response function. For the commensurate AF phase, the low-energy collective excitations are transverse spin fluctuations about the antiferromagnetic ground-state, describing SDWs. In order to see if the Golstone mode is stable for a given doping δ, we need to investigate whether the respective response function diverges at zero frequency. If it does, we have a gapless Goldstone mode and the phase is stable; if not, the phase is unstable. We work at $\omega = 0$ in order to capture gapless modes.

We work in interaction representation, and indicate the spin operators by

$$\hat{\mathbf{S}}(\mathbf{r},t)$$

The retarded transverse spin-spin response is given by:

$$\chi^{-+}(\mathbf{r},t;\mathbf{r}',t') =\equiv i\mathrm{H}(t-t')\langle\Psi|\Big[\hat{S}^-(\mathbf{r},t),\hat{S}^+(\mathbf{r}',t')\Big]|\Psi\rangle\,,$$

being $\mathrm{H}(x)$ the Heaviside function. The argument carried by Singh and Tešanović relies on random phase approximation (RPA). Let χ_0^{-+} be the non-interacting transverse response function. Then

$$\chi^{-+} \simeq \frac{\chi_0^{-+}}{1-U\chi_0^{-+}}\,,\qquad \text{(RPA)}$$

is the random phase approximation (RPA) proper response function.

We move to momentum-frequency space. Within RPA approximation, low-lying spin waves collective excitations at wavevector $\mathbf{q}$ give a stable gapless Goldstone mode if:

$$1 - U\chi_0^{-+}(\mathbf{q},0) = 0\,. \tag{D.7}$$

Commensurate antiferromagnetism is stable if the above equation is satisfied at $\mathbf{q}\to\boldsymbol{\pi}$. Indeed, the wavevector $\boldsymbol{\pi}$ expresses two-sites periodicity in both directions.

Now, as we specified in Sec. 5.1, commensurate antiferromagnetism is described by the simultaneous SSB:

$$\mathrm{SU}(2) \overset{\mathrm{SSB}}{\to} \mathrm{U}^z(1)\,,$$
$$\mathrm{T}^{(x)}_1\otimes\mathrm{T}^{(y)}_1 \overset{\mathrm{SSB}}{\to} \mathrm{T}^{(x)}_2\otimes\mathrm{T}^{(y)}_2\,.$$

Thus translational invariance is halved in both directions. The lattice $\mathcal{L}$ is separated in the two sublattices $\mathcal{S}_1$, $\mathcal{S}_2$ of Fig. 3.1. As a consequence, the non-interacting response function χ_0^{+-} does not depend strictly on $\mathbf{r}$, $\mathbf{r}'$ themselves. Instead, it depends only on which sublattice $\mathbf{r}$ and $\mathbf{r}'$ belong to. Because of this χ_0^{+-} is representable as a 2×2 matrix $\boldsymbol{\chi}_0^{+-}$. Then Eq. (D.7) is actually an eigenvalue equation,

$$\mathbb{I} - U\boldsymbol{\chi}_0^{+-}(\mathbf{q},0) = \mathbf{0}\,,$$

with $\mathbf{0}$ the 2×2 zero matrix. Diagonalising $\boldsymbol{\chi}_0^{+-}(\mathbf{q},0)$ (which is symmetric due to lattice exchange symmetry) we can express the above equation in terms of the diagonal matrix containing its eigenvalues.

Consider the largest eigenvalue $\lambda_\delta(\mathbf{k},\omega)$ of the matrix $[\chi_0^{+-}]$ at doping δ. We work in momentum-frequency space. The phase is unstable if

$$1 - U\lambda_\delta \neq 0\,, \tag{D.8}$$

because, as we said, in that case we do not have a pole for the transverse spin-spin response function. Singh and Tešanović [27] found

$$\lambda_0(\mathbf{k}\to\boldsymbol{\pi},0) = \frac{1}{U}\,, \qquad \text{(half filling)} \tag{D.9}$$

$$\lambda_\delta(\mathbf{k}\to\boldsymbol{\pi},0) = \frac{1}{U} + \frac{2\epsilon_\mathrm{F}}{\Delta}N(\epsilon_\mathrm{F})\,. \qquad \text{(finite doping)} \tag{D.10}$$

The reason for the distinction above is the intrinsic metallic nature of the system for infinitesimal doping. As we discuss at the end of Sec. 5.2.2, the half-filled AF phase is insulating, because the chemical potential lies within the gap. For an insulator Fermi energy is not defined. Instead, at finite filling the phase is metallic because the chemical potential crosses the upper band.

From Eq. (D.8) and (D.9), we see that the half filled antiferromagnet is stable towards transverse spin fluctuations, because χ^{+-} develops a divergence at zero frequency. At any finite doping, since we are describing a metal, a Fermi surface exists and thus $N(\epsilon_\mathrm{F})$ is finite. This makes the phase unstable, and this is the reason we do not discuss the AF phase outside of half-filling.

As we discussed at the beginning of Chap. 5, antiferromagnetism can establish in many ways. For instance, striped antiferromagnets are of particular interest [29]. In order to investigate the most stable physical realisation of an antiferromagnet on the conventional HM, on a pure HF level, we should implement a variational search for the minimum of free energy while varying the AF wavevector $\mathbf{k}$. Notice that at incommensurate wavevectors, $\mathbf{k}\neq\boldsymbol{\pi}$, the unit cell is not as simple as a pair of sites, but can become much larger. Hence, the HF analytical solution would be more complicated than our pseudospin representation of the HM hamiltonian in the commensurate AF channel.

D.4 iHF results

For the conventional HM, antiferromagnetism is described by a single HFP, the magnetisation m. Its self-consistency Eq. (5.52) reads:

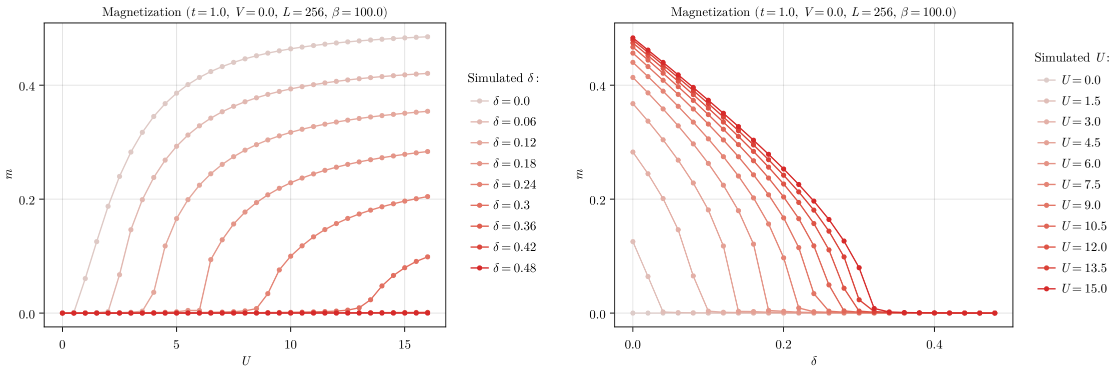
$$m_{i+1} = \frac{U}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{m_i}{2E_\mathbf{k}[\Delta_i]}\left[f(-E_\mathbf{k}[m_i];\beta,\mu) - f(E_\mathbf{k}[m_i];\beta,\mu)\,,\right] \tag{D.11}$$

where we indicated as m_i the value of magnetisation at step i. We ran iHF simulations in the ranges:

$$0\le U\le 16\,,\qquad 0\le\delta\le 0.48\,.$$

We considered also $\delta>0$, although – as we specified in previous section – a doped commensurate antiferromagnet is actually unstable.

In Fig. D.1 magnetisation m at $\beta=100$ is plotted. As is visible in Fig. D.1a, ignoring instability issues, doping disrupts AF ordering; the horizontal asymptote lowers as δ gets bigger. Also, for $\delta>0$, a finite magnetisation m exists only for $U\ge U_\mathrm{c}[\delta]$ (a critical value parametrically dependent on doping). In Fig. D.1b, we show the dependence of the magnetisation on doping varying the local repulsion U. As U grows, the maximum value of δ at which a finite magnetisation is developed gets larger.



a | Local repulsion U/t depencence. b | Doping $\delta = n - 0.5$ dependence.

Figure D.1 | Plots of the zero-temperature AF instability order parameter m as a function of both the local repulsion U/t (Fig. D.1a) and the doping $\delta = n - 1/2$ (Fig. D.1b).

D.5 Analytic expression of self-consistent magnetisation at half-filling

In this subsection, we provide an analytic expression to compute m self-consistently at half-filling. Let us multiply Eq. (5.52) on both sides by U. Since at half filling $\mu = 0$, we have the finite-temperature self-consistent equation

$$\Delta = \frac{U}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{MBZ}} \frac{\Delta}{\sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}} \tanh\left(\frac{\beta}{2}\sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}\right),$$

where we used the relation

$$\mathrm{Th}_{\mathbf{k}}^{(-)}[\beta, 0] = \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right).$$

In thermodynamic limit, $L \to \infty$,

$$\frac{1}{U} = \frac{1}{2} \int_{\mathrm{BZ}} \frac{\mathrm{d}\mathbf{k}}{(2\pi)^2} \frac{1}{\sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}} \tanh\left(\frac{\beta}{2}\sqrt{\epsilon_{\mathbf{k}}^2 + \Delta^2}\right).$$

We substitute the momentum integral with an integral over energies:

$$\frac{2}{U} = \int_{-4t}^{4t} \frac{\mathrm{d}\epsilon \rho(\epsilon)}{\sqrt{\epsilon^2 + \Delta^2}} \tanh\left(\frac{\beta}{2}\sqrt{\epsilon^2 + \Delta^2}\right),$$

being $\rho(\epsilon)$ the 2D square lattice DoS. Its explicit form [41] is given by

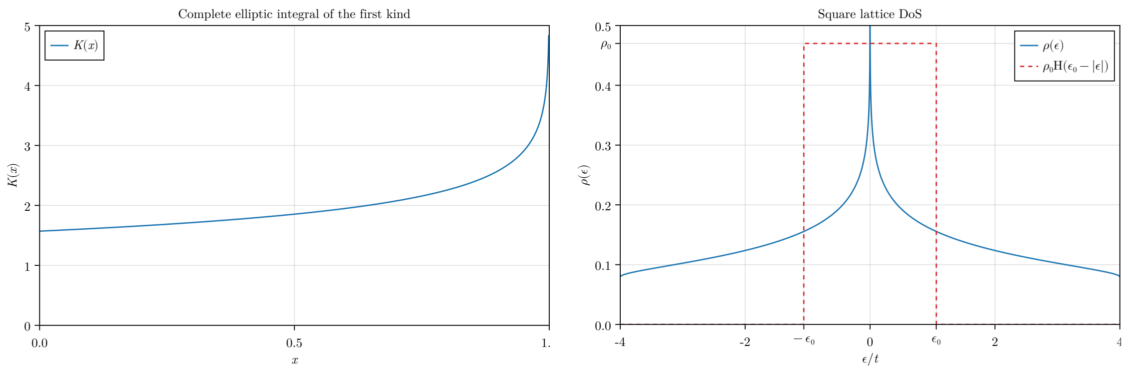
$$\rho(\epsilon) = \frac{1}{2\pi^2} K\left(\sqrt{1 - \left(\frac{\epsilon}{4t}\right)^2}\right) \qquad K(x) \equiv \int_0^{\pi/2} \frac{\mathrm{d}\theta}{\sqrt{1 - x^2 \sin^2\theta}},$$

where $K(x)$ is the complete elliptic integral of the first kind, shown in Fig. D.2a. As is plotted in Fig. D.2b, the DoS $\rho(\epsilon)$ is logarithmically peaked at $\epsilon = 0$, on the MBZ. A common approximation is to substitute the actual DoS with a step function,

$$\rho(\epsilon) \simeq \rho_0 \mathrm{H}\left(\epsilon_0 - |\epsilon|\right) \quad \text{with} \quad 2\rho_0\epsilon_0 = 1.$$

Here, the peak parameters ϵ_0 and ρ_0 are chosen in order to maintain a correct DoS integral, considering also the 2 pre-factor of spin degeneracy. Within this extremely rough approximation, we have

$$\frac{2}{U} \simeq \rho_0 \int_{-\epsilon_0}^{\epsilon_0} \frac{\mathrm{d}\epsilon}{\sqrt{\epsilon^2 + \Delta^2}} \tanh\left(\frac{\beta}{2}\sqrt{\epsilon^2 + \Delta^2}\right).$$



a | Plot of $K(x)$.

b | DoS and its approximation.

Figure D.2 | Plots of the complete elliptic integral of the first kind $K(x)$ and the density of states for the planar square lattice.

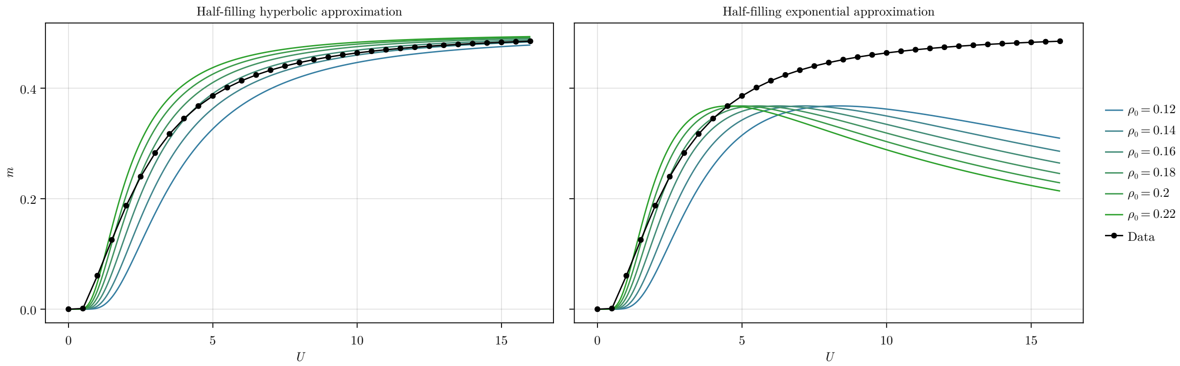


Figure D.3 | Sketch of the analytic expression for m. The left panel shows a family of coloured curves described by Eq. (D.12). The right panel shows a family of coloured curves described by Eq. (D.13), only reasonable at $U \to 0$.

Suppose the system is magnetised, $\Delta > 0$. At low temperature, $\beta \gg 1$, as a first approximation we can substitute the hyperbolic tangent with 1, obtaining

$$\begin{aligned}
\frac{2}{\rho_0 U} &\simeq \int_{-\epsilon_0}^{\epsilon_0} \frac{\mathrm{d}\epsilon}{\sqrt{\epsilon^2 + \Delta^2}} \\
&= \int_{-\epsilon_0/\Delta}^{\epsilon_0/\Delta} \frac{\mathrm{d}x}{\sqrt{1 + x^2}} \\
&= \sinh^{-1}\left(\frac{\epsilon_0}{\Delta}\right) - \sinh^{-1}\left(-\frac{\epsilon_0}{\Delta}\right) \\
&= 2 \sinh^{-1}\left(\frac{\epsilon_0}{\Delta}\right) ,
\end{aligned}$$

thus

$$m \simeq \frac{\epsilon_0}{U \sinh(1/\rho_0 U)} , \tag{D.12}$$

having we used $\Delta = mU$. In the left panel of Fig. D.3 we sketch this function of U varying ρ_0, ϵ_0. The same data of Fig. D.1a, $\delta = 0$ are plotted as black dots. The shown curves are not fits.

At large U, we have

$$\lim_{U \to +\infty} m \simeq \epsilon_0 \rho_0 = \frac{1}{2} ,$$

which gives the correct saturation limit at high local repulsion. In the opposite limit,

$$\begin{aligned}\lim_{U\to 0} m &\simeq \lim_{U\to 0} \frac{2\epsilon_0/U}{\exp(1/\rho_0 U) - \exp(-1/\rho_0 U)} \\ &\simeq \lim_{U\to 0} \frac{2\epsilon_0}{U} \exp\left(-\frac{1}{\rho_0 U}\right),\end{aligned} \tag{D.13}$$

giving a near-0 exponential suppression of magnetisation, typical of MFT self-consistent solutions. In the right panel of Fig. D.3 we sketch this approximation.

Appendix E

The HF hamiltonian in the AF phase

In this appendix, we generalise the analytic expression of the HF hamiltonian for the AF phase to the EHM. Namely, we start from the expression of for the conventional HM of Eq. (5.32), and show how, renormalising all quantities, we can obtain a formally identical expression, Eq. (5.63).

We recall the definitions of Eqs. (5.8), (5.9),

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \,, \tag{E.1}$$

$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{k}\sigma} \,, \tag{E.2}$$

being $\boldsymbol{\pi} = (\pi, \pi)$ the AF vector of Eq. (5.1). We define their EVs as in Eqs. (5.10), (5.11),

$$u_{\mathbf{k}\sigma} \equiv \langle \hat{n}_{\mathbf{k}\sigma} \rangle \,, \tag{E.3}$$

$$v_{\mathbf{k}\sigma} \equiv i\langle \hat{\psi}_{\mathbf{k}\sigma} \rangle \,. \tag{E.4}$$

The i prefactor is useful in later calculations. Using the functions of Tab. 3.5, we define the harmonics as in Eqs. (5.12), (5.13),

$$u^{(\gamma)}_{\sigma} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} u_{\mathbf{k}\sigma} \,, \tag{E.5}$$

$$v^{(\gamma)}_{\sigma} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} v_{\mathbf{k}\sigma} \,. \tag{E.6}$$

Before starting, it is useful to recall the properties of Eqs.(5.22), (5.23) and (5.24):

$$\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}}\hat{\psi}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma} = -\varphi^{(\gamma)}_{\mathbf{k}}\hat{\psi}_{\mathbf{k}\sigma} \,, \tag{E.7}$$

$$\hat{\psi}^{\dagger}_{\mathbf{k}+\boldsymbol{\pi}\sigma} = \hat{\psi}_{\mathbf{k}\sigma} \,, \tag{E.8}$$

$$\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}} = -\varphi^{(\gamma)}_{\mathbf{k}} \,. \tag{E.9}$$

E.1 Generalisation to the EHM

Recall the results of Sec. 5.2. We now include the non-local interaction $\hat{H}_V$. We anticipate here that we are going to obtain the same HF hamiltonian of Eq. (5.26) with renormalised chemical potential, bands and gap,

$$\mu \to \tilde{\mu} \,, \qquad \epsilon_{\mathbf{k}\sigma} \to \tilde{\epsilon}_{\mathbf{k}\sigma} \,, \qquad \Delta_{\mathbf{k}\sigma} \to \tilde{\Delta}_{\mathbf{k}\sigma} \,.$$

The explicit expressions for these renormalised quantities will be given, respectively, in Eqs. (E.24), (E.38) and (E.55). Due to this renormalisation of parameters, the diagonalisation procedure of Sec. 5.2.2 can be generalised.

In diagonal form, the bands $\pm\tilde{E}_{\mathbf{k}\sigma}$ read

$$\tilde{E}_{\mathbf{k}\sigma} \equiv \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + |\tilde{\Delta}_{\mathbf{k}\sigma}|^2} \,.$$

We allow $\tilde{\Delta}_{\mathbf{k}\sigma}$ to be a complex function of $\mathbf{k}$ and σ,

$$\tilde{\Delta}_{\mathbf{k}\sigma} = \mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} + i\,\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}\,.$$

We define the renormalised angle $\tilde{\theta}_{\mathbf{k}\sigma}$, $\tilde{\zeta}_{\mathbf{k}\sigma}$ through the relations:

$$\sin 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{|\tilde{\Delta}_{\mathbf{k}\sigma}|}{\tilde{E}_{\mathbf{k}\sigma}}\,, \qquad \cos 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}}\,, \tag{E.10}$$

and

$$\sin 2\tilde{\zeta}_{\mathbf{k}\sigma} = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|}\,, \qquad \cos 2\tilde{\zeta}_{\mathbf{k}\sigma} = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|}\,. \tag{E.11}$$

A sketch of these angles and the generalised diagonalisation procedure is given in Fig. E.1. For a complex gap function, the new diagonalisation matrix is given by:

$$\begin{aligned}
\tilde{W}_{\mathbf{k}\sigma} &\equiv e^{-i\tilde{\theta}_{\mathbf{k}\sigma}\tau^y} e^{-i\tilde{\zeta}_{\mathbf{k}\sigma}\tau^z} \\
&= \left(\cos\tilde{\theta}_{\mathbf{k}\sigma} - i\sin\tilde{\theta}_{\mathbf{k}\sigma}\tau^y\right)\left(\cos\tilde{\zeta}_{\mathbf{k}\sigma} - i\sin\tilde{\zeta}_{\mathbf{k}\sigma}\tau^z\right) \\
&= \begin{bmatrix} \cos\tilde{\theta}_{\mathbf{k}\sigma} & -\sin\tilde{\theta}_{\mathbf{k}\sigma} \\ \sin\tilde{\theta}_{\mathbf{k}\sigma} & \cos\tilde{\theta}_{\mathbf{k}\sigma} \end{bmatrix} \begin{bmatrix} e^{-i\tilde{\zeta}_{\mathbf{k}\sigma}} & \\ & e^{i\tilde{\zeta}_{\mathbf{k}\sigma}} \end{bmatrix} \\
&= \begin{bmatrix} \cos\tilde{\theta}_{\mathbf{k}\sigma} e^{-i\tilde{\zeta}_{\mathbf{k}\sigma}} & -\sin\tilde{\theta}_{\mathbf{k}\sigma} e^{i\tilde{\zeta}_{\mathbf{k}\sigma}} \\ \sin\tilde{\theta}_{\mathbf{k}\sigma} e^{-i\tilde{\zeta}_{\mathbf{k}\sigma}} & \cos\tilde{\theta}_{\mathbf{k}\sigma} e^{i\tilde{\zeta}_{\mathbf{k}\sigma}} \end{bmatrix}\,,
\end{aligned}$$

and "updates" the diagonalisation matrix of Eq. (5.44).

Pseudospin EVs. The following relations hold (*tilde* sign indicates the renormalised quantity):

$$\begin{aligned}
\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^x \hat{\Psi}_{\mathbf{k}\sigma}\rangle &= -\sin 2\tilde{\theta}_{\mathbf{k}\sigma} \cos 2\tilde{\zeta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma}\rangle\,, \\
\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^y \hat{\Psi}_{\mathbf{k}\sigma}\rangle &= -\sin 2\tilde{\theta}_{\mathbf{k}\sigma} \sin 2\tilde{\zeta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma}\rangle\,, \\
\langle \hat{\Psi}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Psi}_{\mathbf{k}\sigma}\rangle &= \cos 2\tilde{\theta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma}\rangle\,.
\end{aligned}$$

Let us extend Eq. (5.47) and define the renormalised thermal factors,

$$\tilde{\mathrm{Th}}^{(\pm)}_{\mathbf{k}\sigma}[\beta, \tilde{\mu}] \equiv f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \pm f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right)\,, \tag{E.12}$$

which give, in the tilted frame, the z component of the pseudospin

$$\begin{aligned}
\langle \hat{\Gamma}^\dagger_{\mathbf{k}\sigma} \tau^z \hat{\Gamma}_{\mathbf{k}\sigma}\rangle &= f\left(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) - f\left(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}\right) \\
&= -\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\sigma}[\beta, \tilde{\mu}]\,.
\end{aligned} \tag{E.13}$$

At half-filling, we have:

$$\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}\sigma}[\beta, 0] = 1\,, \tag{E.14}$$

$$\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\sigma}[\beta, 0] = \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}\sigma}}{2}\right)\,. \tag{E.15}$$

We will carry out computations at generic density, using these expression only in the last part.

Proof E.1 — Proof of Eqs. (E.14), (E.15).

At half-filling, $\mu = 0$. For a null chemical potential, the Fermi-Dirac distribution is given by

$$f(\epsilon; \beta, 0) = \frac{1}{e^{\beta\epsilon} + 1}\,.$$

Therefore, the following identity holds:

$$\begin{aligned} f(-\epsilon;\beta,0) &= \frac{1}{e^{-\beta\epsilon}+1} \\ &= \frac{e^{\beta\epsilon}}{1+e^{\beta\epsilon}} \\ &= 1-\frac{1}{e^{\beta\epsilon}+1} \\ &= 1-f(\epsilon;\beta,0)\,. \end{aligned}$$

Hence,

$$f(-\tilde{E}_{\mathbf{k}\sigma};\beta,0)+f(\tilde{E}_{\mathbf{k}\sigma};\beta,0)=1$$

which proves Eq. (E.14).

Moreover, we have:

$$\begin{aligned} f(-\epsilon;\beta,0)-f(\epsilon;\beta,0) &= \frac{1}{e^{-\beta\epsilon}+1}-\frac{1}{e^{\beta\epsilon}+1} \\ &= \frac{e^{\beta\epsilon}}{e^{\beta\epsilon}+1}-\frac{1}{e^{\beta\epsilon}+1} \\ &= \frac{e^{\beta\epsilon/2}-e^{-\beta\epsilon/2}}{e^{\beta\epsilon/2}+e^{-\beta\epsilon/2}}\,. \end{aligned}$$

Therefore,

$$f(-\tilde{E}_{\mathbf{k}\sigma};\beta,0)-f(\tilde{E}_{\mathbf{k}\sigma};\beta,0)=\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}\sigma}}{2}\right)$$

which proves Eq. (E.15).

Pseudospin EVs in the three directions are:

$$\langle\hat{s}^x_{\mathbf{k}\sigma}\rangle = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{\tilde{E}_{\mathbf{k}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\sigma}[\beta,\tilde{\mu}]\,, \tag{E.16}$$

$$\langle\hat{s}^y_{\mathbf{k}\sigma}\rangle = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{\tilde{E}_{\mathbf{k}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\sigma}[\beta,\tilde{\mu}]\,, \tag{E.17}$$

$$\langle\hat{s}^z_{\mathbf{k}\sigma}\rangle = -\frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\sigma}[\beta,\tilde{\mu}]\,. \tag{E.18}$$

These equations are an extension of Eqs. (5.49), (5.50), (5.51), obtained for the conventional HM.

Magnetisation and density self-consistency equations. By renormalising all quantities, the self-consistency equation for magnetisation must stay formally the same of Eq. (5.52),

$$m=\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\tilde{\Delta}_{\mathbf{k}\uparrow}}{\tilde{E}_{\mathbf{k}\uparrow}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\uparrow}[\beta,\tilde{\mu}]\,. \tag{E.19}$$

We will be able to prove this statement rigorously in Sec. E.2. Note that, using Eq. (E.15), the half-filling version of Eq. (E.19) reads

$$m=\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\frac{\tilde{\Delta}_{\mathbf{k}\uparrow}}{\tilde{E}_{\mathbf{k}\uparrow}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}\uparrow}}{2}\right). \qquad \text{(half-filling)} \tag{E.20}$$

For the density n an identical argument holds, and Eq. (5.53) is extended to

$$n=\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}\uparrow}[\beta,\tilde{\mu}]\,. \tag{E.21}$$

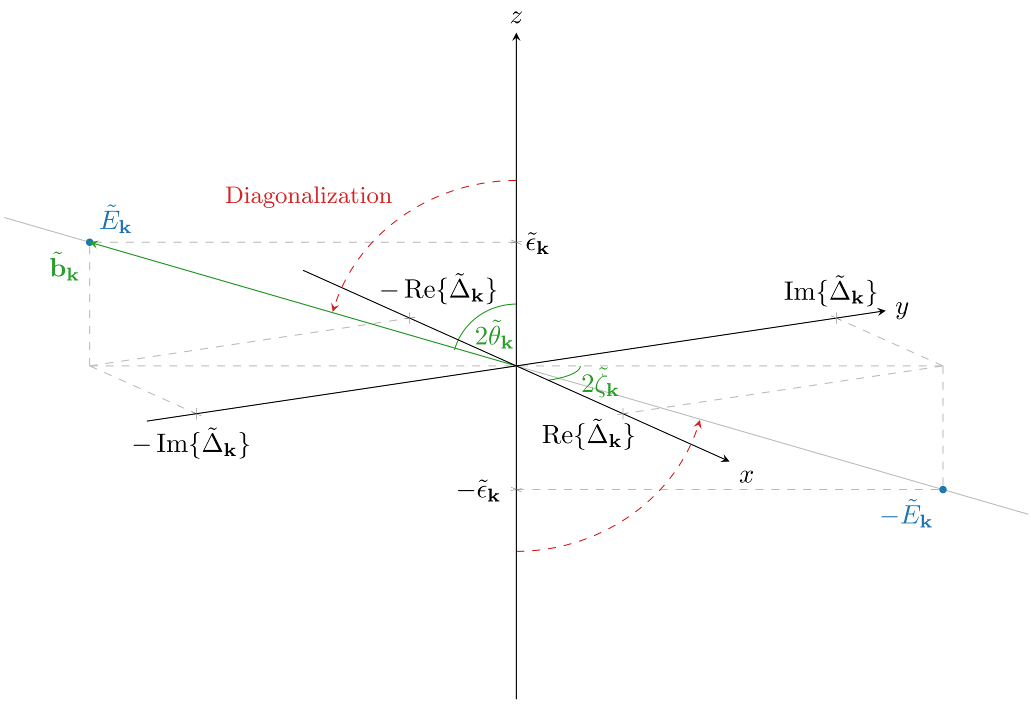


Figure E.1 | Sketch of the diagonalisation procedure for the EHM antiferromagnet. To diagonalise the HF hamiltonian, we align the z axis with the pseudo-field. This is an extension of Fig. 5.2.

E.1.1 Effects of the non-local Hartree terms: chemical potential renormalisation

The approximation to $\hat{H}_V$ in the non-local Hartree channel is given by

$$\hat{H}_V \overset{\text{HF}}{\simeq} -2znV \times \hat{N} - E^{(\text{AF})}_{\text{H},V} + (\text{Fock channel})\,, \tag{E.22}$$

with the contraction energy

$$E^{(\text{AF})}_{\text{H},V} \equiv V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma,\sigma'} \langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle\langle \hat{c}^\dagger_{\mathbf{r}'\sigma'}\hat{c}_{\mathbf{r}'\sigma'}\rangle\,. \tag{E.23}$$

From Eq. (E.22), we see that the Hartree channel of the non-local attraction contributes to the chemical potential shift. Combining Eq. (5.26) with Eq. (E.22), the full shift to chemical potential reads

$$\tilde{\mu} \equiv \mu + n(2zV - U)\,, \tag{E.24}$$

which is identical to what we found for the normal phase, in Eq. (4.8). Eq. (E.24) coincides with Eq. (5.60).

Proof E.2 — Proof of Eq. (E.22).

In Eq. (4.11) we separated the Hartree channel approximation to $\hat{H}_V$ in s.s. and o.s. channels. The same equation is valid in this context. Using the definition of $E^{(\text{AF})}_{\text{H},V}$ given in Eq. (E.23), we obtain

$$\begin{aligned}\hat{H}_V \overset{\text{HF}}{\simeq} &-V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \left(\langle \hat{n}_{\mathbf{r}\sigma}\rangle \hat{n}_{\mathbf{r}'\sigma} + \hat{n}_{\mathbf{r}\sigma}\langle \hat{n}_{\mathbf{r}'\sigma}\rangle\right) \\ &- V \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \left(\langle \hat{n}_{\mathbf{r}\sigma}\rangle \hat{n}_{\mathbf{r}'\overline{\sigma}} + \hat{n}_{\mathbf{r}\sigma}\langle \hat{n}_{\mathbf{r}'\overline{\sigma}}\rangle\right) - E^{(\text{AF})}_{\text{H},V} + (\text{Fock channel})\,.\end{aligned} \tag{E.25}$$

Inserting the Ansatz of Eq. (5.2), we obtain

$$\hat{H}_V \overset{\mathrm{HF}}{\simeq} \overbrace{-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(\hat{n}_{\mathbf{r}'\sigma}+\hat{n}_{\mathbf{r}\sigma}) + mV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{\delta_{\sigma=\uparrow}}\left((-1)^{x+y}\hat{n}_{\mathbf{r}'\sigma}+(-1)^{x'+y'}\hat{n}_{\mathbf{r}\sigma}\right)}^{\text{s.s.}}$$
$$\underbrace{-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}[\hat{n}_{\mathbf{r}'\bar{\sigma}}+\hat{n}_{\mathbf{r}\sigma}] + mV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left((-1)^{x+y+\delta_{\sigma=\uparrow}}\hat{n}_{\mathbf{r}'\bar{\sigma}}+(-1)^{x'+y'+\delta_{\bar{\sigma}=\uparrow}}\hat{n}_{\mathbf{r}\sigma}\right)}_{\text{o.s.}}$$
$$- E^{(\mathrm{AF})}_{\mathrm{H},V} + (\text{Fock channel})\,. \qquad (\mathrm{E}.26)$$

For a square lattice, if $\mathbf{r}=(x,y)$ and $\mathbf{r}'=(x',y')$ are NNs, both the following identities are true

$$(-1)^{x'+y'} = (-1)^{x+y+1}\,,$$
$$(-1)^{\delta_{\bar{\sigma}=\uparrow}} = (-1)^{\delta_{\sigma=\uparrow}+1}\,.$$

Therefore, Eq. (E.26) becomes

$$\hat{H}_V \overset{\mathrm{HF}}{\simeq} \overbrace{-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\sigma})}^{\text{s.s. (density)}} \overbrace{- mV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}(\hat{n}_{\mathbf{r}\sigma}-\hat{n}_{\mathbf{r}'\sigma})}^{\text{s.s. (magnetisation)}}$$
$$\underbrace{-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\bar{\sigma}})}_{\text{o.s. (density)}} \underbrace{+ mV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\bar{\sigma}})}_{\text{o.s. (magnetisation)}}$$
$$- E^{(\mathrm{AF})}_{\mathrm{H},V} + (\text{Fock channel})\,. \qquad (\mathrm{E}.27)$$

In the above expressions we separated in “density” and “magnetisation” terms. Let us deal with them separately.

Density terms. In the sum of s.s. and o.s. density terms of Eq. (E.27) we recognise the operator of Eq. (4.12), obtained for the normal phase. Hence we obtain the same result of Eq. (4.13),

$$-nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\sigma}) - nV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\bar{\sigma}}) = -2znV\times\hat{N}\,. \qquad (\mathrm{E}.28)$$

Magnetisation terms. The magnetisation s.s. and o.s. terms of Eq. (E.27) perfectly cancel out:

$$mV\sum_{\langle \mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}(-1)^{x+y+\delta_{\sigma=\uparrow}}\left[(\hat{n}_{\mathbf{r}\sigma}-\hat{n}_{\mathbf{r}'\sigma})-(\hat{n}_{\mathbf{r}\sigma}+\hat{n}_{\mathbf{r}'\bar{\sigma}})\right]=0\,.$$

Therefore the only contribution to the HF approximation of $\hat{H}_V$, in the Hartree channel, comes from Eq. (E.28). Inserting the latter in Eq. (E.27), Eq. (E.22) is proven.

The contraction energy of Eq. (E.23) reduces to

$$E^{(\mathrm{AF})}_{\mathrm{H},V} = -L_xL_y\times 2zn^2V\,, \qquad (\mathrm{E}.29)$$

which coincides with that for the normal phase, given in Eq. (4.15). This is coherent with the fact that the shift to chemical potential is identical.

Proof E.3 — Proof of Eq. (E.29): non-local Hartree contraction energy.

Separate Eq. (E.23) in s.s. and o.s. sectors,

$$E^{(\mathrm{AF})}_{\mathrm{H},V} = V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \Big(\overbrace{\langle \hat{c}^{\dagger}_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}\sigma} \rangle \langle \hat{c}^{\dagger}_{\mathbf{r}'\sigma} \hat{c}_{\mathbf{r}'\sigma} \rangle}^{\text{s.s.}} + \overbrace{\langle \hat{c}^{\dagger}_{\mathbf{r}\sigma} \hat{c}_{\mathbf{r}\sigma} \rangle \langle \hat{c}^{\dagger}_{\mathbf{r}'\overline{\sigma}} \hat{c}_{\mathbf{r}'\overline{\sigma}} \rangle}^{\text{o.s.}} \Big) . \tag{E.30}$$

s.s. sector contraction energy. The s.s. sector of Eq. (E.30) reduces to

$$\begin{aligned} V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\sigma} \rangle = & \overbrace{V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \Big[(n + (-1)^{x+y} m)(n + (-1)^{x'+y'} m) \Big]}^{\sigma=\uparrow} \\ & + \underbrace{V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \Big[(n - (-1)^{x+y} m)(n - (-1)^{x'+y'} m) \Big]}_{\sigma=\downarrow} . \end{aligned}$$

The mixed terms (those of order $\mathcal{O}(n) \times \mathcal{O}(m)$) cancel out, and since sign factors necessarily alternate on neighbouring sites, the remainder is just

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\sigma} \rangle = V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} 2(n^2 - m^2) . \tag{E.31}$$

o.s. sector contraction energy. Proceeding identically on the o.s. sector of Eq. (E.30), we get

$$\begin{aligned} V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\overline{\sigma}} \rangle = & \overbrace{V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \Big[(n + (-1)^{x+y} m)(n - (-1)^{x'+y'} m) \Big]}^{\sigma=\uparrow} \\ & + \underbrace{V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \Big[(n - (-1)^{x+y} m)(n + (-1)^{x'+y'} m) \Big]}_{\sigma=\downarrow} , \end{aligned}$$

which gives

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\overline{\sigma}} \rangle = V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} 2(n^2 + m^2) . \tag{E.32}$$

Then, summing Eqs. (E.31) and (E.32), we get

$$\begin{aligned} E^{(\mathrm{AF})}_{\mathrm{H},V} &= -2V \sum_{\langle ij \rangle} \sum_{\sigma} n^2 \\ &= -4n^2 V \times \frac{z}{2} L_x L_y , \end{aligned}$$

where we used Eq. (3.22). The above result coincides with Eq. (E.29).

E.1.2 Effects of the Fock terms: gap and bare bands renormalisation

In this subsection, we discuss the Fock contributions to the HF hamiltonian in the AF phase. We start by Eq. (4.17), derived in the context of the normal phase,

$$\begin{aligned} \hat{H}_V \overset{\mathrm{HF}}{\simeq} & \text{ (Hartree channel)} \\ & + \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos(\delta k_x) + \cos(\delta k_y) \right] \langle \hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} - E^{(\mathrm{AF})}_{\mathrm{F},V} , \end{aligned} \tag{E.33}$$

where $E^{(\mathrm{AF})}_{\mathrm{F},V}$, see Eq. (4.18), reads

$$E^{(\mathrm{AF})}_{\mathrm{F},V} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \sum_{\sigma} \left[\cos\left(\delta k_x\right) + \cos\left(\delta k_y\right)\right] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \langle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle \,, \tag{E.34}$$

As we specified in Sec. 5.1.2, the commensurate AF phase is described by two EVs being in non-zero in reciprocal space,

$$\left|\langle \hat{n}_{\mathbf{k}\sigma} \rangle\right| \geq 0 \quad \text{and} \quad \left|\langle \hat{\psi}_{\mathbf{k}\sigma} \rangle\right| \geq 0 \,,$$

This implies only two contributions in Eq. (E.33) are non-zero:

$$\mathbf{k} = -\mathbf{k}' \quad \text{and} \quad \mathbf{k} + \boldsymbol{\pi} = -\mathbf{k}' \,.$$

Then Eq. (E.33) becomes

$$\begin{aligned} \hat{H}_V \overset{\mathrm{HF}}{\simeq} (\text{Hartree channel}) + \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} \left[\cos\left(2k_x\right) + \cos\left(2k_y\right)\right] & \\ \times \Big(\underbrace{\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}}_{\text{Diagonal terms}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}}_{\text{Off-diagonal terms}} \Big) - E^{(\mathrm{AF})}_{\mathrm{F},V} \,, & \end{aligned} \tag{E.35}$$

where we separated the hamiltonian in *diagonal* and *off-diagonal* terms.

Diagonal terms. The diagonal terms of Eq. (E.35) reduce to

$$\begin{aligned} &\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} \left[\cos\left(2k_x\right) + \cos\left(2k_y\right)\right] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\ &\qquad = V \sum_{\mathbf{k},\sigma} \left(u^{(s^*)}_{\sigma} (\cos q_x + \cos q_y) + u^{(d)}_{\sigma} (\cos q_x - \cos q_y) \right) \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \,, \end{aligned} \tag{E.36}$$

where we used Eq. (E.5), where the harmonics of $u_{\mathbf{k}\sigma}$ were defined.

Proof E.4 — Proof of Eq. (E.36).

We define new variables as in Fig. 4.1, $\mathbf{q} \equiv \mathbf{K}+\mathbf{k}$, $\mathbf{q}' \equiv \mathbf{K}-\mathbf{k}$. Applying this change of variables to the diagonal terms of Eq. (E.35), we obtain

$$\begin{aligned} &\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k}} \sum_{\sigma} \left[\cos\left(2k_x\right) + \cos\left(2k_y\right)\right] \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\ &\qquad = \frac{2V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_{\sigma} \left[\cos\left(\delta q_x\right) + \cos\left(\delta q_y\right)\right] \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} \\ &\qquad = \sum_{\gamma \neq s} \sum_{\sigma} \left(\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{n}_{\mathbf{q}\sigma} \rangle \right) \times V \sum_{\mathbf{q}' \in \mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{n}_{\mathbf{q}'\sigma} \\ &\qquad = \sum_{\gamma \neq s} \sum_{\sigma} u^{(\gamma)}_{\sigma} \times V \sum_{\mathbf{q}' \in \mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{n}_{\mathbf{q}'\sigma} \,. \end{aligned} \tag{E.37}$$

We used the decomposition of Eq. (3.52). In the last line, we recognised the harmonics given in Eq. (E.5).

We can re-use the argument that, for the normal phase, led us to Eq. (4.24). We are not breaking inversion symmetry (and as a consequence, neither C_2): this implies that, whatever the shape of the renormalised bands, these are symmetric functions of the moment. Therefore,

$$\tilde{\epsilon}_{\mathbf{q}\sigma} = \tilde{\epsilon}_{-\mathbf{q}\sigma} \quad \Longrightarrow \quad u^{(p_\ell)}_{\sigma} = 0 \quad \text{with} \quad \ell \in \{x, y\} \,.$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = -\dfrac{1}{L_x L_y} \sum\limits_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y) \dfrac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta, \tilde{\mu}]$	Fig. 3.4b
$d_{x^2-y^2}$-wave	$u^{(d)} = -\dfrac{1}{L_x L_y} \sum\limits_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x - \cos k_y) \dfrac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta, \tilde{\mu}]$	Fig. 3.4e

Table E.1 | Symmetry and spin resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$. The shown equations are reformulations of Eq. (E.5), which is equal to Eq. (E.39).

Using this selection rule, Eq. (E.37) reduces to Eq. (E.36).

Apart from the explicit form of the self-consistency equations, this discussion is formally the same of the normal phase, carried out in Sec. 4.2.2. Therefore the bands renormalisation, discussed in Sec. 4.3.1, applies also for the AF phase. Specifically, the renormalised bands are

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u^{(s^*)}_{\sigma} V(\cos k_x + \cos k_y) + u^{(d)}_{\sigma} V(\cos k_x - \cos k_y)\,. \tag{E.38}$$

The d-wave harmonic $u^{(d)}$ is zero if C_4 is unbroken, otherwise is finite.

The self-consistency equation for the harmonics of $u_{\mathbf{k}\sigma}$, given in Eq. (E.5), can be written as

$$u^{(\gamma)}_{\sigma} = -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{k}} \frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\sigma}[\beta, \tilde{\mu}]\,. \tag{E.39}$$

The symmetry-resolved self-consistency equations are reported explicitly in Tab. E.1. Since in Eq. (E.39) all quantities are spin-independent, we omit spin index on these HFPs.

Proof E.5 — Proof of Eq. (E.39): self-consistency equations for $u^{(\gamma)}_{\sigma}$.

Recall the definition of $u_{\mathbf{k}\sigma}$,

$$u_{\mathbf{k}\sigma} = \langle \hat{n}_{\mathbf{k}\sigma} \rangle\,,$$

given in Eq. (E.3). We expand the harmonics decomposition of Eq. (E.5),

$$\begin{aligned}
u^{(\gamma)}_{\sigma} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{n}_{\mathbf{k}\sigma} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \Big(\varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{n}_{\mathbf{k}\sigma} \rangle + \varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}} \langle \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \Big) \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle\,.
\end{aligned}$$

In last line we used Eq. (5.24). From Eq. (5.39), we recognise the z component of the pseudospin operator

$$\begin{aligned}
u^{(\gamma)}_{\sigma} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{k}} \langle \hat{s}^z_{\mathbf{k}\sigma} \rangle \\
&= -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \varphi^{(\gamma)}_{\mathbf{k}} \frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}\sigma}[\beta, \tilde{\mu}]\,,
\end{aligned} \tag{E.40}$$

where, in the last line, Eq. (E.18) has been used. This proves Eq. (E.39).

Similarly to the normal phase, also for the AF phase we can define the symmetry-resolved components of the renormalised hopping amplitude,

$$\tilde{t}^{(s^*)}_{\sigma} \equiv t - \frac{u^{(s^*)}_{\sigma} V}{2} \quad \text{and} \quad \tilde{t}^{(d)}_{\sigma} \equiv -\frac{u^{(d)}_{\sigma} V}{2}\,. \tag{E.41}$$

We re-used the definitions of Eq. (4.28).

Off-diagonal terms. The off-diagonal sector of Eq. (E.35) is reduced to

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos{(2k_x)}+\cos{(2k_y)}]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= V\sum_{\sigma}\sum_{\gamma\neq s}\sum_{\mathbf{k}\in\mathrm{MBZ}}\left(iv^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{k}}\hat{\psi}_{\mathbf{k}\sigma}+\text{h.c.}\right)\,, \quad \text{(E.42)}$$

where we used the harmonics of $v_{\mathbf{k}\sigma}$, defined in Eq. (E.6), and the functions of Tab. 3.5.

Proof E.6 — Proof of Eq. (E.42).

We define new variables as in Fig. 4.1, $\mathbf{q}\equiv\mathbf{K}+\mathbf{k}$, $\mathbf{q}'\equiv\mathbf{K}-\mathbf{k}$. Applying this change of variables to the off-diagonal terms of Eq. (E.35) and using the decomposition of Eq. (3.52), we obtain

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos{(2k_x)}+\cos{(2k_y)}]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= -\sum_{\sigma}\sum_{\gamma\neq s}\left(\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{\psi}^\dagger_{\mathbf{q}\sigma}\rangle\right)\times V\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}'}\hat{\psi}^\dagger_{\mathbf{q}'\sigma}\,. \quad \text{(E.43)}$$

We shift the first sum by an AF vector $\boldsymbol{\pi}$ defining $\mathbf{k}+\boldsymbol{\pi}\equiv\mathbf{q}$. Using Eq. (E.7) as well as reciprocal space periodicity by a shift $\boldsymbol{\pi}$,

$$\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{\psi}^\dagger_{\mathbf{q}\sigma}\rangle = \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}+\boldsymbol{\pi}}\langle\hat{\psi}^\dagger_{\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle$$
$$= -\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}}\langle\hat{\psi}_{\mathbf{k}\sigma}\rangle\,.$$

Recall the definition $v_{\mathbf{k}\sigma}=i\langle\hat{\psi}_{\mathbf{k}\sigma}\rangle$ of Eq. (E.4). Inserting the above equation in Eq. (E.43), we obtain

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos{(2k_x)}+\cos{(2k_y)}]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= \sum_{\sigma}\sum_{\gamma\neq s}\left(\frac{1}{L_xL_y}\sum_{\mathbf{q}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{q}}\langle\hat{\psi}_{\mathbf{q}\sigma}\rangle\right)\times V\sum_{\mathbf{q}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{q}'}\hat{\psi}^\dagger_{\mathbf{q}'\sigma}$$
$$= -i\sum_{\sigma}\sum_{\gamma\neq s}v^{(\gamma)}_\sigma\times V\sum_{\mathbf{q}'}\varphi^{(\gamma)*}_{\mathbf{q}'}\hat{n}_{\mathbf{q}'\sigma}\,, \quad \text{(E.44)}$$

where we used Eq. (E.6). We restrict the sum to the MBZ,

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos{(2k_x)}+\cos{(2k_y)}]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= -iV\sum_{\sigma}\sum_{\gamma\neq s}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left(v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}^\dagger_{\mathbf{q}\sigma}+v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}+\boldsymbol{\pi}}\hat{\psi}^\dagger_{\mathbf{q}+\boldsymbol{\pi}\sigma}\right)\,,$$

and use Eq. (E.7), exchanging terms positions

$$-\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}[\cos{(2k_x)}+\cos{(2k_y)}]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}$$
$$= iV\sum_{\sigma}\sum_{\gamma\neq s}\sum_{\mathbf{q}\in\mathrm{MBZ}}\left(v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}_{\mathbf{q}\sigma}-v^{(\gamma)}_\sigma\varphi^{(\gamma)*}_{\mathbf{q}}\hat{\psi}^\dagger_{\mathbf{q}\sigma}\right)\,. \quad \text{(E.45)}$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$v^{(s^*)} = \dfrac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{MBZ}} (\cos k_x + \cos k_y) \dfrac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}_{\mathbf{k}}[\beta, \tilde{\mu}]$	Fig. 3.4b
p_x-wave	$v^{(p_x)} = \dfrac{i\sqrt{2}}{L_x L_y} \sum_{\mathbf{k}\in\text{MBZ}} \sin k_x \dfrac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}_{\mathbf{k}}[\beta, \tilde{\mu}]$	Fig. 3.4c
p_y-wave	$v^{(p_y)} = \dfrac{i\sqrt{2}}{L_x L_y} \sum_{\mathbf{k}\in\text{MBZ}} \sin k_y \dfrac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}_{\mathbf{k}}[\beta, \tilde{\mu}]$	Fig. 3.4d
$d_{x^2-y^2}$-wave	$v^{(d)} = \dfrac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{MBZ}} (\cos k_x - \cos k_y) \dfrac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}_{\mathbf{k}}[\beta, \tilde{\mu}]$	Fig. 3.4e

Table E.2 Symmetry and spin resolved self-consistency equations for the harmonics of $v_{\mathbf{k}}$. The shown equations are reformulations of Eq. (E.6), which is equal to Eq. (E.47).

Recalling Tab. 3.5, for the four symmetries we are considering, the functions $\varphi^{(\gamma)}_{\mathbf{k}}$ are real-valued for $\gamma = s^*, d$, and purely imaginary for $\gamma = p_x, p_y$. We anticipate the result of Eq. (E.48),

$$v^{(s^*)}_\sigma, v^{(d)}_\sigma \in \mathbb{R}\,, \qquad v^{(p_x)}_\sigma, v^{(p_y)}_\sigma \in i\mathbb{R}\,,$$

that will be obtained later. Then

$$iv^{(\gamma)}_\sigma \varphi^{(\gamma)*}_{\mathbf{q}} \in i\mathbb{R}\,, \tag{E.46}$$

and to conjugate a purely imaginary number it suffices to put a minus sign in front of it,

$$-iv^{(\gamma)}_\sigma \varphi^{(\gamma)*}_{\mathbf{q}} = \left(iv^{(\gamma)}_\sigma \varphi^{(\gamma)*}_{\mathbf{q}}\right)^* .$$

Therefore, substituting this identity in Eq. (E.45), we obtain Eq. (E.42).

The HFPs $v^{(\gamma)}_\sigma$ obey the self-consistency equations

$$\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\text{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\text{Th}}^{(-)}_{\mathbf{q}\sigma}[\beta, \tilde{\mu}]\,. \tag{E.47}$$

These equation are reported, symmetry-resolved, in Tab. E.2. By direct inspection of the equations thereby reported, we see that

$$v^{(s^*)}_\sigma, v^{(d)}_\sigma \in \mathbb{R}\,, \qquad v^{(p_x)}_\sigma, v^{(p_y)}_\sigma \in i\mathbb{R}\,. \tag{E.48}$$

We have used this result before, in Eq. (E.46).

In Tab. E.2 the four derived self-consistency equations are reported, ignoring spin index. Indeed, all quantities in Eq. (E.47) are independent of σ.

Proof E.7 — Proof of Eq. (E.47): self-consistency equations for $v^{(\gamma)}_\sigma$.

We expand Eq. (E.6),

$$\begin{aligned} v^{(\gamma)}_\sigma &= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle\hat{\psi}_{\mathbf{q}\sigma}\rangle \\ &= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\text{MBZ}} \left(\varphi^{(\gamma)}_{\mathbf{q}} \langle\hat{\psi}_{\mathbf{q}\sigma}\rangle + \varphi^{(\gamma)}_{\mathbf{q}+\boldsymbol{\pi}} \langle\hat{\psi}_{\mathbf{q}+\boldsymbol{\pi}\sigma}\rangle\right) \\ &= \frac{i}{L_x L_y} \sum_{\mathbf{q}\in\text{MBZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle\hat{\psi}_{\mathbf{q}\sigma} - \hat{\psi}^\dagger_{\mathbf{q}\sigma}\rangle\,. \end{aligned}$$

In last line we used Eq. (5.24). From Eq. (5.38), we recognise the y component of the pseudospin operator

$$
\begin{aligned}
v_\sigma^{(\gamma)} &= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\Psi}_{\mathbf{q}\sigma}^\dagger \tau^y \hat{\Psi}_{\mathbf{q}\sigma} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\mathrm{Th}}_{\mathbf{q}\sigma}^{(-)}[\beta, \tilde{\mu}] \,,
\end{aligned} \tag{E.49}
$$

where we used Eq. (E.17). Last line coincides with Eq. (E.47).

The contraction energy of Eq. (E.34) reduces to

$$
E_{\mathrm{F},V}^{(\mathrm{AF})} = L_x L_y \times \frac{V}{2} \sum_\sigma \sum_{\gamma\neq s} \left(\left| u_\sigma^{(\gamma)} \right|^2 + \left| v_\sigma^{(\gamma)} \right|^2 \right) \,, \tag{E.50}
$$

Note that, due to spin independency, we can omit the spin index, remove the sum over σ and the $1/2$ prefactor.

Proof E.8 — Proof of Eq. (E.50): Fock contraction energy.

Consider Eq. (E.34). Using the changing variables of Fig. 4.1, we get

$$
E_{\mathrm{F},V}^{(\mathrm{AF})} = \frac{V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_\sigma [\cos(\delta q_x) + \cos(\delta q_y)] \left(\underbrace{\langle \hat{n}_{\mathbf{q}\sigma} \rangle \langle \hat{n}_{\mathbf{q}'\sigma} \rangle}_{\text{Diagonal terms}} - \underbrace{\langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \langle \hat{\psi}_{\mathbf{q}'\sigma} \rangle}_{\text{Off-diagonal terms}} \right) .
$$

Applying the decomposition of Eq. (3.52) and using Eqs. (E.3), (E.4), we obtain:

$$
E_{\mathrm{F},V}^{(\mathrm{AF})} = \underbrace{\frac{V}{2L_x L_y} \sum_\sigma \sum_{\gamma\neq s} \sum_{\mathbf{q},\mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}\sigma} u_{\mathbf{q}'\sigma}}_{\text{Diagonal terms}} + \underbrace{\frac{V}{2L_x L_y} \sum_\sigma \sum_{\gamma\neq s} \sum_{\mathbf{q},\mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma}}_{\text{Off-diagonal terms}} . \tag{E.51}
$$

Diagonal terms. Let us start from the diagonal part of Eq. (E.51). It holds

$$
u_{\mathbf{q}\sigma} = \langle \hat{n}_{\mathbf{q}\sigma} \rangle \quad \Longrightarrow \quad u_{\mathbf{q}\sigma} = u_{\mathbf{q}\sigma}^* \,,
$$

and then

$$
\begin{aligned}
\frac{1}{(L_x L_y)^2} \sum_{\mathbf{q},\mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}\sigma} u_{\mathbf{q}'\sigma} &= \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma} \right) \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}'\sigma}^* \right) \\
&= \left| u_\sigma^{(\gamma)} \right|^2 .
\end{aligned} \tag{E.52}
$$

Off-diagonal terms. Consider the off-diagonal part of Eq. (E.51). Using Eq. (5.23), we get

$$
v_{\mathbf{q}\sigma} = i \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \quad \Longrightarrow \quad v_{\mathbf{q}+\boldsymbol{\pi}\sigma} = -v_{\mathbf{q}\sigma}^* \,,
$$

Since the form factors satisfy $\varphi_{\mathbf{q}+\boldsymbol{\pi}}^{(\gamma)} = -\varphi_{\mathbf{q}}^{(\gamma)}$, we obtain

$$
\begin{aligned}
\frac{1}{(L_x L_y)^2} \sum_{\mathbf{q},\mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma} &= \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} v_{\mathbf{q}\sigma} \right) \left(\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} v_{\mathbf{q}'\sigma} \right) \\
&= v_\sigma^{(\gamma)} \left(\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}+\boldsymbol{\pi}}^{(\gamma)*} v_{\mathbf{k}+\boldsymbol{\pi}\sigma} \right) ,
\end{aligned}
$$

where in the second passage we employed the change of variables $\mathbf{q}' = \mathbf{k} + \boldsymbol{\pi}$ and used BZ periodicity to map the original sum onto a new sum for $\mathbf{k} \in \mathrm{BZ}$. We conclude that

$$\frac{1}{(L_x L_y)^2} \sum_{\mathbf{q},\mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma} = v_\sigma^{(\gamma)} \left(\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)*} v_{\mathbf{k}\sigma}^* \right)$$
$$= \left| v_\sigma^{(\gamma)} \right|^2 . \tag{E.53}$$

Inserting Eqs. (E.52), (E.53) in Eq. (E.51), we obtain the expression of Eq. (E.50).

We examined all contribution to the HF hamiltonian. In next section we collect all results and describe the features of the HF ground-state.

E.2 The HF hamiltonian and its properties

We recollect all results into a single hamiltonian approximation. Summing Eqs. (5.26), (E.36) and (E.42), we obtain the HF hamiltonian

$$\hat{H} - \mu \hat{N} \overset{\mathrm{HF}}{\simeq} \sum_\sigma \sum_{\mathbf{k}\in\mathrm{MBZ}} \tilde{\epsilon}_{\mathbf{k}\sigma} \left(\hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \right)$$
$$- \sum_\sigma \sum_{\mathbf{k}\in\mathrm{MBZ}} \left(\tilde{\Delta}_{\mathbf{k}\sigma} \hat{\psi}_{\mathbf{k}\sigma} + \tilde{\Delta}_{\mathbf{k}\sigma}^* \hat{\psi}_{\mathbf{k}\sigma}^\dagger \right) - \tilde{\mu} \hat{N} + E_{\mathrm{c}}^{(\mathrm{AF})} , \tag{E.54}$$

which coincides with Eq. (5.63). We defined renormalised bands, see Eq. (E.38),

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} + V \sum_{\gamma\in\{s^*,d\}} u_\sigma^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} ,$$

which proves Eq. (5.61). The renormalised gap function is given by

$$\tilde{\Delta}_{\mathbf{k}\sigma} \equiv \Delta_{\mathbf{k}\sigma} - iV \sum_{\gamma\neq s} v_\sigma^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} , \tag{E.55}$$

which coincides with Eq. (5.62). Finally, the total contraction energy is given by the (negative) sum of all contraction energies, see Eqs. (5.31), (E.29) and (E.50),

$$E_{\mathrm{c}}^{(\mathrm{AF})} \equiv - \left(E_{\mathrm{H},U}^{(\mathrm{AF})} + E_{\mathrm{H},V}^{(\mathrm{AF})} + E_{\mathrm{F},V}^{(\mathrm{AF})} \right)$$
$$= L_x L_y \times U(m^2 - n^2) + L_x L_y \times V \left[2zn^2 - \frac{1}{2} \sum_{\gamma\neq s} \sum_\sigma \left(\left| u_\sigma^{(\gamma)} \right|^2 + \left| v_\sigma^{(\gamma)} \right|^2 \right) \right] , \tag{E.56}$$

which coincides with Eq. (5.64).

E.2.1 Gap complexification and AF flavours

Recalling Eq. (E.46), we observe that the correction to the gap function in Eq. (E.55) is purely imaginary. Instead, the HM gap is purely real, $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU \in \mathbb{R}$. We denote this effect as "gap complexification". A direct consequence of the fact that the correction is purely imaginary is that Eq. (E.19), which expresses the self-consistency condition for m, which we postulated, is proven.

Proof E.9 — Proof of Eq. (E.19): self-consistency equation for m.

We know from Eq. (E.16) that the real part of the gap is connected to the x projection of the pseudospin expectation value. Instead, from Eq. (E.17) we know that the imaginary part is related to the y projection. It follows

$$\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = (-1)^{\delta_{\sigma=\downarrow}} mU \in \mathbb{R}$$

Then, the x projection of the pseudospin gives a spin-antisymmetric EV,

$$\langle \hat{s}^x_{\mathbf{k}\uparrow} \rangle = - \langle \hat{s}^x_{\mathbf{k}\downarrow} \rangle$$

Then Eq. (E.19) holds identically, because the proof that led us to Eq. (5.52) for the HM was based on this property.

In this analysis, antiferromagnetism is realised without breaking inversion symmetry. This translates in the requirement that:

$$\tilde{E}_{\mathbf{k}\sigma} \overset{!}{=} \tilde{E}_{-\mathbf{k}\sigma} \,.$$

We know that:

$$\tilde{E}_{\mathbf{k}\sigma} = \sqrt{\tilde{\epsilon}^2_{\mathbf{k}\sigma} + \mathrm{Re}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\} + \mathrm{Im}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\}} \,.$$

Renormalised bare bands as well as the real part of the gap satisfy said inversion symmetry requirement,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \tilde{\epsilon}_{-\mathbf{k}\sigma} \qquad \mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \mathrm{Re}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\} \,,$$

hence inversion symmetry is respected only if

$$\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \pm\, \mathrm{Im}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\} \,.$$

Either the imaginary part of the gap is inversion symmetric, or is antisymmetric. Based on this consideration, we are able two discard two of the four self-consistency equations for the harmonics $v^{(\gamma)}_\sigma$, listed in Tab. E.2. If $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is an even function of momentum, $v^{(p_x)}_\sigma = v^{(p_y)}_\sigma = 0$. If $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is an odd function of momentum, $v^{(s^*)}_\sigma = v^{(d)}_\sigma = 0$. The two options cannot coexist, and separate simulations can be implemented in the two symmetry sectors.

From now on, we use spin degeneracy and omit the spin index σ. We denote the symmetric flavour of the AF phase, described by the harmonics $v^{(s^*)}$ and $v^{(d)}$, as “sAF”. The antisymmetric flavour, described by $v^{(p_x)}$, $v^{(p_y)}$, we denote “aAF”. We report the set of HFPs for each in Tab. 5.2.

E.2.2 The entropy density of the AF ground-state

In this section, we calculate the entropy density of the AF ground state, following the scheme presented for the normal phase in Sec. 4.3.2. The approximated ground-state in the AF phase is given by a gas of free $\hat{\gamma}^{(\pm)}$ fermions populating the two bands $\pm\tilde{E}_{\mathbf{k}}$ following a Fermi-Dirac distribution. Hence, the total entropy density of the ground state is given by that of a Fermi gas [4]:

$$s^{(\mathrm{AF})} = s^{(\mathrm{AF})}_- + s^{(\mathrm{AF})}_+ \,,$$

where the subscript indicates the band,

$$s^{(\mathrm{AF})}_\pm \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \Big[\Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) \ln\Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) \\ + f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \ln f(\pm\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \,. \tag{E.57}$$

Using Eq. (4.41), we can give an alternate formula for the entropy density,

$$s^{(\mathrm{AF})} = -\frac{k_\mathrm{B}}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \ln\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \\ - \frac{k_\mathrm{B}}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \ln\Big(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \,. \tag{E.58}$$

The proof is identical to the one provided for the normal phase in Sec. 4.3.2. Note that, as the gap closes the above expression reduces to the entropy density for the normal phase reported in Eq. (4.40). Indeed, we are summing over the MBZ, and the bare bands $\tilde{\epsilon}_{\mathbf{k}}$ change sign when crossing its boundary. Thus, substituting $\tilde{E}_{\mathbf{k}} \to \tilde{\epsilon}_{\mathbf{k}}$, the first term coincides with half the sum of Eq. (4.40), the second term with the other half.

E.2.3 The free energy density of the AF ground-state

We compute the density of free energy, following the same procedure we used for the normal phase in Sec. 4.3.3.

The general expression of free energy. The free energy density is given by

$$f^{(\mathrm{AF})} \equiv \frac{F^{(\mathrm{AF})}}{L_x L_y}\,, \tag{E.59}$$

being $F^{(\mathrm{AF})}$ the total free energy,

$$F^{(\mathrm{AF})} = E_{\mathrm{HF}}^{(\mathrm{AF})} - \frac{S^{(\mathrm{AF})}}{k_{\mathrm{B}}\beta}\,, \tag{E.60}$$

and $S^{(\mathrm{AF})} \equiv L_x L_y \times s^{(\mathrm{AF})}$ the total entropy. Here $E_{\mathrm{HF}}^{(\mathrm{AF})}$ is the ground-state energy for the HF hamiltonian. In the AF phase, we compute $f^{(\mathrm{AF})}$ using the following formula:

$$\begin{aligned} f^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \Big[\ln\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + \ln\Big(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big)\Big] \\ + U m^2 - V \sum_{\gamma\neq s}\left(\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2\right) + \tilde{\mu} n\,. \end{aligned} \tag{E.61}$$

Proof E.10 — Proof of Eq. (E.61): free energy density for the AF phase.

Using Eqs. (E.59), (E.60), we have

$$f^{(\mathrm{AF})} = \frac{E_{\mathrm{HF}}^{(\mathrm{AF})}}{L_x L_y} - \frac{s^{(\mathrm{AF})}}{k_{\mathrm{B}}\beta}\,,$$

with $s^{(\mathrm{AF})}$ the entropy density computed in Eq. (E.58). Then, $f^{(\mathrm{AF})}$ is obtained by summing the following terms:

1. Quasiparticles energy, taking care of spin degeneracy,

$$e_{\mathrm{g}}^{(\mathrm{AF})} = \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\,.$$

2. Contraction energy,

$$\begin{aligned} e_{\mathrm{c}}^{(\mathrm{AF})} &= \frac{E_{\mathrm{c}}^{(\mathrm{AF})}}{L_x L_y} \\ &= U(m^2 - n^2) + V\left[2zn^2 - \sum_{\gamma\neq s}\left(\left|u^{(\gamma)}\right|^2 + \left|v^{(\gamma)}\right|^2\right)\right]. \end{aligned}$$

The total contraction energy $E_{\mathrm{c}}^{(\mathrm{AF})}$ was given in Eq. (E.56). The above result was obtained by using Eqs. (5.31), (E.29) and (E.50).

3. Entropic energy,

$$\begin{aligned} e_s^{(\mathrm{AF})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \ln\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu}) \\ + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \ln\Big(1 - f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\Big) + \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}};\beta,\tilde{\mu})\,, \end{aligned}$$

obtained by multiplying Eq. (E.58) by $-2/\beta k_\mathrm{B}$.

4. Chemical potential correction,
$$e_n^{(\mathrm{AF})} = \mu n\,.$$

Composing everything,
$$f^{(\mathrm{AF})} \equiv e_\mathrm{g}^{(\mathrm{AF})} + e_\mathrm{c}^{(\mathrm{AF})} + e_s^{(\mathrm{AF})} + e_n^{(\mathrm{AF})}\,,$$
we obtain Eq. (E.61).

Approximation of free energy at half-filling. At the end of Sec. 5.2.2, we discussed the fact that the ground-state we find for the AF phase is stable only at half-filling, where $\tilde{\mu} = 0$. We can derive an approximation to free energy in this condition.

$$f^{(\mathrm{AF})} \simeq -\frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \tilde{E}_\mathbf{k} + Um^2 - V \sum_{\gamma\neq s} \left|u^{(\gamma)}\right|^2. \tag{E.62}$$

This approximation is valid when $\tilde{E}_\mathbf{k} \neq 0$ everywhere. This is always realised at finite magnetisation. Indeed, as is discussed at the end of Sec. 5.3, from Eq. (5.73) we see that the renormalised gap coincide with the conventional AF gap for the HM,

$$\tilde{\Delta}_\mathbf{k} = \Delta_\mathbf{k} = mU\,,$$

which, using Eq. (5.66), implies

$$\tilde{E}_\mathbf{k} = \sqrt{\tilde{\epsilon}_\mathbf{k}^2 + (Um)^2} \geq Um\,. \tag{E.63}$$

At any finite magnetisation m, the bands are gapped. Therefore, at half filling and low temperature, the free energy is well approximated by Eq. (E.62), which describes a completely filled electronic band $-\tilde{E}_\mathbf{k}$.

From a computational point of view, Eq. (E.62) can be useful when calculating the free energy of the ground-state, since the logarithms of Eq. (E.61) were removed.

Proof E.11 — Proof of Eq. (E.62).

We start with the identity:

$$\begin{aligned}\ln\left(1 - \frac{1}{e^x+1}\right) + \ln\left(1 - \frac{1}{e^{-x}+1}\right) &= \ln\left[\frac{e^x}{(e^x+1)^2}\right] \\ &= x - 2\ln(1+e^x)\,.\end{aligned}$$

At $x \to \infty$, we have
$$x - 2\ln(1+e^x) \simeq -x \qquad (x\to\infty)\,.$$

Let now $x = \beta\tilde{E}_\mathbf{k}$. At low temperature for a band with no nodes, we can approximate $\forall\mathbf{k} : \beta\tilde{E}_\mathbf{k} \to +\infty$ on the entire BZ. Hence,

$$\ln\left(1 - \frac{1}{e^{\beta\tilde{E}_\mathbf{k}}+1}\right) + \ln\left(1 - \frac{1}{e^{-\beta\tilde{E}_\mathbf{k}}+1}\right) \simeq -\beta\tilde{E}_\mathbf{k}\,.$$

Substituting in Eq. (E.61), we obtain Eq. (E.62).

Appendix F

Theoretical constraints on the normal and AF HFPs

In this appendix, we derive some theoretical constraints on the HFPs of the normal and AF phase. The derivations presented here are used in the discussions of Chaps. 4 and 5.

F.1 Normal phase

The normal phase is described by two HFPs:

$$\text{Normal HFPs} \quad : \quad u^{(s^*)}, u^{(d)} \,,$$

which obey the self-consistency equations of Tab. 4.2. In this section we show that Eq. (4.45) holds,

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0\,. \tag{F.1}$$

The discussion is divided in three parts: in Sec. F.1.1, we discuss the spatial geometry of bands with a mixed symmetry $s^* \oplus d$. In Sec. F.1.2 we derive Eq. (F.1) by considering the simpler case of unbroken C_4 symmetry (hence setting $u^{(d)} = 0$ "by hand"). In Sec. F.1.3 we break rotational symmetry, $\mathrm{C}_4 \overset{\text{SSB}}{\to} \mathrm{C}_2$, and end the proof.

F.1.1 Mixed $s^* \oplus d$ bands

Before starting, it is useful to spend a few words on the spatial structure of a mixed $s^* \otimes d$-wave band. Let $\tilde{\epsilon}_{\mathbf{k}}$ be a generic mixed band,

$$\tilde{\epsilon}_{\mathbf{k}} = \epsilon^{(s^*)}(\cos k_x + \cos k_y) + \epsilon^{(d)}(\cos k_x - \cos k_y)\,.$$

It is the balance between the two components $\epsilon^{(s^*)}$ and $\epsilon^{(d)}$ that gives the main features of the renormalised band. Consider the s^*-wave and d-wave structure factors already presented in Fig. 3.5, and define the Magnetic Brillouin Zone (MBZ) and Nematic Brillouin Zone (NBZ) as follows:

$$\mathrm{MBZ} \equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi^{(s^*)}_{\mathbf{k}} \geq 0 \right\} \,, \tag{F.2}$$

$$\mathrm{NBZ} \equiv \left\{ \mathbf{k} \in \mathrm{BZ} \mid \varphi^{(d)}_{\mathbf{k}} \geq 0 \right\} \,, \tag{F.3}$$

represented in the top-left panel of Fig. F.1. In the rest of the panels of Fig. F.1 are reported examples of mixed symmetry bands, for positive components. The respective opposite sign plots are obtained by the following rules:

$\epsilon^{(s^*)} < 0$	and	$\epsilon^{(d)} < 0$	$\Longrightarrow$	Invert colours in Fig. F.1
$\epsilon^{(s^*)} > 0$	and	$\epsilon^{(d)} < 0$	$\Longrightarrow$	Rotate plots by $\pi/2$ in Fig. F.1
$\epsilon^{(s^*)} < 0$	and	$\epsilon^{(d)} > 0$	$\Longrightarrow$	Rotate plots by $\pi/2$ and invert colours in Fig. F.1

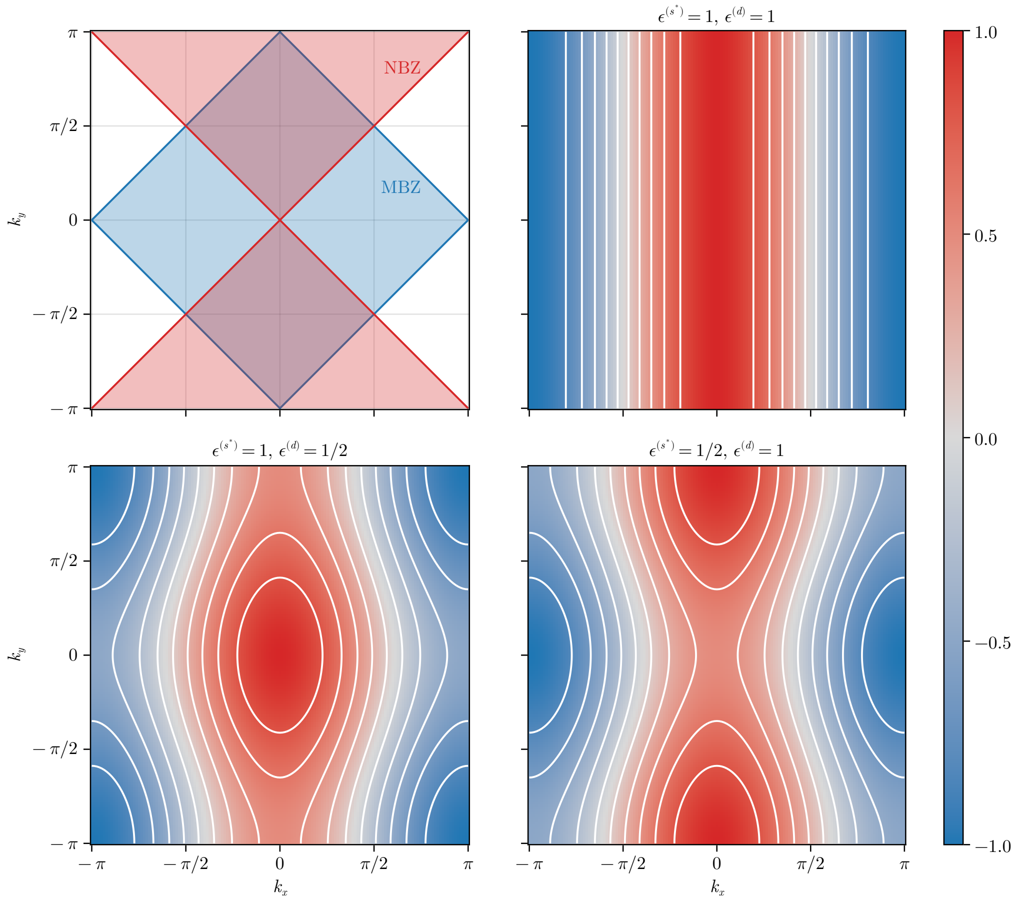


Figure F.1 Theoretical representation of mixed-symmetry nematic bands for the normal phase. The white contour represent level lines. In top-left panel, the MBZ and NBZ are depicted. While these plots only account for positive-defined components, in text is described how to map each one on its counterpart with different sign.

Let us focus on the MBZ and the NBZ. Starting from the intersections of the two zones, both can be factored in two sectors of interest,

$$\mathrm{MBZ} = \mathcal{A} \cup \mathcal{B} \quad \text{and} \quad \mathrm{NBZ} = \mathcal{A} \cup \mathcal{C}\,,$$

being, accordingly to Fig. F.2,

$$\mathcal{A} \equiv \mathcal{A}_1 \cup \mathcal{A}_2\,, \qquad \mathcal{B} \equiv \mathcal{B}_1 \cup \mathcal{B}_2\,, \qquad \mathcal{C} \equiv \mathcal{C}_1 \cup \mathcal{C}_2\,.$$

These domains satisfy:

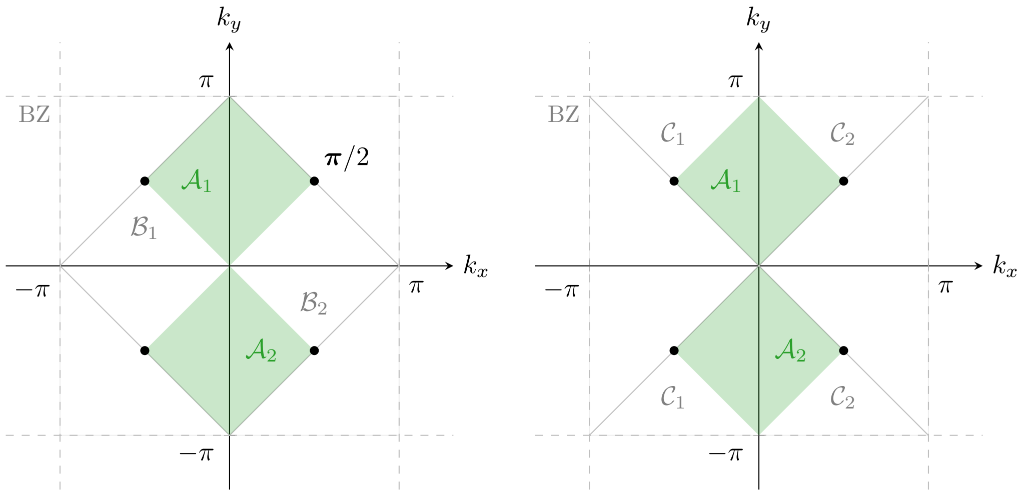
$$\begin{aligned}\mathcal{A} &\equiv \left\{\mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \geq 0\right\} \\ &= \mathrm{MBZ} \cap \mathrm{NBZ}\,,\end{aligned} \tag{F.4}$$

$$\begin{aligned}\mathcal{B} &\equiv \left\{\mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \geq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \leq 0\right\} \\ &= \mathrm{MBZ} \setminus \mathrm{NBZ}\,,\end{aligned} \tag{F.5}$$

$$\begin{aligned}\mathcal{C} &\equiv \left\{\mathbf{k} \in \mathrm{BZ} \mid \varphi_{\mathbf{k}}^{(s^*)} \leq 0 \wedge \varphi_{\mathbf{k}}^{(d)} \geq 0\right\} \\ &= \mathrm{NBZ} \setminus \mathrm{MBZ}\,,\end{aligned} \tag{F.6}$$

and will be used later on. Whatever the shape of the mixed band, the following equation always holds

$$|k_x| = \frac{\pi}{2} \quad \text{and} \quad |k_y| = \frac{\pi}{2} \qquad \Longrightarrow \qquad \tilde{\epsilon}_{(k_x,k_y)} = 0\,,$$



a | MBZ sectors.

b | NBZ sectors.

Figure F.2 | Depiction of the sectors composing the MBZ (Fig. F.2a) and the NBZ (Fig. F.2b), based on the divisions shown in the top-left panel of Fig. F.1. The full green zone composes $\mathcal{A}$, the region where both the s^*-wave and d-wave structure factors are positive defined; its MBZ complementary white counterpart in the left panel gives $\mathcal{B}$, for which the s^*-wave structure factors is positive-definite while the d-wave is negative. Similarly the NBZ complementary white counterpart in the right panel gives $\mathcal{C}$, for which signs are exchanged. The black dots, at the contours intersection, are band nodes.

since only cosines are involved. This identifies four node points, represented as black dots in Fig. F.2.

F.1.2 Bands shrinking effect within unbroken C_4 symmetry

Consider the simpler situation where C_4 symmetry is imposed. This implies that the renormalised bands are given by

$$\tilde{\epsilon}_{\mathbf{k}} \equiv (u^{(s^*)}V - 2t)(\cos k_x + \cos k_y)\,,$$

with a single self-consistency equation, given by the first row of Tab. 4.2:

$$u^{(s^*)} = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x + \cos k_y)f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\,.$$

We have the following constraint on $u^{(s^*)}$

$$\mathrm{C}_4 \quad\Longrightarrow\quad 0 < u^{(s^*)} < 2t/V\,. \tag{F.7}$$

The effective hopping is smaller than the original hopping, therefore this effect is dubbed as "bands shrinking".

Proof F.1 — Proof of Eq. (F.7): bands shrinking effect in C_4.

The proof follows from this chain of deductions:

1. *For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \leq 0$.*

Proof. Suppose instead $u^{(s^*)} \leq 0$. For any function $g_{\mathbf{k}}$ defined on the BZ:

$$\sum_{\mathbf{k}\in\mathrm{BZ}} g_{\mathbf{k}} = \sum_{\mathbf{k}\in\mathrm{MBZ}}\left[g_{\mathbf{k}} + g_{\mathbf{k}+\boldsymbol{\pi}}\right]\,,$$

where we defined the vector $\boldsymbol{\pi} \equiv (\pi, \pi)$ as in Eq. (5.1). The self-consistency equation thus reads

$$\begin{aligned} u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathbf{k}+\boldsymbol{\pi}}; \beta, \tilde{\mu}) \right] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] , \end{aligned} \tag{F.8}$$

having we used the antisymmetry of s^*-wave functions by $\boldsymbol{\pi}$ shifts. Now, if $u^{(s^*)} \leq 0$ the renormalised band $\tilde{\epsilon}_{\mathbf{k}}$ is negative inside the MBZ and positive outside, implying

$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \geq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \,,$$

the equality only being satisfied on the MBZ contour. Then, since inside the MBZ it holds by definition $\cos k_x + \cos k_y \geq 0$, the above equations imply

$$u^{(s^*)} \leq 0 \quad \Longrightarrow \quad u^{(s^*)} \geq 0 \,.$$

To exclude the possibility of $u^{(s^*)} = 0$, it suffices to notice that from the argument above such a situation only arises when the MBZ contour degenerates to the entirety of the MBZ itself – which is, when the band is flat. But that cannot be the case for $u^{(s^*)} = 0$, as long as $t > 0$. Thus we have proven that

$$u^{(s^*)} > 0 \,.$$

2. *For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \geq 2t/V$.*

Proof. We proceed exactly as did for the previous point, up to Eq. (F.8):

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in \mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] .$$

Now, if $u^{(s^*)} \geq 2t/V$ this means the prefactor in front of the renormalised bands is positive. Then $\tilde{\epsilon}_{\mathbf{k}}$ is positive inside the MBZ and negative outside, implying

$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \leq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \,,$$

thus we have

$$u^{(s^*)} \geq 2t/V \quad \Longrightarrow \quad u^{(s^*)} \leq 0 \,,$$

which is absurd. Thus

$$u^{(s^*)} < 2t/V \,.$$

Collecting the two points above, we obtain the constraint of Eq. (F.7).

F.1.3 Bands shrinking effect within broken $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$ symmetry

Let us extend the arguments above to a weaker symmetry, allowing for

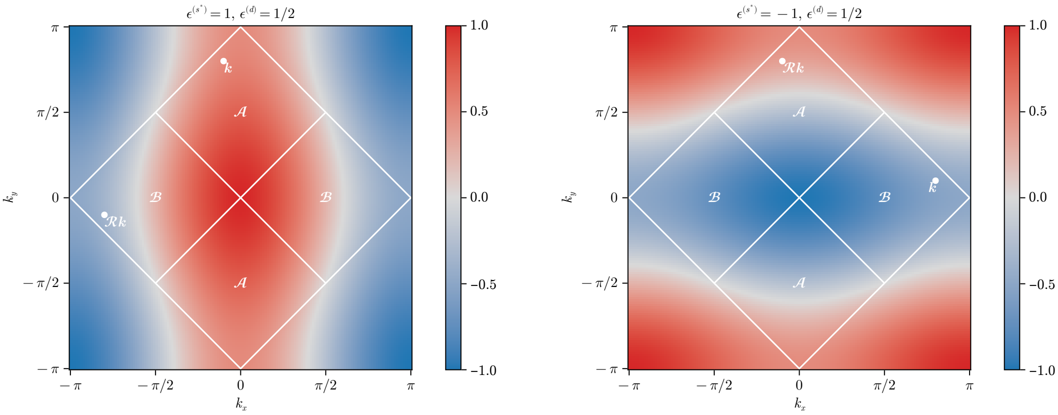
$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2/\mathbb{Z}_2 \,.$$

We work with the full, anisotropic band of Eq. (4.25). In the normal phase, nematic effects are always absent,

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0 \,,$$

which coincides with Eq. (F.1). Therefore, even if we allow for the breaking of planar rotational symmetry, this does not take place.

Proof F.2 — Proof of Eq. (F.1): non-nematic bands shrinking effect in C_2.



a | Same-sign mixed bands. b | Opposite-sign mixed bands.

Figure F.3 | Theoretical example of bands with $u^{(s^*)} > 2t/V$, $u^{(d)} > 0$ (top-left) and $u^{(s^*)} < 0$, $u^{(d)} > 0$ (top right). Two example points are reported, connected by a $\pi/2$ rotation as given in Eqns. (F.10), (F.15)

The proof is organised as a series of sub-proofs. We start by proving the left side of Eq. (F.1), namely that

$$0 < u^{(s^*)} < 2t/V \,.$$

We use everywhere $t, V > 0$.

1. *The following implication holds:*

$$u^{(s^*)} \geq 2t/V \,, \qquad u^{(d)} > 0 \quad \Longrightarrow \quad u^{(s^*)} \leq 0 \,. \tag{F.9}$$

Proof. Let us start by the implication

$$u^{(s^*)} \geq 2t/V > 0 \,, \qquad u^{(d)} > 0 \quad \Longrightarrow \quad \epsilon^{(s^*)} \geq 0 \,, \qquad \epsilon^{(d)} > 0 \,.$$

In Fig. F.3a an example band is shown for $\epsilon^{(s^*)} = 1$, $\epsilon^{(d)} = 1/2$. Note that if $u^{(s^*)} = 2t/V$, the renormalised band is purely d-wave since $\epsilon^{(s^*)} = 0$.

Recall Eq. (F.8), and separate in two sums over $\mathcal{A}$ and $\mathcal{B}$:

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{A}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \\ + \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{B}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \,.$$

Consider the sum over $\mathcal{B}$. Looking at the bottom-left panel of Fig. F.1, such a region contains nodal curves across which bands change sign. Let $\mathcal{R}$ be the clockwise $\pi/2$ rotation: then, any point in $\mathcal{B}$ can be seen as the rotation of a point in $\mathcal{A}$

$$\forall \mathbf{k}' \in \mathcal{B} \quad \exists \quad \mathbf{k} \in \mathcal{A} \mid \mathbf{k}' = \mathcal{R}\mathbf{k} \,. \tag{F.10}$$

A graphical rendition of this idea is given in Fig. F.3a. Then we can rewrite $u^{(s^*)}$ as a single sum over $\mathcal{A}$,

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{A}} (\cos k_x + \cos k_y) \\ \times \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right] \,. \tag{F.11}$$

From Eq. (F.4), $\forall \mathbf{k} \in \mathcal{A}$ the functions $\cos k_x \pm \cos k_y$ are positive by construction. By hypothesis $\epsilon^{(s^*)} \geq 0$ and $\epsilon^{(d)} > 0$. Hence:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}^{\geq 0} + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\geq 0} \,, \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)} (\cos k_x + \cos k_y)}_{\geq 0} - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\geq 0} \,. \end{aligned}$$

Summing the two equations above we have:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} + \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \geq 0 \,,$$

thus $\tilde{\epsilon}_{\mathbf{k}} \geq -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}$. The Fermi-Dirac distribution is monotonically descendent. Then:

$$\forall \mathbf{k} \in \mathcal{A} \quad : \quad \begin{aligned} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\leq f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \,, \\ f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\geq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \,. \end{aligned}$$

This condition, inserted in Eq. (F.11), leads to the absurd of Eq. (F.9).

2. *The following implication holds:*

$$u^{(s^*)} \geq 2t/V > 0 \,, \qquad u^{(d)} < 0 \quad \Longrightarrow \quad u^{(s^*)} \leq 0 \,. \tag{F.12}$$

The proof is identical to the above point, exchanging the roles of $\mathcal{A}$ and $\mathcal{B}$.

3. *The following implication holds:*

$$u^{(s^*)} \geq 2t/V > 0 \quad \Longrightarrow \quad u^{(d)} = 0 \,. \tag{F.13}$$

The proof follows from combining Eqs. (F.9), (F.12).

4. *It is impossible to have $u^{(s^*)} \geq 2t/V$.*

Proof. From Eq. (F.13), we know that if $u^{(s^*)} \geq 2t/V$ then $u^{(d)} = 0$. But then C_4 is unbroken, and we get back to the discussion of Sec. F.1.2. Then, we cannot have $u^{(s^*)} \geq 2t/V$.

5. *The following implication holds*:

$$u^{(s^*)} \leq 0 \,, \qquad u^{(d)} > 0 \quad \Longrightarrow \quad u^{(s^*)} > 0 \,. \tag{F.14}$$

Proof. Let for instance

$$u^{(s^*)} \leq 0 \,, \qquad u^{(d)} > 0 \quad \Longrightarrow \quad \epsilon^{(s^*)} < 0 \,, \qquad \epsilon^{(d)} > 0 \,.$$

In Fig. F.3b an example band is shown for $\epsilon^{(s^*)} = -1$, $\epsilon^{(d)} = 1/2$. Proceed as for the first point of this chain. Similarly to Eq. (F.10), Any point in $\mathcal{A}$ can be obtained by rotating some point in $\mathcal{B}$,

$$\forall \mathbf{k}' \in \mathcal{A} \quad \exists \quad \mathbf{k} \in \mathcal{B} \mid \mathbf{k}' = \mathcal{R}\mathbf{k} \,. \tag{F.15}$$

An equation similar to Eq. (F.11) holds, the sum being performed over $\mathcal{B}$

$$\begin{aligned} u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathcal{B}} &(\cos k_x + \cos k_y) \\ &\times \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) + f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right] \,. \end{aligned} \tag{F.16}$$

From Eq. (F.5), $\forall \mathbf{k} \in \mathcal{B}$ the form factor $\cos k_x + \cos k_y$ is positive by construction; the form factor $\cos k_x - \cos k_y$ is negative by construction. By hypothesis $\epsilon^{(s^*)} < 0$ and $\epsilon^{(d)} > 0$. Hence:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \overbrace{\epsilon^{(s^*)}\left(\cos k_x + \cos k_y\right)}^{\leq 0} + \overbrace{\epsilon^{(d)}\left(\cos k_x - \cos k_y\right)}^{\leq 0} , \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \underbrace{\epsilon^{(s^*)}\left(\cos k_x + \cos k_y\right)}_{\leq 0} - \underbrace{\epsilon^{(d)}\left(\cos k_x - \cos k_y\right)}_{\leq 0} . \end{aligned}$$

Summing the two equations above we have:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} + \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} \leq 0 \, ,$$

thus $\tilde{\epsilon}_{\mathbf{k}} \leq -\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}$. The Fermi-Dirac distribution is monotonically descendent. Then:

$$\forall \mathbf{k} \in \mathcal{B} \quad : \quad \begin{aligned} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\geq f(-\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \, , \\ f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) &\leq f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \, . \end{aligned}$$

This condition, inserted in Eq. (F.11), leads to an absurd:

$$u^{(s^*)} \leq 0 \, , \qquad u^{(d)} > 0 \quad \Longrightarrow \quad u^{(s^*)} > 0 \, .$$

which coincides with Eq. (F.14).

6. *The following implication holds*:

$$u^{(s^*)} \leq 0 \, , \qquad u^{(d)} < 0 \quad \Longrightarrow \quad u^{(s^*)} > 0 \, . \tag{F.17}$$

The proof is identical to the above point, exchanging the roles of $\mathcal{A}$ and $\mathcal{B}$.

7. *The following implication holds:*

$$u^{(s^*)} \leq 0 \quad \Longrightarrow \quad u^{(d)} = 0 \, . \tag{F.18}$$

The proof follows from combining Eqs. (F.14), (F.17).

8. *It is impossible to have* $u^{(s^*)} \leq 0$.

Proof. From Eq. (F.18), we know that if $u^{(s^*)} \leq 0$ then $u^{(d)} = 0$. But then C_4 is unbroken, and we get back to the discussion of Sec. F.1.2. Then we cannot have $u^{(s^*)} \leq 0$.

From the chain of derivations above, we can conclude self-consistently that, also allowing for $C_4 \overset{\text{SSB}}{\to} C_2$, the constraint of Eq. (F.7),

$$0 < u^{(s^*)} < 2t/V \, ,$$

holds. In the s^*-wave channel, partial breaking of rotational symmetry is not enough to modify the main effect identified under C_4 symmetry, "bands shrinking". What is missing is the behaviour of the d-wave channel when said effect takes place.

We move on and prove the right side of Eq. (F.1), namely that

$$u^{(d)} = 0 \, .$$

Continuing the enumeration from the above points,

9. *The following implication holds:*

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} > 0 \quad \Longrightarrow \quad u^{(d)} < 0 \, . \tag{F.19}$$

Proof. Let $0 < u^{(s^*)} < 2t/V$ and $u^{(d)} > 0$. Then $\epsilon^{(d)} > 0$. From Tab. 4.2, the self-consistency equation for $u^{(d)}$ reads

$$\begin{aligned} u^{(d)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{NBZ}} (\cos k_x - \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(\tilde{\epsilon}_{\mathcal{R}\mathbf{k}}; \beta, \tilde{\mu}) \right] , \end{aligned}$$

where we recognised, as is shown in Fig. F.2b, that $\mathrm{BZ} \setminus \mathrm{NBZ} = \mathcal{R}\mathrm{NBZ}$. It holds:

$$\forall \mathbf{k} \in \mathrm{NBZ} \quad : \quad \begin{aligned} \tilde{\epsilon}_{\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) + \overbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}^{\geq 0} , \\ \tilde{\epsilon}_{\mathcal{R}\mathbf{k}} &= \epsilon^{(s^*)} (\cos k_x + \cos k_y) - \underbrace{\epsilon^{(d)} (\cos k_x - \cos k_y)}_{\geq 0} , \end{aligned}$$

thus $\tilde{\epsilon}_{\mathcal{R}\mathbf{k}} < \tilde{\epsilon}_{\mathbf{k}}$, and inserting this fact in the self-consistency equation above we get the absurd of Eq. (F.19).

10. *The following implication holds:*

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} < 0 \quad \Longrightarrow \quad u^{(d)} > 0 \,. \tag{F.20}$$

The proof is identical to the point above.

11. *The following implication holds:*

$$0 < u^{(s^*)} < 2t/V \quad \Longrightarrow \quad u^{(d)} = 0 \,. \tag{F.21}$$

The proof follows from combining Eqs. (F.19), (F.20).

12. *It is impossible to have* $u^{(d)} \neq 0$. The proof follows from combining Eqs. (F.13), (F.18) and (F.21).

Hence we conclude that Eq. (F.1) holds.

F.2 AF phase

As we discuss in App. E, the AF phase divides in two subphases, namely aAF and sAF phases, as reported in Tab. 5.2. The sAF phase is described by five HFPs,

$$\text{sAF HFPs} \quad : \quad m, u^{(s^*)}, u^{(d)}, v^{(s^*)}, v^{(d)} ,$$

while for the aAF phase

$$\text{aAF HFPs} \quad : \quad m, u^{(s^*)}, u^{(d)}, -iv^{(p_x)}, -iv^{(p_y)} .$$

Self-consistency equations are reported in Eq. (5.76) and Tabs. E.1, E.2. We will use all conventions on notations introduced in App. E.

In this section we show that

- Eq. (5.71) holds,

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0 \,. \tag{F.22}$$

Hence also in the AF phase we have a Mott localisation effect;

- Eq. (5.72) holds,

$$\forall \gamma \quad : \quad v^{(\gamma)} = 0 \,. \tag{F.23}$$

Hence the AF gap is not “complexified”.

The discussion is divided in four parts: in Sec. F.2.1, we derive Eq. (F.22) within C_4 symmetry (as in Sec. F.1.2, setting $u^{(d)} = 0$). In Sec. F.2.2 we prove Eq. (F.23) for the sAF phase within C_2 symmetry. In Sec. F.2.3 we do the same for the aAF phase. We conclude, in Sec. F.2.4, by proving Eq. (F.22) within C_2 symmetry.

F.2.1 Bands shrinking effect within unbroken C_4 symmetry

For the normal phase, we saw that when C_4 symmetry is preserved (Sec. F.1.2) we can conclude immediately that bands renormalisation is actually a shrinking. An identical result holds here: if C_4 symmetry is imposed onto the renormalized bands,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv (u^{(s^*)}V - 2t)(\cos k_x + \cos k_y)\,.$$

Among the harmonics $u^{(\gamma)}$, just the s^*-wave channel contributes to renormalisation of bands, with a single self-consistency equation

$$u^{(s^*)} = -\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{MBZ}}(\cos k_x + \cos k_y)\frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta,\tilde{\mu}]\,, \tag{F.24}$$

where the renormalised band $\tilde{E}_{\mathbf{k}}$ is given in Eq. (5.66), and the renormalised thermal factor $\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta,\tilde{\mu}]$ in Eq. (E.12). We now prove that:

$$0 < u^{(s^*)} < 2t/V\,. \tag{F.25}$$

This constraint is identical to Eq. (F.7), obtained for the normal phase. Also in AF phase the bare bands are effectively shrunk by $\hat{H}_V$.

Proof F.3 — Proof of Eq. (F.25).

As we did for the normal phase:

1. *For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \leq 0$.*

   Proof. If $t > 0$ and $u^{(s^*)} \leq 0$

   $$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \leq 0\,,$$

   the equality only holding on the MBZ contour. Looking at the self-consistency Eq. (F.24), all other objects inside the sum are positive-defined. This gives

   $$u^{(s^*)} \leq 0 \quad \Longrightarrow \quad u^{(s^*)} \geq 0\,.$$

   But $u^{(s^*)} \neq 0$ as long as $t > 0$, as can be seen by the self-consistency equation. We conclude necessarily

   $$u^{(s^*)} > 0\,,$$

   which is absurd.

2. *For $t > 0$, there is no self-consistent solution admitting $u^{(s^*)} \geq 2t/V$.*

   Proof. If $t > 0$ and $u^{(s^*)} \geq 2t/V$

   $$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad \tilde{\epsilon}_{\mathbf{k}}\,, \geq 0$$

   the equality only holding on the MBZ contour. Inverting inequalities from point above, we conclude necessarily

   $$u^{(s^*)} < 2t/V\,.$$

   which is absurd.

F.2.2 Selection rules on the magnetic HFPs in sAF phase

In App. E, we apply HF methods to the EHM and obtain two flavours of antiferromagnetism, each of which involves five HFPs listed in Tab. 5.2. The scenario appears to be rich. In this section we present an argument to prove that the actual number of HFPs is much smaller, as many of those vanish.

Let us start from the sAF phase, assuming

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2 \,.$$

The self-consistency problem is described by the five HFPs on the first row of Tab. 5.2 and their self-consistency equations of Tabs. E.1 and E.2.

Let us define the functional

$$F[f(\mathbf{k})] \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta,\tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} f(\mathbf{k}) \,. \tag{F.26}$$

The functional F is linear:

$$F[af(\mathbf{k}) + bg(\mathbf{k})] = aF[f(\mathbf{k})] + bF[g(\mathbf{k})] \,, \tag{F.27}$$

being $a, b \in \mathbb{C}$ and f, g two integrable functions in reciprocal space. Another handy property is the sign-preserving relation for sign-defined functions. Let for instance

$$\forall \mathbf{k} \quad : \quad f(\mathbf{k}) \geq 0 \quad \Longrightarrow \quad F[f(\mathbf{k})] \geq 0 \,. \tag{F.28}$$

The relation above is implied by the fact that both $\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}$ and $\tilde{E}_{\mathbf{k}}$ are positive-defined,

$$\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)} = \frac{1}{e^{\beta(\tilde{E}_{\mathbf{k}}-\mu)}+1} - \frac{1}{e^{\beta(-\tilde{E}_{\mathbf{k}}-\mu)}+1} \geq 0 \quad \text{and} \quad \tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + |\Delta_{\mathbf{k}\sigma}|^2} \geq 0 \,,$$

so, being the functional F of Eq. (F.26) essentially a weighted integral over MBZ, any positive-defined function has a positive image. An identical statement with inverted inequalities holds for negative-defined function.

Consider now the self-consistency Eq. (F.24) for $u^{(s^*)}$: explicitly,

$$\begin{aligned} u^{(s^*)} &= -\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (c_x + c_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta,\tilde{\mu}] \\ &= \frac{\left[2t - Vu^{(s^*)}\right]}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta,\tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} (c_x + c_y)^2 - \frac{Vu^{(d)}}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta,\tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} (c_x^2 - c_y^2) \,, \end{aligned}$$

having we used the notation $c_\ell \equiv \cos k_\ell$ and substituted the full renormalised bands

$$\tilde{\epsilon}_{\mathbf{k}} \equiv -\left[2t - u_\sigma^{(s^*)} V\right] (c_x + c_y) + u_\sigma^{(d)} V (c_x - c_y) \,.$$

In terms of the functional F, we can rewrite:

$$\begin{aligned} u^{(s^*)} &= \left[2t - Vu^{(s^*)}\right] \times F\left[(c_x + c_y)^2\right] - Vu^{(d)} \times F\left[c_x^2 - c_y^2\right] \\ &= \left[2t - Vu^{(s^*)}\right] \times \left\{F\left[c_x^2\right] + 2F\left[c_x c_y\right] + F\left[c_y^2\right]\right\} - Vu^{(d)} \times \left\{F\left[c_x^2\right] - F\left[c_y^2\right]\right\} \,, \end{aligned}$$

having we used the linearity property of Eq. (F.27) in last passage.

Let now:

$$A \equiv F\left[c_x^2\right] \,, \qquad B \equiv 2F\left[c_x c_y\right] \,, \qquad C \equiv F\left[c_y^2\right] \,, \tag{F.29}$$

which allows us to rewrite the self-consistency equation as:

$$u^{(s^*)} = \left[2t - Vu^{(s^*)}\right] (A + B + C) - Vu^{(d)} (A - C)$$

Repeating an identical procedure on the three self-consistency equations for $u^{(d)}$, $v^{(s^*)}$ and $v^{(d)}$, we get the full system:

$$u^{(s^*)} = \left[2t - Vu^{(s^*)}\right](A+B+C) - Vu^{(d)}(A-C)\,, \tag{F.30}$$
$$u^{(d)} = \left[2t - Vu^{(s^*)}\right](A-C) - Vu^{(d)}(A-B+C)\,, \tag{F.31}$$
$$v^{(s^*)} = -Vv^{(s^*)}(A+B+C) - Vv^{(d)}(A-C)\,, \tag{F.32}$$
$$v^{(d)} = -Vv^{(s^*)}(A-C) - Vv^{(d)}(A-B+C)\,. \tag{F.33}$$

These equations define a four-components nonlinear map, whose parameters A, B, C depend on the HFPs themselves, being the functional F dependent on the HFPs. Because of this last point, the four equations are coupled to each other and the map does not decouple in two independent sub-maps. Note that the full map would include another equation for the magnetisation. It holds:

$$v^{(s^*)} = 0 \quad \text{and} \quad v^{(d)} = 0\,. \tag{F.34}$$

Proof F.4 — Proof of Eq. (F.34): all symmetric complexifiers vanish.

We use everywhere $t, V > 0$. Let us start by applying a linear transform on the map: define the new "sum" and "difference" variables

$$S_u \equiv u^{(s^*)} + u^{(d)}\,, \qquad D_u \equiv u^{(s^*)} - u^{(d)}\,, \qquad S_v \equiv v^{(s^*)} + v^{(d)}\,, \qquad D_v \equiv v^{(s^*)} - v^{(d)}\,,$$

whose matrix representation is

$$\begin{bmatrix} S_u \\ D_u \\ S_v \\ D_v \end{bmatrix} \equiv \begin{bmatrix} 1 & 1 & & \\ 1 & -1 & & \\ & & 1 & 1 \\ & & 1 & -1 \end{bmatrix} \begin{bmatrix} u^{(s^*)} \\ u^{(d)} \\ v^{(s^*)} \\ v^{(d)} \end{bmatrix}.$$

The map transform is obtained by summing and subtracting two couples of equations: Eq. (F.30) and (F.31) (the first two) as well as Eq. (F.32) and (F.33) (the second two). By doing this we get the simplified system:

$$(1+2AV)S_u + BVD_u = 2t(2A+B)\,, \tag{F.35}$$
$$(1+2CV)D_u + BVS_u = 2t(B+2C)\,, \tag{F.36}$$
$$(1+2AV)S_v + BVD_v = 0\,, \tag{F.37}$$
$$(1+2CV)D_v + BVS_v = 0\,. \tag{F.38}$$

1. *Either $S_v = 0 \wedge D_v = 0$, or $S_v \neq 0 \wedge D_v \neq 0$.*

   Proof. Let $S_v = 0$. From Eq. (F.38), using $C = F\left[c_y^2\right] \geq 0$ implied by Eq. (F.28):

   $$\underbrace{(1+2CV)}_{>0} D_v = 0 \qquad \Longrightarrow \qquad D_v = 0\,.$$

   Identically, let $D_v = 0$. Then applying the same consideration on Eq. (F.37) with $C \leftrightarrow A$, we conclude $S_v = 0$. Either both are non-zero or both are zero.

2. *Let $S_v \neq 0 \wedge D_v \neq 0$. Then $B \neq 0$.*

   Proof. Consider Eq. (F.37): we are working at $V > 0$, thus

   $$B = -\frac{1+2AV}{V}\frac{S_v}{D_v}\,,$$

which is non-zero because $A = F\left[c_x^2\right] \geq 0$, the latter following from Eq. (F.28). We could have used identically Eq. (F.38).

3. *Let $S_v \neq 0 \wedge D_v \neq 0$. Hence the following relation holds:*

$$(BV)^2 = (1 + 2CV)(1 + 2AV)\,. \tag{F.39}$$

Proof. We multiply Eq. (F.37) by $BV \neq 0$ (see point above),

$$BV(1 + 2AV)S_v + (BV)^2 D_v = 0\,,$$

and substitute from Eq. (F.38) $BVS_v = -(1 + 2CV)D_v$,

$$\left[-(1 + 2CV)(1 + 2AV) + (BV)^2\right] D_v = 0\,.$$

Since $D_v \neq 0$ by hypothesis, Eq. (F.39) is implied.

4. *Let $S_v \neq 0 \wedge D_v \neq 0$. Hence the following relation holds:*

$$B + 2C = V(B^2 - 4AC)\,. \tag{F.40}$$

Proof. We multiply Eq. (F.35) by $BV \neq 0$ (see points above) on both sides,

$$BV(1 + 2AV)S_u + (BV)^2 D_u = 2tBV(2A + B)\,,$$

and substitute from Eq. (F.36) $BVS_v = -(1 + 2CV)D_v + 2t(B + 2C)$,

$$\begin{aligned} -(1 + 2CV)(1 + 2AV)D_v + 2t(B + 2C)(1 + 2AV) + (BV)^2 D_u &= 2tBV(2A + B) \\ 2t(B + 2C)(1 + 2AV) &= 2tBV(2A + B) \\ B + 2ABV + 2C + 4ACV &= 2ABV + B^2 V\,, \end{aligned}$$

Passing from first to second line we used Eq. (F.39), and in the passage below $t > 0$. Last line implies Eq. (F.40).

5. *Let $S_v \neq 0 \wedge D_v \neq 0$. Then $B^2 \neq 4AC$.*

Proof. Let $B^2 = 4AC$. Then from Eq. (F.39),

$$(BV)^2 = 1 + 2V(A + C) + 4ACV^2 \quad \Longrightarrow \quad 1 + 2V(A + C) = 0\,,$$

which is absurd, being $A \geq 0$, $C \geq 0$ from Eq. (F.28).

6. *Let $S_v \neq 0 \wedge D_v \neq 0$. Hence the following relation holds:*

$$V = \frac{B + 2C}{B^2 - 4AC}\,. \tag{F.41}$$

Proof. Use the above point in Eq. (F.40).

7. *Let $S_v \neq 0 \wedge D_v \neq 0$. Then $A = C$.*

Proof. We substitute Eq. (F.41) in Eq. (F.39),

$$(B^2 - 4AC)\left(\frac{B+2C}{B^2-4AC}\right)^2 - 2\frac{B+2C}{B^2-4AC}(A+C) - 1 = 0$$
$$(B+2C)^2 - 2(B+2C)(A+C) - (B^2-4AC) = 0$$
$$B^2 + 4C^2 + 4BC - 2AB - 2BC - 4AC - 4C^2 - B^2 + 4AC = 0\,,$$

Simplifying all recurring terms, we get

$$2B(C-A) = 0\,.$$

From previous points we know $B \neq 0$. Then $A = C$.

8. *It's impossible to have* $S_v \neq 0 \wedge D_v \neq 0$.

Proof. To prove this, assume the contrary and use the point above to reduce Eq. (F.41) to

$$A = C \quad \Longrightarrow \quad V = \frac{B+2C}{B^2-4C^2} = \frac{1}{B-2C}\,.$$

Inserting the latter in Eq. (F.38),

$$\left(1 + \frac{2C}{B-2C}\right)D_v + \frac{B}{B-2C}S_v = 0$$
$$\frac{B}{B-2C}D_v + \frac{B}{B-2C}S_v = 0\,,$$

which implies

$$D_v + S_v = 0 \quad \Longrightarrow \quad v^{(s^*)} = 0\,.$$

Recalling from Tab. E.2 the explicit self-consistency equation for $v^{(d)}$ and setting $v^{(s^*)} = 0$,

$$v^{(d)} = -Vv^{(d)}F\left[(c_x - c_y)^2\right]\,,$$

and using the positivity rule of Eq. (F.28) we get

$$V > 0 \quad \text{and} \quad F\left[(c_x - c_y)^2\right] \geq 0 \quad \Longrightarrow \quad v^{(d)} = 0\,,$$

because no finite positive number can equal any finite negative number, and *viceversa*. Thus

$$S_v \neq 0 \quad \text{and} \quad D_v \neq 0 \quad \Longrightarrow \quad v^{(s^*)} = v^{(d)} = 0 \quad \Longrightarrow \quad S_v = D_v = 0\,.$$

Absurd.

9. Combining the above point with our first point, we get that $S_v = 0 \wedge D_v = 0$.

Eq. (F.34) is a consequence of the last point.

We explored any possibility for S_v, D_v: no stable solution with finite values of the HFPs $v^{(s^*)}$, $v^{(d)}$ exists, thus both must be zero and can be eliminated from the algorithm.

To gain full knowledge on the map fixed point's shape, we should analyse similarly the behaviour of the self-consistency equations for $u^{(s^*)}$, $u^{(d)}$. This is done later in Sec. F.2.4. For now, it suffices to note that – limitedly to the sAF phase – gap complexification is not present. This effect is not a consequence of HF derivation, which remains self-coherent and general; it is instead of the peculiar form of the non-linear map defined by the set of self-consistency equations.

F.2.3 Selection rules on the magnetic HFPs in aAF phase

Let us discuss the aAF phase. It is described by the five HFPs on the second row of Tab. 5.2, as well as the self-consistency equations of Tabs. E.1 and E.2. We continue using the functional F as defined in Eq. (F.26). We also define A, B, C as in Eq. (F.29). Let $s_\ell \equiv \sin k_\ell$, and

$$P \equiv F\left[s_x^2\right] , \qquad Q \equiv 2F\left[s_x s_y\right] , \qquad R \equiv F\left[s_y^2\right] . \tag{F.42}$$

Using these definitions, four of five self-consistency equations are given by

$$u^{(s^*)} = \left[2t - Vu^{(s^*)}\right](A+B+C) - Vu^{(d)}(A-C)\,, \tag{F.43}$$

$$u^{(d)} = \left[2t - Vu^{(s^*)}\right](A-C) - Vu^{(d)}(A-B+C)\,, \tag{F.44}$$

$$v^{(p_x)} = -Vv^{(p_x)}P - Vv^{(p_y)}Q\,, \tag{F.45}$$

$$v^{(p_y)} = -Vv^{(p_x)}Q - Vv^{(p_y)}R\,. \tag{F.46}$$

We have:

$$v^{(p_x)} = 0 \quad \text{and} \quad v^{(p_y)} = 0\,. \tag{F.47}$$

Proof F.5 — Proof of Eq. (F.47): all antisymmetric complexifiers vanish.

We use everywhere $t, V > 0$. Let us write the equation for Q, given in Eq. (F.42), explicitly:

$$\begin{aligned}
Q &= F\left[s_x s_y\right] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} \frac{\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta,\tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} \sin k_x \sin k_y \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta,\tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} \sin k_x \sin k_y \\
&= \frac{1}{2L_x L_y} \sum_{k_x=-\pi}^{+\pi} \sin k_x \sum_{k_y=-\pi}^{+\pi} \frac{\tilde{\mathrm{Th}}_{(k_x,k_y)}^{(-)}[\beta,\tilde{\mu}]}{\tilde{E}_{(k_x,k_y)}} \sin k_y\,.
\end{aligned}$$

Passing from second to third line, we used MBZ periodicity. The k_y sum at last line is null: even if using antiperiodic harmonics for $\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}$, we did not break inversion symmetry neither in x nor y direction. Bogoliubov bands $\tilde{E}_{\mathbf{k}}$ and thermal factors must show said symmetry. The product of a symmetric function $(\tilde{\mathrm{Th}}/\tilde{E})$ with $\sin k_y$ integrates to zero. Then

$$Q = 0\,.$$

Using this result in Eq. (F.45), we get

$$v^{(p_x)} = -Vv^{(p_x)}F\left[s_x^2\right] .$$

Using the positivity rule of Eq. (F.28) we obtain

$$V > 0 \quad \text{and} \quad F\left[s_x^2\right] \geq 0 \qquad \Longrightarrow \qquad v^{(p_x)} = 0\,.$$

Identically, from Eq. (F.46) we get $v^{(p_y)} = 0$. This proves Eq. (F.47).

This ends this short section and shows that also for the aAF phase all gap complexifiers actually vanish. Eqs. (F.34) and (F.47), combined, give Eq. (F.23).

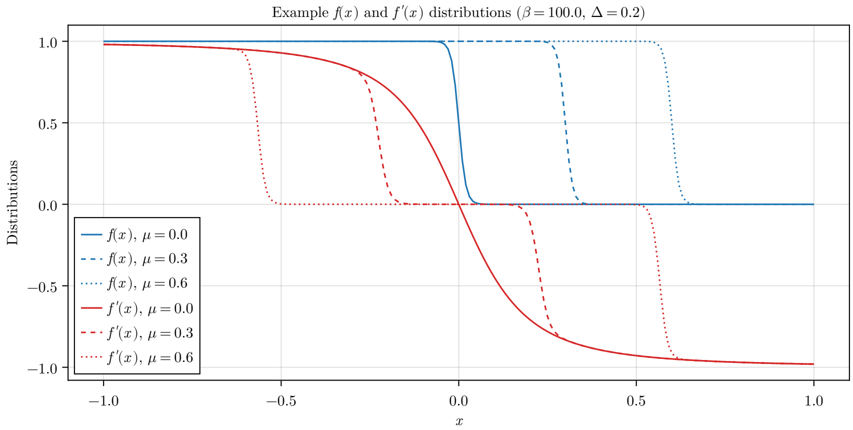


Figure F.4 Sketch of the Fermi-Dirac distribution $f(x)$ and the pseudo-distribution $f'(x)$ described in text for the normal phase. These example data have been plotted arbitrarily at $mU = \Delta = 0.2$ for $\mu = 0.0, 0.3, 0.6$.

F.2.4 Bands shrinking effect within broken $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$ symmetry

A consequence of the two sections above is that the gap function is purely real, non-renormalized and constant across all reciprocal space,

$$\tilde{\Delta}_{\mathbf{k}} = mU \qquad \Longrightarrow \qquad \tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + (mU)^2}\,.$$

Bogoliubov bands maintain renormalisation through the $u^{(\gamma)}$ HFPs channel. For the AF phase, Eq. (F.22) holds,

$$0 < u^{(s^*)} < 2t/V \quad \text{and} \quad u^{(d)} = 0\,.$$

Starting from a rather complicated scenario, we were able to reduce all the AF HFPs down to the pair

$$m, u^{(s^*)}\,.$$

The net effect of the non-local extension of the HM is, once again, the shrinking effect on bands.

Proof F.6 — Proof of Eq. (F.22): non-nematic bands shrinking effect in C_2.

While the proof given in Sec. F.1.3 was long and quite elaborated, this one is shorter. Let us start from recalling the normal phase self-consistency equations from Tab. 4.2,

$$\text{Normal phase:} \qquad \begin{aligned} u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})\,, \\ u^{(d)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})\,. \end{aligned}$$

Using $\boldsymbol{\pi}$ periodicity, for the AF phase we can re-write all equations (from Tab. E.1) doubling the integration domain,

$$\text{AF phase:} \qquad \begin{aligned} u^{(s^*)} &= -\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]\,, \\ u^{(d)} &= -\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]\,. \end{aligned}$$

Let us define the pseudo distribution

$$f'(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \equiv -\frac{\tilde{\epsilon}_{\mathbf{k}}}{2\tilde{E}_{\mathbf{k}}}\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta,\tilde{\mu}]\,,$$

which, substituted in the self-consistency equations, makes them very similar to the ones for the normal phase, upon substituting $f \to f'$. We plot $f(x)$ and $f'(x)$ in Fig. F.4 for various choices of the chemical potential.

The fact that $u^{(d)} = 0$ follows from the simple consideration that in Sec. F.1.3 (when showing the same for the normal phase) we never used the specific form of f; instead, we only made use of it being monotonically descendent as a function of $\epsilon_{\mathbf{k}}$. If f' also is a monotonically descendent function of $\epsilon_{\mathbf{k}}$, we can re-use the entire derivation of Sec. F.1.3. Let us show that using a somewhat lighter notation,

$$\epsilon_{\mathbf{k}} \to x \quad \text{and} \quad mU \to \Delta\,,$$

and omitting parametric arguments. The function f' is given by the product

$$f'(x) = g(x)h(x)\,,$$

with

$$\begin{aligned} g(x) &= -\frac{x}{\sqrt{x^2+\Delta^2}}\,,\\ h(x) &= \left(\frac{1}{e^{\beta(\sqrt{x^2+\Delta^2}-\mu)}+1} - \frac{1}{e^{\beta(-\sqrt{x^2+\Delta^2}-\mu)}+1}\right). \end{aligned}$$

The following points are true:

1. The function is antisymmetric about $x = 0$, because $g(x)$ is antisymmetric, while $h(x)$ is symmetric.

2. $g(x)$ is monotonically descendent. Its derivative is given by

$$\begin{aligned} \frac{\mathrm{d}g}{\mathrm{d}x} &= -\frac{\sqrt{x^2+\Delta^2} - x^2/\sqrt{x^2+\Delta^2}}{x^2+\Delta^2}\\ &= -\frac{\Delta^2}{(x^2+\Delta^2)^{3/2}} < 0\,. \end{aligned}$$

3. The product $g(x)h(x)$ is monotonically descendent at positive finite doping. To prove this, we can compute the derivative of f',

$$\frac{\mathrm{d}f'}{\mathrm{d}x} = \frac{\mathrm{d}g}{\mathrm{d}x}h(x) + g(x)\frac{\mathrm{d}h}{\mathrm{d}x}\,.$$

We have $h(x) > 0$, and as we said $\mathrm{d}g/\mathrm{d}x < 0$, which makes the first term negative. The second is negative too: let

$$E(x) \equiv \sqrt{x^2+\Delta^2} \quad \Longrightarrow \quad \frac{\mathrm{d}E}{\mathrm{d}x} = \frac{x}{E(x)} = -g(x)\,.$$

Then:

$$\begin{aligned} \frac{\mathrm{d}h}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{1}{e^{\beta(-E(x)-\mu)}+1} - \frac{1}{e^{\beta(E(x)-\mu)}+1}\right]\\ &= \beta\frac{\mathrm{d}E}{\mathrm{d}x}\underbrace{\left[\frac{e^{\beta(-E(x)-\mu)}}{\left[e^{\beta(-E(x)-\mu)}+1\right]^2} + \frac{e^{\beta(E(x)-\mu)}}{\left[e^{\beta(E(x)-\mu)}+1\right]^2}\right]}_{k(x)}. \end{aligned}$$

The term in square brackets is positive, $k(x) > 0$. Taking all together, we have

$$g(x)\frac{\mathrm{d}h}{\mathrm{d}x} = -\beta g^2(x)k(x) < 0 \qquad \Longrightarrow \qquad \frac{\mathrm{d}f'}{\mathrm{d}x} < 0\,.$$

Hence f' decreases monotonically.

These points end the proof: as anticipated, f' is a monotonically descendent function of $\epsilon_{\mathbf{k}}$, a feature that allows us to re-use the entire derivation of Sec. F.1.3, and conclude that also for the AF phase

$$u^{(d)} = 0\,,$$

while $u^{(s^*)}$ remains constrained in the limits of Eq. (F.25).

Appendix G

The HF hamiltonian in the SC phase

In this appendix, we derive the analytic expression of the HF hamiltonian for the SC phase, demonstrating all the results mentioned in Sec. 6.2. In particular, we aim to obtain Eqs. (6.10) and (6.12). We start by discussing the SC phase on the conventional HM, and then generalise to the EHM.

Let, as in Eqs. (6.1), (6.2)

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\,, \tag{G.1}$$

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}_{-\mathbf{k}\downarrow}\hat{c}_{\mathbf{k}\uparrow}\,. \tag{G.2}$$

As in Eqs. (6.3), (6.4),

$$u_{\mathbf{k}\sigma} \equiv \langle \hat{n}_{\mathbf{k}\sigma}\rangle\,, \tag{G.3}$$

$$w_{\mathbf{k}} \equiv \langle \hat{\phi}^\dagger_{\mathbf{k}}\rangle\,. \tag{G.4}$$

Using the functions of Tab. 3.5, as in Eqs. (6.5), (6.6) we define the harmonics

$$u^{(\gamma)}_\sigma \equiv \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}}u_{\mathbf{k}\sigma}\,, \tag{G.5}$$

$$w^{(\gamma)} \equiv \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}}w_{\mathbf{k}}\,. \tag{G.6}$$

Before starting, note that translational invariance is unbroken. Therefore, Eqs. (4.1) and (4.2), derived in the context of the normal phase, hold equivalently,

$$\langle \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma'}\rangle = \delta_{\mathbf{k}=\mathbf{k}'}\delta_{\sigma=\sigma'}\langle \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\rangle\,, \tag{G.7}$$

$$\langle \hat{n}_{i\sigma}\rangle = n\,. \tag{G.8}$$

G.1 Superconductivity in the conventional HM: derivation of the HF hamiltonian

The conventional HM is given by $\hat{H}_t + \hat{H}_U$, thus its unique two-body interaction is the local repulsion. In this section, we study superconductivity in the conventional HM, and prove that it is impossible at HF level.

Following Tab. 3.2 and Eq. (2.21), we know that the local repulsion decomposes in Hartree and Cooper channels,

$$\begin{aligned}
\hat{H}_U &= U\sum_{\mathbf{r}\in\mathcal{L}}\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\downarrow}\\
&\overset{\mathrm{HF}}{\simeq} U\sum_{\mathbf{r}\in\mathcal{L}}\Big(\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\langle\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\rangle + \langle\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow} - \langle\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\rangle && \text{(Hartree)}\\
&\qquad + \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\langle\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle + \langle\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\rangle\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow} - \langle\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\rangle\langle\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\Big)\,, && \text{(Cooper)}
\end{aligned}$$

while the Fock channel is identically zero. We deal with each channel separately.

G.1.1 Effects of the local Hartree terms: chemical potential renormalisation

For the Hartree channel, being the density uniform across sites, we can re-use the entirety of Sec. 4.2.1 for the normal phase. Using Eq. (G.8), we get the approximation of Eq. (4.10),

$$\hat{H}_U \overset{\text{HF}}{\simeq} nU \times \hat{N} - E^{(\text{SC})}_{\text{H},U} + (\text{Cooper channel})\,, \tag{G.9}$$

thus chemical potential renormalisation,

$$\tilde{\mu} = \mu - nU \tag{G.10}$$

is present also in the SC phase. Note that $\tilde{\mu}$ will be renormalised again, later, and we will redefine it accordingly. The contraction energy is the same of Eq. (4.14)

$$E^{(\text{SC})}_{\text{H},U} = L_x L_y \times n^2 U\,. \tag{G.11}$$

G.1.2 Effects of the local Cooper terms: isotropic pairing

We move to reciprocal space, through the transformation of Eq. (3.40). Consider the Cooper channel of Eq. (2.21). In reciprocal space we get:

$$\begin{aligned}\hat{H}_U \overset{\text{HF}}{\simeq} (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \Big(&\big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \\ &+ \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle - \big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle \Big)\,. \end{aligned} \tag{G.12}$$

We are not breaking translational invariance, therefore,

$$\big\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}'\sigma'} \big\rangle = \big\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma'} \big\rangle \delta_{\mathbf{k}=-\mathbf{k}'}\,. \tag{G.13}$$

The above equation "completes" Eq. (G.7), and completely specifies the behaviour of fermionic EVs in the translationally invariant SC phase.

Proof G.1 — Proof of Eq. (G.13).

Consider the EV

$$\big\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}'\sigma'} \big\rangle\,,$$

for two momenta $\mathbf{k}$, $\mathbf{k}'$. Moving to real space via an $\mathcal{F}$-transform, we get

$$\big\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}'\sigma'} \big\rangle \overset{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} \sum_{\mathbf{r}'\in\mathcal{L}} e^{-i\left(\mathbf{k}\cdot\mathbf{r}+\mathbf{k}'\cdot\mathbf{r}'\right)} \big\langle \hat{c}^\dagger_{\mathbf{r}\sigma} \hat{c}^\dagger_{\mathbf{r}'\sigma'} \big\rangle\,,$$

where we used the convention of Eq. (3.35). Let now $\mathbf{r} \equiv \mathbf{r}$ and $\Delta\mathbf{r} \equiv \mathbf{r} - \mathbf{r}'$. Translational invariance implies that the function

$$g(\Delta\mathbf{r}, \sigma, \sigma') = \big\langle \hat{c}^\dagger_{\mathbf{r}\sigma} \hat{c}^\dagger_{\mathbf{r}+\Delta\mathbf{r}\sigma'} \big\rangle$$

is independent of $\mathbf{r}$. Thus, we have:

$$\begin{aligned}\big\langle \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}'\sigma'} \big\rangle &\overset{\mathcal{F}}{=} \sum_{\Delta\mathbf{r}\in\mathcal{L}} e^{-i\mathbf{k}'\cdot\Delta\mathbf{r}} g(\Delta\mathbf{r}, \sigma, \sigma') \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} e^{-i\left(\mathbf{k}+\mathbf{k}'\right)\cdot\mathbf{r}} \\ &= \sum_{\Delta\mathbf{r}\in\mathcal{L}} e^{-i\mathbf{k}'\cdot\Delta\mathbf{r}} g(\Delta\mathbf{r}, \sigma, \sigma') \delta_{\mathbf{k}=\mathbf{k}'}\,,\end{aligned}$$

where we used the identity of Eq. (3.36). Then, the above EV vanishes for $\mathbf{k} \neq \mathbf{k}'$. This proves Eq. (G.13).

Using Eq. (G.13) in Eq. (G.12), only $\mathbf{K} = \mathbf{0}$ gives a non-zero contribution, and we obtain

$$\hat{H}_U \overset{\text{HF}}{\simeq} (\text{Hartree channel}) - \sum_{\mathbf{k}\in\text{BZ}} \left(\Delta^{(s)}\hat{\phi}_{\mathbf{k}} + \Delta^{(s)*}\hat{\phi}^\dagger_{\mathbf{k}}\right) - E^{(\text{SC})}_{\text{C},U}\,, \tag{G.14}$$

where we defined

$$\Delta^{(s)} \equiv -U w^{(s)}\,, \tag{G.15}$$

and the contraction energy is given by

$$E^{(\text{SC})}_{\text{C},U} = -\frac{U}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle\langle\hat{\phi}_{\mathbf{k}'}\rangle\,. \tag{G.16}$$

Proof G.2 — Proof of Eq. (G.14): HF approximation of $\hat{H}_U$ in Cooper channel.

Using Eqs. (G.2), (G.12) and (G.13), we obtain

$$\hat{H}_U \overset{\text{HF}}{\simeq} (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \left(\langle\hat{\phi}^\dagger_{\mathbf{k}}\rangle\hat{\phi}_{\mathbf{k}'} + \hat{\phi}^\dagger_{\mathbf{k}}\langle\hat{\phi}_{\mathbf{k}'}\rangle\right) - E^{(\text{SC})}_{\text{C},U}\,, \tag{G.17}$$

From Eq. (G.4), we have

$$\hat{H}_U \overset{\text{HF}}{\simeq} (\text{Hartree channel}) + \left(\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\text{BZ}} w_{\mathbf{k}}\right) \times U \sum_{\mathbf{k}'\in\text{BZ}} \hat{\phi}_{\mathbf{k}'}$$
$$+ \left(\frac{1}{L_x L_y}\sum_{\mathbf{k}'\in\text{BZ}} w^*_{\mathbf{k}'}\right) \times U \sum_{\mathbf{k}\in\text{BZ}} \hat{\phi}^\dagger_{\mathbf{k}}\,.$$

Therefore, from Eq. (G.6), we see that the parenthesised terms coincide, respectively, with $w^{(s)}$ and $w^{(s)*}$. Applying Eq. (G.15), we obtain Eq. (G.14).

The contraction energy of Eq. (G.16) reduces to

$$E^{(\text{SC})}_{\text{C},U} = L_x L_y \times U\left|w^{(s)}\right|^2. \tag{G.18}$$

Proof G.3 — Proof of Eq. (G.18): local Cooper contraction energy.

Consider Eq. (G.16). Substituting Eqs. (G.4), (G.6) we get:

$$E^{(\text{SC})}_{\text{C},U} = L_x L_y \times U \left(\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\text{BZ}} w^*_{\mathbf{k}}\right)\left(\frac{1}{L_x L_y}\sum_{\mathbf{k}'\in\text{BZ}} w_{\mathbf{k}'}\right) \tag{G.19}$$
$$= L_x L_y \times U\left|w^{(s)}\right|^2, \tag{G.20}$$

which proves Eq. (G.18).

In general, $\Delta^{(s)}$ as defined in Eq. (G.15) is a complex number. However, we can always choose $\Delta^{(s)}$ as purely real. This is due to the fact that the hamiltonian is gauge-invariant. Indeed, if we set in Eq. (3.4)

$$\eta \equiv -\frac{\arg w^{(s)}}{2}\,, \tag{G.21}$$

we have $\Delta^{(s)} \in \mathbb{R}$.

Proof G.4 — Choosing the gauge of Eq. (G.21), we obtain a real gap.

Inserting Eq. (G.21) in Eq. (3.4), we obtain

$$\hat{c}^\dagger_{\mathbf{k}\sigma} \quad \to \quad e^{-i\arg w^{(s)}/2}\hat{c}^\dagger_{\mathbf{k}\sigma} \,.$$

Therefore,

$$\hat{\phi}^\dagger_{\mathbf{k}} \quad \to \quad \hat{\phi}^\dagger_{\mathbf{k}} e^{-i\arg w^{(s)}} \qquad \Longrightarrow \qquad w_{\mathbf{k}} \quad \to \quad w_{\mathbf{k}} e^{-i\arg w^{(s)}} \,,$$

where we used Eq. (G.4). It follows, from Eq. (G.6),

$$w^{(s)} \quad \to \quad w^{(s)} e^{-i\arg w^{(s)}} \,.$$

For any complex number z, we have $ze^{-i\arg z} \in \mathbb{R}$. Then, $w^{(s)}$ is purely real under this gauge transform, and so is $\Delta^{(s)}$.

G.1.3 Diagonalisation of the HF hamiltonian

We proceed here as we did for the AF phase in Sec. 5.2.2. We start by giving a matrix representation of the approximated hamiltonian, then map it to a pseudospin problem and finally specify how to diagonalise it to retrieve its ground state.

Matrix representation. Let $\Delta_{\mathbf{k}} = \Delta^{(s)}$. The HM gran-canonical hamiltonian is given by $\hat{H}_t + \hat{H}_U - \mu\hat{N}$. Its HF approximation for the SC phase can be written as

$$\hat{H}_t + \hat{H}_U - \mu\hat{N} \overset{\text{HF}}{\simeq} \sum_{\mathbf{k}\in\text{BZ}} \sum_{\sigma} \hat{\Phi}^\dagger_{\mathbf{k}} h_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} + E^{(\text{SC})}_{\text{a}} - \left(E^{(\text{SC})}_{\text{H},U} + E^{(\text{SC})}_{\text{C},U}\right) \,, \tag{G.22}$$

where we defined the 2×2 matrix:

$$h_{\mathbf{k}} \equiv \begin{bmatrix} \xi_{\mathbf{k}} & -\Delta^*_{\mathbf{k}} \\ -\Delta_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix} \quad \text{with} \quad \xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu} \,. \tag{G.23}$$

In Eq. (G.22), we defined the Nambu spinors

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^\dagger_{-\mathbf{k}\downarrow} \end{bmatrix} \,, \tag{G.24}$$

and the "anticommutation energy" $E^{(\text{SC})}_{\text{a}}$,

$$E^{(\text{SC})}_{\text{a}} \equiv \sum_{\mathbf{k}\in\text{BZ}} \xi_{\mathbf{k}} \,. \tag{G.25}$$

Proof G.5 — Proof of Eq. (G.22).

Summing Eq. (3.38) and Eq. (G.14), we obtain

$$\begin{aligned} \hat{H}_t + \hat{H}_U \overset{\text{HF}}{\simeq} & \sum_{\mathbf{k}\in\text{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \\ & - \sum_{\mathbf{k}\in\text{BZ}} \left(\Delta^{(s)}\hat{\phi}_{\mathbf{k}} + \Delta^{(s)*}\hat{\phi}^\dagger_{\mathbf{k}}\right) + nU \times \hat{N} - \left(E^{(\text{SC})}_{\text{H},U} + E^{(\text{SC})}_{\text{C},U}\right) \,. \end{aligned} \tag{G.26}$$

Inversion symmetry is not broken, then $\xi_{\mathbf{k}} = \xi_{-\mathbf{k}}$. Using Fermi anticommutation rules, the kinetic operator can be written as:

$$\begin{aligned}\hat{H}_t - \mu\hat{N} &= \sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}\sigma}\hat{n}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}} \left(\xi_{\mathbf{k}}\hat{n}_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}}\hat{n}_{-\mathbf{k}\downarrow}\right) \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}}\left(\hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^{\dagger}_{-\mathbf{k}\downarrow} + 1\right) \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}}\left(\hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^{\dagger}_{-\mathbf{k}\downarrow}\right) + E^{(\mathrm{SC})}_{\mathrm{a}}\,.\end{aligned}$$

In last line we used Eq. (G.25). Inserting in Eq. (G.26), we obtain

$$\begin{aligned}\hat{H}_t + \hat{H}_U - \mu\hat{N} \overset{\mathrm{HF}}{\simeq} &\sum_{\mathbf{k}\in\mathrm{BZ}} \epsilon_{\mathbf{k}}\left(\hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{\mathbf{k}\downarrow}\hat{c}^{\dagger}_{\mathbf{k}\downarrow}\right) + E^{(\mathrm{SC})}_{\mathrm{a}} \\ &- \sum_{\mathbf{k}\in\mathrm{BZ}}\left(\Delta_{\mathbf{k}}\hat{\phi}_{\mathbf{k}} + \Delta_{\mathbf{k}}\hat{\phi}^{\dagger}_{\mathbf{k}}\right) - \tilde{\mu}\hat{N} - \left(E^{(\mathrm{SC})}_{\mathrm{H},U} + E^{(\mathrm{SC})}_{\mathrm{C},U}\right)\,, \end{aligned} \tag{G.27}$$

where $\tilde{\mu} = \mu - nU$ and $\Delta^{(s)} = \Delta_{\mathbf{k}}$. The energies $E^{(\mathrm{SC})}_{\mathrm{H},U}$, $E^{(\mathrm{SC})}_{\mathrm{C},U}$ are given respectively in Eqs. (G.11) and (G.18). Using the Nambu spinors defined in Eq. (G.24) and the 2×2 matrix $h_{\mathbf{k}}$ of Eq. (G.23), the HM hamiltonian is reduced to the matrix form of Eq. (G.22).

Pseudospin representation. Let now the pseudospin operator $\hat{\mathbf{s}}_{\mathbf{k}}$ and the pseudofield $\mathbf{b}_{\mathbf{k}}$

$$\hat{s}^{\alpha}_{\mathbf{k}} \equiv \hat{\Phi}^{\dagger}_{\mathbf{k}}\tau^{\alpha}\hat{\Phi}_{\mathbf{k}} \quad \text{with} \quad \hat{\mathbf{s}}_{\mathbf{k}} = \begin{bmatrix}\hat{s}^{x}_{\mathbf{k}}\\ \hat{s}^{y}_{\mathbf{k}}\\ \hat{s}^{z}_{\mathbf{k}}\end{bmatrix}\,, \qquad \mathbf{b}_{\mathbf{k}} \equiv \begin{bmatrix}-\mathrm{Re}\{\Delta_{\mathbf{k}}\}\\ -\mathrm{Im}\{\Delta_{\mathbf{k}}\}\\ \xi_{\mathbf{k}}\end{bmatrix}\,,$$

with τ^{α} the Pauli matrices of Eq. (3.9). As was for the AF phase, the pseudospin operators $\hat{s}^{\alpha}_{\mathbf{k}}$ satisfy a $\mathfrak{su}(2)$ algebra up to a constant, irrelevant in the context of this discussion. They satisfy the identities:

$$\hat{s}^{x}_{\mathbf{k}} = \hat{\phi}_{\mathbf{k}} + \hat{\phi}^{\dagger}_{\mathbf{k}}\,, \tag{G.28}$$

$$\hat{s}^{y}_{\mathbf{k}} = i\left(\hat{\phi}_{\mathbf{k}} - \hat{\phi}^{\dagger}_{\mathbf{k}}\right)\,, \tag{G.29}$$

$$\hat{s}^{z}_{\mathbf{k}} = \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{\mathbf{k}\downarrow} - 1\,. \tag{G.30}$$

Proceeding precisely as for the AF phase, in pseudospin representation the HF hamiltonian of Eq. (G.22) can be reduced to:

$$\hat{H}_t + \hat{H}_U - \mu\hat{N} \overset{\mathrm{HF}}{\simeq} \sum_{\mathbf{k}\in\mathrm{BZ}} \mathbf{b}_{\mathbf{k}}\cdot\hat{\mathbf{s}}_{\mathbf{k}} + E^{(\mathrm{SC})}_{\mathrm{a}} - \left(E^{(\mathrm{SC})}_{\mathrm{H},U} + E^{(\mathrm{SC})}_{\mathrm{C},U}\right)\,. \tag{G.31}$$

Diagonalisation. Let us define the angles $\theta_{\mathbf{k}}$, $\zeta_{\mathbf{k}}$ through the relations

$$\cos 2\theta_{\mathbf{k}} = \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\,, \qquad \sin 2\theta_{\mathbf{k}}\cos 2\zeta_{\mathbf{k}} = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}}\,, \qquad \sin 2\theta_{\mathbf{k}}\sin 2\zeta_{\mathbf{k}} = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}}\,,$$

where we defined the Bogoliubov bands $E_{\mathbf{k}}$,

$$E_{\mathbf{k}} \equiv \sqrt{\xi^2_{\mathbf{k}} + |\Delta_{\mathbf{k}}|^2}\,. \tag{G.32}$$

Recall the Bogoliubov bands for the AF phase, of Eq. (5.43). The key difference, here, is that the bands $E_{\mathbf{k}}$ depend on $\xi_{\mathbf{k}}$, not $\epsilon_{\mathbf{k}}$. We discussed at the end of Sec. 5.2.2 that, in the AF phase, the chemical potential stays the same for electrons and Bogoliubov particles. For the SC phase, being the bands $E_{\mathbf{k}}$ dependent on $\tilde{\mu}$ through $\xi_{\mathbf{k}}$, the chemical potential for the Bogoliubov particles is identically zero. This implies that the chemical potential lies within the SC gap.

For the SC phase, diagonalisation of the HF hamiltonian follows the same procedure as was for the AF phase. Indeed, the diagonalising matrix is given by

$$W_{\mathbf{k}} \equiv e^{-i\theta_{\mathbf{k}}\tau^y} e^{-i\zeta_{\mathbf{k}}\tau^z}\,, \tag{G.33}$$

and the graphic procedure for the diagonalisation is the same of Fig. 5.2, with $\epsilon \to \xi$. We get:

$$d_{\mathbf{k}} = W_{\mathbf{k}} h_{\mathbf{k}} W_{\mathbf{k}}^\dagger \quad \text{with} \quad d_{\mathbf{k}} \equiv \begin{bmatrix} E_{\mathbf{k}} & \\ & -E_{\mathbf{k}} \end{bmatrix}. \tag{G.34}$$

In the tilted frame we have the expression for the Nambu spinors:

$$\hat{\Gamma}_{\mathbf{k}} = W_{\mathbf{k}} \hat{\Psi}_{\mathbf{k}} \quad \text{with} \quad \hat{\Gamma}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{\gamma}_{\mathbf{k}}^{(+)} \\ \hat{\gamma}_{\mathbf{k}}^{(-)} \end{bmatrix}. \tag{G.35}$$

Applying the rotation of Eq. (G.34) to Eq. (G.31), we obtain the free Fermi gas representation of the HF hamiltonian,

$$\hat{H}_t + \hat{H}_U - \tilde{\mu}\hat{N} \overset{\text{HF}}{\simeq} \sum_{\mathbf{k}\in\text{BZ}} \sum_{\sigma} E_{\mathbf{k}} \left(\hat{\gamma}_{\mathbf{k}}^{(+)\dagger} \hat{\gamma}_{\mathbf{k}}^{(+)} - \hat{\gamma}_{\mathbf{k}}^{(-)\dagger} \hat{\gamma}_{\mathbf{k}}^{(-)} \right) + (\cdots)\,,$$

with $(\cdots) = E_{\text{a}}^{(\text{SC})} - \left(E_{\text{H},U}^{(\text{SC})} + E_{\text{C},U}^{(\text{SC})} \right)$ the various energy shifts.

Pseudospin EVs. The EVs of pseudospin operators in the various directions are given by:

$$\begin{aligned}
\langle \hat{\Psi}_{\mathbf{k}}^\dagger \tau^x \hat{\Psi}_{\mathbf{k}} \rangle &= -\sin 2\theta_{\mathbf{k}} \cos 2\zeta_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle\,, \\
\langle \hat{\Psi}_{\mathbf{k}}^\dagger \tau^y \hat{\Psi}_{\mathbf{k}} \rangle &= -\sin 2\theta_{\mathbf{k}} \sin 2\zeta_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle\,, \\
\langle \hat{\Psi}_{\mathbf{k}}^\dagger \tau^z \hat{\Psi}_{\mathbf{k}} \rangle &= \cos 2\theta_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle\,.
\end{aligned}$$

Let us define the thermal factors:

$$\text{Th}_{\mathbf{k}}^{(\pm)}[\beta] \equiv f\left(-E_{\mathbf{k}}; \beta, 0\right) \pm f\left(E_{\mathbf{k}}; \beta, 0\right)\,. \tag{G.36}$$

The only difference with Eq. (G.36) is in the fact that the chemical potential is zero in the Fermi-Dirac distributions, because it is already inside the definition of $E_{\mathbf{k}}$ given in Eq. (G.32). The thermal factors satisfy, at any density,

$$\text{Th}_{\mathbf{k}}^{(+)}[\beta] = 1\,, \tag{G.37}$$

$$\text{Th}_{\mathbf{k}}^{(-)}[\beta] = \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right). \tag{G.38}$$

The proofs are identical to the ones provided for Eqs. (E.14), (E.15).

In the tilted frame, the averaged z projection of pseudospin is:

$$\begin{aligned}
\langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle &= f\left(E_{\mathbf{k}}; \beta, 0\right) - f\left(-E_{\mathbf{k}}; \beta, 0\right) \\
&= -\tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right).
\end{aligned} \tag{G.39}$$

Bringing everything together, we have the EVs for the pseudospin operator:

$$\langle \hat{s}_{\mathbf{k}}^x \rangle = \frac{\text{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right), \tag{G.40}$$

$$\langle \hat{s}_{\mathbf{k}}^y \rangle = \frac{\text{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right), \tag{G.41}$$

$$\langle \hat{s}_{\mathbf{k}}^z \rangle = -\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right). \tag{G.42}$$

s-wave gap self-consistency equation. Te only HFP that we obtained for the SC phase is $w^{(s)}$. It obeys the self-consistency equation

$$w^{(s)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\frac{\Delta_{\mathbf{k}}}{E_{\mathbf{k}}}\tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right). \tag{G.43}$$

which is the "operational" version of Eq. (G.6), for $\gamma = s$.

Proof G.6 — Proof of Eq. (G.43): self-consistency equation for $w^{(s)}$.

Let $\tau^{\pm} \equiv (\tau^x \pm i\tau^y)/2$. From Eq. (G.6), we obtain

$$\begin{aligned} w^{(s)} &= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\langle\hat{\phi}^{\dagger}_{\mathbf{k}}\rangle \\ &= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\langle\hat{\Phi}^{\dagger}_{\mathbf{k}}\tau^{+}\hat{\Phi}_{\mathbf{k}}\rangle \\ &= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\frac{\langle\hat{s}^x_{\mathbf{k}}\rangle + i\,\langle\hat{s}^y_{\mathbf{k}}\rangle}{2} \\ &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\frac{\Delta_{\mathbf{k}}}{E_{\mathbf{k}}}\tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right). \end{aligned}$$

The last line coincides with Eq. (G.43).

Density. Density n can be computed as

$$n = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left[1 - \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right)\right]. \tag{G.44}$$

Proof G.7 — Proof of Eq. (G.44).

In order to calculate density, we can use the z projection of the pseudospin operators. From Eq. (G.30) we know

$$\hat{s}^z_{\mathbf{k}} + 1 = \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{\mathbf{k}\downarrow}\,.$$

Then, density can be compute as:

$$\begin{aligned} n &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\langle\hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow}\rangle \\ &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left(\langle\hat{s}^z_{\mathbf{k}}\rangle + 1\right) \\ &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left[1 - \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right)\right], \end{aligned}$$

which proves Eq. (G.44).

G.1.4 Impossibility of s-wave superconductivity in the HM

At HF level, the HM cannot host superconductivity, i.e.

$$w^{(s)} = 0\,, \tag{G.45}$$

therefore no superconducting gap can develop at HF level. This does not hold for $U < 0$: an attractive model can host s-wave superconductivity.

Proof G.8 — Proof of Eq. (G.45): impossibility of s-wave superconductivity in the conventional HM.

Let

$$U_c[\beta] \equiv \left[\frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\frac{1}{E_\mathbf{k}}\tanh\left(\frac{\beta E_\mathbf{k}}{2}\right)\right]^{-1} . \tag{G.46}$$

By construction $U_c[\beta] > 0$. Using Eq. (G.43) and $\Delta_\mathbf{k} = -Uw^{(s)}$, Eq. (G.46) can be rewritten as:

$$w^{(s)} = -\frac{U}{U_c[\beta]}w^{(s)} .$$

The unique possible self-consistent solution for the equation above is $w^{(s)} = 0$. This implication terminates the argument and proves Eq. (G.45): superconductivity cannot establish self-consistently at HF level in the repulsive HM.

G.2 Generalisation to the EHM

In this section we add to the hamiltonian the non-local attraction $\hat{H}_V$. As we did for the AF phase in Sec. E.1, we look for a solution with an identical mathematical structure as the one here described for the conventional HM. We define the renormalised quantities

$$\xi_\mathbf{k} \to \tilde{\xi}_\mathbf{k} , \qquad \Delta_\mathbf{k} \to \tilde{\Delta}_\mathbf{k} , \qquad \mu \to \tilde{\mu} .$$

The chemical potential was already renormalised once in the HM derivation in Sec. G.2.1. Renormalisation of Bogoliubov bands follows that of bare bands and gap function,

$$\tilde{E}_\mathbf{k} \equiv \sqrt{\tilde{\xi}_\mathbf{k}^2 + |\tilde{\Delta}_\mathbf{k}|^2} .$$

Angles are changed too. We define operatively the angles $\tilde{\theta}_\mathbf{k}$ and $\tilde{\zeta}_\mathbf{k}$ as

$$\sin 2\tilde{\theta}_\mathbf{k} = \frac{|\tilde{\Delta}_\mathbf{k}|}{\tilde{E}_\mathbf{k}} , \qquad \cos 2\tilde{\theta}_\mathbf{k} = \frac{\tilde{\xi}_\mathbf{k}}{\tilde{E}_\mathbf{k}} , \tag{G.47}$$

and

$$\sin 2\tilde{\zeta}_\mathbf{k} = \frac{\mathrm{Re}\{\tilde{\Delta}_\mathbf{k}\}}{|\tilde{\Delta}_\mathbf{k}|} , \qquad \cos 2\tilde{\zeta}_\mathbf{k} = \frac{\mathrm{Im}\{\tilde{\Delta}_\mathbf{k}\}}{|\tilde{\Delta}_\mathbf{k}|} . \tag{G.48}$$

Pseudospin EVs. Using Eqs. (G.47) and (G.48), we compute the pseudospin EVs

$$\langle\hat{\Phi}_\mathbf{k}^\dagger\tau^x\hat{\Phi}_\mathbf{k}\rangle = -\sin 2\tilde{\theta}_\mathbf{k}\sin 2\tilde{\zeta}_\mathbf{k}\langle\hat{\Gamma}_\mathbf{k}^\dagger\tau^z\hat{\Gamma}_\mathbf{k}\rangle , \tag{G.49}$$

$$\langle\hat{\Phi}_\mathbf{k}^\dagger\tau^y\hat{\Phi}_\mathbf{k}\rangle = -\sin 2\tilde{\theta}_\mathbf{k}\cos 2\tilde{\zeta}_\mathbf{k}\langle\hat{\Gamma}_\mathbf{k}^\dagger\tau^z\hat{\Gamma}_\mathbf{k}\rangle , \tag{G.50}$$

$$\langle\hat{\Phi}_\mathbf{k}^\dagger\tau^z\hat{\Phi}_\mathbf{k}\rangle = \cos 2\tilde{\theta}_\mathbf{k}\langle\hat{\Gamma}_\mathbf{k}^\dagger\tau^z\hat{\Gamma}_\mathbf{k}\rangle . \tag{G.51}$$

Renormalisation of the thermal factors and the z pseudospin EV in the tilted frame gives, similarly to Eqs. (G.36), (G.39),

$$\tilde{\mathrm{Th}}_\mathbf{k}^{(\pm)}[\beta,\tilde{\mu}] \equiv f\left(-\tilde{E}_\mathbf{k};\beta,0\right) \pm f\left(\tilde{E}_\mathbf{k};\beta,0\right) . \tag{G.52}$$

$$\langle\hat{\Gamma}_\mathbf{k}^\dagger\tau^z\hat{\Gamma}_\mathbf{k}\rangle = f\left(\tilde{E}_\mathbf{k};\beta,0\right) - f\left(-\tilde{E}_\mathbf{k};\beta,0\right) , \tag{G.53}$$

Eqs. (G.37), (G.38) hold identically. Hence, pseudospin expectation values are:

$$\langle\hat{s}_\mathbf{k}^x\rangle = \frac{\mathrm{Re}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_\mathbf{k}}{2}\right) , \tag{G.54}$$

$$\langle\hat{s}_\mathbf{k}^y\rangle = \frac{\mathrm{Im}\{\tilde{\Delta}_\mathbf{k}\}}{\tilde{E}_\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_\mathbf{k}}{2}\right) , \tag{G.55}$$

$$\langle\hat{s}_\mathbf{k}^z\rangle = -\frac{\tilde{\xi}_\mathbf{k}}{\tilde{E}_\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_\mathbf{k}}{2}\right) . \tag{G.56}$$

These equations are an extension of Eqs. (G.54), (G.55), (G.56) for the EHM.

$w^{(s)}$ and density self-consistency equations. We can rewrite the equations for $w^{(s)}$ and n accordingly,

$$w^{(s)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right), \tag{G.57}$$

$$n = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left[1-\cos 2\tilde{\theta}_{\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right], \tag{G.58}$$

where we used Eqs. (G.43), (G.44).

Contributions of each channel. Recall Tab. 3.3, collecting the selection rules for the non-local attraction $\hat{H}_V$. From Eq. (2.22), for the SC phase we need to account for the following contractions:

$$\begin{array}{ll} \hat{H}_V \text{ (o.s. sector)} & \text{Hartree and Cooper contractions,} \\ \hat{H}_V \text{ (s.s. sector)} & \text{Hartree and Fock contractions.} \end{array}$$

Note that Cooper contractions in the non-local s.s. sector are neglected, because of the selection rule discussed at the end of Sec. 3.2.2. We specify the role of each contraction channel in Tab. 6.2, where also those from $\hat{H}_U$ are included.

G.2.1 Effects of the non-local Hartree terms: chemical potential renormalisation

We start by analysing the non-local Hartree channel,

$$\hat{H}_V \overset{\mathrm{HF}}{\simeq} -V\sum_{\langle ij\rangle}\sum_{\sigma,\sigma'}\Big(\hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\langle\hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'}\rangle + \langle\hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\rangle\hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'} - \langle\hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\rangle\langle\hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'}\rangle\Big) \\ + (\text{Fock channel}) + (\text{Cooper channel}),$$

where we used Eq. (2.22). We have already carried out this computation in Sec. 4.2.1, which led to the renormalisation of the chemical potential. The non-local attraction $\hat{H}_V$ is reduced in the Hartree channel to

$$\hat{H}_V \overset{\mathrm{HF}}{\simeq} -2nzV\times\hat{N} - E^{(\mathrm{SC})}_{\mathrm{H},V} + (\text{Fock channel}) + (\text{Cooper channel}), \tag{G.59}$$

This, together Eq. (G.9), gives the chemical potential full renormalisation, identical to Eq. (4.8),

$$\tilde{\mu} \equiv \mu + n(2zV - U). \tag{G.60}$$

The contraction energy for the non-local sector was derived in Eq. (4.15),

$$E^{(\mathrm{SC})}_{\mathrm{H},V} = -L_xL_y\times 2zn^2V. \tag{G.61}$$

G.2.2 Effects of the Fock terms: bare bands renormalisation

From Eq. (2.22), we have

$$\hat{H}_V \overset{\mathrm{HF}}{\simeq} (\text{Hartree channel}) \\ V\sum_{\langle ij\rangle}\sum_{\sigma,\sigma'}\Big(\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'}\langle\hat{c}^\dagger_{j\sigma'}\hat{c}_{i\sigma}\rangle + \langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'}\rangle\hat{c}^\dagger_{j\sigma'}\hat{c}_{i\sigma} - \langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'}\rangle\langle\hat{c}^\dagger_{j\sigma'}\hat{c}_{i\sigma}\rangle\Big) + (\text{Cooper channel}).$$

Repeating the same derivation of Sec. 4.2.2, with the averages now being computed on the SC thermal state, $\hat{H}_V$ is approximated in the Fock channel by

$$\hat{H}_V \overset{\mathrm{HF}}{\simeq} (\text{Hartree channel}) \\ + V\sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\sigma}\Big(u^{(s^*)}_\sigma\varphi^{(s^*)*}_{\mathbf{k}} + u^{(d)}_\sigma\varphi^{(d)*}_{\mathbf{k}}\Big)\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} - E^{(\mathrm{SC})}_{\mathrm{F},V} \\ + (\text{Cooper channel}). \tag{G.62}$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x + \cos k_y)\left[1-\frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right]$	Fig. 3.4b
$d_{x^2-y^2}$-wave	$u^{(d)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x - \cos k_y)\left[1-\frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right]$	Fig. 3.4e

Table G.1 | Symmetry resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$. The shown equations are reformulations of Eq. (G.5), which is equal to Eq. (G.65).

where, as was for the normal phase,

$$E_{\mathrm{F},V}^{(\mathrm{SC})} = L_xL_y \times \frac{V}{2}\sum_{\sigma}\left(\left|u_{\sigma}^{(s^*)}\right|^2 + \left|u_{\sigma}^{(d)}\right|^2\right). \tag{G.63}$$

The proofs are identical to those presented in Sec. 4.2.2, because the SC phase is translationally invariant as was the normal phase. Renormalisation of bands reads

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}} + u_{\sigma}^{(s^*)}V(\cos k_x + \cos k_y) + u_{\sigma}^{(d)}V(\cos k_x - \cos k_y). \tag{G.64}$$

From now on we omit spin index, using spin rotational invariance. Each harmonic $u^{(\gamma)}$ satisfies the self-consistency Eq. (G.5), which is written identically as:

$$u^{(\gamma)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi_{\mathbf{k}}^{(\gamma)}\left[1-\cos 2\tilde{\theta}_{\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right]. \tag{G.65}$$

In Tab. G.1 we report these equations, writing explicitly the harmonics from Tab. 3.5 and using the angular relations of Eq. (G.47).

Proof G.9 — Proof of Eq. (G.65): self-consistency equations for $u^{(\gamma)}$.

We start by computing explicitly $u_{\mathbf{k}\sigma}$. For $\sigma =\uparrow$, we get the chain of equalities:

$$\begin{aligned} u_{\mathbf{k}\uparrow} &= \langle \hat{c}_{\mathbf{k}\uparrow}^{\dagger}\hat{c}_{\mathbf{k}\uparrow}\rangle \\ &= \left\langle \hat{\Phi}_{\mathbf{k}}^{\dagger}\frac{\mathbb{I}+\tau^z}{2}\hat{\Phi}_{\mathbf{k}}\right\rangle \\ &= \frac{1}{2}\left[\left\langle\hat{\Gamma}_{\mathbf{k}}^{\dagger}\hat{\Gamma}_{\mathbf{k}}\right\rangle + \langle\hat{s}_{\mathbf{k}}^{z}\rangle\right] \\ &= \frac{1}{2}\left[\tilde{\mathrm{Th}}_{\mathbf{k}}^{(+)}[\beta] - \cos 2\tilde{\theta}_{\mathbf{k}}\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta]\right] \\ &= \frac{1}{2}\left[1-\cos 2\tilde{\theta}_{\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right], \end{aligned} \tag{G.66}$$

having we used Eq. (G.52) and sequent. Inserting this result in Eq. (G.5), we get Eq. (G.65) in this spin sector.

Proceeding similarly for $\sigma =\downarrow$,

$$\begin{aligned} u_{-\mathbf{k}\downarrow} &= \langle \hat{c}^\dagger_{-\mathbf{k}\downarrow}\hat{c}_{-\mathbf{k}\downarrow}\rangle \\ &= 1-\langle \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow}\rangle \\ &= 1-\left\langle \hat{\Phi}^\dagger_{\mathbf{k}}\frac{\mathbb{I}-\tau^z}{2}\hat{\Phi}_{\mathbf{k}}\right\rangle \\ &= 1-\frac{1}{2}\left[\left\langle \hat{\Gamma}^\dagger_{\mathbf{k}}\hat{\Gamma}_{\mathbf{k}}\right\rangle - \langle \hat{s}^z_{\mathbf{k}}\rangle\right] \\ &= 1-\frac{1}{2}\left[\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}}[\beta]+\cos 2\tilde{\theta}_{\mathbf{k}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta]\right] \\ &= 1-\frac{1}{2}\left[1+\cos 2\tilde{\theta}_{\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right], \end{aligned} \tag{G.67}$$

We observe that the last line equals Eq. (G.66). Hence

$$u_{\mathbf{k}\uparrow} = u_{-\mathbf{k}\downarrow}\,.$$

Inserting this result in Eq. (G.5), we get Eq. (G.65) for both spin sectors.

G.2.3 Effects of the non-local Cooper terms: generalised pairing

In this section we discuss the effects of Cooper contractions in the HF decomposition of $\hat{H}_V$, shown in Eq. (2.22),

$$\begin{aligned} \hat{H}_V \overset{\mathrm{HF}}{\simeq} &\text{(Hartree channel)} + \text{(Fock channel)} \\ &- V\sum_{\langle ij\rangle}\sum_{\sigma,\sigma'}\left(\hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'}\left\langle\hat{c}_{j\sigma'}\hat{c}_{i\sigma}\right\rangle + \left\langle\hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'}\right\rangle\hat{c}_{j\sigma'}\hat{c}_{i\sigma} - \left\langle\hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma'}\right\rangle\left\langle\hat{c}_{j\sigma'}\hat{c}_{i\sigma}\right\rangle\right). \end{aligned}$$

The HF approximation to $\hat{H}_V$ in Cooper channel is given by

$$\hat{H}_V \overset{\mathrm{HF}}{\simeq} \text{(Hartree channel)} + \text{(Fock channel)} - V\sum_{\gamma\neq s}\sum_{\mathbf{k}\in\mathrm{BZ}}\left[w^{(\gamma)}\varphi^{(\gamma)*}_{\mathbf{k}}\hat{\phi}_{\mathbf{k}}+\mathrm{h.c.}\right] - E^{(\mathrm{SC})}_{\mathrm{C},V}\,, \tag{G.68}$$

where the harmonics $w^{(\gamma)}$ have been defined in Eq. (G.6). The contraction energy is given by:

$$E^{(\mathrm{SC})}_{\mathrm{C},V} \equiv -\frac{V}{L_xL_y}\sum_{\gamma\neq s}\sum_{\mathbf{k},\mathbf{k}'}\varphi^{(\gamma)}_{\mathbf{k}}\varphi^{(\gamma)*}_{\mathbf{k}'}w_{\mathbf{k}}w^*_{\mathbf{k}'}\,. \tag{G.69}$$

In both the above equations, the sum is intended for $\gamma\in\{s^*, p_x, p_y, d_{x^2-y^2}\}$.

Proof G.10 — Proof of Eq. (G.68): HF approximation of $\hat{H}_V$ in Cooper channel.

We know from Eq. (3.26) that the non-local attraction can be separated in o.s. and s.s. channels. As we reported in Tab. 3.3, the only non-zero contribution comes from the o.s. sector.

Let us move to reciprocal space. Applying Eq. (3.42), for the o.s. sector we get

$$\begin{aligned} \hat{H}^{(\mathrm{o.s.})}_V \overset{\mathrm{HF}}{\simeq} &\text{(Hartree channel)} \\ &- \frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos(\delta k_x)+\cos(\delta k_y)\right]\times\Big[\left\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\right\rangle\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \\ &+\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\left\langle\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow}\right\rangle - \left\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\right\rangle\left\langle\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow}\right\rangle\Big]. \end{aligned} \tag{G.70}$$

We know that Cooper pairs have zero net momentum for translationally invariant SC phases.

Hence, only $\mathbf{K} = \mathbf{0}$ contributes. Therefore, the above operator is reduced to:

$$\hat{H}_V^{(\text{o.s.})} \overset{\text{HF}}{\simeq} (\text{Hartree channel}) \\ - \frac{2V}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \times \Big[\langle \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \rangle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \\ + \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \langle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \rangle - \langle \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \rangle \langle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \rangle \Big] .$$

Recall now Eq. (G.4), where we defined $w_{\mathbf{k}} \equiv \langle \phi^\dagger_{\mathbf{k}} \rangle$. Using the decomposition of Eqs. (3.52), (3.53), we end up with:

$$\hat{H}_V^{(\text{o.s.})} \overset{\text{HF}}{\simeq} (\text{Hartree channel}) - \frac{V}{L_x L_y} \sum_{\gamma \neq s} \sum_{\mathbf{k},\mathbf{k}'} \varphi^{(\gamma)}_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}'} w_{\mathbf{k}} \hat{\phi}_{\mathbf{k}'} \\ - \frac{V}{L_x L_y} \sum_{\gamma \neq s} \sum_{\mathbf{k},\mathbf{k}'} \varphi^{(\gamma)*}_{\mathbf{k}} \varphi^{(\gamma)}_{\mathbf{k}'} w^*_{\mathbf{k}} \hat{\phi}^\dagger_{\mathbf{k}'} - E^{(\text{SC})}_{\text{C},V} . \tag{G.71}$$

Sum is intended for $\gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$. We recognised $E^{(\text{SC})}_{\text{C},V}$ as defined in Eq. (G.69). Eq. (G.71) is reduced to

$$\hat{H}_V^{(\text{o.s.})} \overset{\text{HF}}{\simeq} (\text{Hartree channel}) - \sum_{\gamma \neq s} \left(\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} w_{\mathbf{k}} \right) \times V \sum_{\mathbf{k}' \in \text{BZ}} \varphi^{(\gamma)*}_{\mathbf{k}'} \hat{\phi}_{\mathbf{k}'} \\ - \sum_{\gamma \neq s} \left(\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi^{(\gamma)*}_{\mathbf{k}} w^*_{\mathbf{k}} \right) \times V \sum_{\mathbf{k}' \in \text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}'} \hat{\phi}_{\mathbf{k}'} - E^{(\text{SC})}_{\text{C},V} . \tag{G.72}$$

Inserting the expression for $w^{(\gamma)}$ of Eq. (G.6), since we have no s.s. contribution, we obtain Eq. (G.68).

The contraction energy of Eq. (G.69) can be written as follows:

$$E^{(\text{SC})}_{\text{C},V} = -L_x L_y \times V \sum_{\gamma \neq s} \left| w^{(\gamma)} \right|^2 , \tag{G.73}$$

with $\gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$.

Proof G.11 — Proof of Eq. (G.73): non-local Cooper contraction energy.

The contraction energy is formally given by Eq. (G.69). Substituting the harmonics decomposition of Eq. (G.6), we get:

$$\begin{aligned} E^{(\text{SC})}_{\text{C},V} &= -L_x L_y \times V \sum_{\gamma \neq s} \left(\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} w_{\mathbf{k}} \right) \left(\frac{1}{L_x L_y} \sum_{\mathbf{k}' \in \text{BZ}} \varphi^{(\gamma)*}_{\mathbf{k}'} w_{\mathbf{k}'} \right) \\ &= -L_x L_y \times V \sum_{\gamma \neq s} w^{(\gamma)} w^{(\gamma)*} \\ &= -L_x L_y \times V \sum_{\gamma \neq s} \left| w^{(\gamma)} \right|^2 , \end{aligned}$$

which proves Eq. (G.73).

Each harmonic, including s-wave, satisfies the self-consistency Eq. (G.6), which is written equivalently as

$$w^{(\gamma)} = \frac{1}{2 L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \sin 2\tilde{\theta}_{\mathbf{k}} e^{i 2 \tilde{\zeta}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2} \right) . \tag{G.74}$$

Symmetry	Self-consistency equation	Graph
s-wave	$w^{(s)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. 3.4a
s^*-wave	$w^{(s^*)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y)\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. 3.4b
p_x-wave	$w^{(p_x)} = \frac{i\sqrt{2}}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \sin k_x \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. 3.4c
p_y-wave	$w^{(p_y)} = \frac{i\sqrt{2}}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \sin k_y \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. 3.4d
$d_{x^2-y^2}$-wave	$w^{(d)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y)\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. 3.4e

Table G.2 | Symmetry resolved self-consistency equations for the harmonics of $w_{\mathbf{k}}$. The shown equations are reformulations of Eq. (G.6), which is equal to Eq. (G.74).

In Tab. G.2 we report these equations, writing explicitly the harmonics from Tab. 3.5 and using the angular relations of Eq. (G.47).

Proof G.12 — Proof of Eq. (G.74): self-consistency equations for $w^{(\gamma)}$.

Consider Eq. (G.6). For the γ-wave component of the superconducting order parameter, the following chain holds:

$$\begin{aligned} w^{(\gamma)} &\equiv \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)}\langle\hat{\phi}_{\mathbf{k}}^\dagger\rangle \\ &= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)}\langle\hat{\Phi}_{\mathbf{k}}^\dagger\tau^+\hat{\Phi}_{\mathbf{k}}\rangle \\ &= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)}\frac{\langle\hat{s}_{\mathbf{k}}^x\rangle + i\,\langle\hat{s}_{\mathbf{k}}^y\rangle}{2} \\ &= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)}\frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right), \end{aligned}$$

where $\tau^+ = (\tau^x + i\tau^y)/2$. We used Eqs. (G.49), (G.50) and (G.53). Use now Eqs. (G.47), (G.48),

$$\begin{aligned} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} &= \frac{|\tilde{\Delta}_{\mathbf{k}}|}{\tilde{E}_{\mathbf{k}}}\left(\frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} + \frac{i\,\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|}\right) \\ &= \sin 2\tilde{\theta}_{\mathbf{k}}\left(\cos 2\tilde{\zeta}_{\mathbf{k}} + i\sin 2\tilde{\zeta}_{\mathbf{k}}\right) \\ &= \sin 2\tilde{\theta}_{\mathbf{k}} e^{i2\tilde{\zeta}_{\mathbf{k}}}, \end{aligned}$$

and insert in previous equation. This gives Eq. (G.74). The derivation above holds for all symmetries, including s-wave.

With this proof we conclude this cumbersome algebraic derivation. In next section we summarise the obtained results, give the final form of the HF hamiltonian and discuss the symmetries of the gap function.

G.3 The HF hamiltonian and its properties

In this section we take together everything we said so far. Adding $\hat{H}_V$ to the conventional HM, we have seen, produces renormalisation of the bare bands, the gap function and the chemical potential.

We compose Eqs. (G.27), (G.59), (G.62) and (G.68). The final form of the HF gran-canonical hamiltonian reads

$$\hat{H} - \mu\hat{N} \overset{\text{HF}}{\simeq} \sum_{\mathbf{k}\in\text{BZ}} \tilde{\xi}_{\mathbf{k}} \left(\hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow}\right) - \sum_{\mathbf{k}\in\text{BZ}} \left(\tilde{\Delta}_{\mathbf{k}}\hat{\phi}_{\mathbf{k}} + \tilde{\Delta}^*_{\mathbf{k}}\hat{\phi}^\dagger_{\mathbf{k}}\right) + E^{(\text{SC})}_{\text{a}} + E^{(\text{SC})}_{\text{c}}\,, \tag{G.75}$$

which coincides with Eq. (6.10). We included the anticommutation energy defined in Eq. (G.25) and extended to renormalised quantities,

$$E^{(\text{SC})}_{\text{a}} \equiv \sum_{\mathbf{k}\in\text{BZ}} \tilde{\xi}_{\mathbf{k}}\,,$$

and the total contraction energy $E^{(\text{SC})}_{\text{c}}$, which includes all the shifts to energy we encountered. These are given, respectively, in Eqs. (G.11), (G.18), (G.61), (G.63) and (G.73). Namely,

$$\begin{aligned} E^{(\text{SC})}_{\text{c}} &\equiv -\left(E^{(\text{SC})}_{\text{H},U} + E^{(\text{SC})}_{\text{C},U} + E^{(\text{SC})}_{\text{H},V} + E^{(\text{SC})}_{\text{F},V} + E^{(\text{SC})}_{\text{C},V}\right) \\ &= -L_xL_y \times U\left(n^2 + \left|w^{(s)}\right|^2\right) + L_xL_y \times V\left(2zn^2 - \sum_{\gamma}\left|u^{(\gamma)}\right|^2 + \sum_{\gamma}\left|w^{(\gamma)}\right|^2\right)\,, \end{aligned} \tag{G.76}$$

which coincides with Eq. (6.12).

G.3.1 Gap symmetries and anisotropic pairing

We focus on the renormalised quantities. The bare bands are given by:

$$\tilde{\xi}_{\mathbf{k}} = \epsilon_{\mathbf{k}} + V\sum_{\gamma\in\{s^*,d\}} u^{(\gamma)}\varphi^{(\gamma)*}_{\mathbf{k}} - \tilde{\mu}\,, \tag{G.77}$$

which coincides with Eq. (6.8). The full gap function, as in Eq. (6.9), is given by

$$\tilde{\Delta}_{\mathbf{k}} = -Uw^{(s)} + V\sum_{\gamma\neq s} w^{(\gamma)}\varphi^{(\gamma)*}_{\mathbf{k}}\,. \tag{G.78}$$

We can identify the gap harmonics as

$$\tilde{\Delta}^{(s)} = -Uw^{(s)}\,, \qquad \tilde{\Delta}^{(\gamma)} = Vw^{(\gamma)}\,, \quad \text{being} \quad \gamma\in\{s^*,p_x,p_y,d\}\,.$$

We will investigate competition among these components.

We anticipated in Tab. 6.1 that Cooper pairing in the singlet channel requires for the gap wavefunction to be inversion-symmetric. Triplet pairing, instead, requires inversion anti-symmetry. This gives a symmetry selection rule for the symmetries of the gap function: if we work in singlet channel, the gap must be symmetric, thus only three harmonics are allowed:

$$\text{Singlet channel} \quad : \quad \tilde{\Delta}_{\mathbf{k}} = \tilde{\Delta}^{(s)} + \tilde{\Delta}^{(s^*)}(\cos k_x + \cos k_y) + \tilde{\Delta}^{(d)}(\cos k_x - \cos k_y)\,.$$

An inverse statement holds for the triplet pairing, for which

$$\text{Triplet channel} \quad : \quad \tilde{\Delta}_{\mathbf{k}} = i\sqrt{2}\left(\tilde{\Delta}^{(p_x)}\sin k_x + \tilde{\Delta}^{(p_y)}\sin k_y\right)\,.$$

These concept are summaries in Tab. 6.3, where we list the HFPs in singlet and triplet channels.

G.3.2 The entropy density of the SC ground-state

In this section, we calculate the entropy density of the SC ground state, following the scheme presented for the normal phase in Sec. 4.3.2. The approximated ground-state in the SC phase is given by a gas of free fermions populating the two bands $\pm\tilde{E}_{\mathbf{k}}$, with occupations specified by the Fermi-Dirac distribution. Hence, the total entropy density of the ground state is given by that of a Fermi gas [4],

$$s^{(\mathrm{SC})} = s_-^{(\mathrm{SC})} + s_+^{(\mathrm{SC})} ,$$

where the subscript indicates the band,

$$s_\pm^{(\mathrm{SC})} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \Big[\Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,0)\Big) \ln\Big(1 - f(\pm\tilde{E}_{\mathbf{k}};\beta,0)\Big) \\ + f(\pm\tilde{E}_{\mathbf{k}};\beta,0) \ln f(\pm\tilde{E}_{\mathbf{k}};\beta,0)\Big] . \tag{G.79}$$

Note the fact that the Fermi-Dirac distributions are computed at null chemical potential. Because of this, we get:

$$s_-^{(\mathrm{SC})} = s_+^{(\mathrm{SC})} \qquad \Longrightarrow \qquad s^{(\mathrm{SC})} \equiv s_-^{(\mathrm{SC})} + s_+^{(\mathrm{SC})} = 2s_+^{(\mathrm{SC})} . \tag{G.80}$$

Proof G.13 — Proof of Eq. (G.80): Bogoliubov bands $\pm\tilde{E}_{\mathbf{k}}$ give equal entropy densities.

Consider the entropy of the lower band,

$$s_-^{(\mathrm{SC})} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \Big[\Big(1 - f(-\tilde{E}_{\mathbf{k}};\beta,0)\Big) \ln\Big(1 - f(-\tilde{E}_{\mathbf{k}};\beta,0)\Big) \\ + f(-\tilde{E}_{\mathbf{k}};\beta,0) \ln f(-\tilde{E}_{\mathbf{k}};\beta,0)\Big] ,$$

and Eq. (G.37), which implies

$$f(x;\beta,0) = 1 - f(-x;\beta,\mu) .$$

Inserting the latter for $x = \tilde{E}_{\mathbf{k}}$ in the expression for $s_-^{(\mathrm{SC})}$, we get

$$s_-^{(\mathrm{SC})} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \Big[f(\tilde{E}_{\mathbf{k}};\beta,0) \ln f(\tilde{E}_{\mathbf{k}};\beta,0) + \Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) \ln\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) \Big] ,$$

therefore, comparing with Eq. (G.79), we get Eq. (G.80): the two bands give an identical contribution to entropy.

The expression for the entropy density can be simplified by using Eq. (4.41), which implies

$$\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) \ln\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) \\ = \ln\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) - f(\tilde{E}_{\mathbf{k}};\beta,0)\Big(\tilde{E}_{\mathbf{k}}\beta + \ln f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) .$$

It follows that $s^{(\mathrm{SC})} = 2s_+^{(\mathrm{SC})}$ can be written as:

$$s^{(\mathrm{SC})} \equiv -\frac{2k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln\Big(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\Big) + \frac{2k_B\beta}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}} f(\tilde{E}_{\mathbf{k}};\beta,0) . \tag{G.81}$$

We note that, as the gap closes and $\tilde{E}_{\mathbf{k}} \to \tilde{\xi}_{\mathbf{k}}$, the above expression reduces to the entropy density for the normal phase reported in Eq. (4.40).

G.3.3 The free energy density of the SC ground-state

We compute the density of free energy, following the same procedure we used for the normal phase in Sec. 4.3.3. The free energy density is given by

$$f^{(\mathrm{SC})} \equiv \frac{F^{(\mathrm{SC})}}{L_x L_y}\,, \tag{G.82}$$

being $F^{(\mathrm{SC})}$ the total free energy,

$$F^{(\mathrm{SC})} = E_{\mathrm{HF}}^{(\mathrm{SC})} - \frac{S^{(\mathrm{SC})}}{k_{\mathrm{B}}\beta}\,, \tag{G.83}$$

and $S^{(\mathrm{SC})} \equiv L_x L_y \times s^{(\mathrm{SC})}$ the total entropy. Here $E_{\mathrm{HF}}^{(\mathrm{SC})}$ is the ground-state energy for the HF hamiltonian. In the SC phase, we compute $f^{(\mathrm{SC})}$ using the following formula:

$$\begin{aligned} f^{(\mathrm{SC})} = \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\left(\tilde{\xi}_{\mathbf{k}} - \tilde{E}_{\mathbf{k}}\right) + \frac{2}{\beta L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\right) \\ - U\left|w^{(s)}\right|^2 - V\left(\sum_{\gamma}\left|u^{(\gamma)}\right|^2 - \sum_{\gamma}\left|w^{(\gamma)}\right|^2\right) + \tilde{\mu} n\,, \end{aligned} \tag{G.84}$$

which gives the analytic expression for the free energy. Note that, as the gap closes and $\tilde{E}_{\mathbf{k}} \to \tilde{\xi}_{\mathbf{k}}$, the above expression reduces to the entropy density for the normal phase reported in Eq. (4.40).

Proof G.14 — Proof of Eq. (G.84): free energy density for the SC phase.

Free energy density of the SC phase is derived following the procedure used for the other two phases. The five contributions to free energy are:

1. Quasiparticles energy:

$$\begin{aligned} e_{\mathrm{g}}^{(\mathrm{SC})} &= -\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}} \tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)}[\beta] \\ &= -\frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}} \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right). \end{aligned}$$

2. Anticommutation energy, as defined in Eq. (G.25):

$$\begin{aligned} e_{\mathrm{a}}^{(\mathrm{SC})} &\equiv \frac{E_{\mathrm{a}}^{(\mathrm{SC})}}{L_x L_y} \\ &= \frac{1}{L_x L_y}\sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{\xi}_{\mathbf{k}}\,. \end{aligned}$$

3. Contraction energy:

$$\begin{aligned} e_{\mathrm{c}}^{(\mathrm{SC})} &= \frac{E_{\mathrm{c}}^{(\mathrm{SC})}}{L_x L_y} \\ &= -U\left(n^2 + \left|w^{(s)}\right|^2\right) + V\left(2zn^2 - \sum_{\gamma}\left|u^{(\gamma)}\right|^2 + \sum_{\gamma}\left|w^{(\gamma)}\right|^2\right), \end{aligned}$$

where we used Eq. (G.76).

4. Entropic energy

$$
\begin{aligned}
e_s^{(\mathrm{SC})} &= -\frac{s^{(\mathrm{SC})}}{k_{\mathrm{B}}\beta} \\
&= \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\right) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}} f(\tilde{E}_{\mathbf{k}};\beta,0)\,,
\end{aligned}
$$

where we used Eq. (G.81).

5. Chemical potential correction

$$
e_n^{(\mathrm{SC})} = \mu n\,.
$$

Now, bringing everything together,

$$
f^{(\mathrm{SC})} \equiv e_{\mathrm{g}}^{(\mathrm{SC})} + e_{\mathrm{a}}^{(\mathrm{SC})} + e_{\mathrm{c}}^{(\mathrm{SC})} + e_s^{(\mathrm{SC})} + e_n^{(\mathrm{SC})}\,.
$$

In particular, note that $e_{\mathrm{g}}^{(\mathrm{SC})}$ and the second term of $e_s^{(\mathrm{SC})}$ can be composed nicely,

$$
-\frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}} f(\tilde{E}_{\mathbf{k}};\beta,0) = -\frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}}\,.
$$

Indeed, this is proven by noting that

$$
\tanh\left(\frac{x}{2}\right) + \frac{2}{e^x + 1} = 1\,.
$$

which follows from basic algebra.

Taking everything together, in the end we get the following expression for free energy density:

$$
\begin{aligned}
f^{(\mathrm{SC})} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\tilde{\xi}_{\mathbf{k}} - \tilde{E}_{\mathbf{k}}\right] &+ \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\right) \\
&- U\left(n^2 + \left|w^{(s)}\right|^2\right) + V\left(2zn^2 - \sum_{\gamma} \left|u^{(\gamma)}\right|^2 + \sum_{\gamma} \left|w^{(\gamma)}\right|^2\right) + \mu n
\end{aligned}
$$

As was for the normal and AF phases, we recognise the chemical potential shift of Eq. (G.60),

$$
\tilde{\mu} = \mu + n(2zV - U)\,.
$$

Upon substituting this result, we obtain Eq. (G.84).

G.3.4 The energetic constraint on the gap harmonics

In this final section we present a proof of the constraint of Eq. (6.16). We derived the expression of Eq. (G.84), which contains in its second row the HFPs $u^{(\gamma)}$ and $w^{(\gamma)}$. We have:

$$
-U\left|w^{(s)}\right|^2 + V \sum_{\gamma} \left|w^{(\gamma)}\right|^2 = \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right). \tag{G.85}
$$

The right-hand side of Eq.(G.85) is a sum of positive terms, hence is positive (or zero, if the gap is null). So must be the left-hand side. Then,

$$
-U\left|w^{(s)}\right|^2 + V \sum_{\gamma\neq s} \left|w^{(\gamma)}\right|^2 \geq 0\,.
$$

If the superconducting gap is null, the above inequality is satisfied, because $w^{(\gamma)} = 0$ for all γs. If the gap is finite, the above requirement translates to:

$$
\sum_{\gamma\neq s} \left|w^{(\gamma)}\right|^2 \geq M^{(\mathrm{SC})} \tag{G.86}
$$

where we defined the radius:

$$M^{(\mathrm{SC})} \equiv \left|w^{(s)}\right|^2 \frac{U}{V} \,.$$

We recognise Eq. (6.16). For the conventional HM, we know from Sec. G.1.3 that $w^{(s)} = 0$. For the EHM, it can assume non-zero values, as long as Eq. (G.86) is satisfied.

Proof G.15 — Proof of Eq. (G.85).

Recall the explicit form of the conjugate gap function from Eq. (6.9),

$$\tilde{\Delta}^*_{\mathbf{k}} = -Uw^{(s)*} + V \sum_{\gamma} w^{(\gamma)*} \varphi^{(\gamma)}_{\mathbf{k}} \,.$$

Expand the right-hand side of Eq. (G.85) and insert the above equation,

$$\begin{aligned}
\frac{1}{2L_xL_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) &= \frac{1}{2L_xL_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{\Delta}^*_{\mathbf{k}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \\
&= \frac{1}{2L_xL_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left[-Uw^{(s)*} + V \sum_{\gamma\neq s} w^{(\gamma)*} \varphi^{(\gamma)}_{\mathbf{k}}\right] \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) .
\end{aligned}$$

Using the self-consistency equations of Tab. G.2 to recognise the various harmonics. We reduce the right-hand side of Eq. (G.85) to:

$$\begin{aligned}
\frac{1}{2L_xL_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) &= -Uw^{(s)*} \times \overbrace{\frac{1}{2L_xL_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)}^{w^{(s)}} \\
&\quad + V \sum_{\gamma\neq s} w^{(\gamma)*} \times \underbrace{\frac{1}{2L_xL_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)}_{w^{(\gamma)}} ,
\end{aligned}$$

which coincides with the left-hand side of Eq. (G.85).

Acknowledgements

[To be continued…]

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